

Prepladder Version X Qbank

Subject: Anaesthesia

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Introduction and PAC

1. Which of the following is true regarding spinal, epidural, and local anesthesia?

- A. Epidural anesthesia is performed by injecting anesthetic into the intrathecal space.
 - B. Local anesthetics work by blocking calcium channels, preventing nerve impulse propagation.
 - C. Spinal anesthesia is commonly used for procedures above the umbilicus.
 - D. The absolute contraindication for local anesthesia is an allergy to the anesthetic.
-

2. A 65-year-old male patient with a history of severe aortic stenosis is scheduled for elective hip replacement surgery. Which of the following is an absolute contraindication for general anesthesia in this patient?

- A. Patient refusal
 - B. Severe aortic stenosis
 - C. Mallampati class IV airway
 - D. All of the above
-

3. Which of the following is true regarding the history and development of anesthesia?

- A. Ether was introduced by Dr. Joseph O'Dwyer in 1842.
 - B. Chloroform was first used by Dr. Crawford Long in 1847.
 - C. William Morton demonstrated the use of ether as a general anesthetic in 1846.
 - D. Propofol was first introduced as an anesthetic in 1932.
-

4. A 45 year old man with a history of obesity and obstructive sleep apnea presents for elective hernia repair under general anesthesia. The newly joined Junior resident was given a chance to intubate but he could not succeed. Which of the following Mallampati classes is most likely to indicate the above scenario?

- A. Class II
 - B. Class I
 - C. Class 0
 - D. Class III or IV
-

5. A 45 year old male presents for a preoperative evaluation before a scheduled elective surgery. During the airway assessment, the base of the uvula is visible, with the soft palate and hard palate clearly seen. Based on this finding, which Mallampati classification is most appropriate for this patient?

- A. Class I
 - B. Class II
 - C. Class III
 - D. Class I or II
-

6. Which interincisor distance in centimeters is considered an indicator of potential difficulty for intubation?

- A. 3 cm or less
 - B. 5 cm or more
 - C. 6 cm or more
 - D. 8 cm or less
-

7. What position should the patient's head be in to measure the sternomental distance accurately?

- A. Neck flexed, chin down
 - B. Neck extended, mouth open
 - C. Neck fully extended, mouth closed
 - D. Neutral neck position
-

8. A 2 year old child is scheduled for an elective surgery requiring general anesthesia. The child last had a bottle of breast milk 3 hours ago. According to current fasting guidelines for pediatric patients, what is the appropriate next step regarding the child's fasting status before anesthesia?

- A. The child can be taken for surgery immediately since it has been 3 hours since the last feed.
 - B. The child should wait at least 4 hours from the last breast milk feed before undergoing anesthesia.
 - C. The child should not have anything to eat or drink for at least 8 hours before the procedure.
 - D. The child is allowed clear liquids until 4 hours prior to the surgery.
-

9. Which of the following conditions is least likely to be associated with an increased risk of aspiration despite fasting in general anesthesia?

- A. Hiatal hernia
- B. Cerebral palsy
- C. Multiple sclerosis
- D. Patient after Nissen fundoplication surgery

10. Which of the following medications is NOT typically discontinued prior to general anesthesia?

- A. Metformin
 - B. Sulfonylureas
 - C. SGLT2 inhibitors
 - D. Calcium channel blockers
-

11. A 68 year old male with atrial fibrillation, on chronic warfarin therapy for stroke prevention, is scheduled for elective hip replacement surgery. How many days prior to the surgery should the warfarin be stopped?

- A. 2-5 days
 - B. 1 day
 - C. 7 days
 - D. 10 days
-

12. Which of the following drugs can be continued perioperatively?

- A. Antithyroid
 - B. Antiepileptic
 - C. Antidepressants
 - D. All of the above
-

13. A 30 year old primigravida is scheduled for elective cholecystectomy. What ASA category does she belong to?

- A. Class 1
 - B. Class 2
 - C. Class 3
 - D. Class 4
-

14. A 30 year old female patient is undergoing routine laparoscopic appendectomy under general anesthesia. During laryngoscopy, the anesthesiologist can visualize the entire laryngeal aperture. What is the Cormack-Lehane grading of this laryngoscopic view?

- A. Grade 1
 - B. Grade 2
 - C. Grade 3
 - D. Grade 4
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	3
Question 4	4
Question 5	3
Question 6	1
Question 7	3
Question 8	2
Question 9	4
Question 10	4
Question 11	1
Question 12	4
Question 13	2
Question 14	1

Solution for Question 1:

Correct Answer: D) The absolute contraindication for local anesthesia is an allergy to the anesthetic.

Explanation :

The absolute contraindication for local anesthesia is an allergy to the anesthetic.

Local anesthesia	Local anesthesia is a safe, effective pain management method. Works by reversibly binding sodium channels, blocking sodium entry, and inhibiting nerve impulse propagation. (Option B ruled out) Two classes of local anesthetics: Amides : Lidocaine, prilocaine, mepivacaine, bupivacaine. Esters . Widely used in dentistry for pain control. Absolute contraindication: Allergy to anesthetics. (Option D)
Spinal anesthesia	Spinal anesthesia is a neuraxial technique where local anesthetic is injected into the intrathecal (subarachnoid) space. Commonly used for surgeries below the umbilicus , including lower abdomen, pelvis, perineal, and lower extremities. (Option C ruled out) Performed with the patient in sitting or lateral decubitus position. Absolute contraindications: Lack of patient consent, elevated intracranial pressure (ICP). Relative contraindications: Neurological diseases, severe dehydration (hypovolemia), thrombocytopenia/coagulopathy, severe mitral/aortic stenosis, left ventricular outflow obstruction (e.g., hypertrophic obstructive cardiomyopathy)..
Epidural Anesthesia	Epidural anesthesia involves injecting anesthetic into the epidural space to block sensory and motor spinal nerve roots. (Option A ruled out) Covers areas like the thoracic, abdominal, pelvic, and lower extremities. Also used as adjunct pain management intraoperatively or postoperatively. Advantage : Flexible dosing and administration (intermittent or continuous infusion) to fit clinical needs. Primary indications: Obstetric anesthesia during labor, surgical anesthesia for thoracic, major intra-abdominal, or spine surgeries (without need for muscle relaxation). Absolute contraindications: Patient refusal, local infection at puncture site, increased intracranial pressure, traumatic spinal cord injury. Relative contraindications: Hemodynamic instability, obstructive cardiomyopathy, coagulopathy or therapeutic anticoagulation, thrombocytopenia, difficulty maintaining position for epidural placement, spinal abnormalities.
Nerve block	Peripheral nerve blocks inhibit distal nerve impulse transmission, stopping pain signals to the cortex. Uses : Acute pain management (e.g., procedural anesthesia, perioperative analgesia), diagnosing and treating chronic pain. Complications : Nerve injury, infection at catheter site, bleeding, local anesthetic systemic toxicity (LAST).

Solution for Question 2:

Correct Answer: A) Patient refusal

Explanation :

Patient refusal is an absolute contraindication for general anesthesia in this patient.

General anesthesia	General anesthesia induces unconsciousness and blocks pain while controlling autonomic reflexes. Main anesthetic agents: IV anesthetics Inhalational anesthetics IV sedatives Synthetic opioids Neuromuscular blocking drugs. Common process: IV sedatives and analgesics for induction, followed by volatile anesthetics for maintenance. Absolute contraindication : Patient refusal. (Option A) Relative contraindications : Unoptimized medical conditions, difficult airway or significant comorbidities (e.g., severe aortic stenosis, pulmonary disease, CHF), procedures better suited for regional or neuraxial anesthesia to avoid airway manipulation. (Option B & C ruled out) Common side effects: Transient confusion or memory loss Dizziness Urinary retention Nausea and vomiting Chills Sore throat.
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Mallampati class IV airway (Option C): Mallampati class III and IV indicate a difficult airway during intubation, and thus a relative contraindication of general anesthesia

Solution for Question 3:

Correct Answer: C) William Morton demonstrated the use of ether as a general anesthetic in 1846.

Explanation:

History of anesthesia	
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Discovery of Anesthesia	1842: Dr. Crawford Long performed the first ether anesthetic (announced in 1849). 1846: William Morton demonstrated ether as a general anesthetic at Massachusetts General Hospital. (Option C)
Development of Agents	1847: Dr. James Simpson introduced chloroform (used by Queen Victoria during childbirth in 1853). (Option B ruled out) Early anesthetic agents: ether and nitrous oxide.
Airway Management	Late 1800s: Dr. Joseph O'Dwyer described endotracheal tube usage. (Option A ruled out) 1932: Drs. Arthur Guedel and Ralph Waters introduced cuffed endotracheal tubes for positive pressure ventilation. Early 1900s: Dr. Chevalier Jackson developed handheld laryngoscopes. 1937: Dr. Robert Macintosh introduced direct laryngoscopy.
Intravenous Anesthetics	1932: Barbiturates were introduced as IV anesthetics. 1977: Propofol, with faster recovery times, was developed. (Option D ruled out)
Mechanisms of Anesthesia	Early theories focused on anesthetics' lipophilic nature affecting neuronal membranes. Later research identified specific targets like chloride channels and neurotransmitter receptors. Recent theories examine the impact on functional brain connectivity and consciousness

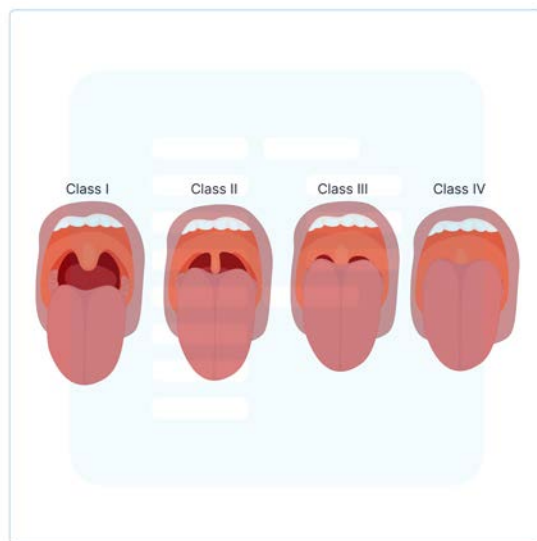
Solution for Question 4:

Correct Answer: D) Class III or IV

Explanation:

Mallampati class III and IV indicate a difficult airway during intubation.

The Mallampati classification is used to predict the ease of intubation based on the visibility of pharyngeal structures during mouth opening and tongue protrusion.



Mallampati Classification:

(Mnemonic - PUSH - Pillars, Uvula, Soft palate, Hard palate)

- Class I: Faucial pillars, uvula, soft palate, and hard palate are visualized.
- Class II: The base of the uvula, soft palate, and hard palate are visualized.
- Class III: Soft palate and hard palate are visualized.
- Class IV: Hard palate only is visualized.
- Class 0: Any part of the epiglottis is visible (easy laryngoscopy with difficult airway management)

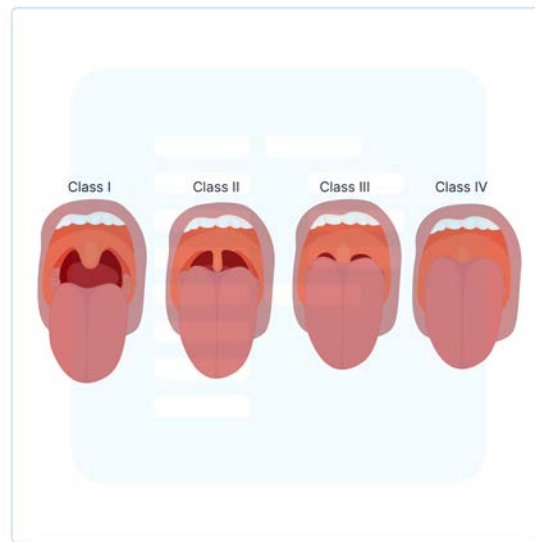
Solution for Question 5:

Correct Answer: B) Class II

Explanation

The scenario is suggestive of Class II of the Mallampati classification as the base of the uvula, soft palate, and Hard palate are visualized on airway assessment.

The Mallampati classification is used to predict the ease of intubation based on the visibility of pharyngeal structures during mouth opening and tongue protrusion.



Mallampati Classification:

- Class I: Faucial pillars, uvula, soft palate, and hard palate are visualized. (Option A ruled out)
- Class II: the uvula but the tip is not visible, soft palate, and hard palate are visualized. (Option B ruled out)
- Class III: Soft palate and hard palate are visualized, and base of uvula. (Option C)
- Class IV: Hard palate only is visualized.
- Class 0: Any part of the epiglottis is visible (easy laryngoscopy with difficult airway management)

Solution for Question 6:

Correct Answer: A) 3 cm or less

Explanation:

Inter-incisor distance refers to the distance between the upper and lower incisors when the mouth is maximally opened.

Measurement	It is typically measured in centimeters or millimeters using a ruler or caliper.
Normal Range	The average interincisor distance for adults is usually around 4-5 cm (40-50 mm).
Significance in Intubation	An interincisor distance of less than 3 cm (2 fingerbreadths) is often considered a potential indicator of difficult intubation. A reduced distance may suggest restricted jaw mobility, making it harder to access the airway.
Related Conditions	Conditions such as temporomandibular joint dysfunction (TMD), jaw trauma, or other musculoskeletal issues can result in a reduced interincisor distance.
Clinical Relevance	An adequate inter-incisor distance is crucial for proper airway management, where intubation may be required.

Solution for Question 7:

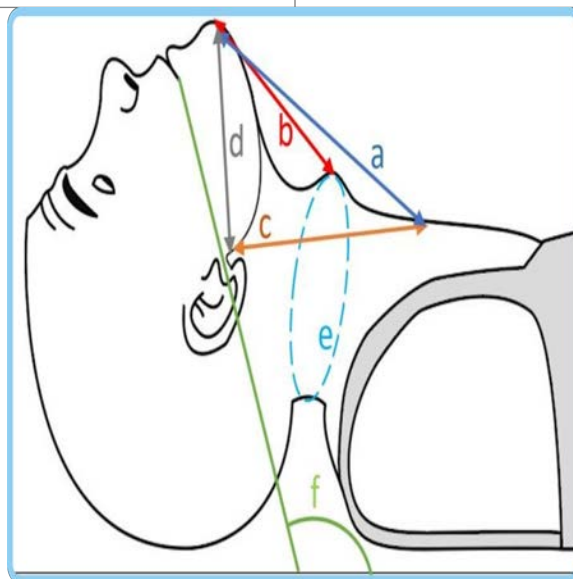
Correct Answer: C) Neck fully extended, mouth closed

Explanation:

The head should be in full extension with the mouth closed to get an accurate measurement of the sternomental distance.

Thyromental Distance	Sternomental Distance
The straight line distance from the thyroid notch (Adam's apple) to the lower border of mentum (tip of the chin).	The straight line distance between the manubrium sterni and bony point of mentum.
A normal thyromental distance is typically greater than 6.5cm. A distance of less than 6.5 cm is often considered a predictor of difficult intubation.	A normal sternomental distance is generally greater than 12.5-13.5 cm. A distance of less than 12 cm is associated with difficult airway management.

Measured with the patient's neck fully extended, mouth closed, and head in a neutral position. Can be measured using a ruler or tape.	Measured with the patient's neck fully extended, mouth closed, and head tilted back. A measuring tape or ruler is used to assess the distance.
A shorter thyromental distance (<6.5 cm) may indicate limited mandibular space and suggest that the intubation could be difficult.	A shorter sternomental distance suggests limited head and neck mobility, potentially indicating difficult intubation.
Commonly used as part of a pre-anesthetic airway assessment.	This measurement helps in predicting airway management challenges, particularly in patients with restricted neck movement or short necks.



Solution for Question 8:

Correct Answer: B) The child should wait at least 4 hours from the last breast milk feed before undergoing anesthesia.

Explanation:

The recommended fasting period after breast milk consumption is 4 hours before anesthesia.

Fasting Guidelines in Children and Adults

Children

Clear Liquids	Fasting period 2 hours before anesthesia. Clear liquids include water, clear juices (without pulp), and clear broths.
Breast Milk	Fasting period 4 hours before anesthesia. Breast milk is digested more quickly than formula or solid foods.
Infant Formula	Fasting period 6 hours before anesthesia. Formula is heavier and takes longer to digest than breast milk.
Non-Human Milk	Fasting period 6 hours before anesthesia. Includes cow's milk and soy milk, which are digested similarly to solid foods.
Light Meal	Fasting period 6 hours before anesthesia. A light meal may include toast and clear liquids but should avoid heavy, fatty, or fried foods.
Solid Foods	Fasting period 8 hours or more before anesthesia. This applies to a full meal, especially one containing fatty or fried foods that take longer to digest.

Adults

Clear Liquids	Fasting period 2 hours before anesthesia. Clear liquids include water, tea, black coffee (no milk), and clear juices.
Light Meal	Fasting period 6 hours before anesthesia. A light meal consists of foods like toast or non-fatty snacks. Avoid heavy foods or meals rich in protein and fat.
Solid Food	Fasting period 8 hours or more before anesthesia.

Clinical Importance of Fasting:

- Aspiration Prevention: Fasting reduces the risk of aspiration during anesthesia, where stomach contents may enter the lungs, leading to serious complications.
- Safe Induction of Anesthesia : Ensuring an empty stomach allows for safer airway management and anesthesia administration.

Solution for Question 9:

Correct Answer: D) Patient after Nissen fundoplication surgery

Explanation:

Nissen fundoplication surgery: This procedure reinforces the lower esophageal sphincter (LES), reducing or eliminating gastroesophageal reflux, and thus lowering the risk of aspiration.

Conditions at High Risk of Aspiration Despite Fasting

Neurological Disorders	Cerebral Palsy (Option B): Impaired motor function affects swallowing coordination. Stroke : Can lead to dysphagia and loss of protective airway reflexes. Multiple Sclerosis (Option C): Neuromuscular impairment affecting swallowing ability. Amyotrophic Lateral Sclerosis (ALS) : Progressive muscle weakness and dysphagia.
Neuromuscular Disorders	Myasthenia Gravis or Muscular Dystrophy
Gastroesophageal Reflux Disease (GERD)	Increased Risk of Reflux: Even during fasting, individuals with GERD can experience reflux, which may lead to aspiration.
Obesity	Increased Abdominal Pressure: Can contribute to gastroesophageal reflux, particularly when lying flat.
Hiatal Hernia (Option 1)	Exacerbate gastroesophageal reflux, increasing the potential for aspiration, especially when lying down.
Pregnancy	Hormonal Changes and Increased Abdominal Pressure: As the pregnancy progresses, these factors can increase the risk of reflux, leading to aspiration, especially in the third trimester.
Diabetes Mellitus with Gastroparesis	Delayed Gastric Emptying: Gastroparesis can lead to increased risk of reflux due to prolonged retention of food in the stomach.
Elderly Population	Age-Related Changes: Result in decreased swallowing efficiency and increased susceptibility to aspiration.

Solution for Question 10:

Correct Answer: D) Calcium channel blockers

Explanation:

Calcium channel blockers are generally continued through the perioperative period

Diabetic Medications:

Medication	Preoperative Management
Metformin (Option A ruled out) Sulfonylureas (Option B ruled out) Repaglinide GLP-1 agonists DPP-4 inhibitors	Stop on the morning of surgery
SGLT2 inhibitors (Option C ruled out)	Stop at least 24 hours before surgery
Insulin	Discontinue while fasting. Exception: Patients on continuous subcutaneous insulin infusion pumps should continue at the lowest basal rate (usually the nighttime fasting rate).
Intermediate-acting or long-acting insulin (Type 1 Diabetes)	Take one third to one half of the usual morning dose of intermediate-acting or long-acting insulin (e.g., lente, isophane) to avoid diabetic ketoacidosis.
Intermediate-acting or long-acting insulin (Type 2 Diabetes)	Either take no insulin or up to one half of the usual morning dose of intermediate-acting, long-acting, or combination insulin (e.g., 70/30 preparations).

Hypertensive Medications:

Medication	Preoperative management
ACE Inhibitors (Enalapril)	Stop 24 hours before surgery
Angiotensin II Receptor Blockers (ARBs) (Telmisartan)	Stop 24 hours before surgery
Calcium Channel Blockers (Amlodipine)	Continue (Option D)

Solution for Question 11:

Correct Answer: A) 2-5 days

Explanation:

Oral anticoagulants are stopped 2-5 days prior to noncardiac surgery

Medication	Continue or stop preoperatively
Oral Anticoagulants	Stop 2-5 days prior to noncardiac surgery Continue in high-patient risk (e.g., high-risk mechanical valves)

Aspirin	Stop preoperatively 7-10 days before surgery Continue for high-risk populations (e.g., CAD, cerebrovascular disease)
Statins	Continue
Thienopyridines (e.g., Clopidogrel)	Discontinue 2 weeks before surgery

Note: Incase of a patient with coronary stent:

- Aspirin should be continued
- P2Y12 Inhibitors (e.g., clopidogrel, ticagrelor, prasugrel) should be discontinued

Solution for Question 12:

Correct Answer: D) All of the above

Explanation:

Drug	Continued or stopped preoperatively
Antithyroid	Continued
Antiepileptic	Continued
Antidepressants	Continued
Smoking	Stop at least 6-8 weeks prior
Steroid	Continued
Anticonvulsants	Continued
Monoamine Oxidase Inhibitors (MAOIs)	Discontinue 2-3 weeks before surgery
Oral Contraceptive Pill	Discontinue at least 6 weeks before
Herbal Supplements	Discontinue at least 2 weeks before surgery

Note:

- Ayurvedic medicine: Stop 2 weeks prior to surgery

- All psychiatric meds continue except Lithium: Stop 48 hrs before surgery MAO Inhibitors/ TCA: Stop 3 Weeks Prior To Surgery
- Lithium: Stop 48 hrs before surgery
- MAO Inhibitors/ TCA: Stop 3 Weeks Prior To Surgery
- Lithium: Stop 48 hrs before surgery
- MAO Inhibitors/ TCA: Stop 3 Weeks Prior To Surgery

Solution for Question 13:

Correct Answer: A) Class 1

Explanation:

Since she is having a pregnancy with no abnormalities mentioned, she belongs to class 2

American Society of Anesthesiologists physical status classification of patients		
Class	Definition	Examples
1	Normal healthy patient	Healthy, non-smoking, no or minimal alcohol use
2	Patient with mild systemic disease (no functional limitations), normal healthy pregnancy.	Mild diseases without functional limitations, including current smoker, social alcohol use, pregnancy, obesity (BMI 30-40), well-controlled DM/HTN, or mild lung disease.
3	Patient with severe systemic disease (some functional limitations)	functional limitations with one or more moderate to severe diseases, such as poorly controlled DM or HTN, COPD, morbid obesity (BMI ≥ 40), active hepatitis, alcohol dependence, pacemaker, moderate reduction in ejection fraction, ESRD on dialysis, or a history of MI, CVA, TIA, or CAD/stents.

4	Patient with severe systemic disease that is a constant threat to life (functionality incapacitated)	Recent (<3 months) MI, CVA, TIA, CAD/stents, ongoing cardiac ischemia, severe valve dysfunction, severe ejection fraction reduction, shock, sepsis, DIC, ARD, or ESRD not on dialysis.
5	Moribund patient who is not expected to survive without the operation	Ruptured abdominal/thoracic aneurysm, massive trauma, intracranial bleed with mass effect, ischemic bowel in the face of significant cardiac pathology or multiple organ/system dysfunction
6	Brain-dead patient whose organs are being removed for donor	
E	If the procedure is an emergency, the physical status is followed by "E" (e.g., "2E")	

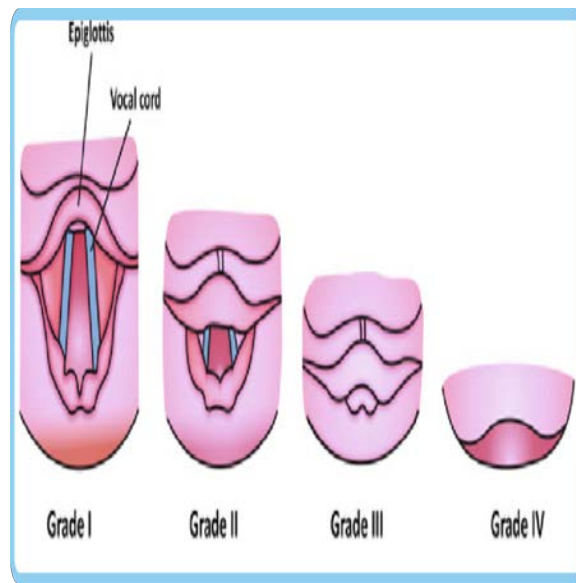
Solution for Question 14:

Correct Answer: A) Grade 1

Explanation:

Since the entire laryngeal aperture is visible, this is a Grade 1 view

Cormack-Lehane grading system for laryngoscopic view:



Grade 1: Entire laryngeal aperture visible (best view)

Grade 2: Posterior portion of laryngeal aperture visible

Grade 3: Only the epiglottis visible

Grade 4: No structures visible (the most difficult view)

Previous Year Questions

1. Which of the following comes under ASA grade 1?

- A. Healthy patient
 - B. Mild disease
 - C. Moderate disease
 - D. Morbid patient
-

2. A patient scheduled for elective hip surgery is currently taking aspirin, enalapril, a multivitamin, and metoprolol. The surgery is planned in 5 days. What is the appropriate perioperative management for this patient?

- A. Stop enalapril
 - B. Stop aspirin to minimize perioperative bleeding risk
 - C. Stop metoprolol to increase cardiac output
 - D. Increase aspirin dosage for additional analgesic benefit
-

3. What is the appropriate preoperative fasting recommendation?

- A. 4 hours for solids in adults
 - B. 2 hours for liquids in adults
 - C. 2 hours for breast milk in infants
 - D. 4 hours for solids in infants
-

4. Which of the following is the drug of choice for preoperative antibiotic prophylaxis in a patient undergoing cardiac surgery?

- A. Penicillin G
 - B. Erythromycin
 - C. Azithromycin
 - D. Cefazolin
-

5. Which investigation yields the most precise prognostic information for predicting the risk of perioperative cardiac complications among the options provided?

- A. Exercise ECG testing
 - B. Dobutamine stress echocardiography
 - C. Myocardial perfusion scintigraphy
 - D. Coronary angiography
-

6. When assessing a patient before anesthesia, the oral cavity should be examined. The image provided below displays the examined cavity. It illustrates__



- A. Mallampati class IV
 - B. Mallampati Class II
 - C. Mallampati class III
 - D. Mallampati class I
-

7. Which of the following is contraindicated in patients with acute intermittent porphyria?

- A. Thiopentone
 - B. Propofol
 - C. Etomidate
 - D. Ketamine
-

8. Which surgery should be avoided in a patient with a fibroma and moderate to severe cardiac disease?

- A. Laparoscopic hysterectomy
 - B. Abdominal hysterectomy
 - C. Total hysterectomy
 - D. Sub total hysterectomy
-

9. In adults, at which level does the spinal cord terminate?

- A. L3-L4
 - B. L1-L2
 - C. T12-L1
 - D. L2-L3
-

10. Glycopyrrolate is used as a preanesthetic agent for

- A. Reducing the secretions
 - B. Sedative effect
 - C. Skeletal muscle relaxant
 - D. Anxiolytic
-

11. Preoperative assessment of oral cavity for endotracheal intubation in an adult is done by:

- A. Mallampatti score
- B. Conmack lehane
- C. ASA grading
- D. AHA grading

12. What is the most effective approach for verifying the accuracy of intubation during laryngoscopy?

- A. Capnography
 - B. Air entry on auscultation
 - C. Ultrasonography
 - D. Chest movement
-

13. What is the recommended position for inserting a Ryle's tube?

- A. Supine with neck flexed
 - B. Sitting with neck flexed
 - C. Supine with neck extended
 - D. Sitting with neck extended
-

14. Which of the following corresponds to ASA grade III?

- A. Moribund patient
 - B. Patient having systemic illness which is under control or without any functional limitations
 - C. Patient having systemic illness which is not under control or with functional limitations
 - D. Patient having systemic illness which is a constant threat to life
-

15. In a 45-year-old patient undergoing a preoperative assessment for a surgical procedure, which classification of Mallampati score is assigned based on the following description: Hard palate, soft palate, uvula, tip of the uvula, and tonsil pillars are visible. No oropharyngeal structures are obstructing the view.

- A. Grade I
 - B. Grade II
 - C. Grade III
 - D. Grade IV
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	2
Question 4	4
Question 5	2
Question 6	3
Question 7	1
Question 8	3
Question 9	2
Question 10	1
Question 11	1
Question 12	1
Question 13	2
Question 14	3
Question 15	1

Solution for Question 1:

Correct Option A - Healthy patient:

- This classification is used to describe patients who are completely healthy and free from systemic illness, without any comorbidities. Such patients are typically considered at minimal risk for anesthesia-related complications.

Incorrect Options:

Option B - Mild disease: Patients with mild systemic disease that does not significantly affect their daily activities fall under ASA Grade 2

Option C - Moderate disease: Patients with moderate systemic disease that poses a constant threat to their life are classified as ASA Grade 3.

Option D - Morbid patient: Patients having systemic illness come under ASA4.

Solution for Question 2:

Correct Answer: A) Stop enalapril

Explanation:

ACE inhibitors such as enalapril should typically be stopped 24 hours before elective surgery. This recommendation is based on evidence that continuing ACE inhibitors perioperatively can lead to clinically significant hypotension during anesthesia, particularly when combined with the vasodilatory effects of anesthetic agents.

Medication discontinuation before surgery	
Drug	Continued or stopped preoperatively
Antithyroid	Continued
Antiepileptic	Continued
Antidepressants	Continued
Aspirin	Continued
Beta Blockers	Continued
Smoking	Stop at least 6-8 weeks prior

Steroid	Continued
Anticonvulsants	Continued
Monoamine Oxidase Inhibitors (MAOIs)	Discontinue 2-3 weeks before surgery
Oral Contraceptive Pill	Discontinue at least 6 weeks before
Herbal Supplements	Discontinue at least 2 weeks before surgery
ACE Inhibitors (Enalapril)	Stop 24 hours before surgery
Angiotensin II Receptor Blockers (ARBs) (Telmisartan)	Stop 24 hours before surgery
Calcium Channel Blockers (Amlodipine)	Continue

Stop aspirin to minimize perioperative bleeding risk (Option B) may be considered for neurosurgery or ocular procedures, but current guidelines generally recommend continuing aspirin for most orthopedic procedures including hip surgery. The modest increase in surgical bleeding with aspirin continuation is typically manageable and outweighed by the potential cardiovascular risks of discontinuation.

Stop metoprolol to increase cardiac output (Option C) would be inappropriate and potentially dangerous. Abrupt beta-blocker discontinuation perioperatively significantly increases the risk of sympathetic hyperactivity, tachycardia, hypertension, myocardial ischemia, and mortality.

Increasing aspirin dosage for additional analgesic benefit (Option D) has no supporting evidence of additional benefit and would unnecessarily increase bleeding risk.

Solution for Question 3:

Correct Option B: 2 hours for liquids in adults

- A preoperative fasting period is a time prescribed before any procedure that involves general anesthesia, sedation, or regional anesthesia.
- The collective fasting recommendation is two hours for clear and light liquids, four hours for breast milk, and six hours for solid foods, infant formula, light meals, and non-human milk.

Incorrect Options:

Option A: 4 hours for solids in adults

- Eight hours of preoperative fasting is recommended for solid food in adults.

Option C: 2 hours for breast milk in infants

- Four hours of preoperative fasting is recommended for breast milk.

Option D: 4 hours for solids in infants

- It is recommended 6 hours for solid foods in preoperative fasting before anesthesia. It should be kept in mind that cereal and baby food are solid foods.

Solution for Question 4:

Correct Option D - Cefazolin:

- Cefazolin is a first-generation cephalosporin antibiotic commonly used for preoperative antibiotic prophylaxis in patients undergoing cardiac surgery. It works by inhibiting the growth and multiplication of bacteria that can cause surgical infections.
- administering cefazolin before surgery can significantly reduce the risk of surgical infections.

Incorrect Options:

Option A - Penicillin G: Penicillin G isn't often used in cardiac surgery prophylaxis due to its limited spectrum of activity, which doesn't cover all potential infection-causing bacteria.

Option B - Erythromycin: Erythromycin isn't commonly used in cardiac surgery prophylaxis because it's less effective than other antibiotics in preventing surgical infections.

Option C - Azithromycin: Azithromycin isn't commonly used in cardiac surgery prophylaxis because it's ineffective against all bacteria that can cause surgical infections.

Solution for Question 5:

Correct Option B - Dobutamine stress echocardiography:

- Dobutamine stress echocardiography evaluates the heart's response to stress, detecting coronary artery disease (CAD) and cardiac function. It assesses structure, function, inducible ischemia, and contractility, providing more accurate predictions of perioperative cardiac issues than exercise ECG testing.

Incorrect Options:

Option A - Exercise ECG testing:

- Exercise ECG tests focus on heart electrical activity during exercise but may miss structural or functional issues.

Option C - Myocardial perfusion scintigraphy:

- While informative about perfusion abnormalities, Myocardial perfusion scintigraphy, may not predict perioperative cardiac complications as precisely as dobutamine stress echocardiography.

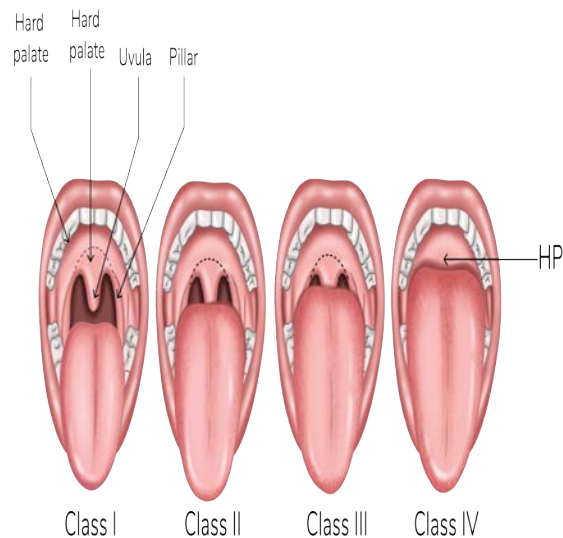
Option D - Coronary angiography:

Coronary angiography, the gold standard for diagnosing coronary artery disease, isn't typically used to predict perioperative cardiac complications as it offers anatomical, not functional, data.

Solution for Question 6:

Correct Option C - Mallampati class III:

The picture shows Mallampati Class III as only the base of the uvula is visible.



Incorrect Options: A, B & D

Solution for Question 7:

Correct Option A - Thiopentone:

- Thiopentone is contraindicated in patients with acute intermittent porphyria. Thiopentone is a barbiturate anesthetic agent that can potentially trigger an acute porphyria attack. The administration of barbiturates, including Thiopentone, can induce hepatic enzyme activity, which can disrupt heme synthesis and trigger the accumulation of porphyrin precursors, worsening the symptoms of AIP.

Incorrect Options:

Options B, C & D - Propofol, Etomidate & Ketamine : On the other hand, Propofol, Etomidate, and Ketamine are considered safer options for anesthesia in patients with acute intermittent porphyria. These agents do not significantly affect heme synthesis or induce hepatic enzyme activity, making them less likely to precipitate an acute attack in patients with AIP.

Solution for Question 8:

Correct Option C - Total hysterectomy:

Total hysterectomy, removing the uterus and cervix, is contraindicated in patients with moderate to severe cardiac disease due to increased surgical stress on the cardiovascular system, which can lead to complications.

Incorrect Options:

Option A - Laparoscopic hysterectomy:

- It is generally considered a suitable option for patients with cardiac disease because it involves smaller incisions, less blood loss, and faster recovery compared to open surgery.

Option B - Abdominal hysterectomy:

- Abdominal hysterectomy remains an option for patients with moderate to severe cardiac disease after careful evaluation and consideration of risks and benefits.

Option D - Subtotal hysterectomy:

- Subtotal hysterectomy may be a preferred option for patients with moderate to severe cardiac disease because it is less invasive compared to total hysterectomy.
-

Solution for Question 9:

Correct Option B - L1-L2:

- In adults, the spinal cord typically ends at the level of the first lumbar vertebra (L1) or the second lumbar vertebra (L2). This termination point is known as the conus medullaris. It's important to note that the spinal cord does not extend down to the end of the vertebral column. Instead, beyond the conus medullaris, the spinal cord tapers into a bundle of nerve roots called the cauda equina.

Incorrect Options:

Option A - L3-L4: This option is commonly seen as the termination level of the dural sac, which contains the cerebrospinal fluid (CSF) surrounding the spinal cord and nerve roots.

Option C - T12-L1: The thoracic spinal cord extends down to approximately the level of T12, and the conus medullaris is located in the lumbar region rather than the thoracic region.

Option D - L2-L3: Similar to option a), this level is commonly associated with the termination of the dural sac, not the spinal cord itself.

Solution for Question 10:

Correct Option A-Reducing the secretions:

- Glycopyrrolate, an anticholinergic, is commonly used before surgery to reduce excessive secretions, particularly in the respiratory tract.
- By blocking acetylcholine effects, it inhibits saliva, sweat, and other secretions, helping maintain a clear airway and prevent complications like aspiration pneumonia..

Incorrect Options

Option B: Sedative effect

- While it may have some mild sedative properties, its main purpose is to reduce secretions rather than induce sedation.

Option C: Skeletal muscle relaxant

- Glycopyrrolate is not a skeletal muscle relaxant. It does not directly affect muscle relaxation or neuromuscular transmission.

Option D: Anxiolytic

- Its main function is to reduce secretions rather than alleviate anxiety or provide calming effects.
-

Solution for Question 11:

Correct Option A: Mallampatti score

- The Mallampatti score is a classification system used to assess the visibility of the structures in the oral cavity and predict the ease of intubation.
- It grades visibility of the soft palate, uvula, and tonsillar pillars, ranging from Class I (full visibility) to Class IV (no visibility). Higher scores indicate more difficult intubation.

Incorrect Options

Option B: Cormack-Lehane

- The Cormack-Lehane classification is used to assess the view of the vocal cords during laryngoscopy, not specifically the oral cavity.

Option C: ASA grading

- The ASA grading system assesses a patient's physical status before surgery but doesn't specifically evaluate the oral cavity for intubation.

Option D: AHA grading

- There is no commonly known anesthesia grading system called AHA grading.

Solution for Question 12:

Correct Option A - Capnography:

- Capnography is the optimal method for confirming correct intubation during laryngoscopy. It measures exhaled carbon dioxide (CO₂) concentration, providing immediate confirmation of proper tube placement in the trachea.

Incorrect Options:

Option B - Air entry on auscultation: While auscultation for air entry is a common method for confirming tube placement, it may not be as reliable as capnography.

Option C - Ultrasonography: Ultrasonography can be used to visualize the structures of the airway, but it is not typically used to confirm tube placement during laryngoscopy.

Option D - Chest movement: Chest movement alone is not a reliable indicator of correct intubation and may not provide definitive confirmation.

Solution for Question 13:

Correct Option B - Sitting with neck flexed:

- When inserting a Ryle's tube, which is a type of nasogastric tube, the ideal position for the patient is sitting with the neck flexed. This position helps to straighten the nasal passage and facilitate the passage of the tube into the stomach. It allows for a more straightforward and safer insertion process.

Incorrect Options:

Option A - Supine with neck flexed: This position may make the insertion of the Ryle's tube more challenging as the neck flexion is limited in the supine position.

Option C - Supine with neck extended: Neck extension can cause discomfort and hinder Ryle's tube insertion by obstructing the nasal passage, making tube passage more difficult.

Option D - Sitting with neck extended: This position may cause discomfort and can increase the risk of the tube going into the respiratory tract instead of the stomach.

Solution for Question 14:

Correct Option C - Patient having systemic illness which is not under control or with functional limitations:

ASA classification:

ASA 1	Free from systemic illness
ASA 2	Patient having systemic illness which is under control or without any functional limitations
ASA 3	Patient having systemic illness which is not under control or with functional limitations
ASA 4	Patient having systemic illness which is a constant threat to life
ASA 5	Moribund patient
ASA 6	Brain dead patient

Incorrect Options: Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 15:

Correct Option A- Grade I:

- In the scenario described, the entire soft palate, uvula, pillars, and tonsillar region can be seen without any obstruction. The correct classification is Grade I. This indicates an optimal airway visualization and ease of intubation.

Incorrect Options:

Option B- Grade II: Hard palate, soft palate, uvula, and tonsil pillars are visible. The tip of the uvula is not visible.

Option C- Grade III: Hard palate, soft palate, base of the uvula are visible.

Option D- Grade IV: Only hard palate is visible.

Local Anaesthetics

1. Which of the following drugs is not metabolized by plasma esterases?

- A. Procaine
 - B. Chloroprocaine
 - C. Tetracaine
 - D. Cocaine
-

2. Patient who is a known case of alcoholic liver disease is being planned for a dental extraction procedure. Which of the following local anesthetic can be safely administered in this patient?

- A. Lignocaine
 - B. Articaine
 - C. Bupivacaine
 - D. Mepivacaine
-

3. What is the mechanism of action of the given drug?



- A. Blocks Na⁺ channels
- B. Blocks K⁺ channels
- C. Both (1) & (2)
- D. Blocks Cl⁻ channels

4. All among the following are true about adding Adrenaline with Lignocaine except:

- A. Increases the duration of action
- B. Decreases blood loss
- C. Contraindicated in patients with cardiovascular disease
- D. Increases the rate of systemic absorption

5. Patient following an RTA presented with multiple lacerations over the forearm and knee regions, which were sutured under LA. Few minutes after suturing, he complained of lightheadedness and dizziness, which was followed by an episode of seizure. On examination, his vitals were: HR-58 bpm; BP-88/50mmHg. What is

the definitive management for this patient?

- A. Perform urgent CT head
 - B. Administer IV Lorazepam
 - C. Give 20% intralipid solution
 - D. Airway management and ventilation
-

6. Anesthetist decides to administer Lignocaine with adrenaline for epidural analgesia. Choose the correct pair of concentration of Lignocaine and the concentration ratio when used with adrenaline.

- A. 1% - 1:100,000
 - B. 1% - 1:200,000
 - C. 5% - 1:200,000
 - D. 5% - 1:100,000
-

7. Cocaine exerts its local anesthetic action by blocking which ion channel?

- A. Potassium channels
 - B. Calcium channels
 - C. Sodium channels
 - D. Chloride channels
-

8. What is the major concern associated with bupivacaine toxicity?

- A. Liver toxicity
- B. Nephrotoxicity

- C. Cardiotoxicity
 - D. Neurotoxicity
-

9. What is the primary function of pseudocholinesterase?

- A. Breaks down acetylcholine at neuromuscular junctions
 - B. Hydrolyzes ester-based compounds in the plasma
 - C. Converts succinylcholine into its active form
 - D. Metabolizes amide local anesthetics
-

10. In comparison to the retrobulbar block, which statement is true regarding the peribulbar block?

- A. It is performed deeper than the retrobulbar block.
 - B. It has a higher risk of optic nerve injury.
 - C. It generally provides a wider area of anesthesia.
 - D. It is more likely to cause complications compared to the retrobulbar block.
-

11. What are the two active ingredients in EMLA cream?

- A. 2.5% Lidocaine and 2.5% Procaine
 - B. 3.5% Prilocaine and 3.5% Tetracaine
 - C. 2.5% Lidocaine and 2.5% Prilocaine
 - D. 4.5% Bupivacaine and 4.5% Lidocaine
-

12. A 50 year old female patient is undergoing a minor surgical procedure on her left forearm using intravenous regional anesthesia (IVRA) for pain control. During the procedure, the patient suddenly develops seizures and exhibits signs of cardiovascular instability. Which of the following is the most appropriate immediate management step for this patient?

- A. Administer intravenous calcium gluconate
 - B. Administer intravenous magnesium sulphate
 - C. Initiate lipid emulsion therapy
 - D. Give an additional bolus of lidocaine
-

13. Which of the following is the correct order of susceptibility of nerve fibers to local anesthetics (LA)?

- A. A alpha = A beta > A gamma = A delta > B > C
 - B. B > A alpha = A beta > A gamma = A delta > C
 - C. A gamma = A delta > A alpha = A beta > B > C
 - D. A gamma = A delta > B > A alpha = A beta > C
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	4
Question 5	3
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	3
Question 11	3
Question 12	3
Question 13	3

Solution for Question 1:

Correct Answer: D) Cocaine

Explanation:

Cocaine is metabolized by hepatic carboxylesterase

Amino Ester Local Anesthetic	Metabolism	Allergy Potential
Procaine	Metabolized by plasma esterases	Aminoester causes more allergies in comparison to aminoamides since they are derivatives of p-aminobenzoic acid (allergenic)
Chloroprocaine	Metabolized by plasma esterases	
Tetracaine	Metabolized by plasma esterases	
Cocaine	Metabolized predominantly by hepatic carboxylesterase	

Solution for Question 2:

Correct answer: B) Articaine

Explanation:

- All the following drugs (in options) belong to Aminoamides group of local anesthetics.
- All aminoamides undergo Enzymatic degradation in Liver except Articaine - undergoes metabolism by Plasma cholinesterase.
- In a patient with Liver disease, aminoamide that can be safely administered is Articaine, which is also usually preferred for short dental procedures.

Aminoesters & Aminoamides:

	Aminoesters (1 "i" in names)	Aminoamides (2 "i" in names)
Drugs	Procaine Chloroprocaine Tetracaine Cocaine	Lidocaine Mepivacaine Prilocaine Bupivacaine Ropivacaine
Duration of action & stability	Short; stable	Long; unstable in solution
Metabolism	Plasma cholinesterase Exception: Cocaine (metabolized in liver)	Enzymatic degradation in Liver Exception: Articaine (undergoes metabolism by plasma cholinesterase)

Metabolism rate	Rapidly metabolized: Chloroprocaine	Rapidly metabolized rate: Lignocaine > Mepivacaine > Bupivacaine
Allergic potential	Maximum with Prilocaine and Benzocaine	-

Solution for Question 3:

Correct answer: A) Blocks Na⁺ channels

Explanation:

Image shown is Bupivacaine, a local anesthetic. All local anesthetic acts by inhibiting Na⁺ channels.



Mechanism of action & absorption:

- Site: acts on nodes of ranvier
- Blocks transmission of pain impulse from nociceptor to spinal cord.
- Acts on intracellular side of Na channels at inactivation gate blocking the channel.
- LA doesn't affect RMP of nerves.
- Blocks inactivation gates of Na⁺ channels → Disrupts depolarisation → No action potential generated → No pain sensation.

- Highest systemic absorption (routes of administration): IV route > Tracheal > Intercostal > Brachial plexus > Epidural.
- Form for onset of LA action (to reach target nerve) - Unionized form.
- Active form at site of block of Na channels- Ionized form.
- Higher doses will act for longer duration and lower doses for a relatively shorter duration.

Duration of onset for different local anesthetics (infiltration anesthesia):

Short Duration	Procaine Chloroprocaine
Moderate Duration	Lidocaine Prilocaine
Long Duration	Bupivacaine Ropivacaine

- Shortest acting- Chloroprocaine
- Longest acting- Dibucaine
- Fastest onset- Chloroprocaine (low potency so high concentration used) > Lignocaine (pka is near physiological pH)

Solution for Question 4:

Correct answer: D) Increases the rate of systemic absorption

Explanation:

Adding Adrenaline (Epinephrine) with Lignocaine decreases the rate of systemic absorption of Lignocaine and thereby decreases the effects of systemic toxicity of Lignocaine.

Additives of Local Anesthetics:

Epinephrine	Effects: Increases the duration of action (Option A) Decreases the rate of systemic absorption (Option D) Decreases toxicity Decreases blood loss (Option B) Increases the risk of arrhythmia (increases peripheral vascular resistance)
Contraindication: Hypertension, Ischaemic heart disease/ cardiovascular diseases (Option C)	
Can be added with most of the local anesthetics except a few eg. Ropivacaine	

Alternative: Felypressin - no increase in heart rate and blood pressure	
Alkalinization	By adding Sodium bicarbonate (NaHCO ₃) - increases the speed of onset. By increasing unionized form of LA which reaches the target quickly.
Opioids	All opioids can be added safely, exception: Remifentanyl Commonly used: Fentanyl, Alfentanil, Sufentanyl
Can be added with most of the local anesthetics, exception: Chloroprocaine	
Action: synergistic analgesia Side effects: Respiratory depression, most common - pruritus	
α 2-adrenergic agonists	Action: synergistic analgesia (in peripheral administration) ; additive effect (in systemic administration) Commonly used: Clonidine
Steroids	Effect: prolongs the duration of action Commonly used: Dexamethasone

Solution for Question 5:

Correct answer: C) Give 20% intralipid cardiac solution

Explanation:

- The clinical scenario is suggestive of Local anesthetic associated Systemic toxicity (LAST) that has led to CNS (dizziness, seizures) and cardiac effects (bradycardia, hypotension).
- The definitive management would be to administer 20% intralipid solution .
- Although other options: Administer IV Lorazepam (Option B) and Airway management (Option D) is also a part of management protocol, but is not the definitive management in this case.
- Perform Urgent CT Head (Option A) is not required at this stage as there is no history of trauma to the head and the symptoms started a few minutes after administration of local anesthesia.

Local Anesthetic Systemic Toxicity (LAST):

Clinical Features	CNS effects: Early and less severe. First symptom: circumoral paraesthesia First sign: nystagmus At low plasma concentration: mild sensory disturbances. Increased concentration: CNS excitation, seizure activity. Higher concentration: CNS depression, coma, respiratory depression, arrest.
Cardiac effects: Usually late but severe form of toxicity, may lead to death. Hypotension, dysrhythmias, myocardial depression. Potent agents (bupivacaine, ropivacaine, levobupivacaine) linked to cardiovascular collapse, heart block.	
Safety profile	Bupivacaine has more risk to cause LAST than Lidocaine Most potent highly lipid soluble drugs are more likely to cause
Prevention	Risk recognition and dose reduction. Careful aspiration before injection. Incremental administration (3-5 mL) with continuous heart rate monitoring. Use ultrasonographic guidance to confirm anesthetic deposit. Use a vasoactive agent like epinephrine to reduce systemic absorption and serve as an intravascular indicator.
Treatment	Recognize LAST (Local Anesthetic Systemic Toxicity) early and call for help. Airway management: Secure airway, ventilate with 100% oxygen. Administer 20% Lipid Emulsion: (Option C) Bolus: 12 mL/kg over 2-3 min. Infusion: at 0.25 mL/kg/min-0.5ml/kg/min for 15-20 mins Maximum initial dose: 12 ml/kg Administration of antiepileptics: for seizure control, benzodiazepines such as Lorazepam is preferred. Cardiopulmonary resuscitation: if necessary, monitor the patient at least for initial 2-6 hours following resuscitation.

Solution for Question 6:

Correct answer: B) 1% - 1:200,000

Explanation:

Ideal concentration of Lignocaine when given epidurally is 1-2% and the concentration ratio with Adrenaline for Lignocaine/Bupivacaine/Prilocaine is 1:200,000 .

Concentrations of Lignocaine:

Route of administration	Concentration	
Intravenous regional anesthesia (IVRA)	0.5%	
Nerve blocks	Minor block:	1%
Major block:	1%	
Epidural	1-2% (Option B)	
Topical	Solution:	2-4%
EMLA:	2.5%	
Aerosol/suppository:	10%	
Spinal	5%	

Concentration of adrenaline (epinephrine):

Anaphylactic shock	1:1000 IM 0.5 mg IM or SC
Cardiac arrest	1:10,000 IV 1mg IV
Local anesthetic	1:200,000 (Option B)

Solution for Question 7:

Correct Answer: C) Sodium channels

Explanation:

Cocaine blocks sodium channels, which inhibits the propagation of action potentials in nerve cells, producing its local anesthetic effect.

Cocaine :

- Classification : Ester local anesthetic and CNS stimulant.
- Mechanism of Action: Blocks sodium channels to inhibit nerve impulses and inhibits reuptake of dopamine, norepinephrine, and serotonin, resulting in CNS stimulant effects.

- Medical Use: Used for topical anesthesia especially due to its vasoconstrictive properties.
- Vasoconstriction : Causes vasoconstriction, which helps reduce bleeding and prolong anesthesia during procedures.
- Duration of Action: Short half life (approx. 1 hour)
- Metabolism : Metabolized predominantly by hepatic carboxylesterase.
- CNS Effects: Acts as a strong CNS stimulant, leading to euphoria, increased energy, and potential addiction.
- Side Effects: Includes hypertension, tachycardia, arrhythmias, seizures, and addiction.
- Contraindications : Contraindicated in patients with heart disease, hypertension, and seizure disorders because of its cardiovascular and CNS stimulant properties.

Chloroprocaine

- Classification : Ester local anesthetic.
- Mechanism of Action: Blocks sodium channels, preventing the propagation of nerve impulses to produce local anesthesia.
- Primarily used for local, epidural, and spinal anesthesia, especially in obstetric procedures(labor and delivery).
- No Vasoconstriction
- Duration of Action: Short duration of action due to rapid metabolism by plasma cholinesterase, making it ideal for short surgical procedures.
- Metabolism : Metabolized quickly by plasma cholinesterase, resulting in a lower risk of systemic toxicity.
- CNS Effects: Lacks CNS stimulant properties and does not cause euphoria or have any potential for addiction.
- Side Effects: Generally well-tolerated; possible side effects include allergic reactions due to its ester structure. Minimal systemic effects.
- Contraindications : Contraindicated in patients allergic to ester-type local anesthetics.

Solution for Question 8:

Correct Answer: C) Cardiotoxicity

Explanation:

Bupivacaine is known for its potential cardiotoxic effects , particularly when administered in high doses or if accidentally injected into a blood vessel, which can lead to severe cardiac arrhythmias and cardiac arrest.

Bupivacaine

Classification	Bupivacaine is an amide-type local anesthetic
Mechanism of Action	It works by blocking sodium channels in nerve membranes, preventing the propagation of nerve impulses, which results in local anesthesia.
Clinical Uses	Commonly used for spinal, epidural, and peripheral nerve blocks. Widely used in labor and delivery for epidural anesthesia and in long-duration surgeries.
Duration of Action	Longer duration of action compared to other local anesthetics like lidocaine. It can provide anesthesia for several hours, making it ideal for prolonged procedures.
Onset of Action	Has a slower onset (10-15 minutes) compared to faster-acting anesthetics like lidocaine.
Cardiotoxicity	Major concern with bupivacaine is its potential for cardiotoxicity, especially at higher doses or when inadvertently injected intravenously, leading to severe arrhythmias or cardiac arrest.
Metabolism	Metabolized primarily in the liver by cytochrome P450 enzymes and excreted by the kidneys.
Enantiomers	The S-enantiomer of bupivacaine, known as levobupivacaine, is associated with reduced cardiotoxicity and is often used as a safer alternative.
Lipid Solubility & Protein Binding	Bupivacaine has high lipid solubility and high protein binding, which contributes to its prolonged duration of action.
Contraindications	Should be avoided in patients with severe cardiac arrhythmias, and caution should be used in patients with liver disease due to its hepatic metabolism.
Common Side Effects	In addition to cardiotoxicity, other potential side effects include CNS toxicity, such as dizziness, seizures, and lightheadedness, especially with accidental intravascular injection.

Toxicity	Direct intravenous injection or rapid vascular uptake of the drug is the most common reason for bupivacaine toxicity, which has an upper limit of 2.5 to 3.5 mg/kg upper limit of dosing. Treatment for toxicity: 20% intralipid solution.
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Solution for Question 9:

Correct Answer: B) Hydrolyzes ester-based compounds in the plasma

Explanation :

Pseudocholinesterase plays a critical role in metabolizing ester-based local anesthetics (chloroprocaine, procaine, tetracaine) and neuromuscular blocking agents (succinylcholine, mivacurium).

Pseudocholinesterases : Serine hydrolase enzyme primarily produced in the liver that catalyzes the hydrolysis of choline esters.

Drugs Metabolized by Pseudocholinesterase

Succinylcholine	Use : Neuromuscular blocking agent (commonly used for rapid sequence intubation or short surgical procedures). Mechanism : Succinylcholine is hydrolyzed by pseudocholinesterase in the plasma.
Mivacurium	Use : Non-depolarizing neuromuscular blocking agent. Mechanism : Mivacurium, like succinylcholine, is metabolized by pseudocholinesterase.
Chloroprocaine	Use : Local anesthetic (commonly used in obstetric anesthesia). Mechanism : Chloroprocaine is an ester local anesthetic rapidly metabolized by pseudocholinesterase.
Procaine	Use : Local anesthetic (used in dental procedures and minor surgeries). Mechanism : Procaine, another ester-type local anesthetic, is hydrolyzed by pseudocholinesterase into para-aminobenzoic acid (PABA) and diethylaminoethanol.
Tetracaine	Use : Local anesthetic (used in spinal and topical anesthesia). Mechanism : Tetracaine is also an ester local anesthetic metabolized by pseudocholinesterase.

Breaks down acetylcholine at neuromuscular junctions (Option A) :This function is carried out by acetylcholinesterase.

Converts succinylcholine into its active form (Option C): Pseudocholinesterase breaks down succinylcholine, rather than activating it.

Metabolizes amide local anesthetics (Option D): Amide local anesthetics (like lidocaine) are metabolized by the liver, not by pseudocholinesterase.

Solution for Question 10:

Correct Answer: C) It generally provides a wider area of anesthesia.

Explanation:

Peribulbar block provides a wider area of anesthesia.

Aspect	Retrobulbar Block	Peribulbar Block
Anatomical Location	Behind the globe within the orbit	Around the globe, in the peribulbar space
Depth of Injection	Deeper (often near the optic nerve)	Shallower (more superficial to the globe) (Option A)
Indications	Cataract surgery, vitrectomy	Cataract surgery
Area of Anesthesia	Primarily targets extraocular muscles	Provides a wider area of anesthesia (Option C)
Onset of Action	Rapid onset	Slower onset than retrobulbar
Risk of Complications	Higher risk of serious complications with increased risk of optic nerve injury.	Lower risk of serious complications (Option B & D)
Patient Position	Supine with head turned	Supine or sitting position

Solution for Question 11:

Correct Answer: C) 2.5% Lidocaine and 2.5% Prilocaine

Explanation:

EMLA cream contains a eutectic mixture of 2.5% lidocaine and 2.5% prilocaine, which work together to provide local anesthesia.

EMLA Cream: Eutectic Mixture of Local Anesthetics cream is a topical anesthetic used to numb the skin before medical procedures.

Active Ingredients	Contains 2.5% lidocaine and 2.5% prilocaine
Mechanism of Action	Works by blocking sodium channels in the nerve membranes, preventing nerve impulse transmission and resulting in localized anesthesia.
Common Uses	Primarily used for Venipuncture (insertion of IV lines) Minor surgical procedures Dermatological procedures Numbing before vaccinations
Onset of Action	Takes approximately 20-30 minutes to achieve effective anesthesia when applied properly.
Age Considerations	Generally safe for use in adults and children over 3 months old
Contraindications	Should not be applied to broken, inflamed, or infected skin. Caution in patients with a history of hypersensitivity to local anesthetics.
Side Effects	Common side effects may include skin irritation, redness, or erythema at the application site.

Solution for Question 12:

Correct Answer: C) Initiate lipid emulsion therapy

Explanation :

Intravenous Regional Anesthesia (IVRA): Intravenous Regional Anesthesia (IVRA), also known as a Bier block , is a technique that provides local anesthesia to a specific region of the body,

typically an extremity, through intravenous administration of a local anesthetic.

Indications	Suitable for minor surgical procedures on the upper or lower extremities (hand surgeries, forearm procedures, and foot surgeries). Used when general anesthesia is not necessary and for patients with contraindications to other forms of anesthesia.
Anesthetic Agents	Commonly used local anesthetics include lidocaine (most frequent) and prilocaine.
Procedure Steps	Tourniquet Application: Apply a tourniquet to the proximal part of the extremity to prevent blood flow and keep the local anesthetic localized. Venous Access: Establish intravenous access in the selected limb. Administration of Anesthetic agent: Inject the local anesthetic into the IV line, typically in a volume of 40-50 mL.
Tourniquet Management	The tourniquet should be inflated to a pressure that is approximately 50 mmHg above systolic blood pressure. It is typically safe to keep the tourniquet inflated for 30-60 minutes; prolonged inflation can increase the risk of complications.
Advantages	Provides effective analgesia with minimal systemic absorption. Allows for a bloodless surgical field. Quick onset of action and ease of administration.
Disadvantages and Complications	Local Anesthetic Systemic Toxicity (LAST) Can occur if the local anesthetic enters the systemic circulation. Tourniquet Pain Nerve Damage
Contraindications	Infection at the injection site or over the tourniquet. Severe peripheral vascular disease. Patient refusal or inability to cooperate.
Lipid Emulsion Therapy	Used as a treatment for Local anesthesia systemic toxicity, acts by sequestering the local anesthetic from circulation and mitigating toxic effects. Administer a 20% lipid emulsion intravenously if toxicity occurs.

Solution for Question 13:

Correct Answer: 3) $A_{\gamma} = A_{\delta} > A_{\alpha} = A_{\beta} > B > C$

Explanation:

Nerve fibers have the following order of susceptibilities to

- LA : $A \text{ gamma} = A \text{ delta} > A \text{ alpha} = A \text{ beta} > B > C$
 - Local anesthetics block nerve conduction by inhibiting sodium channels.
 - Small, myelinated fibers (like A gamma and A delta) are more susceptible than larger, myelinated fibers (A alpha and A beta).
 - Unmyelinated C fibers are the least susceptible, while B fibers fall between A and C fibers.
 - Susceptibility of nerve fibres to Pressure : $A > B > C$
 - Susceptibility of nerve fibres to Hypoxia : $B > A > C$
 - Controversial: Nerve fibre most susceptible to LA: As per Katzung Pharmacology 16th edition page no. 728- $B=C>A \text{ delta}>A \text{ gamma}>A \text{ beta}>A \text{ alpha}$ As per Ganongs Physiology 26th edition page no. 93- $C>B>A$
 - As per Katzung Pharmacology 16th edition page no. 728- $B=C>A \text{ delta}>A \text{ gamma}>A \text{ beta}>A \text{ alpha}$
 - As per Ganongs Physiology 26th edition page no. 93- $C>B>A$
 - As per Katzung Pharmacology 16th edition page no. 728- $B=C>A \text{ delta}>A \text{ gamma}>A \text{ beta}>A \text{ alpha}$
 - As per Ganongs Physiology 26th edition page no. 93- $C>B>A$
-

Regional Anaesthesia: Spinal, Epidural And Caudal

1. A 25-year-old primigravida is scheduled for a caesarean section. After discussing the risks, she chooses epidural anaesthesia. Which sign is not seen in correct placement of the epidural needle in the epidural space?

- A. CSF drops following guidewire removal
- B. Loss of resistance
- C. Aspiration of solution in the needle hub.
- D. Absence of knee-jerk

2. A 68-year-old male undergoing elective orthopaedic surgery under spinal anaesthesia suddenly becomes unresponsive. The anesthesiologist suspects total spinal anaesthesia. Which of the following set of statements is true? Sensory loss typically follows the caudal spread Profound motor weakness due to cephalad spread Respiratory depression Severe hypotension and bradycardia due to sympathetic blockade.

- A. 1,2,3
- B. 2,3,4
- C. 1,3,4
- D. 1,2,3,4

3. A 71-year-old man with COPD, atrial fibrillation, and osteoporosis is admitted for revision hip surgery. The surgical team chooses continuous spinal anaesthesia (CSA) to manage pain and reduce respiratory risks, considering his desire to stay awake. Which of the following is most commonly associated with CSA?

- A. Conus medullaris syndrome

- B. Lumbar spinal stenosis
 - C. Syringomyelia
 - D. Cauda equina syndrome
-

4. A 43-year-old man with scoliosis is scheduled for inguinal hernia mesh repair. During the pre-anaesthetic assessment, spinal anaesthesia was chosen. What would be the most appropriate approach for inducing anaesthesia in this patient?

- A. Midline approach
 - B. Paramedian approach
 - C. Caudal approach
 - D. Taylor's approach
-

5. A 35-year-old man is set to undergo a hemorrhoidectomy, and the anaesthetist intends to utilise a saddle block for anaesthesia. Which of the following is the main benefit of employing a saddle block in this procedure?

- A. Provides deep sedation
 - B. Longer duration of action
 - C. Complete motor blockade.
 - D. Rapid recovery of motion
-

6. A 2-year-old child weighing 10 kg is scheduled for a procedure that requires caudal anaesthesia. The anaesthetist plans to use a 0.1% lidocaine solution, with the recommended dose set at 0.5 mL/kg. What total volume of lidocaine should be administered?

- A. 3 mL

- B. 4 mL
 - C. 7 mL
 - D. 5 mL
-

7. A 3-year-old boy is scheduled for circumcision and is being assessed by an anaesthetist during the pre-anaesthesia checkup. What would be the most preferred type of anaesthesia for this procedure?

- A. Spinal anaesthesia
 - B. General anaesthesia
 - C. Local anaesthesia
 - D. Caudal anaesthesia
-

8. A 37-year-old male is undergoing inguinal hernia repair with spinal anaesthesia. During the procedure, the patient experiences cardiac arrest. Which anaesthetic drug is likely responsible for this?

- A. Ropivacaine
 - B. Lidocaine
 - C. Chloroprocaine
 - D. Bupivacaine
-

9. A 48-year-old male is undergoing an open inguinal hernia repair. The anesthesiologist decides to use epidural anaesthesia instead of spinal anaesthesia for better postoperative pain management. What is an additional advantage of epidural anaesthesia over spinal in this patient?

- A. Complete sensory and motor block

- B. Less volume of anaesthesia is required
 - C. Immediate onset of anaesthesia
 - D. Longer duration of anaesthesia
-

10. A 67-year-old female with a history of osteoarthritis, significantly affecting her quality of life, is scheduled for knee replacement surgery. After exploring her options, she decides to opt for epidural anaesthesia. All of the following are the disadvantages of epidural anaesthesia, except?

- A. High risk of post-dural headache
 - B. Late onset of anaesthesia
 - C. Patchy level of blockade
 - D. Difficult technique
-

11. A 75-year-old male with osteoarthritis is undergoing hip replacement surgery. After obtaining proper consent and following necessary precautions, the catheter was successfully inserted. What is the most likely factor that could hinder the administration of epidural anaesthesia in this patient?

- A. Increase in epidural venous pressure
 - B. Decrease in epidural venous pressure
 - C. Narrowing of epidural space
 - D. Widening of epidural space
-

12. A 30-year-old primigravida presents to the hospital with progressively intense contractions over the past few hours. After discussing her pain management options, she chooses epidural anaesthesia for labour. What is the dermatome level

associated with the first stage of labour?

- A. S2 to S4
 - B. T10 to L1
 - C. L1 to S4
 - D. T10 to S1
-

13. A 37-year-old man is on the surgical table for inguinal hernia repair. At what level of the spinal cord is spinal anaesthesia administered to achieve the preferred level of the block?

- A. L1 - L2
 - B. L5 - S1
 - C. S1 - S2
 - D. L3 - L4
-

14. A 52-year-old male is scheduled for a prostatectomy under regional anaesthesia. The anesthesiologist decides to perform epidural anaesthesia for postoperative pain control. Which of the following needles is typically used for administering epidural anaesthesia?

- A.
 - B.
 - C.
 - D.
-

15. A 65-year-old male with a history of aortic stenosis, hypertension, and recent cerebral venous thrombosis on anticoagulant therapy is scheduled for lower limb

surgery. The anesthesiology team is evaluating the use of spinal anaesthesia. Which of the following is not a contraindication for spinal anaesthesia in this patient?

- A. Aortic stenosis
 - B. Cerebral venous thrombosis
 - C. Hypertension
 - D. Vertebral osteomyelitis
-

16. A 65-year-old male scheduled for foot and ankle surgery under spinal anaesthesia is unable to raise his leg or move his knee but can move his feet. To ensure adequate sensory block, at which dermatomal level should the spinal anaesthesia be achieved, and what is his motor blockade score according to the Modified Bromage Scale?

- A. L2-L3, Score: 2
 - B. L4-L5, Score: 1
 - C. L2-L3, Score: 3
 - D. L4-L5, Score: 2
-

17. A 40-year-old male undergoing hernia repair under spinal anaesthesia is being assessed by the anesthesiologist for factors influencing the level of anaesthesia. Which of the following factors is least likely to affect the level of spinal anaesthesia?

- A. Baricity of the local anaesthetic
 - B. Volume of cerebrospinal fluid
 - C. Age of the patient
 - D. Type of spinal needle
-

18. A 30-year-old female undergoes spinal anaesthesia for a caesarean section. Twenty hours later, she develops a severe occipital headache that worsens when standing up and improves on lying down. Which of the following is the most definitive treatment for her condition?

- A. Oral caffeine
 - B. Supine positioning
 - C. Epidural blood patch
 - D. Intravenous fluids
-

19. A 45-year-old woman undergoes a caesarean section under spinal anaesthesia. During her recovery, she is monitored for potential complications associated with the anaesthesia. Which of the following is not related to spinal anaesthesia?

- A. Transient Neurologic Symptoms
 - B. Post-Dural Puncture Headache
 - C. Respiratory Depression
 - D. Malignant Hyperthermia
-

20. A 58-year-old man undergoing urological surgery under spinal anaesthesia develops hypotension that does not improve with fluid resuscitation. Which of the following should be administered to manage his hypotension?

- A. Metoprolol
 - B. Atropine
 - C. Pethidine
 - D. Ephedrine
-

21. A 63-year-old man with benign prostatic hyperplasia (BPH) is scheduled for prostate surgery. During the pre-anaesthesia evaluation, spinal anaesthesia is selected. To ensure adequate anaesthesia for this procedure, what is the required level of the sensory block?

- A. Block till L1
 - B. Block till L2
 - C. Block till T4
 - D. Block till T10
-

22. A 32-year-old woman is on the surgical table for a lower segment caesarean section (LSCS). During the administration of spinal anaesthesia, the anaesthetist suddenly feels a "snap." Which structure is most likely indicated by this?

- A. Arachnoid mater
 - B. Infraspinal Ligament
 - C. Supraspinous Ligament
 - D. Ligamentum Flavum
-

23. A 27-year-old male diagnosed with testicular torsion is scheduled for emergency surgery. During the pre-anesthesia assessment, the type of anaesthesia was decided. Which of the following about the needle (in the image below) used for administering the necessary anaesthesia is incorrect?



- A. It creates a smaller puncture
 - B. Less chances of Post dural puncture headache
 - C. Used commonly in spinal anaesthesia
 - D. It is a dura-cutting needle
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	4
Question 4	4
Question 5	4
Question 6	4
Question 7	4
Question 8	4
Question 9	4
Question 10	1
Question 11	3
Question 12	2
Question 13	4
Question 14	4
Question 15	3
Question 16	1
Question 17	4
Question 18	3

Question 19	4
Question 20	4
Question 21	4
Question 22	4
Question 23	4

Solution for Question 1:

Correct Answer: A) CSF drops following guidewire removal

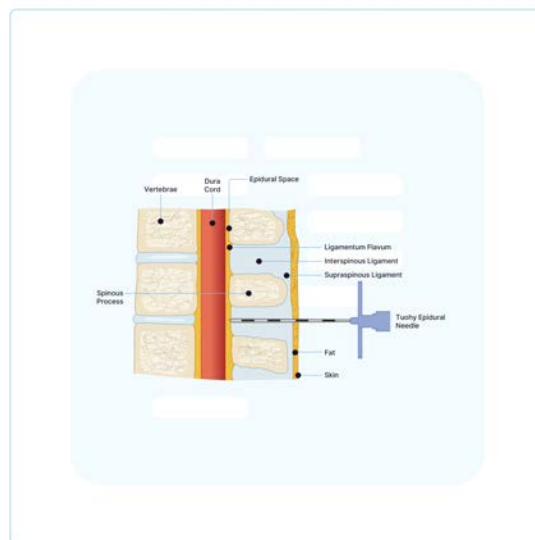
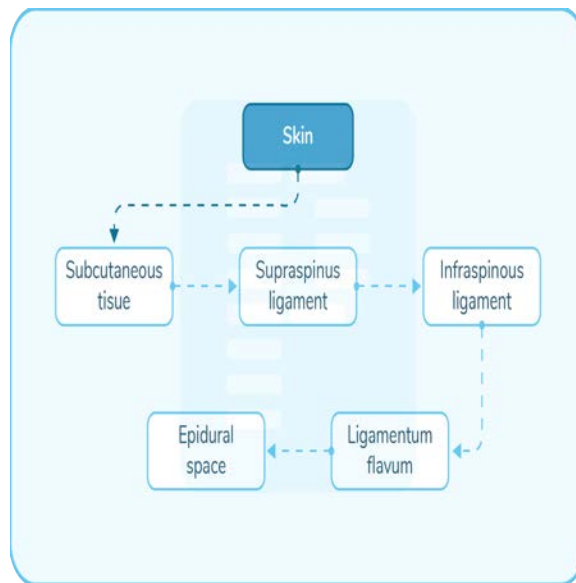
Explanation:

Cerebrospinal fluid (CSF) drops following guidewire removal are observed in spinal anaesthesia, not epidural.

Signs for identification of epidural space:

Loss of Resistance	As the tip of the needle enters the epidural space, there is a sudden loss of resistance. (Option B)
Hanging drop test	The hub of the needle should be filled with solution, allowing a drop to hang from its external opening, which is sucked in due to high negative pressure in the epidural space). (Option C)
Duran sign	If the catheter is placed in the epidural space then there is no resistance for the drug to pass (drug passes easily)
Westphal sign	Absence of PULSATING MOVEMENTS of csf in the catheter confirms correct needle placement in the epidural space.

Structures pierced to reach epidural space:



Solution for Question 2:

Correct Answer: B) 2,3,4

Explanation:

Total spinal block is a condition in which local anaesthetic spreads to cranial levels, leading to profound motor and sensory block, respiratory compromise, and significant hypotension.

Total Spinal Block	
Cause	If the patient is kept in the Trendelenburg position for a longer duration, the cervical and intracranial segments are blocked through CSF during the spinal block. It also happens due to accidental intrathecal injection during caudal and epidural blocks.
Symptoms	Severe hypotension, Bradycardia, Respiratory distress (due to phrenic nerve involvement) Sensory loss and profound motor weakness due to Cephalad spread of local anaesthetic. Loss of consciousness
Management	Oxygen supplementation IV Fluids Intubation and mechanical ventilation, if needed Phenylephrine for Hypotension Atropine for Bradycardia Adrenaline in case of Cardiac arrest

Solution for Question 3:

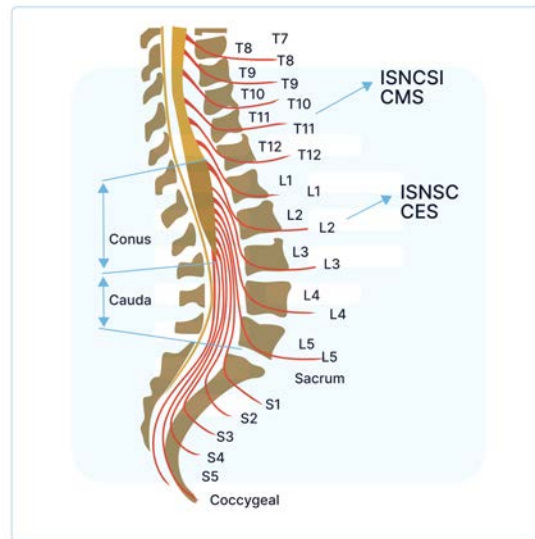
Correct Answer: D) Cauda equina syndrome

Explanation:

- Cauda Equina Syndrome occurs when the cauda equina, a bundle of spinal nerves at the lower end of the spinal cord, is compressed.
- Continuous Spinal Anesthesia (CSA) can potentially lead to this condition if a complication, such as a hematoma or local anaesthetic spread, affects the nerve roots.

Continuous Spinal Anesthesia (CSA)	
Advantages	Disadvantages
Better hemodynamic stability (because of predictable titration of the block) Controls arterial blood pressure (in patients with severe aortic stenosis or pregnant women with complex cardiac disease). It can be used in cases of morbid obesity (as previous spinal surgery may hinder epidural spread) Preferred in a few cases of laparotomies (where general anaesthesia may be a risk).	Postdural puncture headache (PDPH) Cauda equina syndrome: Spinal cord nerve root compression leading to: Difficulty with micturition Loss of anal sphincter tone or faecal incontinence Saddle anaesthesia Gait disturbance Pain, numbness or weakness affecting one or both legs.

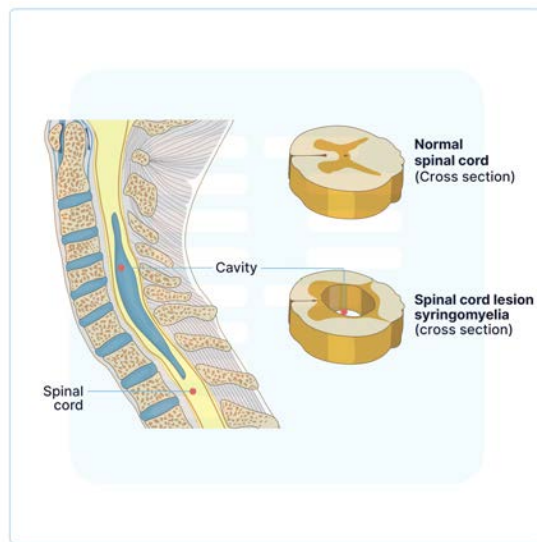
Conus medullaris syndrome (Option A) affects the conus medullaris at the end of the spinal cord and is less commonly associated with CSA. It is characterised by lower limb weakness, increased knee reflex, decreased ankle reflex, and extensor plantar reflex.



Lumbar spinal stenosis (Option B) occurs with a congenitally narrowed lumbar spinal canal, exacerbated by the degenerative changes that commonly occur with age, characterised by exercise-induced weakness and paraesthesia in the legs, but is not directly associated with CSA.

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Syringomyelia (Option C) is a condition characterised by the formation of a cyst (syrinx) within the spinal cord. It leads to pain and numbness in the neck, shoulders, arm, or back and is not specifically associated with CSA.

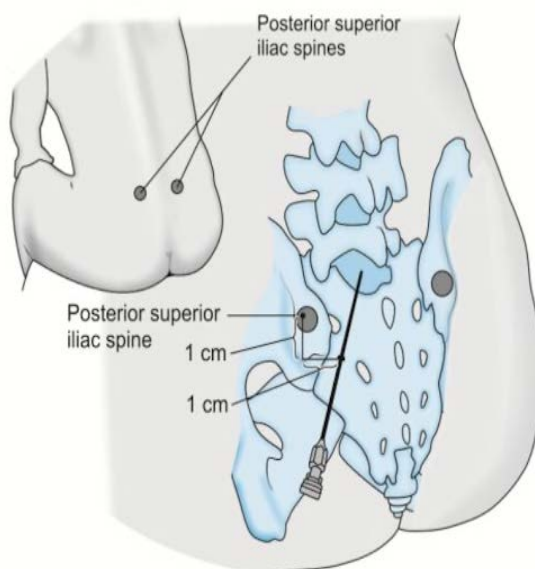


Solution for Question 4:

Correct Answer: D) Taylor's approach

Explanation:

- Taylor's approach, also called a modified Paramedian approach, is done between L5 to S1 space.
- It is used in patients with trauma who cannot tolerate sitting position or in people with deformed spines.
- The needle is inserted 1 cm medial and 1 cm inferior to the posterior superior iliac spine, angled medially and cephalad at 45 to 55 degrees.



Technique	Midline approach	Paramedian approach	Caudal approach
Used in	Spinal & Epidural block (Option A)	Spinal & Epidural block (Option B)	Spinal & Epidural block (mostly in paediatric age groups) (Option C)
Site of puncture	L2-L3 L3-L4 L4-L5	Similar to midline block	S1 - S2
Method	Inserting the needle in the midline.	Inserting the needle 1-2 cm lateral to the midline.	Inserting the needle 45 degrees to the sacrum`

Solution for Question 5:

Correct Answer: D) Rapid recovery of motor function.

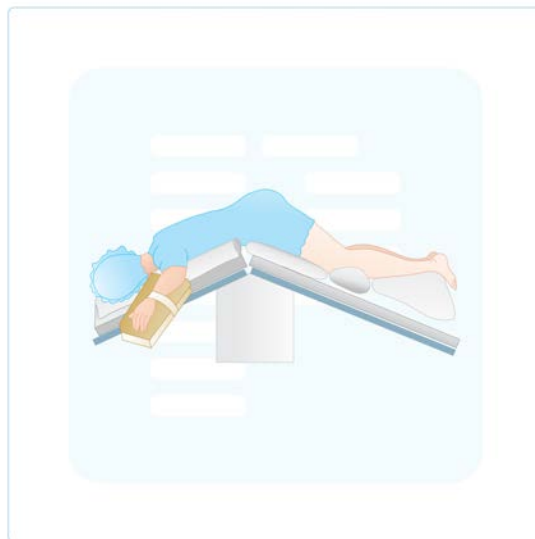
Explanation:

- Saddle block is a type of spinal anaesthesia achieved by keeping the patient in a sitting position for 3 to 5 minutes after the injection. This allows for the blockade of only the lower lumbar and sacral nerves.
- This gives benefits in obstetric, urological, anorectal procedures, and perioperative pain management: Allows the patient to be awake through the procedure (Option A) Reduced

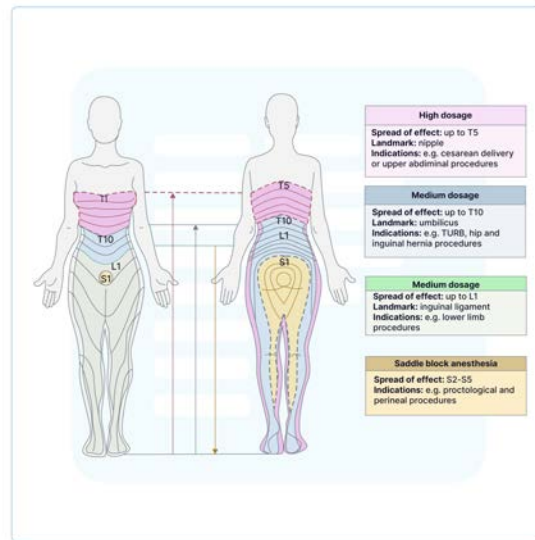
systemic side effects Rapid onset and recovery (Option B) Less post-op nausea
Minimal motor block (Option C)

- Allows the patient to be awake through the procedure (Option A)
- Reduced systemic side effects
- Rapid onset and recovery (Option B)
- Less post-op nausea
- Minimal motor block (Option C)
- Allows the patient to be awake through the procedure (Option A)
- Reduced systemic side effects
- Rapid onset and recovery (Option B)
- Less post-op nausea
- Minimal motor block (Option C)

Complications like urinary retention can be prevented by changing the patient to a jack-knife position after maintaining the initial sitting posture for only one minute.



This has a low incidence of hypotension as it does not induce excessive sympathetic blockade compared to spinal block.



Solution for Question 6:

Correct Answer: D) 5 mL

Explanation:

- The Armitage formula is employed to determine the correct dose of local anaesthetic for caudal anaesthesia, taking into account the patient's weight and the desired concentration.
- This formula reduces the risk of complications while ensuring effective pain relief during procedures.

$$\text{Volume} = \text{Weight (kg)} \times \text{Effective dose (mL/kg)} = 10 \text{ kg} \times 0.5 \text{ mL/kg} = 5 \text{ mL}$$

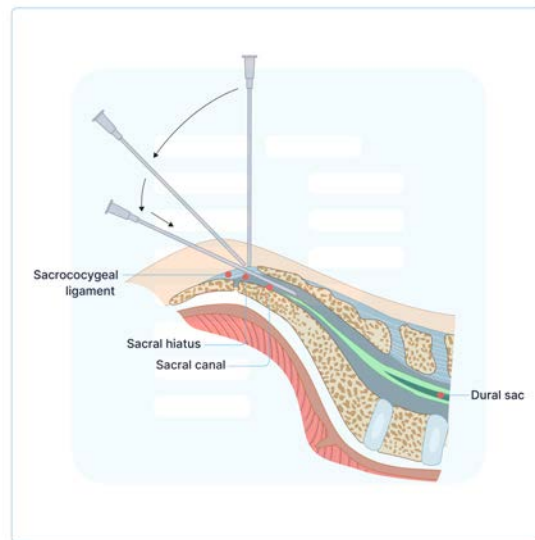
Armitage formula	
Dermatomes to be blocked	Required volume of anaesthesia
All sacral dermatomes	0.5 mL/Kg
All sacral and Lumbar dermatomes	1 mL/Kg
Block till mid-thoracic	1.25 mL/Kg

Solution for Question 7:

Correct Answer: D) Caudal anaesthesia

Explanation:

- Caudal epidural anaesthesia is a common practice in paediatric age groups .
- It involves the insertion of a needle or catheter through the sacrococcygeal ligament, which covers the sacral hiatus formed by the un-fused laminae of S4 and S5.
- It is frequently employed for procedures located below the diaphragm , such as urogenital, rectal, inguinal, and lower extremity surgeries.
- In adults , it is employed for anorectal procedures , as it provides dense sacral sensory blockade with minimal upward spread .



Spinal anaesthesia (Option A) involves injecting anaesthetic into the cerebrospinal fluid but is generally not preferred for young children due to risks of hypotension and prolonged motor blockade.

General anaesthesia (Option B) induces unconsciousness and provides complete analgesia, but it carries risks such as airway complications and is often more than necessary for simple procedures like circumcision.

Local anaesthesia (Option C) consists of infiltrating anaesthetic agents at the surgical site to achieve localised analgesia. While effective for minor procedures, it may not provide adequate

pain relief in paediatric patients who can be uncooperative during surgery.

Solution for Question 8:

Correct Answer: D) Bupivacaine

Explanation:

- Bupivacaine is a long-acting local anaesthetic known for its potency and potential for cardiotoxicity. It can lead to severe cardiac complications, including cardiac arrest, especially when used in high doses or inadvertently injected into a blood vessel.
- CPR and immediate infusion of 20% Intralipid (ACLS Protocol) along with incremental doses of epinephrine has been effective in reversing bupivacaine-induced cardiac toxicity
- It is crucial to ensure that the CSF is free of blood during barbotage (aspirating CSF into a syringe) before administering spinal anaesthesia.
- In Obstetric use, very dilute concentrations of Bupivacaine are often combined with fentanyl for labour analgesia and postoperative pain management.

Ropivacaine (Option A) is also a long-acting local anaesthetic that has a lower risk of cardiovascular toxicity and severe central nervous system effects compared to bupivacaine.

Both can cause transient neurologic symptoms and injection site reactions, but ropivacaine is generally safer for obstetric and outpatient procedures.

Lidocaine (Option B) is a widely used local anaesthetic with a rapid onset and moderate duration. While it can lead to cardiovascular effects at high doses, it is generally well-tolerated and not commonly associated with cardiac arrest in typical clinical use.

Chloroprocaine (Option C) is an ultra-short-acting anaesthetic with a low risk of systemic toxicity. Its rapid metabolism and lower potency significantly decrease the likelihood of severe cardiovascular complications, including cardiac arrest.

Previous formulations of chloroprocaine containing preservatives caused cauda equina syndrome when inadvertently injected in large volumes intrathecally.

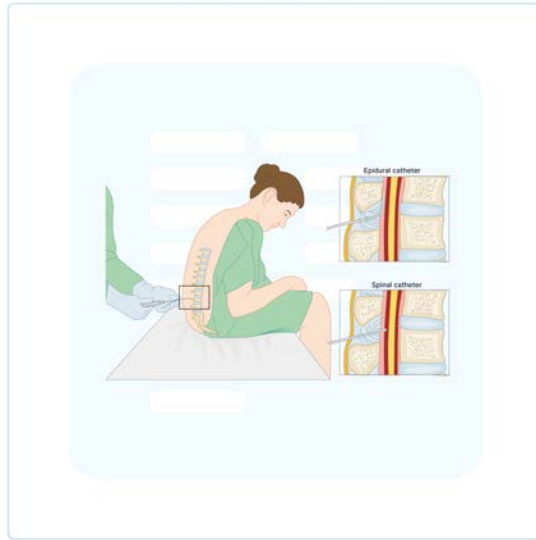
Solution for Question 9:

Correct Answer: D) Longer duration of anaesthesia

Explanation:

- Epidural anaesthesia is a better option for the given condition, as it provides selective, segmental blockade of sympathetic fibres, preserves some sympathetic function, helps maintain vascular tone, and reduces the risk of hypotension.
- It also offers a longer duration of analgesia due to the larger volume of anaesthetic used and allows for continuous infusion.

Criteria	Epidural anaesthesia	Spinal anaesthesia
Site of puncture	Thoracic or lumbar region	Lumbar region
Needle used	Tuohy's needle (thick bore needle)	Sprotte's needle or Whitacre needle (Pencil point tip)
Space where anaesthesia is administered	Epidural space	Sub-arachnoid space
The volume of drugs needed	more	Less (Option B)
The needle position is confirmed by	Hanging drop test Dural sign Westphal sign	Needle drops of CSF, on removing the guide wire Barbotage
Time on onset	Gradual	Faster (Option C)
Duration of anaesthesia	Longer duration due to catheter insertion	Shorter duration
Post-Op pain relief	Fast	Slow
Technique of anaesthesia administration	Comparatively difficult	Simple
PDPH (Post-dural Puncture Headache)	Rare	Common
Complications (Hypotension, Bradycardia, Respiratory distress)	Rare	Common
Quality of block	Patchy or incomplete	Dense (Complete block) (Option A)



Solution for Question 10:

Correct Answer: A) High risk of post-dural headache

Explanation:

Epidural anaesthesia involves a challenging technique, may produce a patchy level of the blockade, has a delayed onset of action, and presents a very low risk of post-dural headache.

Advantages and disadvantages of epidural anaesthesia:

Advantages	Disadvantages
It can be used for long-duration surgeries (catheter) Low risk of Post dural headache (PDH) (Option A) The level of anaesthesia can be altered (drug volume) Hemodynamic stability	Difficult technique (Option D) Patchy (inadequate) level of blockade (due to the presence of fat) (Option C) Accidental dural puncture (severe PDH) Bupivacaine toxicity: Cardiac arrhythmias Lignocaine toxicity: Seizures Not used for emergency surgeries (due to the late onset of anaesthesia) (Option B)

Solution for Question 11:

Correct Answer: C) Narrowing of epidural space

Explanation:

Age-related alterations, such as intervertebral disc degeneration and osteophyte formation, can narrow the epidural space, making it more challenging to place the catheter properly.

Factors affecting epidural anaesthesia:

- Drug factor The larger volume of drugs leads to a higher level of blockade. Typically, 1 to 2 mL of the solution is administered per segment for blockade.
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- Typically, 1 to 2 mL of the solution is administered per segment for blockade.
- The larger volume of drugs leads to a higher level of blockade.
- Typically, 1 to 2 mL of the solution is administered per segment for blockade.
- Patient factor Old age: Narrowing of epidural space (higher level of blockade with the same amount of drug) (Option C) Pregnancy: Increase in epidural venous pressure (decrease in volume of drug required). (Option A ruled out)
- Old age: Narrowing of epidural space (higher level of blockade with the same amount of drug) (Option C)
- Pregnancy: Increase in epidural venous pressure (decrease in volume of drug required). (Option A ruled out)
- Old age: Narrowing of epidural space (higher level of blockade with the same amount of drug) (Option C)
- Pregnancy: Increase in epidural venous pressure (decrease in volume of drug required). (Option A ruled out)
- Procedure factor The level of injection affects the spread as:
 - The level of injection affects the spread as:
 - The level of injection affects the spread as:

Site of injection	Spread
Upper cervical	Caudal
Midthoracic	Cephalad & caudal (both equally)
Low thoracic	Cephalad (primarily)

- Effect of additives: Addition of epinephrine prolongs the duration of the following anaesthetics: Chloroprocaine Lidocaine Mepivacaine Bupivacaine Ropivacaine Etidocaine Levobupivacaine
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- Bupivacaine
- Ropivacaine
- Etidocaine
- Levobupivacaine
- Chloroprocaine
- Lidocaine
- Mepivacaine
- Bupivacaine
- Ropivacaine
- Etidocaine
- Levobupivacaine

Solution for Question 12:

Correct Answer: B) T10 to L1

Explanation

The dermatome for the first stage of labour spans from T10 to L1.

Epidural Anesthesia in Labor:

Site	L2-L3 and L4-L5.
Purpose	Provides a sensory block while maintaining intact motor function, allowing the mother to participate actively in labour.
Dermatome level	1st Stage of labour: T10 to L1 (Option B) 2nd Stage of Labor: S2 to S4 (Option A ruled out)
Benefits	Decreased maternal catecholamines Effective pain relief Increased patient satisfaction Ability to quickly achieve surgical anaesthesia (emergency caesarean delivery).
Dose	Bupivacaine: 0.0625% to 0.125% Ropivacaine: 0.1% to 0.2%
Onset and Duration	The onset of action typically occurs within 10-20 minutes, and depending on the concentration and volume used, the effects can last for several hours.
Monitoring	Continuous maternal and foetal monitoring is essential during the administration of epidural anaesthesia.

Solution for Question 13:

Correct Answer: D) L3 - L4

Explanation:

- Spinal anaesthesia is ideally administered below the termination of the spinal cord, known as the conus medullaris, which typically ends at the L1 - L2 vertebral level in adults. Therefore, the preferred spaces for administering spinal anaesthesia in adults are between L2 and L3, L3 and L4, or L4 and L5.
- The L3 - L4 interspace is commonly chosen due to its accessibility and the presence of cerebrospinal fluid (CSF).

- In infants, the preferred space for spinal anaesthesia is between L4 and L5, as their conus medullaris is situated lower in the vertebral column.
- The L4 - L5 space is easily identified by the intercrystal line (Tuffier's line), an imaginary line drawn between the highest points of the iliac crests, which typically corresponds to the L4 spinous process.

Positioning for Spinal Anesthesia

Spinal anaesthesia can be administered in two positions based on the patient's condition:

- Lateral decubitus position : Allows easier access to the lumbar interspaces and is often more comfortable.
- Sitting position: Facilitates the spread of the anaesthetic and is used when lateral positioning is not suitable.

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Solution for Question 14:

Correct Answer: D)

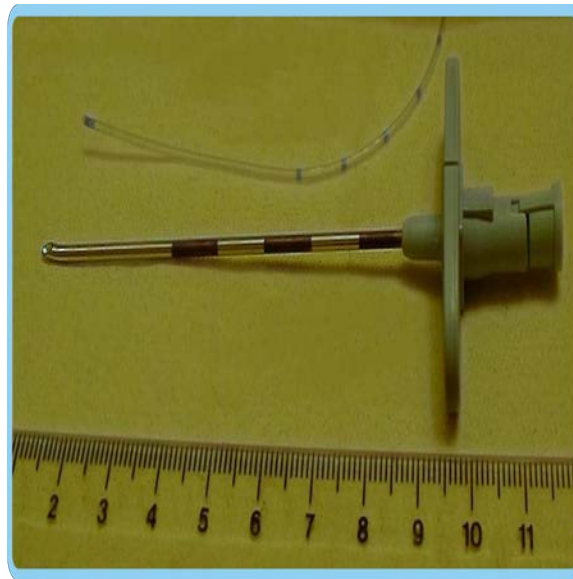


Explanation:

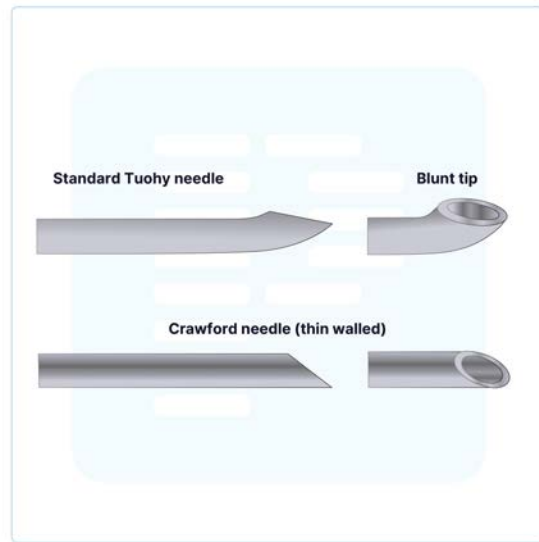
The needle depicted in the image is a Tuohy needle (Option D), which is preferred for epidural anaesthesia. This involves delivering local anaesthetic into the epidural space, typically with the insertion of a catheter to extend the duration of pain relief.

Epidural needles

- Tuohy Epidural Needle: A thick-bore needle is used in epidural anesthesia (to minimise damage to the dura and facilitate the insertion of the epidural catheter). On reaching the epidural space, the guidewire is removed, and the catheter is then placed. The needle has alternating dark and light bands at 1 cm intervals to aid in depth measurement during insertion. Specifications: Gauge: 18G Size: 9 to 10 cm (3.5 inches long)
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- Gauge: 18G
- Size: 9 to 10 cm (3.5 inches long)



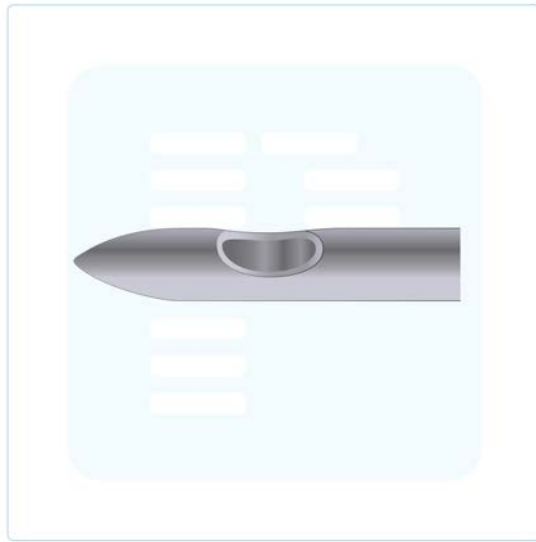
- Crawford needle Straight needles without a curved tip Greater incidence of dural puncture
- Straight needles without a curved tip
- Greater incidence of dural puncture
- Straight needles without a curved tip
- Greater incidence of dural puncture



It is the image of the Quincke Needle (Dura cutting) used in spinal anaesthesia. (Option A):



It is the image of Sprotte Needle (conical, pencil-point tip; Dura splitting) is used in spinal anaesthesia. (Option B):



It is the image of Whitacre Needle with a conical, pencil-point tip (Dura splitting) used mainly in spinal anaesthesia. (Option C):



Solution for Question 15:

Correct Answer: C) Hypertension

Explanation:

- Hypertension is not a contraindication for spinal anaesthesia.
- Spinal anaesthesia can be safely used in patients with well-controlled hypertension.
- It is essential to monitor blood pressure closely during and after the procedure, as spinal anaesthesia can cause hypotension due to sympathetic blockade.

Contraindications of Spinal Anesthesia	
Absolute Contraindications	Patient refusal Localized sepsis Allergy to planned medications Patient's inability to remain still Raised intracranial pressure (ICP)
Relative Contraindications	Myelopathy or Peripheral Neuropathy Spinal Stenosis Previous spine surgery Neural Tube Defects Inherited Coagulopathy System infections Cardiac valvular stenosis Multiple sclerosis Anticoagulant medication

Aortic stenosis (Option A) decreases the heart's ability to maintain adequate blood flow. The sudden drop in blood pressure from spinal anaesthesia could lead to dangerously low coronary perfusion and cardiac ischemia.

Cerebral venous thrombosis (Option B), combined with anticoagulant therapy, is a contraindication for spinal anaesthesia due to the risk of elevated intracranial pressure and bleeding complications.

Vertebral osteomyelitis (Option C): Introducing a needle into an infected area carries a high risk of spreading the infection to the spinal canal, potentially leading to meningitis or epidural abscess.

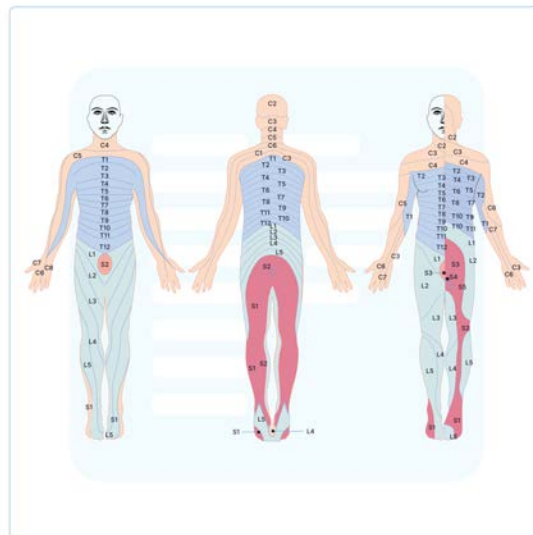
Solution for Question 16:

Correct Answer: A) L2-L3, Score: 2

- Adequate sensory block for foot and ankle surgery typically requires a dermatomal level of L2-L3.
- A Bromage scale score of 2 indicates partial motor block, where the patient is unable to raise an extended leg or move a knee but can move feet.

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A Bromage scale score of 2 indicates partial motor block, where the patient is unable to raise an extended leg or move a knee but can move feet.



Dermatomal Level Required for Common Surgical Procedures	
Type of Surgery	Dermatomal Level
Upper abdominal surgery	T4
Caesarean delivery	
Transurethral resection of prostate	T10
Hip surgery	
Foot and ankle surgery	L2
Block Monitoring	
Motor block - Modified Bromage Scale	Sensory block
0: No motor block 1: Inability to raise extended leg; able to move knees and feet 2: Inability to raise extended leg and move knee; able to move feet 3: Complete block of motor limb	Cold sensation Pinprick Light touch

Solution for Question 17:

Correct Answer: D) Type of spinal needle.

Explanation:

The type of needle used for the spinal injection does not have a significant impact on the level of anesthesia achieved but it may affect the risk of complications, such as post-dural puncture headache.

Factors Affecting Level of Spinal Anesthesia		
Factors	Effect	
Drug factors	Dose	Higher the dose, the higher the level of anaesthesia

Baricity (Option A)	Hypobaric drug floats in CSF, leading to a higher level of anaesthesia Hyperbaric drug settles down, resulting in a lower level of anaesthesia.	
Patient factors	CSF volume (Option B)	Higher cerebrospinal fluid (CSF) volume dilutes the anaesthetic, reducing its spread and resulting in a lower level of spinal anaesthesia.
Age (Option C)	Advanced age is associated with ↓CSF volume, leading to a higher level of spinal anaesthesia.	
Pregnancy	Pregnancy increases intra-abdominal pressure, leading to a higher level of spinal anaesthesia due to enhanced cephalad spread of anaesthetic.	
Procedural factors	Patient position	In the sitting position, hyperbaric drugs settle lower in the spinal column, resulting in a lower level of anaesthesia. In the Trendelenburg position, hyperbaric drugs move cephalad (toward the neck), leading to a higher level of anaesthesia.
Epidural Injection Post-Spinal Anesthesia	It increases the level of anaesthesia by compressing the subarachnoid space and enhancing drug spread.	

Solution for Question 18:

Correct Answer: C) Epidural blood patch

Explanation: The patient's symptoms are indicative of a post-dural puncture headache (PDPH), which occurs due to a cerebrospinal fluid (CSF) leak following a puncture of the dura mater during procedures like spinal anaesthesia or epidural catheter placement.

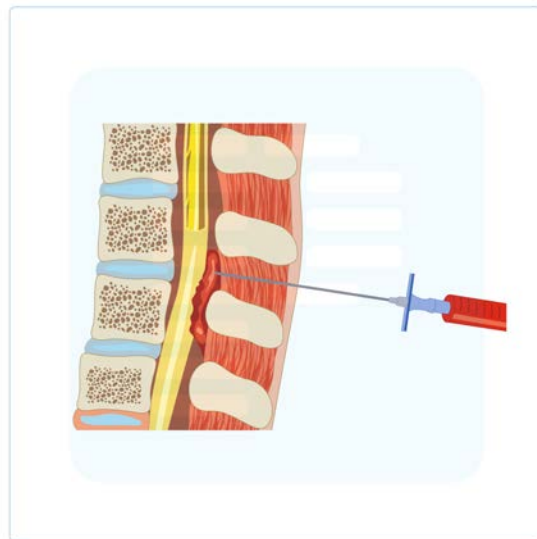
- The onset is usually after 12-72 hours following the procedure and the pain is bilateral, frontal, retro orbital, or occipital and extends into the neck

The most definitive treatment for PDPH is an epidural blood patch.

Management of post-dural puncture headache (PDPH):

- Conservative management: A supine position (Option B) can help reduce the headache by decreasing the pressure gradient and allowing the cerebrospinal fluid (CSF) to stabilize.
- Hydration (Option C) : Ensuring adequate fluid intake can help replenish lost CSF and alleviate symptoms.
- Oral caffeine (Option A) (500mg) can constrict blood vessels and increase CSF production, providing symptomatic relief.
- Analgesics such as acetaminophen or non-steroidal anti-inflammatory drugs (NSAIDs) like ibuprofen may help manage pain associated with PDPH.
- A supine position (Option B) can help reduce the headache by decreasing the pressure gradient and allowing the cerebrospinal fluid (CSF) to stabilize.
- Hydration (Option C) : Ensuring adequate fluid intake can help replenish lost CSF and alleviate symptoms.
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- Analgesics such as acetaminophen or non-steroidal anti-inflammatory drugs (NSAIDs) like ibuprofen may help manage pain associated with PDPH.
- Definitive Management: An epidural blood patch is ideally performed 24 hours after the dural puncture. It involves injecting 15-20 mL of autologous blood into the epidural space to create a clot that seals the dural puncture, providing relief from the headache. If the first blood patch does not provide sufficient relief, a second epidural blood patch can be administered 24 to 48 hours later.
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- If the first blood patch does not provide sufficient relief, a second epidural blood patch can be administered 24 to 48 hours later.
- Prevention: by using a smaller gauge needle (Pencil Point needles) for spinal procedures and minimizing the number of punctures. Ultrasound guidance for lumbar puncture.

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- If the first blood patch does not provide sufficient relief, a second epidural blood patch can be administered 24 to 48 hours later.
- by using a smaller gauge needle (Pencil Point needles) for spinal procedures and minimizing the number of punctures.
- Ultrasound guidance for lumbar puncture.



Solution for Question 19:

Correct Answer: D) Malignant Hyperthermia

Explanation:

- Malignant Hyperthermia is more likely associated with general anaesthesia, not spinal anaesthesia.

- It is primarily triggered by volatile anaesthetics (like halothane, isoflurane, and sevoflurane) and the depolarising muscle relaxant (succinylcholine).
- It presents as tachypnea, tachycardia, muscle rigidity, hyperthermia, elevated blood pressure, and raised end-tidal CO₂ levels.

Transient Neurological Symptoms (Option A) typically occur after spinal anaesthesia, presenting as pain in the back radiating to the legs without any neurological deficits.

Post-dural puncture headache (Option B) presents as a severe headache that occurs due to a CSF leak through the puncture in the dura mater following spinal anaesthesia.

Respiratory Depression (Option C) occurs following a higher level of spinal anaesthesia or in patients with preexisting respiratory disease.

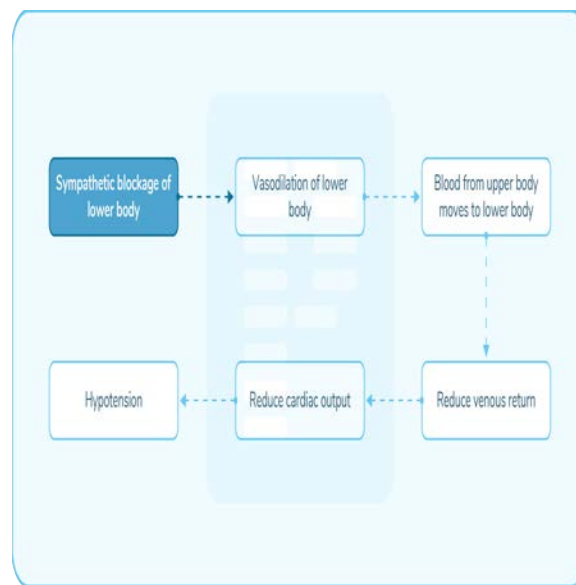
Post-Op complications of Spinal Anesthesia	
Neurological	Nerve injury Epidural hematoma Transient Neurologic Symptoms Postdural puncture headache Arachnoiditis Meningitis
Cardiovascular	Hypotension Bradycardia Cardiac arrest
Other complications	Urinary retention (Most common) Respiratory depression Backache Nausea & vomiting

Solution for Question 20:

Correct Answer: D) Ephedrine

Explanation:

Hypotension is the most commonest intraoperative complication of spinal anaesthesia due to,



- It can also happen due to adrenal gland blockage.
- Preloading with IV fluids , typically crystalloids , before spinal anaesthesia can help prevent Hypotension.
- If this occurs mid-surgery, the patient can initially be resuscitated with fluids; if the condition does not improve, it can be managed by Vasopressors like Phenylephrine (the drug of choice in Pregnancy) or Ephedrine.
- Ephedrine acts by increasing both cardiac output and systemic vascular resistance.

Other intraoperative complications of Spinal anaesthesia are

- Bradycardia can be managed by Atropine (Option B) and shivering with Pethidine (Option C) .
- Respiratory distress can happen primarily due to total spinal anaesthesia (block of cervical and intracranial segment due to prolonged Trendelenburg position) leading to phrenic nerve paralysis or high spinal anaesthesia in which the block is achieved above the desired level. This should be managed by laryngoscopy and intubation .
- Apnea can happen more likely due to prolonged hypotension or medullary hypoperfusion than phrenic nerve paralysis.
- Cardiac arrest is a potential possibility preceded by hypotension and bradycardia, which can be avoided by appropriate treatment of such complications and immediate management by Adrenaline .

Metoprolol (Option A) : Administering beta-blockers would worsen the hypotension , as it would further decrease cardiac contractility and slow the heart rate, which is contraindicated in hypotensive states caused by spinal anaesthesia.

Solution for Question 21:

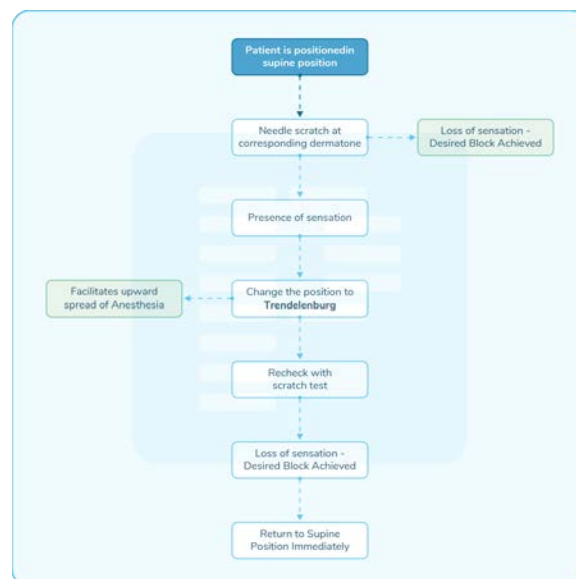
Correct Answer: D) Block till T10

Explanation:

- Spinal anaesthesia is generally preferred for surgeries involving areas below the umbilicus and for procedures lasting between 2 to 3 hours.
- For prostate surgery, a spinal block is administered up to the T10 level; however, the required level can vary for different procedures.

Type of surgery	Preferred dermatome for spinal block
Prostate Surgery	Block till T10
Lower segment caesarean section (LSCS)	Block till T4 or T6 (Option C)
Diabetic foot	L1 or L2 (Option A and B)

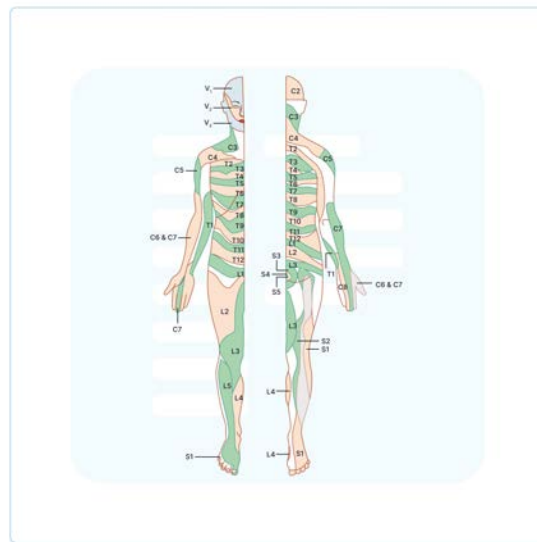
Assessing the level of block:



- Failure to return the patient to the supine position quickly may cause the anaesthesia to ascend beyond the intended level, leading to high spinal anaesthesia.

- If the anaesthesia reaches the cervical or intracranial segments, it can cause total spinal anaesthesia, increasing the risk of hypotension, bradycardia, and respiratory distress.

Dermatome level	Landmark Area
T4	Nipples
T6	Sternum
T8	Area between Xiphoid and Umbilicus
T10	Umbilicus
T12	Pubic symphysis



Solution for Question 22:

Correct Answer: D) Ligamentum Flavum

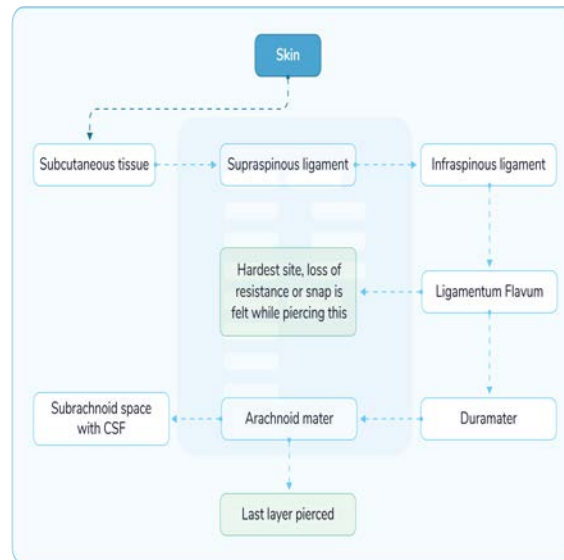
Explanation:

Ligamentum flavum is the toughest ligament and is easily identifiable during a lumbar puncture for spinal anaesthesia, as it presents as a sudden loss of resistance or a snap.

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Order of structures pierced during lumbar puncture:



Solution for Question 23:

Correct Answer: D) It is a dura-cutting needle

Explanation:

- The image depicts a Whitacre needle, a dura-separating needle (not a dura-cutting needle) commonly used in spinal anaesthesia. (Option C ruled out)
- It has a conical, pencil-point tip with a small orifice, which is useful in reducing the incidence of Post dural puncture headache (PDPH)

Types of needles used in Spinal anaesthesia:

Dura Separating Needle	Dura Cutting needle
Sprotte's needle and Whitacre's needle	Quincke's needle
The pencil point tip creates a smaller puncture. (Option A ruled out)	A sharp tip makes a larger puncture
Reduced chances of PDPH (Option B ruled out)	Highly prone to PDPH due to CSF leak

25, 26 or 27 G are regularly used	Rarely used these days

Previous Year Questions

1. Which of the following local anesthetics causes hypertension?

- A. Cocaine
 - B. Lignocaine
 - C. Bupivacaine
 - D. Prilocaine
-

2. A patient presented with headache, vomiting, and fever. Meningitis was suspected. While performing a lumbar puncture, which is the last structure you will pierce just before entering the lumbar cistern?

- A. Ligamentum flavum
 - B. Dura mater
 - C. Arachnoid membrane
 - D. Pia mater
-

3. A young male patient was given a regional block with 0.25% bupivacaine for a minor surgical procedure. Shortly after the injection, the patient suddenly became unresponsive and the pulse was not recordable. Which of the following would be the best course of management for this patient?

- A. CPR with 20% intralipid
- B. CPR with sodium bicarbonate
- C. CPR with dobutamine
- D. CPR with calcium

4. What is the mechanism of action of local anesthetics?

- A. Block sodium channels
 - B. Block potassium channels
 - C. Block calcium channels
 - D. Block chloride channels
-

5. During resuscitation, a patient received a 20% intralipid injection. What could have been the most probable diagnosis?

- A. Local anaesthesia toxicity
 - B. Asphyxia due to drowning
 - C. Electrocution
 - D. Pulseless electrical activity
-

6. A smoker who was given an interscalene block for an arthroscopic acromioclavicular joint reconstruction 2 days ago was followed up by a chest X-ray which showed as below. He is asymptomatic at present and is comfortably resting on the bed. What is the next step in management?



- A. Insert chest drain as soon as possible
 - B. Do thoracostomy
 - C. Elective intubation and IPPV
 - D. Wait and watch and give supplementary oxygen if needed
-

7. Which of the following structures are not pierced during epidural anesthesia?

- A. Supraspinous ligament
 - B. Interspinous ligament
 - C. Ligamentum flavum
 - D. Dura
-

8. The best type of anesthesia for pre eclampsia is:

- A. Spinal anesthesia
- B. Epidural anesthesia

- C. General anesthesia
 - D. Caudal block
-

9. Spinal anaesthesia in an adult is given at this level:

- A. T12-L1
 - B. L1-L2
 - C. L3-L4
 - D. L5-S1
-

10. The duration of spinal anaesthesia is based directly on:

- A. Dose
 - B. Height
 - C. Age
 - D. Total body fat
-

11. Which is the anesthesia of choice in a daycare dental surgery?

- A. General anesthesia
 - B. Spinal anesthesia
 - C. Epidural anesthesia
 - D. Infiltration anesthesia
-

12. Duration of action of Proparacaine:

- A. 2 min
 - B. 5 min
 - C. 40 min
 - D. 20 min
-

13. In epidural anesthesia, Which of the following layers is least likely to be punctured during the procedure?

- A. Arachnoid mater
 - B. Ligamentum flavum
 - C. Supraspinatus
 - D. Infraspinalis
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	1
Question 4	1
Question 5	1
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	1
Question 11	4
Question 12	4
Question 13	1

Solution for Question 1:

Correct Option A - Cocaine:

- Cocaine, a potent local anesthetic, has vasoconstrictive properties that can lead to hypertension and cardiovascular complications. It blocks catecholamine reuptake, increasing

neurotransmitter levels and stimulating alpha-adrenergic receptors in vascular smoothmuscle, causing vasoconstriction and elevated blood pressure. Additionally, cocaine induce tachycardia and arrhythmias, worsening cardiovascular effects.

Incorrect Options:

Options B, C and D: Lignocaine, bupivacaine, and prilocaine are common local anesthetics. Though they may cause cardiovascular effects like lowered blood pressure or heart rate, they are not known to induce hypertension.

Solution for Question 2:

Correct Option C - Arachnoid membrane:

- While entering the lumbar cistern, the last structure you puncture is the arachnoid membrane. This is because it is the innermost layer.

Incorrect Options:

Option A - Ligamentum flavum: The ligamentum flavum is the middlemost layer that you shall see while doing a lumbar puncture.

Option B - Dura mater: The Dura mater is the second last layer, which lies above the arachnoid membrane.

Option D - Pia mater: The Pia mater is not the last layer observed during the lumbar puncture.

Solution for Question 3:

Correct Option A - CPR with 20% intralipid:

- The symptoms exhibited by the patient are of local anesthesia systemic toxicity , LAST, which is best treated with CPR with 20% intralipid.

Incorrect Options:

Option B - CPR with sodium bicarbonate: CPR with sodium bicarbonate is a preferred line of action in case of metabolic acidosis.

Option C - CPR with dobutamine: CPR with dobutamine is preferred, with the necessary action to increase the heart rate. However, it doesn't help with LAST.

Option D - CPR with calcium: CPR with calcium can definitely help in managing certain types of cardiac arrest problems. However, it cannot be the only line of action in the case of LAST.

Solution for Question 4:

- Correct Options A- Block sodium channels:
- The mechanism of action of local anaesthetics primarily involves the blockage of sodium channels in nerves.

Correct Options A- Block sodium channels:

Local anesthetics block sodium channels in nerve fibers, preventing the influx of sodium ions responsible for initiating action potentials and transmitting pain signals. Their binding to sodium channels is voltage-dependent, meaning they target channels in open and inactivated states, preferentially blocking pain signal transmission while sparing non-painful sensory or motor nerve activity.

Incorrect Options:

- Option b. Block potassium channels: While potassium channels regulate the repolarization phase of the action potential, their blockade is not an important mechanism of action for local anaesthetics.
 - Option c. Block calcium channels: Blockade of calcium channels is more commonly associated with other drugs, such as calcium channel blockers.
 - Option d. Block chloride channels: While some local anesthetics may impact chloride channel function by altering membrane potential, their primary action is sodium channel blockade.
-

Solution for Question 5:

Correct Option A-Local Anaesthesia Toxicity:

Intralipid is a lipid emulsion commonly used to treat local anaesthetic systemic toxicity (LAST). Local anaesthetic agents, such as lidocaine or bupivacaine, can cause systemic toxicity when they enter the bloodstream excessively. Intralipid is a rescue therapy for LAST because it helps sequester the local anaesthetic agent, reducing its toxic effects.

Incorrect Options:

Option B: Asphyxia due to drowning: While drowning can cause significant hypoxia and require resuscitation, the administration of intralipid during resuscitation is not directly related to drowning.

Option C: Electrocution: Although resuscitation may be required in cases of Electrocution, the administration of intralipid is not a specific intervention for electrocution-related injuries.

Option D: Pulseless electrical activity (PEA): Intralipid can treat certain cardiac arrest rhythms like LAST-induced ventricular arrhythmias, but it's not specific to PEA.

Solution for Question 6:

Correct Option D -Wait and watch and give supplementary oxygen if needed:

- This is the most appropriate option in this case. Since the patient is asymptomatic and resting comfortably, close observation is warranted. If any signs of respiratory distress or hypoxia develop, supplementary oxygen can be provided as needed.

Incorrect Options:

Option A: Insert chest drain as soon as possible

- Since the patient is asymptomatic and comfortably resting, there is no indication for immediate chest drain insertion.

Option B: Do thoracostomy

- Thoracostomy is not necessary in this case as there are no indications of a pneumothorax or hemothorax.

Option C: Elective intubation and IPPV

- Elective intubation and invasive positive pressure ventilation (IPPV) are not warranted in this scenario as the patient is asymptomatic and comfortably resting.
-

Solution for Question 7:

Correct Option D - Dura:

- Dura is not pierced during epidural anesthesia. The local anesthetic is injected using a Tuohy needle into the epidural space which lies between dura and ligamentum flavum and the site of action is the nerve roots.

Incorrect Options:

Option A - Supraspinous ligament: Supraspinous ligament is pierced during epidural anesthesia.

Option B - Interspinous ligament: Interspinous ligament is pierced during epidural anesthesia.

Option C - Ligamentum flavum: Ligamentum flavum is pierced during epidural anesthesia.

Solution for Question 8:

Correct Option B - Epidural anesthesia:

- Epidural anesthesia is often considered the best choice for anesthesia in pre-eclampsia because because in preeclampsia there is risk of airway edema that can lead to difficult intubation, therefore continuous epidural anesthesia avoid that need of intubation

Incorrect Options:

Option A - Spinal anesthesia: Spinal anesthesia may cause a sudden drop in blood pressure, which can be problematic in pre-eclampsia where blood pressure control is important.

Option C - General anesthesia: it can be used in pre-eclampsia, it is not necessarily the best choice solely based on the risk of airway edema, but indicated in severe preeclampsia and coagulopathy.

Option D - Caudal block: This option is incorrect.

Solution for Question 9:

Correct Option C - L3-L4:

- Spinal anesthesia is a regional anesthesia technique used to block sensation and motor function in a specific area of the body. The level at which spinal anesthesia is typically administered in adults is the L3-L4 vertebral level. The anesthetic agent is injected into the cerebrospinal fluid (CSF) in the subarachnoid space at this level, allowing for selective anesthesia below that level.

Incorrect Options:

Option A - T12-L1: This level is higher than the usual level for spinal anesthesia and is not commonly used.

Option B - L1-L2: This level is also higher than the usual level for spinal anesthesia and is not commonly used.

Option D - L5-S1: This level is lower than the usual level for spinal anesthesia and is not typically used for this technique.

Solution for Question 10:

Correct Option A - Dose:

- The duration of spinal anesthesia, which refers to the period of anesthesia achieved by administering local anesthetics into the subarachnoid space, is primarily based on the dose of the anesthetic agent used. The higher the dose, the longer the duration of anesthesia.

Incorrect Options:

Option B - Height: The duration of spinal anesthesia is not directly influenced by a person's height.

Option C - Age: The duration of spinal anesthesia is not directly determined by a person's age.

Option D - Total body fat: The duration of spinal anesthesia is not directly related to a person's total body fat.

Solution for Question 11:

Correct Option D -Infiltration anesthesia :

- Infiltration anesthesia is a form of local anesthetic giving anesthesia only to the applied region and the patient is fully awake. This type of anesthesia is preferred for dental / minor procedures.

Incorrect Options:

Option A - General anesthesia: General anesthesia is not preferred for daycare dental procedures..

Option B -Spinal anesthesia: Spinal anesthesia is a form of regional anesthesia given at L3-L4 OR L4-L5 level, preferred in fracture femur/inguinal hernia/umbilical hernia surgeries.

Option C - Epidural anesthesia: Epidural anesthesia is given in pregnancy.

Solution for Question 12:

Correct Option D - 20 min:

- The duration of action of Proparacaine is relatively short, around 20 minutes. This means that its effects will typically last for approximately 20 minutes after application. It provides temporary and localized anesthesia to the eye, allowing for the performance of necessary procedures without causing significant discomfort to the patient.

Incorrect Options:

Option A - 2 min: This option suggests an extremely short duration of action, which is not consistent with the typical effects of Proparacaine.

Option B - 5 min: This option suggests a slightly longer duration of action compared to option A, but it is still shorter than the correct answer

Option C - 40 min: This option suggests a significantly longer duration of action, which is not consistent with the properties of Proparacaine.

Solution for Question 13:

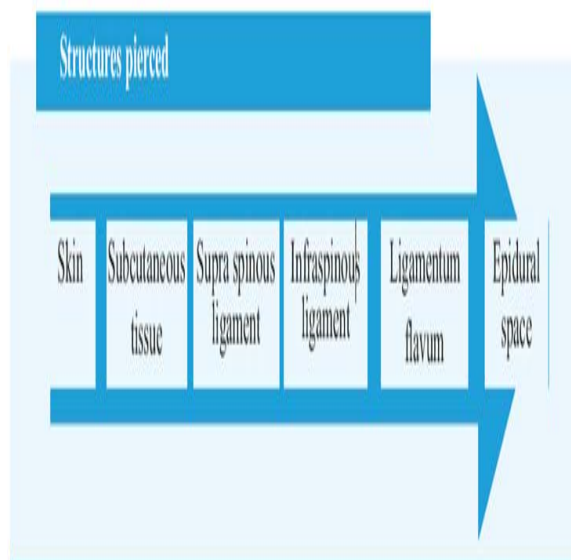
Correct Option A - Arachnoid mater :

- It is least likely to be punctured in epidural anesthesia

Incorrect Options:

Option B, C, and D

- Epidural anesthesia is used for longer duration below umbilicus surgery, painless labor



Muscle Relaxants

1. A patient undergoing rapid sequence induction of anaesthesia receives succinylcholine as a muscle relaxant. Shortly after administration, the anesthesiologist notices muscle fasciculations followed by flaccid paralysis. What is the mechanism responsible for these observed effects?

- A. Prolonged depolarization of the muscle membrane
 - B. Competitive inhibition of Ach at the neuromuscular junction
 - C. Rapid hydrolysis by acetylcholinesterase
 - D. Blockade of presynaptic nicotinic receptors
-

2. Which muscle is the first to regain function after muscle relaxation?

- A. Triceps
 - B. Adductor pollicis
 - C. Quadriceps
 - D. Diaphragm
-

3. What type of muscle relaxant works by blocking acetylcholine at the neuromuscular junction?

- A. Depolarizing muscle relaxant
 - B. Non-depolarizing muscle relaxants
 - C. Centrally acting muscle relaxants
 - D. Cholinesterase inhibitor
-

4. During surgery, a patient receives a long-acting steroidal non-depolarizing muscle relaxant that is mainly excreted by the kidneys. Which drug is likely used?

- A. Vecuronium
 - B. Rocuronium
 - C. Pancuronium
 - D. Gallamine
-

5. A patient undergoing prolonged surgery is given the most effective muscle relaxant that undergoes Hofmann elimination. Which of the following benzyliisoquinolinium muscle relaxant has been administered?

- A. Alcuronium
 - B. d-Tubocurarine
 - C. Atracurium
 - D. Cisatracurium
-

6. A 45-year-old woman with a known history of myasthenia gravis is undergoing an emergency appendicectomy. The anesthesiologist plans to use muscle relaxants for rapid tracheal intubation. The patient is currently on pyridostigmine. Which of the following is the most appropriate muscle relaxant of choice for this patient?

- A. Succinylcholine at a standard dose
 - B. Rocuronium with sugammadex available for reversal
 - C. Succinylcholine at a reduced dose
 - D. Pancuronium for longer paralysis duration
-

7. A 30-year-old male is brought to the emergency department after a motor vehicle accident and requires rapid tracheal intubation. The patient has a history of spinal cord injury from a previous accident 6 months ago. Which of the following muscle relaxants should be used with utmost caution?

- A. Rocuronium
 - B. Suxamethonium
 - C. Vecuronium
 - D. Atracurium
-

8. A 40 y/o man undergoes surgery and receives suxamethonium for rapid sequence induction. After surgery, he experiences prolonged apnea. Genetic testing reveals he has a homozygous atypical pseudocholinesterase enzyme. What is the most likely explanation for his prolonged apnea?

- A. Increased acetylcholinesterase activity
 - B. Increased sensitivity to suxamethonium due to genetic predisposition
 - C. Interaction with a non-depolarizing muscle relaxant
 - D. Hyperkalemia caused by suxamethonium
-

9. A 45-year-old male is undergoing surgery and is given repeated doses of suxamethonium for muscle relaxation. After some time, his muscle response starts to fade despite continuous stimulation. This is most likely due to the development of a phase II block. Which of the following is not a characteristic feature of phase II block with suxamethonium?

- A. It is associated with a fade in muscle response during repetitive nerve stimulation
- B. It can be influenced by prolonged exposure to suxamethonium

- C. It responds predictably to cholinesterase inhibitors like neostigmine
 - D. Fast-twitch muscle fibres are more susceptible to phase II block than slow-twitch fibres
-

10. A 30-year-old male with a history of extensive burns sustained two weeks ago requires rapid sequence intubation for surgery. The anaesthetist considers using suxamethonium for muscle relaxation. Which of the following is the most appropriate reason to avoid using suxamethonium in this patient?

- A. Risk of increased intracranial pressure
 - B. Risk of phase II block
 - C. Risk of hyperkalemia
 - D. Risk of myalgia
-

11. A 28-year-old man with a history of asthma is undergoing elective surgery. He is induced with isoflurane and given succinylcholine for muscle relaxation. Within minutes, the surgical team observes that he has developed severe muscle rigidity, a rapid heart rate, and a notable increase in body temperature. Which drug/s is/are most likely responsible for the onset of these symptoms in this patient?

- A. Isoflurane only
 - B. Succinylcholine only
 - C. Both isoflurane and succinylcholine
 - D. Neither isoflurane nor succinylcholine
-

12. During a routine orthopaedic surgery, a 35-year-old female patient develops muscle rigidity, tachycardia, and elevated body temperature shortly after being administered halothane and succinylcholine. Which of the following actions should the anesthesiologist take first to manage this patient's condition?

- A. Administer dantrolene and bicarbonate immediately.
 - B. Stop halothane and succinylcholine, and hyperventilate with 100% oxygen.
 - C. Administer dantrolene and continue the surgery with halothane only.
 - D. Administer cold IV fluids with dantrolene.
-

13. Which of the following factors will not potentiate the effect of neuromuscular blockade?

- A. Hypercalcemia, Alkalosis
 - B. Hypermagnesemia, Hypothermia
 - C. Lithium, Tetracycline
 - D. Local anesthetics, Clindamycin
-

14. A 45-year-old female presents to the emergency department with altered mental status, flushed skin, and tremors after starting a new antidepressant. On examination, there is hyperreflexia in the lower extremities, muscle rigidity, and clonus. Which syndrome should be considered in her case?

- A. Malignant Hyperthermia
 - B. Serotonin Syndrome
 - C. Neuroleptic Malignant Syndrome
 - D. Heat Exhaustion
-

15. A 30-year-old female patient is scheduled for laparoscopic cholecystectomy under daycare anaesthesia. Which muscle relaxant would be appropriate to ensure rapid recovery while minimising residual effects?

- A. Succinylcholine

- B. Cisatracurium
 - C. Rocuronium with Sugammadex
 - D. Pancuronium
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	2
Question 4	3
Question 5	4
Question 6	2
Question 7	2
Question 8	2
Question 9	3
Question 10	3
Question 11	3
Question 12	2
Question 13	1
Question 14	2
Question 15	3

Solution for Question 1:

Correct answer: A) Prolonged depolarization of the muscle membrane

Explanation:

- Succinylcholine is a depolarizing neuromuscular blocker. It mimics acetylcholine (ACh) and binds to nicotinic receptors at the neuromuscular junction, causing prolonged depolarization of the muscle membrane.
- This initial depolarization results in muscle fasciculations, followed by flaccid paralysis because the muscle fibers cannot repolarize, leading to the observed effects.

Competitive inhibition of ACh at the neuromuscular junction (Option B): This is characteristic of nondepolarizing muscle relaxants like vecuronium. These agents prevent acetylcholine from binding to nicotinic receptors, without causing depolarization, and thus do not induce fasciculations.

Rapid hydrolysis by acetylcholinesterase (Option C): Unlike ACh, which is rapidly hydrolyzed by acetylcholinesterase (AChE), succinylcholine is not quickly broken down. It primarily depends on plasma cholinesterase (butyrylcholinesterase) for metabolism. This rapid metabolism accounts for the short duration of succinylcholine's action but does not explain the muscle fasciculations.

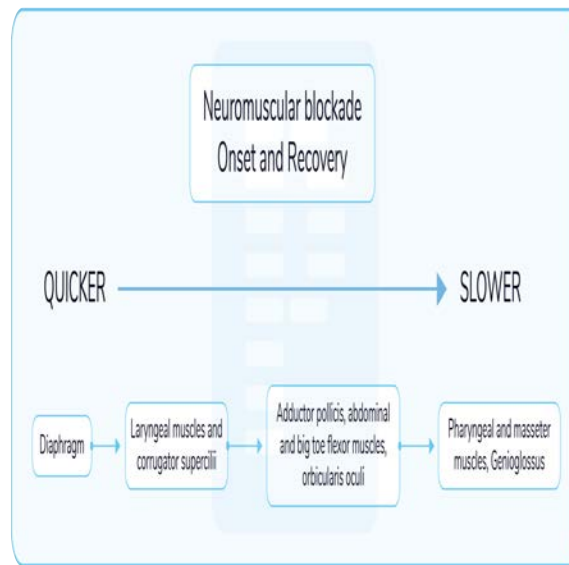
Blockade of presynaptic nicotinic receptors (Option D): Succinylcholine acts directly on postsynaptic receptors, causing continuous depolarization and fasciculations. It does not act on presynaptic receptors.

Solution for Question 2:

Correct answer: D) Diaphragm

Explanation:

The diaphragm, a central muscle involved in breathing, receives a high blood supply, making it one of the first to recover after muscle relaxants wear off. The fast recovery of the diaphragm is important for restoring normal respiratory function following surgery.



Triceps (Option A): The triceps, located in the upper arm, are peripherally situated and recover later than centrally located muscles. The slower return of function in muscles like the triceps is due to their lower blood supply and distant location from central circulation.

Adductor pollicis (Option B): Located in the hand, this peripheral muscle tends to recover more slowly because it has lower blood flow compared to central muscles. It is useful for monitoring complete neuromuscular recovery.

Quadriceps (Option C): Quadriceps exhibit a slower recovery because they are more distant from central circulation and have less blood flow compared to muscles like the diaphragm.

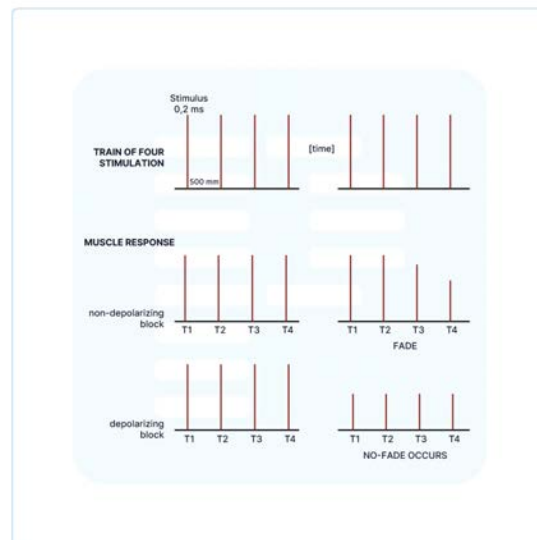
Solution for Question 3:

Correct answer: B) Nondepolarizing muscle relaxants

Explanation:

Skeletal muscle relaxants		
	Depolarizing Muscle Relaxants	Non-depolarizing Muscle Relaxants

Mechanism	Mimic acetylcholine, binds to the Acetylcholine receptors, causing continuous depolarization of the muscle membrane. (Option A ruled out)	Block acetylcholine by competitively binding to nicotinic acetylcholine receptors at the neuromuscular junction, preventing depolarization. (Option B)
Effect	Causes an initial phase of muscle contractions (fasciculations), followed by paralysis.	Causes muscle paralysis without initial muscle contraction.
Action	Not reversed by cholinesterase inhibitors but effects wear off rapidly due to fast metabolism by butyrylcholinesterase.	Reversed by cholinesterase inhibitors that increase acetylcholine levels.
Examples	Succinylcholine, decamethonium	Vecuronium, Rocuronium, Pancuronium



Train-of-four (TOF) stimulation involves four supramaximal stimuli delivered at 2 Hz over 2 seconds, with responses measured as a TOF count (0-4) or a TOF ratio (0-1 or 0-100%).

TOF in Depolarizing Blockade : In depolarizing blockade, the amplitude of all four muscle responses decreases equally, resulting in a TOF ratio of 1.0, meaning no fade is observed.

TOF in Non-Depolarizing Blockade : In non-depolarizing blockade, the TOF ratio decreases as the block intensifies, with 1-2 muscle responses indicating a 90-95% block, and a fade occurs between the fourth and first responses.

Centrally acting muscle relaxants (Option C): These drugs work within the central nervous system and are primarily used to reduce muscle spasticity or tension in conditions like multiple sclerosis or spinal cord injuries. They do not directly interfere with acetylcholine at the neuromuscular junction.

Cholinesterase inhibitor (Option D): Cholinesterase inhibitors, like neostigmine, increase the levels of acetylcholine at the neuromuscular junction by inhibiting the enzyme that breaks it down. They do not block acetylcholine; instead, they enhance its action.

Solution for Question 4:

Correct answer: C) Pancuronium

Explanation:

Pancuronium is an aminosteroid muscle relaxant with a long duration of action. It is primarily excreted through the kidneys, making it suitable for patients with normal kidney function. Its limited metabolism means it maintains its effectiveness throughout the surgical procedure.

Vecuronium (Option A): This is an intermediate-acting aminosteroid muscle relaxant. Unlike pancuronium, vecuronium undergoes significant hepatic metabolism.

Rocuronium (Option B): It is an intermediate-acting steroidal muscle relaxant and its primary elimination occurs in the liver and therefore does not undergo renal excretion.

Gallamine (Option D): It is now rarely used in clinical practice due to its side effects. Gallamine blocks both the acetylcholine recognition site and the ion channel of the Nicotinic acetylcholine receptors (nAChR). Gallamine is categorized as a benzyloquinoline compound, distinguishing it from steroidal agents.

Solution for Question 5:

Correct answer: D) Cisatracurium

Explanation:

- Cisatracurium is the most potent benzyloquinolinium muscle relaxant. It is a single isomer of atracurium and is metabolised through Hofmann elimination.

- Due to its high potency, it is administered in lower doses compared to atracurium, making it highly effective in prolonged surgeries.
- Its breakdown is independent of kidney or liver function, reducing the risk of accumulation and toxicity in patients with organ dysfunction.

Alcuronium (Option A): Its pharmacokinetics and metabolic pathways are less understood and not as well-documented as those of atracurium or cisatracurium. It is not commonly used in contemporary clinical practice, especially in prolonged surgeries.

d-Tubocurarine (Option B): d-Tubocurarine is a long-acting, non-depolarizing muscle relaxant that is excreted unchanged by the kidneys. It has largely been replaced by more modern agents due to its prolonged effects and higher risk of side effects, such as histamine release and ganglionic blockade.

Atracurium (Option C): Atracurium is also metabolized via Hofmann elimination but is less potent than cisatracurium. The byproduct laudanosine, which results from Hofmann elimination, can cause seizures in critically ill patients, limiting its use in prolonged procedures.

Solution for Question 6:

Correct Answer: B) Rocuronium with sugammadex available for reversal

Explanation:

Myasthenia Gravis (MG) involves reduced acetylcholine (ACh) receptors at neuromuscular junctions due to an autoimmune process, causing muscle weakness and fatigue. The patient is on pyridostigmine, which is an anticholinesterase drug to help with her MG.

Rocuronium is a steroidal neuromuscular blocker, and sugammadex provides a safer, effective reversal option in MG patients, especially in those receiving anticholinesterase treatment.

- Succinylcholine in MG : Larger doses (1.5-2.0 mg/kg) may be needed for rapid tracheal intubation due to fewer ACh receptors. However, anticholinesterase treatment prolongs the effect due to decreased cholinesterase activity. (Options A&C; ruled out)
- Neuromuscular blockade monitoring should be done as it is essential for guiding the administration and reversal of muscle relaxants.
- Sugammadex : Offers safer and more effective reversal of steroidal neuromuscular blockers (e.g., rocuronium, vecuronium), especially in MG patients, where neuromuscular recovery can be more complex.

Pancuronium for longer paralysis duration (Option D): Pancuronium is a non-depolarizing muscle relaxant. It is generally not used in patients with MG as it is unpredictable, due to uneven muscle weakness in these patients.

Solution for Question 7:

Correct Answer: B) Suxamethonium

Explanation:

- Suxamethonium (succinylcholine) is a depolarizing muscle relaxant. In patients with a history of spinal cord injury, especially several months after the injury, there is an increased risk of hyperkalemia due to upregulation of acetylcholine receptors in the muscle.
- This can lead to excessive potassium release when succinylcholine is administered, potentially causing life-threatening cardiac arrhythmias. Hence, it should be used with utmost caution.

Depolarizing Muscle Relaxants

- Examples: Suxamethonium, decamethonium
- They act as agonists at nicotinic acetylcholine receptors (nAChRs), mimicking acetylcholine by binding and activating these receptors.

Mechanism of Blockade :

- Unlike acetylcholine, depolarizing agents are not rapidly broken down by acetylcholinesterase, hence causing prolonged depolarization.
- Sustained depolarization leads to : The desensitisation of nAChRs, d/t continued exposure
Inactivation of voltage-gated sodium channels , blocking further muscle contractions.
Increased potassium permeability , potentially increasing serum potassium levels, especially in patients with upregulated AChRs
- The desensitisation of nAChRs, d/t continued exposure
- Inactivation of voltage-gated sodium channels , blocking further muscle contractions.
- Increased potassium permeability , potentially increasing serum potassium levels, especially in patients with upregulated AChRs
- The desensitisation of nAChRs, d/t continued exposure
- Inactivation of voltage-gated sodium channels , blocking further muscle contractions.

- Increased potassium permeability , potentially increasing serum potassium levels, especially in patients with upregulated AChRs
- Phase I Block : Initial depolarization causes muscle fasciculations (involuntary contractions).
- Phase II Block : After depolarization, acetylcholine receptors become desensitized, and the end plate repolarizes, but the receptors remain blocked, leading to paralysis.

Succinylcholine	
Side Effects	Contraindications
Bradycardia and ventricular dysrhythmias Hyperkalemia Increased Intracranial Pressure Myalgia Anaphylaxis Malignant Hyperthermia	Metabolic acidosis Burns Hypovolemia Muscular Dystrophies Upper and Lower motor denervation Neuromuscular disease

Rocuronium and vecuronium & Atracurium (Option A, C & D) are non-depolarizing muscle relaxants that do not cause significant potassium release (leading to cardiac arrhythmias), like how succinylcholine does.

Solution for Question 8:

Correct Answer: B) Increased sensitivity to suxamethonium due to genetic predisposition

Explanation:

Suxamethonium (succinylcholine) is broken down by pseudocholinesterase (plasma cholinesterase).

- In individuals with a genetic variant that leads to atypical pseudocholinesterase, the breakdown of suxamethonium is significantly slower, resulting in prolonged muscle paralysis and apnea.
- This condition is more common in individuals with a homozygous atypical variant of the pseudocholinesterase enzyme, causing them to be more sensitive to the drug and leading to a prolonged duration of action.

Genetic predisposition :

- Genetic variants (e.g., homozygous atypical enzyme) reduce enzyme activity, slowing suxamethonium breakdown and increasing the risk of prolonged apnea. (Option B)
- The dibucaine number test can assess pseudocholinesterase function, indicating susceptibility to prolonged apnea.

- The prolonged paralysis and apnea are treated with continued mechanical ventilation and sedation until muscle function returns to normal.

Dibucaine Number	
Definition	The dibucaine number is a test used to measure butyrylcholinesterase enzyme activity, which breaks down muscle relaxants like succinylcholine. Dibucaine inhibits normal butyrylcholinesterase more effectively than the abnormal variant. This test measures the extent of enzyme inhibition and helps determine the presence of the abnormal genetic variant.
Activity	Normal butyrylcholinesterase is inhibited by 80% (dibucaine number of 80). Atypical butyrylcholinesterase is inhibited by 20% (dibucaine number of 20), indicating reduced enzyme function.
Clinical Importance	Helps identify individuals with atypical enzyme variants who may experience prolonged paralysis after succinylcholine use. Useful for guiding anaesthesia decisions to prevent complications.
Enzyme Concentration	The test does not measure the enzyme concentration, which can be influenced by various factors like diseases or medications.

Increased acetylcholinesterase activity (Option A) : Acetylcholinesterase does not significantly metabolize suxamethonium.

Interaction with a non-depolarizing muscle relaxant (Option C) : Suxamethonium can interact with other muscle relaxants (e.g., mivacurium), prolonging their effects and potentially exacerbating respiratory muscle paralysis but is not the most likely cause in this scenario as there is the presence of atypical pseudocholinesterase.

Hyperkalemia caused by suxamethonium (Option D) : While suxamethonium can cause hyperkalemia in certain conditions, the main cause of prolonged apnea here is genetic pseudocholinesterase deficiency.

Solution for Question 9:

Correct Answer: C) It responds predictably to cholinesterase inhibitors like neostigmine

Explanation:

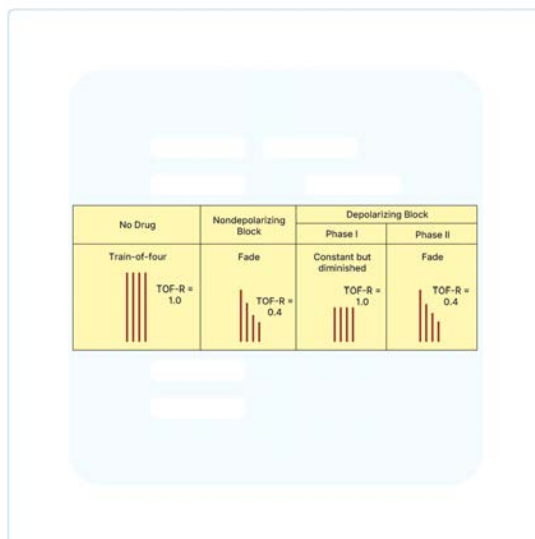
The response to cholinesterase inhibitors is unpredictable and may even worsen the block.

Phase II Block (accommodation):

It occurs with continuous or repeated exposure to depolarizing muscle relaxants, particularly suxamethonium, and is characterised by a gradual fade in muscle response. (Option A ruled out)

Mechanisms Behind Phase II Block:

- Prejunctional effects : High concentrations of suxamethonium can block prejunctional acetylcholine receptors (AChRs), affecting acetylcholine synthesis and release.
- Postjunctional effects : Fade may also stem from changes in postjunctional AChR sensitivity or kinetics, resulting in reduced response despite the continued presence of the relaxant.



TOF of Phase II depolarizing block

Factors Influencing Phase II Block:

- Duration of exposure : Prolonged exposure to suxamethonium increases the chance of developing Phase II block. (Option B ruled out)
- Drug concentration : Higher suxamethonium concentrations make Phase II block more likely.
- Muscle type : Fast-twitch muscle fibres are more susceptible to Phase II block than slow-twitch fibres. (Option D ruled out)
- Interaction with other agents : Anesthetics or other drugs can affect the development of Phase II block.

Reversal of Phase II Block:

- Cholinesterase inhibitors (e.g., neostigmine) : Not recommended as the response is unpredictable and could worsen the block. (Option C) .

Solution for Question 10:

Correct Answer: C) Risk of hyperkalemia

Explanation:

Suxamethonium can cause a dangerous release of potassium in patients with burns or muscle trauma due to upregulated acetylcholine receptors, leading to life-threatening hyperkalemia and cardiac arrhythmias.

Side effects:

Bradycardia	2. Hyperkalemia: Increases serum potassium (0.5 mEq/dL) due to potassium efflux from muscle cells. Exaggerated in burns, muscle trauma, and neuromuscular diseases, leading to life-threatening hyperkalemia.
Myalgia	4. Masseter Muscle Rigidity An increase in the tone of the masseter muscle may be an early indicator of malignant hyperthermia. Malignant hyperthermia can be triggered in susceptible individuals by suxamethonium.

Risk of increased intracranial pressure (Option A) : While suxamethonium can increase intracranial pressure, this is not the primary concern in patients with recent burns.

Risk of Phase II block (Option B) : Phase II block can occur with repeated or prolonged use, but it is not the main concern in burn patients.

Risk of myalgia (Option D) : Myalgia is a known side effect of suxamethonium, but it is not the primary contraindication in this scenario.

Solution for Question 11:

Correct Answer: C) Both isoflurane and succinylcholine

Explanation:

- In this case, the diagnosis of malignant hyperthermia is typically made based on the presence of characteristic clinical signs—such as muscle rigidity, tachycardia, and hyperthermia—following exposure to certain anaesthetic agents.
- Both isoflurane (an inhaled anaesthetic) and succinylcholine (a depolarizing muscle relaxant) are well-known triggers for malignant hyperthermia.
- The combination of these agents heightens the risk of developing this serious condition, which is indicated by the rapid onset of symptoms in the patient after their administration.

Triggers of Malignant Hyperthermia:

- Inhaled general anesthetics Ether Halothane Enflurane Isoflurane Desflurane Sevoflurane
- Ether
- Halothane
- Enflurane
- Isoflurane
- Desflurane
- Sevoflurane
- Depolarizing muscle relaxant Succinylcholine
- Succinylcholine
- Prior exercise, even just an hour before anaesthesia induction, intensifies the severity and accelerates the onset of rigidity.
- Ether
- Halothane
- Enflurane
- Isoflurane
- Desflurane
- Sevoflurane
- Succinylcholine

Initial Indicators of Malignant Hyperthermia (MH) During Anesthesia:

- Muscle rigidity
- Masseter spasm
- Tachycardia

- Unexplained hypercarbia
- Increased temperature

Other Features: Cyanosis, metabolic acidosis, hypertension, fever, sweating, Hyperkalemia, Hyperphosphatemia, Myoglobinemia, Myoglobinuria, Cardiac Arrest, DIC, etc.

Solution for Question 12:

Correct Answer: B) Stop halothane and succinylcholine, and hyperventilate with 100% oxygen.

Explanation:

The priority in managing malignant hyperthermia is to immediately discontinue the triggering agents—halothane and succinylcholine—and then hyperventilate the patient with 100% oxygen. This helps to counteract hypoxia and stabilize the patient's condition.

Diagnosis of malignant hyperthermia

- Best screening test- Creatine kinase test
- Gold Standard Diagnostic test- Halothane-Caffeine Test

Treatment for malignant hyperthermia

Administer dantrolene and bicarbonate immediately. (Option A): While dantrolene is critical, it should not be the first action. Stopping the triggers and ensuring oxygenation takes priority.

Administer dantrolene and continue the surgery with halothane only. (Option C): Continuing halothane use after administering dantrolene can worsen the condition.

Administer cold IV fluids with dantrolene. (Option D): While cold IV fluids and dantrolene are important for treatment, they should not be the first actions. Stopping the anesthetics and providing oxygen must come first for immediate stabilization.

Solution for Question 13:

Correct Answer: A) Hypercalcemia, Alkalosis

Explanation:

Both hypercalcemia and alkalosis can reduce the effectiveness of neuromuscular blockers thus shortening their effects.

Factors That Potentiate the Effect of Neuromuscular Blockade

Solution for Question 14:

Correct answer: B) Serotonin Syndrome

Explanation:

	Malignant Hyperthermia (Option A ruled out)	Serotonin Syndrome (Option B)	Neuroleptic Malignant Syndrome (Option C ruled out)
Definition	A life-threatening condition triggered by certain anaesthetics. Often linked to mutations in the ryanodine receptor (RyR1) gene.	A potentially life-threatening condition resulting from excessive serotonergic activity.	A rare, potentially fatal condition linked to dopamine receptor antagonists, primarily antipsychotics.
Triggers	Volatile anaesthetics (e.g., halothane, isoflurane) and succinylcholine.	Use of serotonergic medications, antidepressants, drug interactions, or overdose (e.g., SSRIs, tramadol).	High-potency antipsychotics (e.g., haloperidol), rapid withdrawal of dopaminergic drugs.
Mechanism	Dysregulation of calcium release in skeletal muscle leads to hypermetabolic state.	Increased serotonin synthesis, impaired metabolism, or enhanced receptor activation.	D2 receptor blockade leads to decreased dopaminergic activity and calcium handling issues.

Symptoms	Rapidly increasing body temperature (up to 1°C every 5 minutes) Muscle rigidity Metabolic acidosis	Altered mental status (e.g., agitation, confusion) Autonomic instability (e.g., tachycardia, hypertension) Neuromuscular signs (e.g., tremors, clonus) Hyperreflexia, positive Babinski sign, muscle rigidity, and flushed skin.	Severe muscle rigidity Hyperthermia Altered mental status (e.g., confusion, agitation) Autonomic dysfunction (e.g., fluctuations in blood pressure)
Management	Administer dantrolene to reduce calcium release and muscle rigidity.	Discontinue serotonergic agents. Supportive care, including IV fluids and benzodiazepines	Discontinue neuroleptics and provide supportive care. Consider bromocriptine or dantrolene in severe cases.

Heat Exhaustion (Option D): While this condition can cause hyperthermia and altered mental status, it does not explain the muscle rigidity and the other symptoms observed in this patient.

Solution for Question 15:

Correct answer: C) Rocuronium with Sugammadex

Explanation:

- An ideal muscle relaxant used in daycare surgery should have a rapid onset, minimal side effects, and a rapid and reliable recovery to avoid postoperative residual curarization (PORC).
- Rocuronium, when used with sugammadex, allows for rapid onset of muscle relaxation and a reliable and fast reversal of the neuromuscular blockade. This is ideal for day-care procedures where a quick recovery and discharge are essential.
- Sugammadex binds rocuronium directly, providing a rapid and safe reversal of muscle paralysis.

Succinylcholine (Option A): This muscle relaxant offers a rapid onset and an ultrashort duration of action. However, it is often avoided in daycare anaesthesia due to the risk of severe side effects, including malignant hyperthermia, hyperkalemia, and postoperative muscle pain, which can complicate recovery.

Cisatracurium (Option B): Cisatracurium is an intermediate-acting muscle relaxant that is also safe and effective. However, its onset is slower than rocuronium, and it lacks a specific reversal

agent as fast as sugammadex, which can prolong the time to discharge in daycare settings.

Pancuronium (Option D): This long-acting muscle relaxant is known for significant residual neuromuscular effects, which can lead to extended recovery times. It also increases the risk of postoperative pulmonary complications, which is inappropriate for daycare anaesthesia.

Previous Year Questions

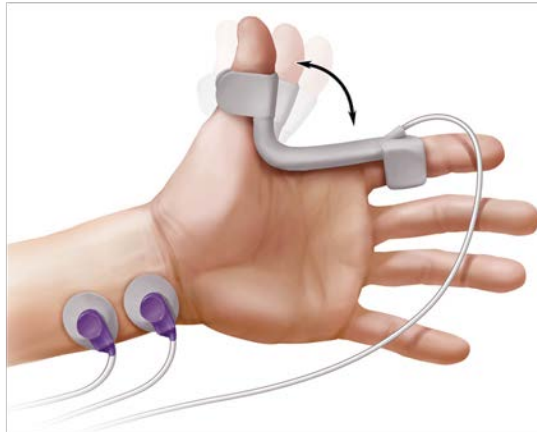
1. A 45-year-old patient undergoing surgery was administered 85 mg of rocuronium to induce anesthesia, but the induction failed. Which drug should be administered for immediate reversal of the rocuronium?

- A. Neostigmine
 - B. Sugammadex
 - C. Glycopyrrolate
 - D. Edrophonium
-

2. Which of the following conditions is not reversed with the administration of atropine?

- A. Bradycardia
 - B. Muscle weakness due to neuromuscular blockade
 - C. Excessive salivation
 - D. Bronchoconstriction
-

3. The image given below shows neuromuscular monitoring of the patient after anesthesia. What is the most commonly used nerve for monitoring?



- A. Ulnar nerve
 - B. Median nerve
 - C. Radial nerve
 - D. Metacarpal nerve
-

4. Which method is considered the most effective for evaluating the extent of muscle relaxation?

- A. Tetanic stimulation
 - B. Double-burst stimulation
 - C. Train of four
 - D. Electromyography
-

5. A patient is given a nicotinic receptor antagonist as a muscle relaxant. Which drug is given postoperatively to recover from muscle weakness?

- A. Physostigmine
 - B. Neostigmine
 - C. Carbachol
 - D. Succinylcholine
-

6. What is the mechanism of action of the curare group of muscle relaxants?

- A. Persistently depolarizing at neuromuscular junction
 - B. Competitively blocking the binding of ACh to its receptors
 - C. Repetitive stimulation of ACh receptors on muscle endplate
 - D. Inhibiting the calcium channels on presynaptic membrane
-

7. A patient who received a supraclavicular brachial plexus block with 0.5% bupivacaine develops hypotension and arrhythmias. What is the first-line treatment for this situation?

- A. Invasive ventilation
 - B. 20% intralipid infusion
 - C. Sodium bicarbonate
 - D. Cardiopulmonary resuscitation
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	2

Solution for Question 1:

Correct Option B - Sugammadex:

- Sugammadex is a modified gamma-cyclodextrin that is used to reverse steroidal non-depolarizing neuromuscular blocking drugs rocuronium and vecuronium.
- It works by rapidly encapsulating free rocuronium in the plasma, which decreases its concentration. This creates a concentration gradient that drives more rocuronium out of the neuromuscular junction and into the plasma, where it is then encapsulated by sugammadex.
- It increases the speed of neuromuscular blockade reversal and greatly reduces the risk of residual neuromuscular paralysis.

Incorrect Options:

Options A , C, and D are less effective than sugammadex for reversing rocuronium.

Solution for Question 2:

Correct Answer: B) Muscle weakness due to neuromuscular blockade

Explanation:

Muscle weakness due to neuromuscular blockade is not reversed by atropine administration.

Atropine is an antimuscarinic (anticholinergic) agent that competitively blocks muscarinic acetylcholine receptors. It has no effect on nicotinic acetylcholine receptors at the neuromuscular junction, which are responsible for skeletal muscle contraction. Therefore, atropine cannot reverse muscle weakness caused by neuromuscular blocking agents.

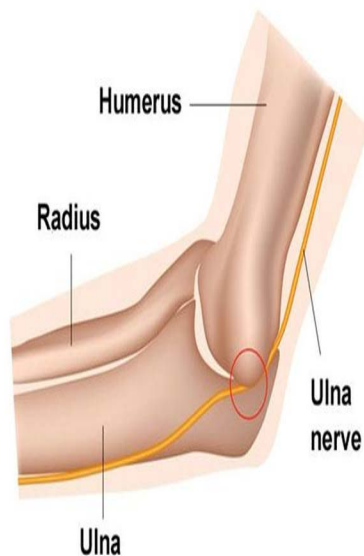
Atropine	
Classification	Anticholinergic drug (Muscarinic receptor antagonist) Competitive antagonist at muscarinic receptors
Mechanism of Action	Competitive inhibition of postganglionic acetylcholine muscarinic receptors Blocks all subtypes of muscarinic receptors (M ₁ , M ₂ , M ₃) Direct vagolytic action leading to parasympathetic inhibition Allows preexisting sympathetic stimulation to predominate
Pharmacological Actions	
1. CNS Effects	Overall CNS stimulant action Stimulates medullary centers (vagal, respiratory, vasomotor) Depresses vestibular excitation (antimotion sickness property) Suppresses tremor and rigidity in parkinsonism High doses: cortical excitation, restlessness, hallucinations, delirium Very high doses: respiratory depression and coma
2. CVS	Heart: Tachycardia (primary effect) - blocks M ₂ receptors on SA node (Option A Ruled Out) Initial transient bradycardia may occur with IM/SC injection Abbreviates refractory period of A-V node Facilitates A-V conduction Shortens P-R interval Blood Pressure: No consistent/marked effect on BP Tachycardia and vasomotor center stimulation tend to raise BP Histamine release and direct vasodilator action (at high doses) tend to lower BP Blocks vasodepressor action of cholinergic agonists

3. Eye	Mydriasis (pupil dilation) Cycloplegia (paralysis of accommodation) Abolition of light reflex Effects last 7-10 days with topical application May increase intraocular tension, especially in narrow angle glaucoma Has mild local anesthetic action on cornea
4. Smooth Muscles	Relaxes all visceral smooth muscles with parasympathetic innervation (M ³ blockade) Reduces tone and amplitude of GI contractions Slows passage of chyme; may cause constipation Causes bronchodilation , reduces airway resistance (Option D Ruled Out) Relaxes ureter and urinary bladder Relaxation of biliary tract (less marked) Minimal effect on uterus
5. Glandular Secretions	Markedly decreases sweat, salivary , tracheobronchial and lacrimal secretions (M ³ blockade) (Option C Ruled Out)
6. Body Temperature	Higher doses can cause rise in body temperature Due to inhibition of sweating and stimulation of temperature regulating center Children are highly susceptible to atropine fever
Relative Sensitivity of Organs	Decreasing order of sensitivity: Saliva, sweat, bronchial secretion > eye, bronchial muscle, heart > smooth muscle of intestine, bladder > gastric glands and smooth muscle

Solution for Question 3:

Correct Option A - Ulnar nerve:

- The ulnar nerve, accessed easily at the wrist, is commonly utilized for neuromuscular monitoring during anesthesia due to its reliability and resistance to muscle relaxants. Monitoring is typically done through the adductor pollicis muscle, which it innervates, providing a clear indication of neuromuscular blockade levels.



Incorrect Options:

Option B - Median nerve:

- Anesthetic substances may impact the median nerve, obstructing precise neuromuscular monitoring.

Option C - Radial nerve:

- The radial nerve can sometimes be hard to find and stimulate, which might result in erroneous monitoring.

Option D - Metacarpal nerve:

- The metacarpal nerve is less reliable than the ulnar nerve because it cannot detect minute variations in neuromuscular blocking.

Solution for Question 4:

Correct Option C: Train-of-four:

- The train-of-four (TOF) method is considered the gold standard for assessing the degree of muscle relaxation during anesthesia.
- It involves stimulating a motor nerve at a high frequency and observing the resulting muscle twitch responses.

- By comparing the amplitude of the fourth twitch to the first, anesthesiologists can accurately quantify neuromuscular blockade and tailor drug dosages to achieve optimal relaxation while minimizing the risk of residual paralysis..

Incorrect Options A, B, and D are not the most effective for evaluating the extent of muscle relaxation.

Option A: Tetanic stimulation:

- Unlike tetanic stimulation, which induces sustained muscle contraction, TOF delivers a series of four brief stimuli, allowing for the assessment of fade in muscle twitch responses as neuromuscular blockade increases.

Option B: Double burst stimulation:

- While double burst stimulation delivers two bursts of electrical impulses in rapid succession, TOF provides four individual stimuli, allowing for a more detailed assessment of fade in muscle twitch responses.

Option D: Electromyography:

- EMG measures muscle electrical activity, which may be influenced by factors other than neuromuscular blockade, such as muscle fatigue or electrode placement.

Solution for Question 5:

Correct Option B: Neostigmine:

- Nicotinic receptor antagonists, also known as non-depolarizing neuromuscular blocking agents, are commonly used during anesthesia to induce muscle relaxation and facilitate surgical procedures.
- However, their effect can persist after surgery, leading to muscle weakness and impaired neuromuscular function.
- Neostigmine is an acetylcholinesterase inhibitor commonly used for the reversal of nondepolarizing neuromuscular blockade.
- It acts by inhibiting the breakdown of acetylcholine, thereby increasing its availability and facilitating neuromuscular transmission.

Incorrect Options: Options A, C, and D are incorrect.

Option A: Physostigmine:

- Physostigmine is an acetylcholinesterase inhibitor that increases the availability of acetylcholine in the synaptic cleft.
- While it can be used to treat anticholinergic toxicity or certain cases of central nervous system depression, it is not the appropriate choice for reversing the effects of a nicotinic receptor antagonist-induced muscle weakness.

Option C: Carbachol:

- Carbachol is a cholinergic agonist that stimulates both muscarinic and nicotinic receptors.
- It is not typically used as a reversal agent for neuromuscular blockade.

Option D: Succinylcholine:

- Succinylcholine, also known as suxamethonium, is a depolarizing neuromuscular blocking agent.
 - It is not used as a reversal agent but rather as an induction agent during rapid sequence intubation or for short procedures requiring muscle relaxation.
-

Solution for Question 6:

Correct Option B - Competitively blocking the binding of ACh to its receptors:

- These drugs competitively bind to the nicotinic acetylcholine receptors at the neuromuscular junction, preventing the binding of acetylcholine (ACh) and thereby inhibiting muscle contraction.

Incorrect Options: Options A, C, and D are incorrect.

Solution for Question 7:

Correct Option B - 20% intralipid infusion:

- Bupivacaine is the most cardiotoxic drug
- It should never be administered intravenously

- Treatment of Bupivacaine toxicity includes: For arrhythmia: Anti-arrhythmic drugs (bretylium, amiodarone, procainamide) 20% intralipid solution: 1.5 ml/kg bolus followed by 0.5 ml/kg infusion
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- For arrhythmia: Anti-arrhythmic drugs (bretylium, amiodarone, procainamide) 20% intralipid solution: 1.5 ml/kg bolus followed by 0.5 ml/kg infusion
- 20% intralipid solution: 1.5 ml/kg bolus followed by 0.5 ml/kg infusion
- 20% intralipid solution: 1.5 ml/kg bolus followed by 0.5 ml/kg infusion

Incorrect Options:

Option A and D - Invasive ventilation and Cardiopulmonary resuscitation:

- This is indicated only there is a cardiac arrest in Bupivacaine toxicity
- Since the question stem indicates hypotension and arrhythmia, this is not the correct answer.

Option C - Sodium bicarbonate :

- This is given in sodium channel blockers induced toxicity
- Bupivacaine is a sodium channel blocker, yet the first line of treatment for Bupivacaine toxicity is 20% intralipid infusion.

Intravenous Anaesthetic Agents

1. A patient is in long-term sedation with propofol due to a neurological condition. The patient develops rhabdomyolysis, hyperkalemia, and lactic acidosis. Which of the following steps would not be useful in managing this scenario?

- A. Hemodialysis
- B. Sodium bicarbonate infusion
- C. Lowering the dose of propofol
- D. Fluid management and monitoring

2. Match the following anesthetic drug with its systemic effect

Drug	Effect
1. Thiopental	a. No effect on HR, ■ MAP
2. Dexmedetomidine	b. ■ HR, ■ MAP
3. Propofol	c. ■ HR, ■ ICP
4. Ketamine	d. ■ HR, ■ ICP

Drug	Effect
1. Thiopental	a. No effect on HR, ■ MAP
2. Dexmedetomidine	b. ■ HR, ■ MAP
3. Propofol	c. ■ HR, ■ ICP
4. Ketamine	d. ■ HR, ■ ICP

- A. 1-c, 2-b, 3-a, 4-d
- B. 1-b, 2-a, 3-d, 4-c
- C. 1-a, 2-d, 3-c, 4-b
- D. 1-d, 2-a, 3-b, 4-c

3. A 24-year-old male with no significant medical history is scheduled for a minor surgical procedure. The anesthesiologist administers remimazolam for sedation. What is the primary advantage of remimazolam over midazolam in this setting?

- A. More rapid and consistent recovery profile.
 - B. Reduced risk of respiratory depression.
 - C. Stronger anxiolytic properties.
 - D. Longer duration of action.
-

4. A 75-year-old male with a history of chronic liver disease presents with severe anxiety and insomnia. The patient has been prescribed benzodiazepines in the past, but his physician is concerned about potential drug accumulation and adverse effects due to impaired liver function. Which of the following benzodiazepines is the most appropriate choice for this patient?

- A. Diazepam
 - B. Clonazepam
 - C. Lorazepam
 - D. Flurazepam
-

5. A 60-year-old female with Parkinson's disease is undergoing total knee replacement surgery. The anesthesiologist is considering giving droperidol as part of the anesthesia plan to prevent PONV. The patient is currently on levodopa-carbidopa therapy and has well-controlled symptoms. What should the anesthesiologist consider before administering droperidol?

- A. It is contraindicated in patients with Parkinson's disease as it may exacerbate motor symptoms.
- B. It will not interfere with Parkinson's medications and can be safely administered.
- C. It should only be given in higher doses to prevent complications in Parkinson's patients.
- D. It is a first-line antiemetic in Parkinson's disease and should be used without caution.

6. A 40-year-old male with hypertension undergoes fiberoptic intubation. After receiving a loading dose of dexmedetomidine for sedation, his heart rate drops to 45 bpm, and his blood pressure increases slightly. What explains these changes?

- A. Initial increase in BP followed by reflex bradycardia
 - B. Direct SA node depression causing bradycardia
 - C. Decrease in BP and HR due to alpha-1 agonism
 - D. Bradycardia from beta-blocker-like activity
-

7. During induction of anesthesia, a patient receiving etomidate shows an increase in blood pressure and heart rate. How does etomidate primarily achieve cardiovascular stability?

- A. By activating the sympathetic nervous system
 - B. By affecting the baroreceptor function
 - C. Through α_2 -adrenoceptor activation
 - D. By enhancing myocardial oxygen demand
-

8. A 47-year-old male patient with a history of adrenal carcinoma and hypercortisolemia is scheduled for a surgical procedure. The anesthesiology team is evaluating induction agents that suppress adrenal function. Which of the following agents can be used in this patient?

- A. Propofol
- B. Ketamine
- C. Etomidate
- D. Thiopental

9. A 60-year-old female patient with a history of chronic migraines and increased intracranial pressure due to a recent stroke is scheduled for a minor procedure requiring anesthesia. Which anesthetic agent should be avoided in this patient?

- A. Thiopentone
 - B. Midazolam
 - C. Ketamine
 - D. Sevoflurane
-

10. Which of the following statements about ketamine is not true in the context of its pharmacological properties and clinical use?

- A. It primarily acts as an antagonist at N-methyl-D-aspartate (NMDA) receptors, inducing dissociative anesthesia.
 - B. The S(+) isomer is less potent as an analgesic than the R(-) isomer.
 - C. It has high lipid solubility, allowing for rapid brain uptake.
 - D. It is metabolized in the liver, and its elimination half-life is approximately 2 hours .
-

11. A 12-year-old child is scheduled for surgery and has a history of asthma. Which preoperative medication should be considered to reduce airway secretions and minimize the risk of bronchospasm?

- A. Midazolam
- B. Glycopyrrolate
- C. Ondansetron
- D. Fentanyl

12. A 40-year-old man is given propofol for anesthesia. Which of the following statements regarding the effects of propofol is false?

- A. Inhibits ventilatory drive by depressing hypercarbic and hypoxic responses
 - B. Causes profound respiratory depression leading to apnea
 - C. It provides antipruritic and antiemetic effects
 - D. Increases the cerebral perfusion rate and cerebral blood volume
-

13. A 4-year-old patient in the ICU requires sedation while on mechanical ventilation. The medical team considers using Propofol. Which of the following is a key pharmacokinetic feature of Propofol that supports its use in this scenario?

- A. Long onset of action.
 - B. Minimal hangover effects after recovery.
 - C. High risk of prolonged sedation.
 - D. Primarily excreted via feces.
-

14. A patient is sedated using methohexital for surgery. What is false about methohexital?

- A. It belongs to the barbiturates group of drugs
 - B. Acts by binding to GABA (A) receptors
 - C. Can be used to briefly control grand-mal seizures
 - D. Produces muscle relaxation and analgesia
-

15. A 45-year-old male is undergoing an elective surgical procedure Thiopental sodium is administered for induction. Shortly after the injection, the anesthesiologist notices that the patient's arm is becoming pale and cold, indicating possible intra-arterial injection of the anesthetic. What is the appropriate sequence of steps to manage this situation?

- A. Administer 5 ml of 2% procaine or lignocaine into the affected artery, then consider a brachial plexus block.
 - B. Perform an early arteriotomy, then start full heparinization.
 - C. Administer 5 ml of 2% procaine or lignocaine, then provide deep anesthesia.
 - D. Start full heparinization immediately, then assess for the need for an arteriotomy.
-

16. A 65-year-old patient undergoing neurosurgery is administered thiopental sodium as an induction agent. Which of the following statements regarding its effects is true?

- A. It increases cerebral blood flow and oxygen consumption.
 - B. It causes peripheral vasodilation and increased heart rate.
 - C. It enhances respiratory rate and tidal volume during awakening.
 - D. It enhances hepatic blood flow and reduces the metabolism of other drugs.
-

17. A 65-year-old patient is scheduled for elective surgery requiring general anesthesia. The anesthesiologist decides to use thiopental sodium as an induction agent. Which of the following statements is true about thiopental sodium?

- A. It has a long duration of action due to extensive renal metabolism.
- B. Its rapid onset is primarily due to its high potency and redistribution.
- C. It appears as a white milky liquid.

D. Its elimination half-life is approximately 1 hour.

18. Which of the following statements is correct regarding intravenous anesthetic drugs?

- A. Barbiturates primarily block NMDA receptors to induce anesthesia.
 - B. Barbiturates can be safely mixed with lactated Ringer's solution.
 - C. Thiopental is mixed with anhydrous sodium carbonate.
 - D. Ketamine primarily acts on GABA-A inhibitory receptors.
-

19. A 38-year-old male is being taken up for emergency surgery following a trauma. The anesthesiologist decides to give medications as prophylaxis for aspiration. Which of the following drugs is not used as a prophylactic agent?

- A. Bicitra
 - B. Metoclopramide
 - C. H₂-antagonists
 - D. Rocuronium
-

20. A 35-year-old woman undergoes a laparoscopic hysterectomy under general anesthesia. After 5 minutes of preoxygenation with 100% oxygen, anesthesia is induced and maintained. What is the main purpose of preoxygenation in this case?

- A. To increase CO₂ retention
- B. To ensure the patient's lungs are fully saturated with oxygen before induction
- C. To reduce anesthesia depth requirement
- D. To reduce respiratory rate during anesthesia

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	1
Question 4	3
Question 5	1
Question 6	1
Question 7	3
Question 8	3
Question 9	3
Question 10	2
Question 11	2
Question 12	4
Question 13	2
Question 14	4
Question 15	1
Question 16	2
Question 17	2
Question 18	3

Question 19	4
Question 20	2

Solution for Question 1:

Correct Answer: C) Lowering the dose of propofol

Explanation:

The clinical scenario points to the diagnosis of propofol infusion syndrome, which is treated with immediate cessation of propofol infusion and not just by lowering the dose.

Propofol Infusion Syndrome (PRIS)

- A rare but serious condition associated with prolonged use of propofol for sedation, particularly in critically ill patients.

Symptoms:

- Lipemia
- Metabolic acidosis
- Myopathy
- Cardiac dysfunction due to impairment of cardiovascular contractility
- Lactic acidosis
- Rhabdomyolysis
- Hyperkalemia
- Hepatomegaly
- Acute renal failure
- Eventually, mortality in most cases

Risk Factors:

- Long-term sedation (especially in pediatric and neurocritical care patients).
- High doses of propofol.
- Use of propofol in patients with significant metabolic stress or poor organ function and critical illnesses
- Increased serum catecholamines

- Steroid therapy
- Obesity
- Young age (significantly below three years)
- Depleted carbohydrate stores in the body
- Increased serum lipids.

Pathophysiology:

- The mechanism is not fully understood but may involve mitochondrial dysfunction, leading to impaired fatty acid oxidation and ATP synthesis.

Management:

- Immediate cessation of propofol infusion.
- Supportive care, including fluid management and monitoring of metabolic status. (Option D)
- Addressing electrolyte imbalances and other complications as they arise.
- Sodium bicarbonate infusion, hemodialysis, and hemofiltration for metabolic and lactic acidosis. (Options A & B)
- Oxygenation and circulatory support.

Prevention:

- Limit the duration of propofol use in high-risk populations.
- Consider alternative sedation strategies for prolonged use.



Complications	Contraindications
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Cardiovascular Hypotension Bradycardia Impaired baroreflex response	Hypersensitivity to propofol or its components Severe cardiac dysfunction Significant hypotension Allergy to egg lecithin (in some cases) Severe hepatic impairment Severe allergies to soy products Inability to maintain a sterile technique
Respiratory Apnea Depressed hypoxic ventilatory drive	
Neurological Excitatory phenomena (e.g., twitching) Propofol Infusion Syndrome (PRIS)	
Gastrointestinal Nausea and vomiting (though antiemetic)	
Infection Risk Sepsis from contaminated preparations	

Solution for Question 2:

Correct Answer: D) 1-c, 2-b, 3-a, 4-d

Explanation:

- Thiopental: ■ HR, ■ ICP
- Dexmedetomidine: ■ HR, ■ MAP
- Propofol: No effect on HR, ■ MAP
- Ketamine: ■ HR, ■ ICP

	Agent	Cardiovascular	Respiratory	Cerebral				
HR	MAP	Vent	B'dil	CBF	CMRO■	ICP		
1. Barbiturates	Thiopental	↑↑	↓↓	↓↓↓	↓	↓↓↓	↓↓↓	↓↓↓
Methohexital	↑↑	↓↓	↓↓↓	0	↓↓↓	↓↓↓	↓↓↓	
2. Benzodiazepines	Diazepam	0/ ↑	↓	↓↓	0	↓↓	↓↓	↓↓
Lorazepam	0/ ↑	↓	↓↓	0	↓↓	↓↓	↓↓	
Midazolam	↑	↓↓	↓↓	0	↓↓	↓↓	↓↓	

3. Ketamine (Phencyclidine derivative)	↑↑	↑↑	↓	↑↑↑	↑↑	↑	↑↑	
4. Etomidate (Imidazole derivative)	0	↓	↓	0	↓↓↓	↓↓↓	↓↓↓	
5. Propofol (Alkylphenol derivative)	0	↓↓	↓↓↓	0	↓↓↓	↓↓↓	↓↓↓	
6. Dexmedetomidine	↓	↓	0	?	↓↓	↓↓	↓↓	

Key:

B'dil: Bronchodilation

CBF: Cerebral blood flow

CMRO₂: Cerebral oxygen consumption (cerebral metabolic rate of oxygen)

HR: Heart rate

ICP: Intracranial pressure

MAP: Mean arterial pressure

Vent: Ventilatory drive

Changes:

0 = no effect

↑ = increase

↓ = decrease

? = unknown effect

Solution for Question 3:

Correct answer: A) More rapid and consistent recovery profile.

Explanation:

Remimazolam has a significant advantage over midazolam due to its rapid and consistent recovery profile. It is an ultra-short-acting benzodiazepine that is metabolized quickly by tissue esterases, resulting in faster clearance from the body. This leads to quicker recovery times.

While both remimazolam and midazolam enhance GABA-A receptor activity, Remimazolam's shorter duration of action makes it more predictable in terms of recovery, reducing the chance of prolonged sedation.

Effects of benzodiazepines on various systems

System	Effects of Benzodiazepines
CNS	Primary Effects : Hypnotic, sedative, anxiolytic, amnesic, anticonvulsant, and muscle-relaxant properties. GABA Modulation : Enhances GABA neurotransmitter effects, leading to inhibition and neuroprotection. Receptor Binding : High-affinity binding to GABA-A receptors, with effects mediated by subunits ($\alpha 1$ for sedation, $\alpha 2$ for anxiolysis). Increases the threshold for seizures Midazolam has a fast onset and shorter duration, while lorazepam has a slower onset but prolonged effects.
Respiratory System	Causes dose-dependent respiratory depression by reducing muscular tone and flattening the CO ₂ response curve. Synergistic Effect: Increases respiratory depression when combined with opioids.
CVS	Modest reduction in arterial blood pressure due to decreased systemic vascular resistance. Stable hemodynamics are maintained through homeostatic reflexes.

- Midazolam : Common for sedation, anxiolysis, and amnesia with minimal respiratory effects in adults.
- Common for sedation, anxiolysis, and amnesia with minimal respiratory effects in adults.
- Diazepam : Adults: 5-10 mg (oral/IV) Used for sedation and anxiolysis. Dosage depends on patient factors like age and surgery type.
- Adults: 5-10 mg (oral/IV)
- Used for sedation and anxiolysis. Dosage depends on patient factors like age and surgery type.
- Lorazepam : Adults: 2-4 mg (oral) Primarily for prolonged anxiolysis (e.g., cardiac surgery). Administered 2 hours before surgery.
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- Primarily for prolonged anxiolysis (e.g., cardiac surgery). Administered 2 hours before surgery.
- Common for sedation, anxiolysis, and amnesia with minimal respiratory effects in adults.

- Adults: 5-10 mg (oral/IV)
- Used for sedation and anxiolysis. Dosage depends on patient factors like age and surgery type.
- Adults: 2-4 mg (oral)
- Primarily for prolonged anxiolysis (e.g., cardiac surgery). Administered 2 hours before surgery.
- Midazolam : Rapid onset (2-3 minutes) with titration. Used in minor surgeries and diagnostic procedures. Slower recovery than propofol but preserves respiratory function.
- Rapid onset (2-3 minutes) with titration.
- Used in minor surgeries and diagnostic procedures.
- Slower recovery than propofol but preserves respiratory function.
- Remimazolam : Provides quicker and more consistent recovery than midazolam, especially in endoscopic procedures.
- Provides quicker and more consistent recovery than midazolam, especially in endoscopic procedures.
- Rapid onset (2-3 minutes) with titration.
- Used in minor surgeries and diagnostic procedures.
- Slower recovery than propofol but preserves respiratory function.
- Provides quicker and more consistent recovery than midazolam, especially in endoscopic procedures.
- Midazolam : Adults: 0.1-0.3 mg/kg (IV) Induction within 30-60 seconds. Often combined with opioids for enhanced effects, lower doses required in elderly patients.
- Adults: 0.1-0.3 mg/kg (IV)
- Induction within 30-60 seconds.
- Often combined with opioids for enhanced effects, lower doses required in elderly patients.
- Adults: 0.1-0.3 mg/kg (IV)
- Induction within 30-60 seconds.
- Often combined with opioids for enhanced effects, lower doses required in elderly patients.
- Midazolam : Adults: 0.075 mg/kg (IV) Pediatrics: 0.05 mg/kg (IV) Effective in reducing postoperative nausea and vomiting (PONV), especially when combined with dexamethasone.
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- Pediatrics: 0.05 mg/kg (IV)
- Effective in reducing postoperative nausea and vomiting (PONV), especially when combined with dexamethasone.
- Adults: 0.075 mg/kg (IV)

- Pediatrics: 0.05 mg/kg (IV)
- Effective in reducing postoperative nausea and vomiting (PONV), especially when combined with dexamethasone.
- Respiratory depression , particularly with lorazepam and diazepam.
- Possible venous irritation (lorazepam, diazepam).
- Prolonged sedation and amnesia.
- Flumazenil can reverse residual effects .

Reduced risk of respiratory depression (Option B) : Both remimazolam and midazolam have a similar risk of respiratory depression due to their effects on GABA-A receptors.

Stronger anxiolytic properties (Option C) : Both drugs offer similar anxiolytic effects through the same mechanism, with no significant difference in potency.

Longer duration of action (Option D) : Remimazolam has a shorter duration of action, not longer, which is why it allows for faster recovery compared to midazolam.

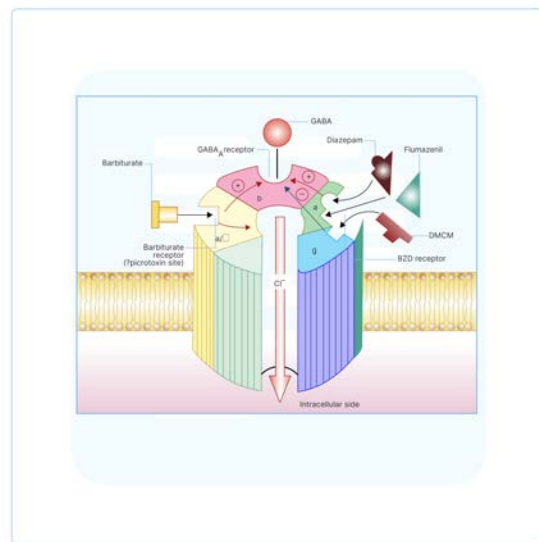
Solution for Question 4:

Correct Answer: C) Lorazepam

Explanation:

Lorazepam is metabolized directly through glucuronidation (Phase 2) without producing active metabolites, making it a safer option in patients with liver impairment.

MOA:



- Benzodiazepines act by enhancing GABA-mediated inhibition through binding to the GABA-A receptor-Cl⁻ channel complex.
- Its site of action lies between α and γ subunits . They increase GABA binding and the frequency of Cl⁻ channel opening , facilitating inhibition without directly mimicking GABA.
- Different subunits ($\alpha 1$, $\alpha 2$, etc.) of the GABA-A receptor modulate different BZD effects. Effects mediated via $\alpha 1$ (sedation, amnesia, anticonvulsant) and $\alpha 2$ (anxiolysis, muscle relaxation) subunits

Classification of benzodiazepine

Hypnotic	Antianxiety	Anticonvulsant	Muscle relaxant
Diazepam Flurazepam Nitrazepam Alprazolam Lorazepam Temazepam Triazolam	Diazepam Chlordiazepoxide Oxazepam Lorazepam Alprazolam Clonazepam	Clonazepam Clobazam Diazepam Lorazepam	Diazepam

Metabolism:

Class A drugs

Here, drugs are metabolized both in phase 1 (by CYP3A4 enzymes in the liver), and their active metabolites undergo Phase 2 (Glucuronidation) reaction, which is then excreted in the urine.

Examples: All benzodiazepines except 4 class B drugs (Lorazepam, Oxazepam, Temazepam, Estazolam).

Class B drugs

Metabolized by phase 2 or direct glucuronidation . Therefore, no active metabolite, so it is safer to use in elderly and liver failure patients.

Examples:

- Lorazepam
- Oxazepam
- Temazepam
- Estazolam

Diazepam (Option A) and Clonazepam (Option B) are metabolized by Phase 1 hepatic enzymes (CYP3A4), producing active metabolites that could accumulate in patients with liver disease.

Flurazepam (Option D) also undergoes Phase 1 metabolism and has long-acting active metabolites.

Solution for Question 5:

Correct answer: A) It is contraindicated in patients with Parkinson's disease as it may exacerbate motor symptoms.

Explanation:

Droperidol is contraindicated in Parkinson's disease because it blocks dopamine receptors, worsening motor symptoms and counteracting levodopa-carbidopa therapy. Alternatives like ondansetron are safer.

Droperidol	
Mechanism of Action	Acts on dopamine, norepinephrine, and serotonin receptors in the CNS. Inhibits GABA receptors and affects neurotransmission.
Antiemetic Effect Blocks GABA receptors in the chemoreceptor trigger zone.	
Pharmacokinetics	Metabolized in the liver. Two-compartment plasma decay model.
CNS Effects	Causes cerebral vasoconstriction (40% reduction in blood flow). Minimal impact on cerebral metabolic rate.
Respiratory Effects	Minimal respiratory impact Slight reduction in the respiratory rate at certain doses.

Cardiovascular Effects	QT prolongation , risk of torsades de pointes (dose-dependent). Causes vasodilation and decreased blood pressure due to α -adrenergic blockade.
Perioperative Use	Primarily used as an antiemetic (10-20 μ g/kg, 0.6-1.25 mg for a 70 kg individual). It has equal antiemetic efficacy with similar side effects, but more cost-effective than ondansetron. Combination with serotonin antagonists enhances efficacy.
Other Uses	Effective in treating pruritus from opioids. Epidural use reduces nausea but increases sedation (not approved for this route).

It will not interfere with Parkinson's medications and can be safely administered (Option B) : Droperidol can interfere with dopaminergic medications, worsening Parkinson's symptoms.

It should only be given in higher doses to prevent complications in Parkinson's patients (Option C) : Higher doses would increase the risk of exacerbating symptoms rather than preventing complications.

It is a first-line antiemetic in Parkinson's disease and should be used without caution (Option D) : Droperidol is not a first-line antiemetic for Parkinson's patients due to its potential to worsen motor symptoms; safer alternatives exist.

Solution for Question 6:

Correct answer: A) Initial increase in BP followed by reflex bradycardia

Explanation:

Dexmedetomidine is an α_2 -adrenergic agonist that causes a transient increase in blood pressure due to vasoconstriction , followed by reflex bradycardia. This explains the hemodynamic changes seen in the patient. The bradycardia is not due to a direct depressant effect on the heart and the drug does not have beta-blocker-like activity.

- Dexmedetomidine is an α_2 -adrenergic agonist , similar to clonidine , primarily used for sedation, anxiolysis, and analgesia. It is often combined with other agents to produce anesthesia and is used to prolong regional blocks when combined with local anesthetics.

Dexmedetomidine	
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Uses	Procedural sedation (e.g., awake craniotomies, fiberoptic intubation) ICU sedation (e.g., ventilated patients recovering from cardiac surgery) Supplement to general anesthesia (to reduce intraoperative opioid needs or emergence delirium, especially in children) Treatment for alcohol withdrawal and cocaine intoxication side effects.
Pharmacokinetics	Absorption: Approved for IV injection; Initial loading dose: 1 mcg/kg over 5-10 minutes; Maintenance infusion: 0.2-1.4 mcg/kg/h. For premedication, nasal (1-2 mcg/kg) or oral (2.5-4 mcg/kg) routes are used in children.
Distribution: Rapid redistribution (minutes); elimination half-life <3 hours.	
Biotransformation : Metabolized in the liver via CYP450 system and glucuronidation ; to be used cautiously in patients with severe liver disease.	
Excretion : Metabolites are primarily excreted in the urine.	
Cardiovascular	May cause transient increase in blood pressure and reflex bradycardia . Intraoperative use results in dose-dependent reduction of mean arterial pressure and heart rate. Can cause hypertension, hypotension, or bradycardia based on dose. Avoid rapid bolus to minimize side effects.
Respiratory	No respiratory depression; ideal for sedation during mechanical ventilation weaning or awake tracheal intubations.
CNS	Dose-dependent sedation; opioid-sparing; reduces the need for general anesthetics; preferred for awake craniotomies.
Drug Interactions	Dexmedetomidine may cause exaggerated bradycardia in patients on beta-blockers. It has additive effects with sedative-hypnotic agents.

Direct SA node depression causing bradycardia (Option B) :The bradycardia is not a direct effect on the SA node but a reflex response to vasoconstriction and increased blood pressure.

Decrease in BP and HR due to alpha-1 agonism (Option C) : Dexmedetomidine is an α_2 -agonist, not α_1 , and initially increases blood pressure, followed by reflex bradycardia.

Bradycardia from beta-blocker-like activity (Option D) : Dexmedetomidine does not act on beta receptors but reduces sympathetic outflow through α_2 -receptor activation, leading to bradycardia.

Solution for Question 7:

Correct answer: C) Through α 2-adrenoceptor activation

Explanation:

Etomidate activates α 2-adrenoceptors, which increases blood pressure and contributes to cardiovascular stability by preventing excessive sympathetic stimulation.

Effects of Etomidate on the Cardiovascular System

- Hemodynamic Stability : Etomidate maintains hemodynamic stability by not affecting the sympathetic nervous system or baroreceptor function. (Option A & B)
- Etomidate maintains hemodynamic stability by not affecting the sympathetic nervous system or baroreceptor function. (Option A & B)
- Blood Pressure Effects : Activation of α 2-adrenoceptors leads to increased blood pressure, contributing to cardiovascular stability during anesthesia induction.
- Activation of α 2-adrenoceptors leads to increased blood pressure, contributing to cardiovascular stability during anesthesia induction.
- Comparison with Other Anesthetics : Minimal cardiovascular effects distinguish etomidate from other rapid-onset anesthetics, making it suitable for patients with heart conditions.
- Minimal cardiovascular effects distinguish etomidate from other rapid-onset anesthetics, making it suitable for patients with heart conditions.
- Patient Suitability : Effective for patients with valvular or ischemic heart disease and those with poor cardiac function during non cardiac surgeries.
- Effective for patients with valvular or ischemic heart disease and those with poor cardiac function during non cardiac surgeries.
- Hemodynamic Responses : Etomidate may lead to more hypertension and tachycardia compared to propofol during induction.
- Etomidate may lead to more hypertension and tachycardia compared to propofol during induction.
- Oxygen Supply and Demand : Maintains a stable myocardial oxygen supply-to-demand ratio . (Option D)
- Maintains a stable myocardial oxygen supply-to-demand ratio . (Option D)
- Analgesic Considerations : Lacks analgesic properties; should be combined with opioids to manage hemodynamic changes during laryngoscopy and intubation.

- Lacks analgesic properties; should be combined with opioids to manage hemodynamic changes during laryngoscopy and intubation.
- Advantages in Hemorrhagic Shock : In cases of hemorrhagic shock, etomidate's pharmacodynamics and pharmacokinetics remain largely unchanged, making it advantageous for induction.
- In cases of hemorrhagic shock, etomidate's pharmacodynamics and pharmacokinetics remain largely unchanged, making it advantageous for induction.
- Etomidate maintains hemodynamic stability by not affecting the sympathetic nervous system or baroreceptor function. (Option A & B)
- Activation of α_2 -adrenoceptors leads to increased blood pressure, contributing to cardiovascular stability during anesthesia induction.
- Minimal cardiovascular effects distinguish etomidate from other rapid-onset anesthetics, making it suitable for patients with heart conditions.
- Effective for patients with valvular or ischemic heart disease and those with poor cardiac function during non cardiac surgeries.
- Etomidate may lead to more hypertension and tachycardia compared to propofol during induction.
- Maintains a stable myocardial oxygen supply-to-demand ratio . (Option D)
- Lacks analgesic properties; should be combined with opioids to manage hemodynamic changes during laryngoscopy and intubation.
- In cases of hemorrhagic shock, etomidate's pharmacodynamics and pharmacokinetics remain largely unchanged, making it advantageous for induction.

Solution for Question 8:

Correct answer: C) Etomidate

Explanation:

Etomidate inhibits 11 β -hydroxylase in the adrenal cortex, which is essential for cortisol synthesis. Therefore, it effectively decreases cortisol levels , making it suitable for patients with hypercortisolemia.

Etomidate	
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Mechanisms of Action	Depresses the reticular activating system. Mimics GABA's inhibitory effects by binding to GABA A receptors , particularly the R(+) isomer.
Structure-Activity Relationships	Contains a carboxylated imidazole The imidazole ring provides water solubility in acidic solutions and lipid solubility at physiological pH.
Pharmacokinetics	Absorption : Only available for IV use , primarily for induction of anesthesia.
Distribution: Rapid onset due to high lipid solubility; redistribution lowers plasma concentration post-awakening.	
Biotransformation: Rapidly hydrolyzed by hepatic microsomal enzymes to an inactive metabolite .	
Excretion : Hydrolysis products are excreted in urine.	
CNS Effects	Induces hypnosis via GABA A receptor modulation (0.3 mg/kg). Reduces cerebral blood flow and cerebral metabolic rate of oxygen without affecting MAP. Decreases intracranial pressure (ICP) by 50% in patients with increased ICP.
Respiratory Effects	Minimal impact on ventilation; does not release histamine. May cause brief hyperventilation and mild apnea, with slight PaCO ₂ increase.
Cardiovascular Effects	Minimal impact on sympathetic tone and myocardial function. It may cause mild arterial blood pressure decline without significant changes in cardiac output or contractility.
Endocrine Effects	Inhibits 11 β -hydroxylase , reducing cortisol and mineralocorticoid production, leading to adrenal suppression.
Clinical Uses	Induction dose : 0.2 to 0.6 mg/kg IV. Suitable for patients with cardiovascular issues and Increased ICP Effective in treating endogenous hypercortisolemia .
Side Effects	Associated with postoperative nausea and vomiting (PONV), Pain on injection, Myoclonic movements , and Hiccups. Pre-medication with lidocaine can reduce pain ; myoclonus can be lowered with midazolam or magnesium.

Novel Derivatives	Methoxycarbonyl etomidate (MOC) : Rapidly metabolized, minimal adrenal suppression, similar hypnotic potency. Carboetomidate: Reduces adrenal suppression potency, maintains hypnotic effects, minimal hemodynamic changes. Methoxycarbonyl-carboetomidate: Combines benefits of etomidate without adrenal suppression, longer action duration. CPMM and DMMM: Newer derivatives showing higher potency and rapid recovery post-infusion, promising in animal studies.
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Propofol (Option A) , Ketamine (Option B) , and Thiopental (Option C) do not have specific effects on adrenal hormone synthesis and are not known to inhibit 11 β -hydroxylase. They are primarily used for their anesthetic and sedative properties.

Solution for Question 9:

Correct Answer: C) Ketamine Explanation:

Ketamine causes increased cerebral blood flow , which can lead to elevated intracranial pressure . This is especially problematic in patients with conditions such as stroke, where ICP management is crucial to prevent further neurological damage.

Systemic effects, uses and C/I of Ketamine

System	Effects
CNS	It is associated with increased cerebral oxygen consumption , blood flow , and intracranial pressure. It causes functional disorganization by selectively depressing neuronal activity in the cortex and thalamus while stimulating the limbic system, leading to disrupted pain perception and emotional processing. Emergence reactions , including vivid dreams, hallucinations, and illusion, occurs due to depression of auditory and visual relay nuclei. It provides analgesia , amnesia , and unconsciousness , making it a "complete" anesthetic.

CVS	Increases arterial blood pressure , heart rate , and cardiac output, particularly after rapid bolus injections. Stimulates the sympathetic nervous system and inhibits norepinephrine reuptake, increasing pulmonary artery pressure and myocardial workload. Direct myocardial depression can occur with sympathetic blockade or catecholamine depletion.
Respiratory	Racemic ketamine is a strong bronchodilator , useful in asthmatic patients . Upper airway reflexes remain intact. Excessive salivation can be managed with anticholinergics (e.g., glycopyrrolate).
Muscle	It increases muscle tone

Uses:

- Induction & Maintenance : Effective for unstable cardiovascular patients (e.g., hypovolemia, trauma, sepsis).
- Combines with propofol or midazolam to reduce side effects (tachycardia, hypertension) and stabilize hemodynamics.
- Useful in cardiac conditions (congenital heart disease(Right to Left shunt) , septic shock , restrictive pericarditis).
- IV induction agent of choice in bronchial asthma
- It has prothyroid property: IV induction agent of choice myxedema coma patients
- Preventive Analgesia : Small doses (20-60 mg) reduce postoperative pain and opioid consumption.
- Used for chronic pain (e.g., cancer pain, neuropathic pain, fibromyalgia)
- Procedures & Emergency Use: Common in emergency departments and short painful procedures .

Contraindications:

- Increased ICP (Head injury, neuro surgery)
- Epilepsy
- Ischemic heart disease, vascular aneurysms
- Open eye injuries, and
- Schizophrenia.

Thiopentone (Option A) : It is the IV induction agent of choice in increased ICP.

Midazolam (Option B) : A benzodiazepine that provides sedation and amnesia without significantly affecting ICP.

Sevoflurane (Option D) : An inhalational anesthetic that is typically well-tolerated and does not have a direct effect on increasing ICP, making it a safer option in this scenario.

Solution for Question 10:

Correct answer: B) The S(+) isomer is less potent as an analgesic compared to the R(-) isomer.

Explanation:

The S(+) isomer of ketamine is more potent as an analgesic than the R(-) isomer.

Mechanisms of Action

- Inhibits N-methyl-D-aspartate receptor (NMDA) receptors (Option A) , affecting the CNS.
- Induces dissociative anesthesia , making patients appear conscious but unable to process sensory input.
- Influences endogenous analgesic pathways.
- It is an analog of phencyclidine.
- It is a racemic mixture of R(-) ketamine and S(+) ketamine .
- Molecular weight: 238 kD
- High lipid solubility (5 to 10 times that of thiopental) (Option C) and low protein binding (12%).
- S(+) isomer (Ketanest) is 3 to 4 times more potent as an analgesic with fewer psychomimetic effects.
- S(+) ketamine causes cognitive impairment and reduced reaction time despite its analgesic properties.
- S(+) isomer has higher NMDA affinity and potency.

Pharmacokinetics

- Absorption: Primarily given intravenously or intramuscularly; peak levels in 10-15 minutes.
- Distribution: Highly lipid-soluble; rapid brain uptake with a 10-15 minute distribution half-life.
- Biotransformation: Metabolized in the liver to active metabolites; tolerance develops with repeated use. (Option D)
- Excretion: Renally excreted ; elimination half-life of about 2 hours.

Solution for Question 11:

Correct answer: B) Glycopyrrolate

Explanation:

Glycopyrrolate is an anticholinergic medication that reduces airway secretions and minimizes bronchospasm by blocking acetylcholine at muscarinic receptors. It is especially important for asthmatic patients at risk of increased secretions and bronchoconstriction.

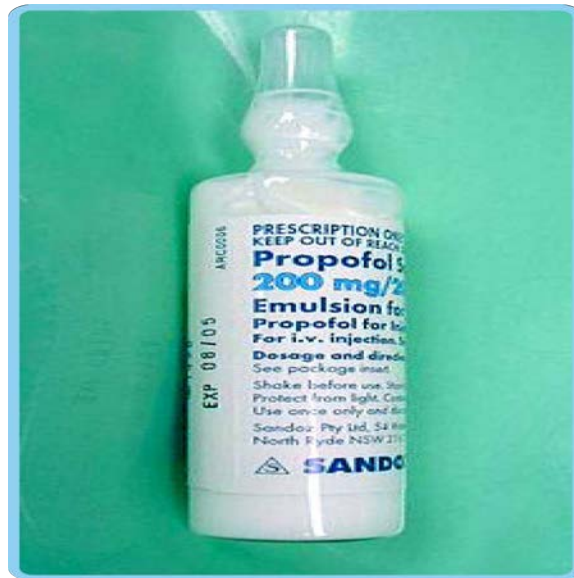
Class of Medication	Common Drugs	Use
Anxiolytics	Midazolam, (Option A) Lorazepam, Diazepam, Ketamine (pediatrics)	Preoperative anxiety reduction Midazolam is preferred for adults due to short half-life and potency. Avoid premedication in pediatric patients < 6 months.
Antiemetics	Ondansetron, (Option C), Dexamethasone, Metoclopramide, Droperidol, Transdermal Scopolamine	Prevention of postoperative nausea and vomiting (PONV)
Aspiration Prophylaxis	H2-Receptor Blockers (Ranitidine), Proton Pump Inhibitors (PPIs)	Reduce risk of pulmonary aspiration Effective in reducing gastric acidity and volume. Recommended for patients with conditions like GERD, obesity, or non-fasted status
Antibiotics		Infection prevention at surgical sites
Anticholinergic	Glycopyrrolate	To reduce secretions
Analgesics	Fentanyl (Option D), NSAIDs (Celecoxib)	Pain management, opioid dependency Fentanyl is preferred for rapid clearance and potency. - NSAIDs may be given preoperatively to reduce pain.

Solution for Question 12:

Correct Answer: D) Increases the cerebral perfusion rate and cerebral blood volume

Explanation:

Propofol is an anesthetic agent that causes a reduction in cerebral blood volume, thus reducing the cerebral perfusion rate .



Cardiovascular Effects:

- Hypotension : decreases arterial blood pressure by reducing systemic vascular resistance, preload, and cardiac contractility.
- Reflex Bradycardia: may occur due to decreased cardiac filling (Bezold–Jarisch reflex).
- Impaired Baroreflex: markedly impairs normal response to hypotension.
- Myocardial Oxygen Supply/Demand Mismatch: increases coronary sinus lactate production in some patients.

Respiratory Effects:

- Respiratory Depression: Profound depression, often causing apnea after induction doses. (Option B)
- Inhibition of Ventilatory Drive: Depresses hypoxic and hypercarbic responses, necessitating qualified personnel for administration. (Option A)
- Upper Airway Reflex Depression: Exceeds that of thiopental, facilitating procedures like intubation without neuromuscular blockade.

Cerebral Effects:

- Decreased Cerebral Blood Flow: reduces cerebral blood volume and intracranial pressure.

- Critical CPP Reduction : Can cause a critical reduction in cerebral perfusion pressure (<50 mm Hg) in patients with elevated intracranial pressure.
- Cerebral Protection: comparable to thiopental during experimental focal ischemia.
- Antipruritic and antiemetic properties: Provides unique benefits for outpatient anesthesia. (Option C)
- Anticonvulsant Effects: Can terminate status epilepticus; safe for epileptic patients.
- Decreased intraocular pressure: Beneficial in ocular surgeries.

Other effects:

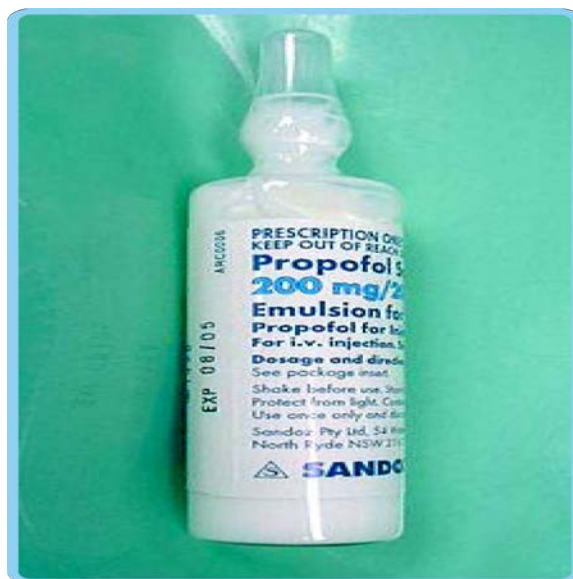
- Excitatory Phenomena: May cause muscle twitching, spontaneous movement, or hiccupping during induction.
- Low Addiction Potential: Uncommon for physical dependence or addiction, but inappropriate use has led to fatalities.

Solution for Question 13:

Correct answer: B) Minimal hangover effects after recovery.

Explanation:

Propofol is known for its rapid onset and quick recovery , with minimal residual sedation effects after administration. This characteristic is particularly beneficial for patients in the ICU, facilitating timely assessment and care.



Propofol

- Appearance: Clear, white emulsion, available as a 1% aqueous preparation (10 mg/mL) for IV administration.

Composition of Propofol

- Chemical structure: phenol ring with two isopropyl groups .
- Solubility: not water-soluble; available as a 1% aqueous oil-in-water emulsion.
- Emulsion ingredients: contains soybean oil, glycerol, and egg lecithin.
- Storage requirements: this should be administered within 6 hours of opening due to the risk of bacterial contamination, and a strict sterile technique is required
- 0.005% disodium edetate or 0.025% sodium metabisulfite helps slow bacterial growth.

Pharmacokinetics of Propofol

- Administration Route: only available for intravenous use.
- Onset of Action: rapid onset (Option A) ; effects typically observed for 2–8 minutes after administration.
- Distribution: short initial distribution half-life leads to quick awakening . (Option C)
- Recovery: faster recovery compared to other anesthetics, minimal "hangover" effects .
- Biotransformation: high clearance exceeding hepatic blood flow (possible extrahepatic metabolism); metabolism occurs in the liver, resulting in inactive metabolites.
- Excretion : Primarily eliminated via urine; unaffected by end-stage kidney disease . (Option D)

Fospropofol

- Appearance: water-soluble prodrug of propofol.
- It is metabolized to propofol, phosphate, and formaldehyde in vivo.
- Studies showed that it produced more complete amnesia and better conscious sedation (for endoscopy).
- Pharmacokinetics: Slower onset and recovery than propofol; not favoured over propofol for anesthesia.
- Composition: similar to propofol but formulated as a prodrug for intravenous administration.
- Induction of Anesthesia : Suitable for patients aged 3 years and older ; can be used in younger children with IV access.
- Maintenance of Anesthesia : For patients over 2 months old.
- Sedation : During monitored anesthesia for various procedures. For intubated, mechanically ventilated ICU patients.
- During monitored anesthesia for various procedures.
- For intubated, mechanically ventilated ICU patients.
- During monitored anesthesia for various procedures.
- For intubated, mechanically ventilated ICU patients.
- Management of refractory status epilepticus in both children and adults.
- Treatment of refractory postoperative nausea and vomiting.

Solution for Question 14:

Correct Answer: D) Produces muscle relaxation and analgesia

Explanation: Methohexital, which is a barbiturate, does not cause muscle relaxation and does not affect pain perception . It also causes involuntary skeletal muscle contractions.

Methohexital

- Type : A short-acting barbiturate used for induction of anesthesia. (Option A)
- Use : commonly administered for procedural sedation and electroconvulsive therapy (ECT).

Mechanism of Action

- GABA(A) Receptor (Option B) : Binds to GABA(A) receptors, enhancing GABA's effects by increasing the duration of chloride-specific ion channel opening.

- **CNS Effects** : Depresses the reticular activating system, leading to decreased consciousness.

Pharmacokinetics

- **Absorption** : typically administered intravenously; can also be given rectally in children.
- **Distribution** : rapidly crosses the blood-brain barrier due to high lipid solubility; effects seen within 30 seconds.
- **Redistribution** : duration of action is primarily determined by redistribution rather than metabolism; typically, plasma levels drop to 10% of peak within 20-30 minutes.
- **Biotransformation** : primarily metabolized by the liver to inactive, water-soluble metabolites; clears faster than thiopental.
- **Excretion** : excreted mainly in feces; renal excretion is limited.

Clinical Effects

- **CNS depression** : induces sedation to unconsciousness based on dosage; can briefly control grand mal seizures (50-100 mg IV). (Option C)
- **Taste sensation** : some patients report a garlic or onion-like taste during induction.
- **No pain impairment** : does not impair pain perception.
- **Does not cause muscle relaxation** but may cause involuntary muscle contractions .

Effects on Organ Systems

- **Cardiovascular** : can cause hypotension and tachycardia; effects vary based on administration rate and patient condition.
- **Respiratory** : may depress respiratory function; apnea can occur following induction.
- **Cerebral** : reduces cerebral blood flow and intracranial pressure; decreases oxygen consumption.
- **Renal**: renal blood flow and GFR are reduced.
- **Hepatic**: hepatic blood flow is reduced

Drug Interactions

- **CNS Depressants** : effects potentiated by ethanol, opioids, and antihistamines.
- **Protein Binding** : drugs that occupy protein-binding sites can displace methohexital, increasing free drug levels.

Considerations

- **Dosing** : requires careful dosing, particularly in older patients or those with compromised cardiovascular status.

- Adverse Reactions : sulfonamides reduce the effect of barbiturates, ethanol, opioids, antihistamines and other CNS depressants increase the sedative effect of barbiturates.
-

Solution for Question 15:

Correct Answer: A) Administer 5 ml of 2% procaine or lignocaine into the affected artery, then consider a brachial plexus block.

Explanation:

This option starts with the immediate administration of a local anesthetic to address the intra-arterial injection, which is crucial to prevent tissue damage. Considering a brachial plexus block afterward may provide additional pain control.

Complications of thiopental sodium:

Intra arterial injection: Induces severe inflammatory and necrotic reaction, threatening limb survival.

- Treatment: Inject 5 ml of 2% procaine or lignocaine into the affected artery , ideally through the existing needle . General Anesthesia: To decrease sympathetic tone in the arm. Avoid deep anesthesia that could lower blood pressure and cardiac output (Option C) Brachial plexus block or stellate ganglion block is considered an alternative. Anticoagulation: Full heparinization is recommended if surgery is not performed. Surgical Treatment: Early arteriotomy may be considered.
- Inject 5 ml of 2% procaine or lignocaine into the affected artery , ideally through the existing needle .
- General Anesthesia: To decrease sympathetic tone in the arm. Avoid deep anesthesia that could lower blood pressure and cardiac output (Option C)
- Brachial plexus block or stellate ganglion block is considered an alternative.
- Anticoagulation: Full heparinization is recommended if surgery is not performed.
- Surgical Treatment: Early arteriotomy may be considered.
- Prevention: Use a more dilute solution of thiopental sodium (e.g., 2.5% concentration) to reduce the risk of complications.
- Laryngospasm:
- Inject 5 ml of 2% procaine or lignocaine into the affected artery , ideally through the existing needle .

- General Anesthesia: To decrease sympathetic tone in the arm. Avoid deep anesthesia that could lower blood pressure and cardiac output (Option C)
- Brachial plexus block or stellate ganglion block is considered an alternative.
- Anticoagulation: Full heparinization is recommended if surgery is not performed.
- Surgical Treatment: Early arteriotomy may be considered.
- Occur if respiratory secretions or other irritants are present or during light anesthesia intubation.
- Prevention: Atropine premedication and succinylcholine administration after thiopentone can help prevent it.

Postoperative Restlessness:

- Postoperative pain can cause restlessness.
- Therefore, adequate analgesia should be provided.

Contraindications of Thiopental Sodium:

The immediate need is to manage the local tissue effects first before considering surgical intervention (Option B).

Starting heparinization is not the first step; the immediate priority is to address the local complication before considering systemic treatment options (Option D).

Solution for Question 16:

Correct Answer: B) It causes peripheral vasodilation and increased heart rate.

Explanation:

Effect on Cardiovascular System:

- Causes peripheral vasodilation , leading to the pooling of blood in the venous system, which reduces cardiac output.
- Increases heart rate due to a baroreceptor-mediated sympathetic reflex compensating for reduced cardiac output and blood pressure.
- Contraindication: Coronary artery disease patients: Increase in heart rate induced by thiopental increases myocardial oxygen consumption, which can be harmful. Patients at risk of ventricular dysrhythmias or with prolonged QT intervals. Hypovolemic patients: It reduces

cardiac output and arterial blood pressure.

- Coronary artery disease patients: Increase in heart rate induced by thiopental increases myocardial oxygen consumption, which can be harmful.
- Patients at risk of ventricular dysrhythmias or with prolonged QT intervals.
- Hypovolemic patients: It reduces cardiac output and arterial blood pressure.
- Coronary artery disease patients: Increase in heart rate induced by thiopental increases myocardial oxygen consumption, which can be harmful.
- Patients at risk of ventricular dysrhythmias or with prolonged QT intervals.
- Hypovolemic patients: It reduces cardiac output and arterial blood pressure.

Effect on Central Nervous System:

- Used for narcoanalysis.
- Induction of anesthesia in neurosurgery.
- Cerebral Vasoconstriction: Barbiturates constrict cerebral vasculature, reducing cerebral blood flow (CBF), cerebral blood volume, and intracranial pressure (ICP). ICP decreases more than arterial blood pressure, often leading to increased cerebral perfusion pressure (CPP).
- Barbiturates constrict cerebral vasculature, reducing cerebral blood flow (CBF), cerebral blood volume, and intracranial pressure (ICP).
- ICP decreases more than arterial blood pressure, often leading to increased cerebral perfusion pressure (CPP).
- Reduction in Cerebral Oxygen Consumption: Reduce cerebral oxygen consumption by up to 50% without causing harm due to the parallel decrease in CBF. (Option A)
- Reduce cerebral oxygen consumption by up to 50% without causing harm due to the parallel decrease in CBF. (Option A)
- Potential Neuroprotection: Protect the brain from focal ischemia (e.g., cerebral embolism).
- Protect the brain from focal ischemia (e.g., cerebral embolism).
- Dose-Dependent CNS Depression: Range from mild sedation to unconsciousness, depending on the dose.
- Range from mild sedation to unconsciousness, depending on the dose.
- Sensory Effects: Some patients experience a taste sensation (e.g., garlic, onions, or pizza) during thiopental induction.
- Some patients experience a taste sensation (e.g., garlic, onions, or pizza) during thiopental induction.
- Pain Perception: Not impaired

- Muscle Effects: Do not cause muscle relaxation and may induce involuntary skeletal muscle contractions.
- Do not cause muscle relaxation and may induce involuntary skeletal muscle contractions.
- Anticonvulsant Effects: Small doses of thiopental (50-100 mg IV) can rapidly control grand mal seizures briefly.
- Small doses of thiopental (50-100 mg IV) can rapidly control grand mal seizures briefly.
- Barbiturates constrict cerebral vasculature, reducing cerebral blood flow (CBF), cerebral blood volume, and intracranial pressure (ICP).
- ICP decreases more than arterial blood pressure, often leading to increased cerebral perfusion pressure (CPP).
- Reduce cerebral oxygen consumption by up to 50% without causing harm due to the parallel decrease in CBF. (Option A)
- Protect the brain from focal ischemia (e.g., cerebral embolism).
- Range from mild sedation to unconsciousness, depending on the dose.
- Some patients experience a taste sensation (e.g., garlic, onions, or pizza) during thiopental induction.
- Do not cause muscle relaxation and may induce involuntary skeletal muscle contractions.
- Small doses of thiopental (50-100 mg IV) can rapidly control grand mal seizures briefly.

Effects on Respiratory System:

- Causes respiratory depression.
- Double Apnea Pattern: An initial brief apnea upon administration, followed by a few breaths with adequate tidal volume, and then a more prolonged apnea (about 25 seconds).
- During awakening, both tidal volume and respiratory rate are reduced. (Option C)
- Do not fully suppress airway reflexes during laryngoscopy and intubation, making airway instrumentation potentially problematic: In asthmatic patients, airway manipulation can induce bronchospasm. Laryngospasm may occur in lightly anesthetized patients.
- In asthmatic patients, airway manipulation can induce bronchospasm.
- Laryngospasm may occur in lightly anesthetized patients.
- In asthmatic patients, airway manipulation can induce bronchospasm.
- Laryngospasm may occur in lightly anesthetized patients.

Effect on Kidney:

- Reduce renal blood flow and GFR in proportion to a fall in BP.

Effect on Liver:

- Cause a reduction in hepatic blood flow.

- Chronic exposure induces hepatic enzymes, leading to an increased rate of drug metabolism. (Option D)
 - Precipitate acute intermittent porphyria or variegate porphyria in susceptible individuals.
-

Solution for Question 17:

Correct Answer: B) Its rapid onset is primarily due to its high potency and redistribution.

Explanation: Thiopental sodium has a rapid onset due to its high lipid solubility and high nonionized fraction, allowing for quick redistribution from the bloodstream to the brain.

Thiopentone Sodium:

- Use: Intravenous induction agent.
- Prepared concentration: 2.5% solution
- Form : Yellowish-white crystalline powder or pale greenish hygroscopic powder (Option C)
- Odor: Garlic-like odor.
- Formulation : Thiopental sodium for injection is a sterile mixture of thiopental sodium and anhydrous sodium carbonate (buffer).
- Should not be reconstituted with lactated Ringer's solution or mixed with acidic solutions, as this reduces alkalinity and may cause precipitation of the drug as free acid.
- pH: 11
- Melting Point : The free acid (thiopental) melts at 158–160°C.
- Solubility : Water : Soluble in 1.5 parts of water at 20°C.
- Water : Soluble in 1.5 parts of water at 20°C.
- Water : Soluble in 1.5 parts of water at 20°C.
- Alcohol : Partially soluble.
- Insoluble : Practically insoluble in ether, benzene, and petroleum ether.
- Pharmacokinetics: Higher potency. Rapid onset and short duration of action. Dose: 3-6 mg/kg Duration of onset: 15-20 seconds Duration of action: 6-10 minutes Elimination T_{1/2}: 12 hours (Option D) Exhibits first-order kinetics. Unique property: Duration of induction dose is determined by redistribution. Metabolism and excretion site: Liver (Option A)
- Higher potency.
- Rapid onset and short duration of action.

- Dose: 3-6 mg/kg
 - Duration of onset: 15-20 seconds
 - Duration of action: 6-10 minutes
 - Elimination T_{1/2}: 12 hours (Option D)
 - Exhibits first-order kinetics.
 - Unique property: Duration of induction dose is determined by redistribution.
 - Metabolism and excretion site: Liver (Option A)
 - Higher potency.
 - Rapid onset and short duration of action.
 - Dose: 3-6 mg/kg
 - Duration of onset: 15-20 seconds
 - Duration of action: 6-10 minutes
 - Elimination T_{1/2}: 12 hours (Option D)
 - Exhibits first-order kinetics.
 - Unique property: Duration of induction dose is determined by redistribution.
 - Metabolism and excretion site: Liver (Option A)
-

Solution for Question 18:

Correct Answer: C) Thiopental is mixed with anhydrous sodium carbonate.

Explanation:

- Barbiturates (thiopental , methohexital, pentobarbital) are formulated as sodium salts, mixed with 6% by weight anhydrous sodium carbonate (increases alkalinity and therefore increases shelf life)
- Reconstituted with water, 5% glucose, or normal saline to produce: 2.5% solution of thiopental 2% solution of thiamylal 1% solution of methohexital
- 2.5% solution of thiopental
- 2% solution of thiamylal
- 1% solution of methohexital
- Barbiturates should not be reconstituted with lactated Ringer's solution or mixed with acidic solutions, as this reduces alkalinity and may cause precipitation of the drug as free acid.

(Option B)

- 2.5% solution of thiopental
- 2% solution of thiamylal
- 1% solution of methohexital

Classification of Intravenous Anesthetics:

Feature	Propofol	Thiopentone	Ketamine	Etomidate	Benzodiazepine
Color	Emulsion, milky white	Sodium salts	Clear aqueous solution	Clear solution	Midazolam-clear aqueous solution
Concentration	1%, 2% soln. Medium, long chain triglycerides	Yellow amorphous powder 500mg, 1g	1%, 2%, 10% solution	2mg/ml in 35% propylene glycol	
Lipid solubility	Only in emulsion form	Highly soluble	Very Lipid soluble	Acidic PH lipid soluble	Lipid soluble midazolam
Water solubility	Insoluble	Readily soluble	Soluble	Poorly water soluble	Water soluble
pH	4.5-6.4	10.5	3.5-5.5	6.9	3.5
Pka	11	7.6	7.5	4.2	6.1
Chirality	Non chiral	Racemic	Racemic	Single isomer R+	

Barbiturates primarily block NMDA receptors to induce anesthesia (Option A): While NMDA receptor antagonism can play a role in some anesthetics (like ketamine), barbiturates mainly enhance GABA-A receptor activity.

Ketamine primarily acts on GABA-A inhibitory receptors (Option D): Ketamine primarily acts as an NMDA receptor antagonist, not GABA-A inhibitors.

Solution for Question 19:

Correct Answer: D) Rocuronium

Explanation: Rocuronium is an intermediate-acting nondepolarizing neuromuscular blocking drug that acts as a muscle relaxant that helps in faster and more efficient intubation.

- Any patient taken up for an emergency surgery or trauma is considered to have a “full stomach”.
- If the patient has consumed food and it is a case of elective surgery, it must be postponed. But if it cannot be delayed, regional or local anesthesia is preferred.
- When these methods are impractical, the airway must be secured and precautions are taken to prevent aspiration as explained below.

Risk Factors for Aspiration:

- Full stomach
- Symptomatic gastroesophageal reflux disease (GERD)
- Hiatal hernia
- Presence of a nasogastric tube
- Morbid obesity
- Diabetic gastroparesis
- Pregnancy

Aspiration Prophylaxis:

- Goal: Decrease gastric volume and increase gastric fluid pH.
- Commonly used agents: Nonparticulate antacids (e.g., Bicitra) (Option A) Promotility drugs (e.g., metoclopramide) (Option B) H2-receptor antagonists (Option C)
- Nonparticulate antacids (e.g., Bicitra) (Option A)
- Promotility drugs (e.g., metoclopramide) (Option B)
- H2-receptor antagonists (Option C)
- Can be used alone or in combination.
- Nonparticulate antacids (e.g., Bicitra) (Option A)
- Promotility drugs (e.g., metoclopramide) (Option B)
- H2-receptor antagonists (Option C)

Anesthetic Management and Rapid Sequence Induction (RSI)

Preoxygenation	Always pre-oxygenate patients before induction. Patients with lung disease require 3 to 5 minutes of preoxygenation.
Preparation	Have a variety of laryngoscope blades, video laryngoscopes, intubation bougies, and endotracheal tubes ready and immediately available.

Cricoid Pressure (Sellick Maneuver)	An assistant applies firm pressure over the cricoid cartilage before induction to compress the esophagus and reduce the risk of aspiration. Excessive pressure during active regurgitation can cause esophageal rupture; the effectiveness of this maneuver remains uncertain.
Induction Agent Administration	Administer a bolus dose of propofol ; modify the dose if cardiovascular instability is present. Alternative rapid-acting agents include etomidate, ketamine, and methohexital.
Neuromuscular Blockade	Administer succinylcholine (1.5 mg/kg) or rocuronium (0.9–1.2 mg/kg) immediately after the induction dose, even if the patient is still conscious
Ventilation	Do not use bag-mask ventilation to avoid inflating the stomach and increasing aspiration risk. After loss of muscle response, rapidly intubate the patient while maintaining cricoid pressure until the cuff is inflated and the tube position is confirmed. Gentle ventilation may be allowed if cricoid pressure is maintained.
Difficult Intubation	If intubation is difficult, continue cricoid pressure and gently ventilate with oxygen until another attempt can be made. If unsuccessful, allow spontaneous ventilation and consider awake intubation. Sugammadex can be used to reverse rocuronium-induced muscle relaxation.
Postoperative Management	Keep the patient intubated until airway reflexes and consciousness return.

Indications for Tracheal Intubation :

- Full stomach or increased risk for aspiration of gastric secretions or blood.
- Critically ill patients.
- Significant lung abnormalities (e.g., low lung compliance, high airway resistance).
- Patients requiring lung isolation.
- Patients undergoing surgeries where SGAs would interfere.
- Patients needing postoperative ventilatory support.
- Failed SGA placement.
- Surgical requirement for NMBDs.
- Predicted difficult airway.
- Prolonged procedures.

Emergency Situations :

- Protect patients from pulmonary aspiration while managing intraocular pressure (IOP) during procedures.
- Succinylcholine can cause a modest increase in IOP; its use is controversial but may be acceptable based on current evidence.

Alternative Options :

- Use a higher dose of Rocuronium (0.9-1.2 mg/kg) and reverse with Sugammadex.
- Ensure adequate relaxation using a peripheral nerve stimulator before intubation.

Solution for Question 20:

Correct answer: B) To ensure the patient's lungs are fully saturated with oxygen before induction

Explanation:

Preoxygenation involves administering 100% oxygen for about 5 minutes before anesthesia induction, which increases lung and blood oxygen levels. This process creates an oxygen reserve that is crucial during the apnea period that follows induction and intubation, helping to prevent hypoxemia (low blood oxygen levels) during this critical time.

Phases of general anesthesia

Phase	Procedure	Details
Preoxygenation	Administer 100% O ₂ for 3 to 5 minutes	Can be done using a bag & mask or normal oxygen mask.
Induction	Induction to render the patient unconscious	IV induction is used in 99% of patients. Inhalation induction is commonly used in children (M/C agent: Sevoflurane).
Muscle relaxation	Short-acting muscle relaxants (e.g., Scoline) should be used before intubation. Long-acting muscle relaxants are administered after intubation.	
Intubation	Intubation is performed after muscle relaxation	

Maintenance	Maintain unconsciousness	Use of inhalation agents to keep the patient unconscious.
Maintain muscle relaxation	Use of long-acting muscle relaxants to maintain muscle relaxation	
Reversal Phase	Restore consciousness and contract muscles	Stop inhalational agents to make the patient conscious. Reversal agents (e.g., Neostigmine, Sugammadex, Pyridostigmine, Edrophonium) are used against muscle relaxants. Calabation 2 with 82 times affinity of sugammadex
Reverse muscle relaxation	L-cysteine for gantacurium reversal. Glycopyrrolate or Atropine is combined with reversal agents to prevent salivation and bradycardia.	
Extubation	Perform extubation after confirming signs of adequate reversal	
Signs of adequate reversal	Sustained head lift >5 seconds Moving all limbs. Obeying commands. TOF ratio > 0.9 (overall best sign).	

Increasing CO₂ retention is not the goal of preoxygenation (Option A). In fact, it may help to improve gas exchange.

Preoxygenation does not affect the depth of anesthesia required (Option C) . It focuses on increasing oxygen reserves.

Preoxygenation does not directly influence the respiratory rate during anesthesia (Option D).

Previous Year Questions

1. Which of the following is the most common day care drug used in day care surgery/TIVA?

- A. Sodium thiopentone
 - B. Propofol
 - C. Halothane
 - D. Ketamine
-

2. Which of the following intravenous anesthetics functions as an NMDA receptor antagonist?

- A. Ketamine
 - B. Propofol
 - C. Thiopentone sodium
 - D. Midazolam
-

3. Which of the following is most suitable for intravenous regional anesthesia?

- A. Bupivacaine
 - B. Mepivacaine
 - C. Procaine
 - D. Lidocaine
-

4. Which of the following is the most potent local anesthetic agent?

- A. Halothane
 - B. Isoflurane
 - C. Lidocaine
 - D. Bupivacaine
-

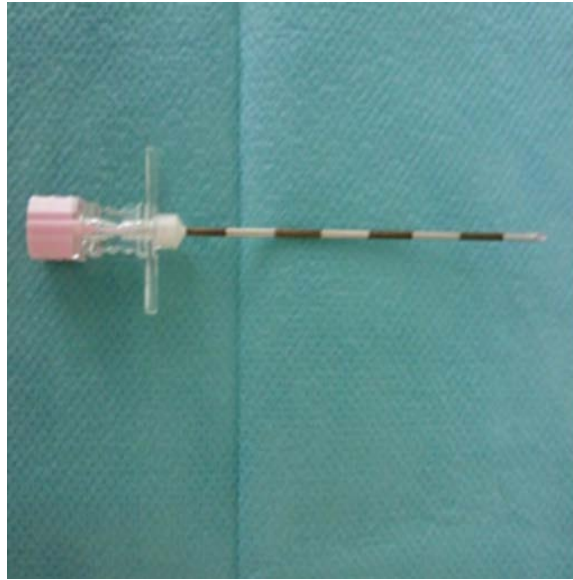
**5. Which of the following is required for the cannulation of a newborn? 24G needle
16G needle 5-mL syringe Adhesive tape Normal saline**

- A. 1, 3, 4, 5
 - B. 2, 3, 4, 5
 - C. 2, 4
 - D. 1, 4
-

6. Which of the following agents does not elicit pain on injection?

- A. Ketamine
 - B. Propofol
 - C. Etomidate
 - D. Thiopentone
-

7. What is the needle shown in the picture?



- A. Whitacre needle
 - B. Sprotte needle
 - C. Tuohy needle
 - D. Quincke-Babcock needle
-

8. Which of the following is an absolute contraindication to the use of morphine?

- A. Bronchial asthma
 - B. Renal failure
 - C. Emphysema
 - D. Head Injury
-

9. Which of the following is the shortest acting local anesthetic?

- A. Procaine
- B. Cocaine

- C. Dibucaine
 - D. Chloroprocaine
-

10. Intravenous induction agent of choice for day care anesthesia is?

- A. Ketamine
 - B. Propofol
 - C. Etomidate
 - D. Thiopentone
-

11. During recovery from anesthesia, a patient started experiencing hallucinations and vivid dreams. Which inducing agent is most likely responsible?

- A. Ketamine
 - B. Propofol
 - C. Thiopentone
 - D. Etomidate
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	4
Question 4	4
Question 5	1
Question 6	1
Question 7	3
Question 8	4
Question 9	4
Question 10	2
Question 11	1

Solution for Question 1:

Correct Option B - Propofol:

- Propofol is the preferred choice for daycare surgery and Total Intravenous Anesthesia (TIVA) due to its rapid onset and quick recovery profile. Additionally, Propofol is associated with a lower incidence of postoperative nausea and vomiting (PONV)

Incorrect Options:

- Options A, C and D are less commonly chosen for daycare surgeries

Solution for Question 2:

Correct Option A - Ketamine:

- Ketamine is an NMDA (N-methyl-D-aspartate) receptor antagonist.

Incorrect Options:

- Options B, C, and D act on GABA receptors
-

Solution for Question 3:

Correct Option D: Lidocaine

- Intravenous local anesthesia provides short-term anesthesia and analgesia during surgery on an extremity.
- Lignocaine is more suitable for intravenous regional anesthesia.

Incorrect Options:

Option A: Bupivacaine:

- Bupivacaine is the most cardiotoxic drug and it should never be administered intravenously.

Option B: Mepivacaine

- Mepivacaine has the same anesthetic potency as lidocaine but also has a milder dilating ability, leading to a longer duration of anesthesia without a vasoconstrictor.

Option C: Procaine

- Procaine can be used as an alternative if allergic to lignocaine.
 - It can cause methemoglobinemia.
 -
-

Solution for Question 4:

Correct Option D - Bupivacaine:

- Bupivacaine is 3-4 times more effective than lidocaine and with a long duration of action due to its high solubility in lipids and binding ability to proteins.

Incorrect Options:

Options A and B are not local anesthetics. Bupivacaine is more potent than lidocaine, hence option C is incorrect.

Solution for Question 5:

Correct Option A - 1, 3, 4, 5:

- Cannulation is inserting a thin tube or catheter into a vein or artery to administer fluids or medication or to obtain blood samples. It is a common procedure performed in newborns, especially those who require intensive care.
- To cannulate a newborn, several items are required. These include a needle, syringe, adhesive tape, and normal saline. The correct size of the needle and syringe will depend on the size and age of the newborn. Generally, a 24G needle is recommended for cannulation in newborns.
- Adhesive tape is used to secure the catheter or tube after insertion, preventing it from dislodging. Normal saline is used to flush the catheter before and after use to maintain patency and prevent blockages.

Incorrect Options:

Option B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 6:

Option A: Ketamine

- Ketamine is known for its unique pharmacological properties, including its analgesic effects. It has a less painful injection compared to other intravenous agents.

Option B: Propofol

- Propofol is associated with pain on injection. It can cause a burning or stinging sensation at the injection site. To mitigate this, propofol is often mixed with lidocaine or administered through a larger vein or with the use of a local anesthetic block.

Option C: Etomidate

- Etomidate is also associated with pain on injection. It can cause discomfort and burning at the injection site. Similar to propofol, methods like dilution, premedication with lidocaine, or administration through larger veins can reduce the pain.

Option D: Thiopentone

- Thiopentone can cause pain on injection. It is a barbiturate anesthetic and may lead to a burning or stinging sensation at the injection site. Techniques such as dilution or premedication with lidocaine may be employed to minimize the pain.
-

Solution for Question 7:

Correct Option C: Tuohy needle:

- The Tuohy needle is a large-bore needle with a curved tip and a relatively long bevel.
- It is commonly used for epidural anesthesia and pain management procedures.
- The curved tip helps in threading a catheter through the needle into the epidural space, and the long bevel allows for effective tissue penetration.

Incorrect Options:

Option A: Whitacre needle:

- The Whitacre needle is a type of spinal needle known for its pencil-point tip design.
- It has a short, sharp bevel and a small side port near the tip.

- The pencil-point tip is less likely to cause tissue damage or post-dural puncture headaches compared to other needle designs.
- It is commonly used for spinal anesthesia and lumbar punctures.



Option B: Sprotte needle:

- The Sprotte needle is another type of spinal needle with an atraumatic design.
- It features a small side port and a blunt tip, reducing the risk of tissue trauma during insertion.
- The Sprotte needle is used for spinal anesthesia and is known for its low incidence of post-dural puncture headaches.



Option D: Quincke-Babcock needle:

- The Quincke-Babcock needle is a type of spinal needle that features a sharp, triangular beveled tip.
- It is used for performing lumbar punctures and spinal anesthesia.
- The sharp bevel allows for easier penetration of the dura mater during the procedure.



Solution for Question 8:

Correct Option D - Head injury:

- Head Injury is an absolute contraindication for giving morphine.
- As it further increases the intracranial pressure.
- It masks the useful diagnostic & prognostic signs.
- The effects of morphine such as respiratory depression, miosis & mental clouding also interfere with the diagnosis.

Incorrect Options:

Option A - Bronchial asthma:

- Bronchial asthma is not an absolute contraindication to the use of morphine.

- However, caution should be exercised because morphine can cause respiratory depression and may exacerbate bronchospasm in susceptible individuals.
- If necessary, alternative analgesic options may be considered for patients with severe bronchial asthma or a known hypersensitivity to opioids.

Option B - Renal failure:

- Renal failure is not a contraindication to the use of morphine.
- However, in patients with renal impairment or end-stage renal disease, dosage adjustments may be necessary due to the potential for accumulation of morphine and its metabolites.
- Close monitoring of renal function and titration of the dose are recommended in such cases.

Option C - Emphysema:

- Emphysema is not a contraindication to the use of morphine.
- However, caution should be exercised because morphine can depress the respiratory drive and may worsen respiratory function in patients with severe emphysema.
- Adjustments to the dosage or consideration of alternative analgesics may be necessary in these situations.

Solution for Question 9:

Correct Option D - Chloroprocaine:

- Chloroprocaine is a local anesthetic with a short duration of action.
- It is often used for short procedures and is suitable for regional anesthesia techniques where a rapid onset and short duration of anesthesia are desired.
- Longest to shortest acting local anesthetics: Dibucaine (Longest) > Tetracaine > Bupivacaine > Lignocaine > Procaine > Chloroprocaine (Shortest)

Incorrect Options: Options A, B, and C are incorrect. Refer to the explanation of the correct answer.

Solution for Question 10:

Option B: Propofol

- Propofol is an intravenous anesthetic agent of choice for day care surgery because it does not have a residual effect.
- Propofol has rapid onset of action and smooth recovery, hence preferred for day care surgeries.

Incorrect options: Options A, C, and D are not the intravenous induction agent of choice for day care anesthesia.

Solution for Question 11:

Correct Option A: Ketamine:

- Ketamine is a non opioid intravenous anesthetic agent that causes emergence delirium/ hallucinations and vivid dreams due to thalamo-cortical disassociation.
- The most common side effect of ketamine is hallucinations.

Incorrect options: Options B, C, and D are incorrect.

Option B: Propofol:

- Propofol is an intravenous anesthetic agent of choice for day care surgery because it does not have a residual effect.

Option C: Thiopentone:

- The most common adverse effect of thiopentone sodium is laryngospasm.

Option D: Etomidate:

- The most common adverse effect of etomidate is post operative nausea and vomiting.
-

Inhalational Anaesthetics

1. Which of the following statements about Minimal Alveolar Concentration (MAC) is true?

- A. MAC is the lowest for nitrous oxide
 - B. MAC is used to compare the potency of inhaled anesthetics
 - C. MAC is a measure of the blood concentration of anesthetic agents
 - D. MAC does not vary with age
-

2. During a routine surgical procedure requiring general anesthesia, an anesthesiologist is determining the appropriate dosage of sevoflurane for a 45-year-old patient. Several factors could influence the Mean Alveolar Concentration (MAC) needed to achieve adequate anesthesia. The anesthesiologist considers the following factors: Patient Age: The patient is 45 years old. Body Temperature: The patient has a normal body temperature of 37°C. Use of Opioids: The patient has received a dose of fentanyl prior to induction. Which factor is likely to decrease the MAC value for sevoflurane in this patient?

- A. Height
 - B. Gender
 - C. Use of opioids
 - D. Thyroid status
-

3. A 45-year-old male is undergoing general anesthesia for a routine surgery. The anesthesiologist uses two different inhalational agents: one with a high blood-gas partition coefficient and another with a low blood-gas partition coefficient. Which of the following is TRUE regarding the blood-gas partition coefficient and its

clinical implications?

- A. An inhalational agent with a high blood-gas partition coefficient will achieve anesthesia more quickly than one with a low coefficient.
 - B. An inhalational agent with a low blood-gas partition coefficient will require a higher concentration in the alveoli to maintain anesthesia.
 - C. An inhalational agent with a high blood-gas partition coefficient will have a slower onset and offset of action compared to one with a low coefficient.
 - D. The blood-gas partition coefficient has no effect on the pharmacokinetics of inhalational anesthetics.
-

4. Which of the following anesthetics is ordered from lowest to highest in terms of Minimum Alveolar Concentration (MAC) values (indicating potency)?

- A. Desflurane, Isoflurane, Halothane
 - B. Halothane, Isoflurane, Desflurane
 - C. Sevoflurane, Nitrous Oxide, Desflurane
 - D. Nitrous Oxide, Sevoflurane, Halothane
-

5. All the following statements about the systemic effects of inhalational agents are correct except:

- A. Isoflurane can be safely used in patients with head injury
 - B. Halothane is the anesthetic agent of choice in patients with ischaemic heart disease
 - C. Almost all inhalational agents cause bronchodilation
 - D. Sevoflurane is the anesthetic of choice in bronchial asthma
-

6. A 45-year-old woman is undergoing elective cholecystectomy under general anesthesia. The anesthesiologist is using halothane as the primary inhalational anesthetic. During the procedure, the surgeon encounters unexpected bleeding and requests local infiltration with adrenaline (epinephrine) for hemostasis. What is the most appropriate response of the anesthesiologist to the surgeon's request for adrenaline infiltration?

- A. Agree to the use of adrenaline without any concerns
 - B. Refuse the use of adrenaline completely due to high-risk
 - C. Allow the use of adrenaline, but advise using a dilute solution and limiting the total dose
 - D. Suggest switching to a different inhalational anesthetic before using adrenaline
-

7. A 68-year-old patient with a history of liver disease is scheduled for a prolonged liver transplant surgery. Which of the following inhalational anesthetics would be most appropriate to use in this case?

- A. Halothane
 - B. Desflurane
 - C. Methoxyflurane
 - D. Isoflurane
-

8. A 45-year-old woman with a history of moderate alcohol use undergoes general anesthesia for an emergency appendectomy. The anesthesiologist uses halothane for induction and maintenance. Two weeks after the surgery, the patient presents with fatigue, nausea, jaundice, and tender hepatomegaly. Laboratory tests reveal the following: AST (Aspartate Aminotransferase): 850 U/L (normal range: 5-40 U/L) ALT (Alanine Aminotransferase): 920 U/L (normal range: 7-56 U/L) Total Bilirubin: 4.8 mg/dL (normal range: 0.1-1.2 mg/dL) Direct Bilirubin: 3.2 mg/dL (normal range: 0.0-0.3 mg/dL) Which of the following best explains the patient's condition?

- A. Direct hepatotoxicity of halothane

- B. Immune-mediated hepatotoxicity of halothane
 - C. Alcohol-induced liver failure precipitating due to surgical stress
 - D. Postoperative sepsis
-

9. A 60-year-old man with hypertension and coronary artery disease is undergoing general anesthesia for abdominal surgery. The anesthesiologist chooses to use isoflurane for the maintenance of anesthesia. Which of the following is a correct statement about isoflurane's effects and properties?

- A. It causes significant myocardial depression
 - B. It increases systemic vascular resistance
 - C. It is highly metabolized in the liver
 - D. It provides cardio-protection through ischemic preconditioning
-

10. A 55-year-old female is scheduled for elective surgery. The anesthesiologist is reviewing the anesthesia plan, which includes the use of sevoflurane. Which of the following statements regarding sevoflurane is TRUE?

- A. It significantly increases heart rate during anesthesia.
 - B. can be given to the patient with raised intracranial pressure.
 - C. It is known to cause significant renal toxicity.
 - D. It doesn't reverse bronchospasm.
-

11. Which of the following statements about the systemic effects of desflurane is true?

- A. Increase in systemic vascular resistance.
- B. Severe nephrotoxicity.

- C. Increases cerebral blood flow (CBF) while decreasing cerebral oxygen consumption.
 - D. No effect on respiratory rate.
-

12. A 30-year-old male with a family history of severe reactions to anesthesia is scheduled for surgery under general anesthesia. The anesthetic plan includes nitrous oxide (N₂O) along with other agents. Given his history, which of the following statements about nitrous oxide is FALSE?

- A. Nitrous oxide is a colorless and odorless gas.
 - B. It can trigger malignant hyperthermia.
 - C. It is contraindicated in patients with Pneumothorax.
 - D. It causes MI in patients with CAD.
-

13. Which of the following statements best describes the second gas effect of Nitrous oxide when administered along with Isoflurane?

- A. Decreased concentration of Isoflurane in the alveoli
 - B. Increased concentration of Isoflurane in the alveoli
 - C. Increase concentration of Nitrous oxide in alveoli
 - D. No effect on the concentration of gases in the alveoli
-

14. A 35-year-old patient is recovering from anesthesia after a dental procedure that involved the use of nitrous oxide (N₂O). As the anesthesiologist prepares to discontinue N₂O, they consider the potential risks associated with it. To prevent this complication in the patient, which of the following actions should be taken?

- A. Administer room air immediately after stopping N₂O.
- B. Administer 100% oxygen for 5 to 10 minutes after stopping N₂O.

- C. Continue N₂O for an additional 10 minutes.
 - D. Reduce the concentration of N₂O gradually over the next 15 minutes.
-

15. Which among the following statements explains the concept of Mayer Overton's rule?

- A. The lipid solubility of the agent is directly proportional to the potency of the agent
 - B. The rule asserts that all anesthetics act through the same mechanism of action regardless of their structure.
 - C. The rule indicates that anesthetics with lower boiling points are more potent.
 - D. None of the above
-

16. During a surgical procedure, the anesthesiologist is monitoring the patient and observes signs of complete respiratory depression, total loss of reflexes, and dilated, non-reactive pupils. According to Guedel's stages of anesthesia, which stage is the patient most likely in, and what are the implications for airway management?

- A. Stage I (Analgesia)
 - B. Stage II (Excitement)
 - C. Stage III (Surgical Anesthesia)
 - D. Stage IV (Overdose)
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	2
Question 6	3
Question 7	2
Question 8	2
Question 9	4
Question 10	2
Question 11	3
Question 12	2
Question 13	2
Question 14	2
Question 15	1
Question 16	4

Solution for Question 1:

Correct Answer: B) MAC is used to compare the potency of inhaled anesthetics.

Explanation:

Minimal Alveolar Concentration (MAC) is a standard measure used in anesthesia to assess the potency of inhaled anesthetics.

- It represents the concentration of anesthetic in the alveoli needed to prevent movement in 50% of patients exposed to a surgical/painful stimulus.
- A lower MAC value indicates a more potent anesthetic, meaning less is needed to achieve the desired anesthetic effect.
- Methoxyflurane has the least MAC
- Nitrous oxide has the highest MAC

MAC 95:

- Minimal alveolar concentration at which 95% of patients do not respond to any stimuli.
- $MAC\ 95 = 1.3 \times MAC$

MAC awake:

- MAC at which 50% of patients become awake.
- $MAC\ awake = 0.3 \times MAC$

MAC is a measure of the blood concentration of anesthetic agents (Option C) : Mean Alveolar Concentration (MAC) is not actually a measure of the blood concentration of anesthetic agents; rather, it refers specifically to the concentration of inhaled anesthetics in the alveoli of the lungs.

MAC does not vary with age (Option D): MAC value is inversely proportional to age.

Solution for Question 2:

Correct Answer: C) Use of opioids

Explanation:

Opioids decrease MAC value.

- Opioids like fentanyl have a synergistic effect with anesthetics, meaning they can enhance the anesthetic effect. This allows for a lower MAC to achieve the same anesthetic depth, effectively decreasing the required concentration of sevoflurane.

Factors increasing MAC	Factors decreasing MAC	Factors not affecting MAC
Young patient (maximum at 6 months) Chronic alcoholic Acute amphetamine/cocaine user Hypernatremia Hyperthermia Red hair	Elderly person Acute alcoholic Chronic amphetamine user Hyponatremia Hypothermia Hypoxia Hypercarbia Anemia (low hemoglobin) Pregnancy Local anaesthetics I.V. induction	Gender (Option B) Height (Option A) Thyroid status (Option D)

Solution for Question 3:

Correct Answer: C) An inhalational agent with a high blood-gas partition coefficient will have a slower onset and slower recovery than one with a low coefficient.

Explanation:

Anesthetic agents having a high blood-gas partition coefficient (B/G) will have a high solubility in blood leading to a low alveolar concentration, slow induction, and slow recovery.

Blood gas coefficient:

- Indicates the speed of onset and recovery of inhalational agent
- Indicates solubility of the inhalational agent in the blood
- Inversely proportional to the speed of onset
- High B/G has a slow onset (Option A)
- Low B/G has a fast onset
- The inhalational agent enters the lungs → blood vessels → brain

An inhalational agent with a low blood-gas partition coefficient will require a higher concentration in the alveoli to maintain anesthesia (Option B): This is misleading. While low solubility agents reach higher concentrations in the alveoli more quickly, they do not necessarily require higher concentrations to maintain anesthesia.

The blood-gas partition coefficient has no effect on the pharmacokinetics of inhalational anesthetics (Option D): This is false. The blood-gas partition coefficient is critical in determining how quickly an anesthetic will act and how it will be eliminated from the body.

Solution for Question 4:

Correct Answer: B) Halothane, Isoflurane, Desflurane

Explanation:

- Halothane has the lowest MAC and a higher blood-gas partition coefficient, making it the most potent of the three.
- Isoflurane has a moderate MAC and partition coefficient.
- Desflurane has the highest MAC, indicating it is the least potent of the three.

MAC values of different drugs	Most potent to least potent
N ₂ O: 105%	Methoxyflurane: 0.16%
Xenon: 70%	Halothane: 0.75%
Desflurane: 6%	Isoflurane: 1.15%
Sevoflurane: 2%	Sevoflurane: 2%
Isoflurane: 1.15%	Desflurane: 6%
Halothane: 0.75%	
Methoxyflurane: 0.16%	

Mnemonic for most potent to least potent: MaHI S traight D rive - M ethoxyflurane, H alothane, I soflurane, S evoflurane, D esflurane.

Solution for Question 5:

Correct Answer: B) Halothane is the anesthetic agent of choice in patients with ischaemic heart disease.

Explanation:

Halothane is contraindicated in patients with ischaemic heart disease.

- Halothane decreases cardiac output, BP, and PR
- Halothane causes hypotension by direct myocardial depression
- Halothane causes bradycardia by blunting carotid body receptor response
- Halothane is a common inhalational agent used in the respiratory system (due to maximum bronchodilation) while Isoflurane is the common inhalational agent used in CNS and CVS

Central Nervous System:

- Inhalational agents are CNS depressant
- They reduce cerebral metabolic rate
- Commonly used inhalational agent is isoflurane
- They increase the cerebral blood flow thereby increasing the ICP, so they are avoided in head injury and neurosurgery (maximum with halothane)
- Isoflurane is an inhalational agent of choice in head injury as it decreases the ICP (Option A)
- Most epileptogenic agent: Enflurane (increases ICP and also causes epilepsy)

Respiratory system

- Almost all inhalational agents cause bronchodilation (Option C)
- Reduces the Respiratory rate
- Causes decreased mucociliary function (except ether)
- Maximum bronchodilation is seen with halothane, therefore 2nd best inhalational agent for bronchial asthma is halothane.
- Best inhalational agent for bronchial asthma: Sevoflurane (Option D)

Cardiovascular system

- Reduces heart rate, BP, and systemic vascular resistance
- Halothane increases the sensitivity of the myocardium toward catecholamines
- Halothane, isoflurane, enflurane - prolong QT interval

Gastrointestinal system:

- Reduces the liver blood flow
- Halothane hepatitis: maximally metabolized produces trifluoroacetic acid as an end product that reacts with liver cells (immunologically) causing acute fulminant hepatic necrosis.

Uterus:

- All agents are good uterine relaxants so there is a high chance of PPH
-

Solution for Question 6:

Correct Answer: C) Allow the use of adrenaline, but advise using a dilute solution and limiting the total dose

Explanation: This scenario highlights the important interaction between halothane and adrenaline (epinephrine) The key points to consider are:

- Halothane sensitizes the myocardium to the arrhythmogenic effects of catecholamines, including adrenaline.
 - This interaction can potentially lead to severe cardiac arrhythmias , particularly ventricular arrhythmias.
 - However, the risk is dose-dependent and can be mitigated by using appropriate precautions.
 - Completely avoiding adrenaline may compromise surgical hemostasis, which is also important for patient safety.
 - Modern practice allows for the cautious use of adrenaline with halothane under certain conditions.
-

Solution for Question 7:

Correct Answer: B) Desflurane

Explanation: Desflurane is the least metabolized inhalational agent in the liver , with <0.01% metabolized in the liver and 99.9% in the lungs. This minimal hepatic metabolism makes it safer for patients with liver disease.

Metabolism of inhalational anesthetic agents

Anesthetic	Metabolism	Key Characteristics	Specific Indications/Contraindications
Halothane	20% in liver (maximum)	Most hepatotoxic	Can cause greatest reduction in cardiac output
Desflurane	<0.01% in liver, 99.9% in lungs	Least metabolized in the liver Minimally metabolized in the body Readily excreted by exhalation	Choice for liver and renal transplant Induction or maintenance of anesthesia in adults Maintenance of anesthesia in pediatric patients after induction with other agents
Methoxyflurane	70% in tissues	Most potent and slow onset Most nephrotoxic Nephrotoxic effect attributed to inorganic fluoride produced by it	

Solution for Question 8:

Correct Answer: B) Immune-mediated hepatotoxicity of halothane

Explanation: The patient's presentation is consistent with halothane-induced hepatitis, also known as Type-2 halothane hepatotoxicity. This is an immune-mediated reaction that typically occurs 1-2 weeks after exposure to halothane, especially in patients with previous exposure or risk factors such as moderate alcohol use.

Halothane

Category	Details
Chemical Properties	- Clear, heavy, colorless liquid - Sweet, non-irritating odor, photosensitive (stored in an amber bottle with thymol preservative) - Structure of an alkane - Highly soluble in blood and tissues, hence can cause delayed recovery
Systemic Effects	- Cardiovascular: Myocardial depression, sensitization to catecholamines - Respiratory: Respiratory depression - Central Nervous System: General anesthesia, increased cerebral blood flow

Uses	- Induction and maintenance of general anesthesia - Less commonly used now due to side effect profile - Still used in some resource-limited settings
Side Effects	- Type 1 Hepatotoxicity: Mild, self-limiting liver injury (20-30% of patients) - Type 2 Hepatotoxicity (Halothane Hepatitis): Severe, potentially fatal (1 in 35,000 after single exposure) - Cardiorespiratory instability - Arrhythmias, especially in children - Malignant hyperthermia (rare)
Metabolism	- 20-45% metabolized in the liver - Produces trifluoroacetyl (TFA) protein adducts which are immunogenic and are responsible for causing halothane hepatitis.

Solution for Question 9:

Correct Answer: D) It provides cardio-protection through ischemic preconditioning

Explanation: Isoflurane is a volatile anesthetic used for the induction and maintenance of general anesthesia.

- It provides cardio-protection through ischemic preconditioning , which can help reduce ischemia-reperfusion injury to the heart. This makes it a good choice for patients with coronary artery disease.

Category	Details
Chemical Properties	- Halogenated ether compound - Non-flammable - Strong, pungent odor
Cardiovascular Effects	- Minimal impact on left ventricular function - Dose-dependent decrease in systemic vascular resistance - Mild increase in heart rate - Coronary vasodilation - Coronary steal syndrome - (HIND Drugs - Hydralazine, Isoflurane, Nitroprusside, Dipyridamole) - Cardioprotection through ischemic preconditioning
Respiratory Effects	- Decrease in tidal volume - Minimal increase in respiratory rate - Overall decrease in minute ventilation - Increased PaCO ₂

CNS Effects	- Increased cerebral blood flow at >1 MAC - Increased intracranial pressure at >1 MAC - Decreased cerebral metabolic rate
Uses	- Induction of general anesthesia (less commonly used due to odor, irritation, and laryngospasm) - Maintenance of general anesthesia - agent of choice in cardiothoracic and neurosurgery.
Side Effects	- Dose-dependent hypotension - Potential trigger for malignant hyperthermia
Metabolism	- Minimal hepatic metabolism (0.2%) - Metabolized to trifluoroacetic acid - No significant renal or hepatic toxicity at clinical doses

It causes significant myocardial depression (Option A): Isoflurane has minimal impact on left ventricular function.

It increases systemic vascular resistance (Option B): Isoflurane has a dose-dependent decrease in systemic vascular resistance.

It is highly metabolized in the liver (Option C): Isoflurane undergoes minimal hepatic metabolism (only about 0.2%).

Solution for Question 10:

Correct Answer: D) It doesn't reverse bronchospasm.

Explanation:

Sevoflurane and isoflurane when compared to other inhalational agents will cause less increase in cerebral blood flow and intracranial pressure.

Sevoflurane	
Properties	Sevoflurane is a fluorinated ether with low blood solubility (λ_b/g 0.65), allowing for rapid induction and emergence. Its non-pungency makes it suitable for inhalation induction, especially in pediatric patients. It is used with conventional vaporizers.

Uses	Ideal for smooth and rapid inhalation induction, particularly in daycare procedures (4-8% concentration with nitrous oxide and oxygen). It is commonly used for both pediatric and adult anesthesia.
C/I	Should be avoided in cases of severe hypovolemia, susceptibility to malignant hyperthermia, .
Systemic Effects	
Cardiovascular	Mild myocardial depression with a slight reduction in systemic vascular resistance and blood pressure. It causes a minimal heart rate increase and may prolong the QT interval. (Option A)
Hepatic	Decreases portal vein flow but increases hepatic artery flow, maintaining overall hepatic perfusion.
Renal	Slightly decreases renal blood flow but has no known association with significant renal toxicity. (Option C) Compound A , a degradation product of sevoflurane, is nephrotoxic in animals but has no known adverse effects on humans.
Neuromuscular	Provides adequate muscle relaxation for intubation but is often combined with other agents or muscle relaxants.
Cerebral	Slight increases in cerebral blood flow (CBF) and intracranial pressure. High concentrations may impair CBF autoregulation. Seizures have not been reported.
Respiratory	Depresses respiration and helps reverse bronchospasm. Thus, agent of choice in asthmatics.

Solution for Question 11:

Correct Answer: C) Increases cerebral blood flow (CBF) while decreasing cerebral oxygen consumption.

Explanation:

Desflurane increases cerebral blood flow and intracranial pressure while decreasing cerebral oxygen consumption.

Desflurane	
Properties	Desflurane is a fluorinated ether with high vapor pressure (681 mm Hg at 20°C) and very low blood solubility (λ_b/g 0.42), allowing rapid induction and emergence. It requires a special vaporizer (tec 6) for precise anesthetic control. TEC 6
Uses	Desflurane is ideal for rapid induction and emergence due to its low blood solubility, but it is not suitable for inhalation induction due to its pungency and airway irritation.
C/I	It should be avoided in patients with severe hypovolemia, malignant hyperthermia, and intracranial hypertension.
Systemic Effects	
Cardiovascular	Desflurane decreases systemic vascular resistance and arterial blood pressure. (Option A) It causes moderate increases in heart rate, central venous pressure, and pulmonary artery pressure, particularly during rapid concentration increases, which can be managed with fentanyl, esmolol, or clonidine.
Respiratory	Causes decreased tidal volume and increased respiratory rate. It is pungent and can cause airway irritation, making it unsuitable for inhalation induction. (Option D)
Hepatic	Minimal metabolism and risk of anesthetic-induced hepatitis, with maintained hepatic oxygen delivery.
Renal	Minimal nephrotoxicity, but decreases in urine output and glomerular filtration occur with reduced cardiac output. (Option B)
Neuromuscular	Desflurane causes a dose-dependent decrease in neuromuscular response, enhancing the effects of nondepolarizing neuromuscular blockers.
Cerebral	Increases cerebral blood flow (CBF), intracranial pressure, and cerebral blood volume but decreases cerebral oxygen consumption. CBF can be lowered through hyperventilation.

Solution for Question 12:

Correct Answer: B) It can trigger malignant hyperthermia.

Explanation:

Nitrous oxide does not trigger malignant hyperthermia, which is a severe reaction typically associated with certain volatile anesthetics and succinylcholine.

Nitrous Oxide	
Properties	Nitrous oxide (N ₂ O) is a colorless, odorless, nonflammable gas, stored as a liquid under pressure. (Option A ruled out) It supports combustion and is a relatively inexpensive anesthetic. It is an NMDA receptor antagonist. Stored in blue color cylinders at 745 psi pressure.
Uses	Commonly used in combination with volatile anesthetics to reduce their required dose via the second gas effect. It provides analgesia in dental surgery, labor, and minor surgical procedures. Nitrous oxide has minimal muscle relaxation effects and doesn't trigger malignant hyperthermia.
Contraindications	Should be avoided in patients with air-containing cavities because it can cause rapid expansion of these air cavities (e.g., pneumothorax, air embolism, intestinal obstruction, pneumocephalus, intraocular air, tympanic membrane grafting). (Option C ruled out) It is unsuitable for patients needing high oxygen concentrations and is typically avoided in early pregnancy due to possible teratogenic effects.
Side Effects	Can cause myocardial depression in patients with coronary artery disease or hypovolemia. (Option D ruled out) It may increase pulmonary vascular resistance, lead to nausea and vomiting, and impair the body's response to infection. Chronic exposure may result in bone marrow suppression and neurological deficits due to vitamin B12 enzyme inhibition .

Solution for Question 13:

Correct Answer: B) Increased concentration of Isoflurane in the alveoli

Explanation:

As N₂O is rapidly absorbed into the blood, it decreases its own concentration in the alveoli (Option C ruled out), which allows for a relatively higher concentration of the second anaesthetic gas (Isoflurane in this case).

Second Gas Effect

The second gas effect occurs when a second lesser soluble Inhalational anaesthetic is given alongside nitrous oxide (N₂O), leading to a concentration effect.

(Example: 2% of a less soluble potent anaesthetic is administered with 70% N₂O and 28% oxygen)

↓

N₂O partitions into the blood quickly due to its high partial pressure.

As N₂O is absorbed, its concentration in the alveoli decreases.

Increase in concentration of less soluble potent anaesthetic in the alveoli.

(Remaining gases in alveoli: 35 parts N₂O, 28 parts oxygen, 2 parts potent anesthetic)

Potent anesthetic concentration increases to 3.1%.

- As N₂O partitions into the blood, it leaves behind a higher relative concentration of the potent anesthetic in the alveoli. This results in an increased concentration of the anesthetic gas, although the clinical impact on speeding induction is minimal.

Decreased concentration of Isoflurane in the alveoli (Option A): The second gas effect actually leads to an increased concentration of Isoflurane in the alveoli.

No effect on the concentration of gases in the alveoli (Option D): The second gas effect directly influences the concentration of the anaesthetic gas in the alveoli by increasing the concentration of Isoflurane in the alveoli.

Solution for Question 14:

Correct Answer: B) Administer 100% oxygen for 5 to 10 minutes after stopping N₂O.

Explanation:

The possible risk associated with N₂O stoppage is diffusion hypoxia.

Administering 100% oxygen for 5 to 10 minutes after discontinuing nitrous oxide helps to counteract the dilution of alveolar gases, preventing diffusion hypoxia and ensuring adequate oxygenation.

- Diffusion hypoxia occurs during recovery from anesthesia due to the washout of high concentrations of N₂O, leading to lower O₂ and CO₂ levels in the alveoli, which is called the fink effect.
- This can cause hypoxemia (low blood oxygen) and hypocarbia (low carbon dioxide), which may suppress respiratory drive and worsen hypoxemia.
- To prevent diffusion hypoxia, it is recommended to administer 100% O₂ for 5 to 10 minutes after stopping nitrous oxide.
- The alveolar membrane is the main route for eliminating inhalation anesthetics.
- Factors that speed up induction also enhance recovery, including high fresh gas flows, low circuit volume, and increased ventilation.

Administer room air immediately after stopping N₂O (Option A) can lead to further dilution of alveolar O₂ and CO₂, increasing the risk of hypoxemia and respiratory depression.

Continuing N₂O for an additional 10 minutes (Option C) would not resolve the risk of diffusion hypoxia and would prolong the potential adverse effects.

Reducing the concentration of N₂O gradually over the next 15 minutes (Option D) does not effectively prevent diffusion hypoxia, and the immediate administration of 100% oxygen is necessary to counteract its effects.

Solution for Question 15:

Correct Answer: A) The lipid solubility of the agent is directly proportional to the potency of the agent

Explanation:

- Meyer Overton rule: Lipid solubility of the agent is directly proportional to the potency of the agent
- Potency is measured by MAC and is inversely proportional to MAC value.
- While anesthetics may share some mechanisms, they can have vastly different structures and effects, which the rule does not address (Option B)

The boiling point does not correlate with anesthetic potency in the context of the Meyer-Overton rule (Option C)

Solution for Question 16:

Correct Answer: D) Stage IV (Overdose)

Explanation:

Stage IV of Guedel's staging represents overdose/ medullary paralysis and presents with:

- Respiratory arrest
- Cardiovascular collapse
- Dilated, non-reactive pupils

Guedel's Staging:

- Stage I (Analgesia) (Option A): In this stage, the patient is conscious and can respond to verbal commands. Airway management is typically not necessary as they can maintain their airway.
- Stage II (Excitement) (Option B): Patients may exhibit erratic breathing and are often agitated. Close monitoring and potential airway interventions may be needed to prevent airway obstruction
- Stage III (Surgical Anesthesia) (Option C): This is the stage where the patient is deeply unconscious, and controlled ventilation is essential. While airway management is critical, reflexes may still be present.

Stage IV (Overdose): Indicates severe respiratory depression and potential cardiovascular instability. Immediate airway intervention and resuscitation are required, as the patient may not be breathing adequately or may be unresponsive.

	Respiration		Ocular Movements	Pupils No Premedication	Eye reflexes	Secretion of tears	Laryngeal and pharyngeal reflexes	Respiratory response to skin incision	Muscular tone
	Inter-costal	Diaphragm							
Stage 1			Voluntary control			Normal			Normal
Stage 2					Eyebash				
Stage 3 (Plane I)					Lid		Swallowing retching vomiting		Normal struggling
Stage 3 (Plane II)					Corneal				
Stage 3 (Plane III)					Pupillary				
Stage 3 (Plane IV)					Lid reflex				
Stage 4									

Previous Year Questions

1. Which among the following is the induction anesthesia of choice in pediatric age group?

- A. Sevoflurane
 - B. Desflurane
 - C. Halothane
 - D. Isoflurane
-

2. Inhalational induction of choice in children is:

- A. Sevoflurane
 - B. Desflurane
 - C. Halothane
 - D. Isoflurane
-

3. Which of the following induction agents is most appropriate for a patient with cardiac disease?

- A. Ketamine
 - B. Etomidate
 - C. Propofol
 - D. Thiopentone
-

4. Which of the following can be used for induction of anesthesia, maintenance and can be safely used in endotracheal intubation?

- A. Halothane
 - B. Desflurane
 - C. Sevoflurane
 - D. Isoflurane
-

5. Which of the following is not used for induction of anesthesia in pediatric patients?

- A. Halothane
 - B. Sevoflurane
 - C. Desflurane
 - D. Nitrous oxide
-

6. A patient was given halothane for a surgical procedure. After a few minutes, the anesthetist noticed that he developed rigidity and hyperthermia. Which of the following ions is the reason for the above symptoms?

- A. Sodium
 - B. Calcium
 - C. Potassium
 - D. Magnesium
-

7. Which inhalational agent is best suited for patients with compromised cardiac function?

- A. Halothane
 - B. Isoflurane
 - C. Desflurane
 - D. Sevoflurane
-

8. Which of the following Anaesthesia agent is hepatotoxic:

- A. Sevoflurane
 - B. Isoflurane
 - C. Halothane
 - D. Procaine
-

9. Which of the following drugs is used for hypotensive anesthesia?

- A. Isoflurane
 - B. Nitroglycerin
 - C. Both 1 & 2
 - D. Dantrolene
-

10. Which of the following inhalation agents has a blood-gas partition co-efficient similar to nitrous oxide?

- A. Isoflurane
- B. Desflurane
- C. Sevoflurane
- D. Halothane

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	2
Question 4	3
Question 5	3
Question 6	2
Question 7	3
Question 8	3
Question 9	3
Question 10	2

Solution for Question 1:

Correct Option A - Sevoflurane:

- Sevoflurane is the preferred induction agent in pediatric anesthesia due to its rapid onset, pleasant smell, and minimal airway irritation, making it well-tolerated by children.

Incorrect Options:

Option B - Desflurane: Desflurane causes airway irritation and laryngospasm, making it less suitable for pediatric induction.

Option C - Halothane: Halothane is less favored as it is the most hepatotoxic anesthetic agent and can also lead to halothane hepatitis, malignant hyperthermia, halothane shivering, and heart diseases.

Option D - Isoflurane: Isoflurane is effective but has a strong pungent odor and can cause airway irritation, making it less ideal for pediatric use.

Solution for Question 2:

Correct Option A - Sevoflurane:

- The preferred method for inducing anesthesia through inhalation in children is Sevoflurane .
- Sevoflurane: Sevoflurane is the preferred inhalational anesthetic for inducing anesthesia in children.
- It has several advantages, including: Rapid Onset and Offset: Sevoflurane has a rapid onset and emergence, which is particularly important for children who might be anxious or resistant to mask induction. Smooth Induction: Sevoflurane is well-tolerated by the respiratory system, leading to smooth induction without causing significant airway irritation. Minimal Irritation: It is less pungent and causes less airway irritation compared to other agents. Low Blood-Gas Solubility: Sevoflurane has a low blood-gas partition coefficient, allowing for quick uptake and emergence from anesthesia.
- Rapid Onset and Offset: Sevoflurane has a rapid onset and emergence, which is particularly important for children who might be anxious or resistant to mask induction.
- Smooth Induction: Sevoflurane is well-tolerated by the respiratory system, leading to smooth induction without causing significant airway irritation.
- Minimal Irritation: It is less pungent and causes less airway irritation compared to other agents.
- Low Blood-Gas Solubility: Sevoflurane has a low blood-gas partition coefficient, allowing for quick uptake and emergence from anesthesia.
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- Smooth Induction: Sevoflurane is well-tolerated by the respiratory system, leading to smooth induction without causing significant airway irritation.

- Minimal Irritation: It is less pungent and causes less airway irritation compared to other agents.
- Low Blood-Gas Solubility: Sevoflurane has a low blood-gas partition coefficient, allowing for quick uptake and emergence from anesthesia.

Incorrect Options: Option B, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 3:

Correct Option B - Etomidate:

- It is also an α_2 -receptors agonist therefore it maintains the vascular tone and myocardial contractility, thus maintaining cardiovascular haemodynamics which makes it an anesthetic of choice in cardiac disease.

Incorrect Options:

Option A - Ketamine:

- Ketamine is contraindicated in patients with uncontrolled hypertension as it tends to increase blood pressure.
- It also increases heart rate, therefore it is contraindicated in patients with ischemic heart disease.

Option C and D - Sodium thiopentone and Propofol:

- Both act via GABA receptors modulation but they have a direct myocardial depressant action leading to peripheral vasodilation causing an immediate reduction in mean arterial Pressure (MAP) after induction of anesthetic effect.
-

Solution for Question 4:

Correct Option C - Sevoflurane:

- Sevoflurane is a volatile inhalation anesthetic that can be used for induction of anesthesia, maintenance, and endotracheal intubation.
- Induction of anesthesia: Sevoflurane is commonly used for the induction of anesthesia in both adult and pediatric patients. It has a pleasant odor and a low blood-gas solubility, which means it is rapidly taken up and eliminated from the body. This allows for a smooth and rapid induction of anesthesia.
- Sevoflurane is commonly used for the induction of anesthesia in both adult and pediatric patients.
- It has a pleasant odor and a low blood-gas solubility, which means it is rapidly taken up and eliminated from the body.
- This allows for a smooth and rapid induction of anesthesia.
- Maintenance of anesthesia: Sevoflurane is also frequently used for the maintenance of anesthesia during surgical procedures. It provides a good depth of anesthesia and allows for precise control of the anesthetic depth by adjusting its concentration. Sevoflurane is well-suited for use with various anesthesia techniques, including endotracheal intubation, regional anesthesia, and balanced anesthesia.
- Sevoflurane is also frequently used for the maintenance of anesthesia during surgical procedures.
- It provides a good depth of anesthesia and allows for precise control of the anesthetic depth by adjusting its concentration.
- Sevoflurane is well-suited for use with various anesthesia techniques, including endotracheal intubation, regional anesthesia, and balanced anesthesia.
- Endotracheal intubation: Sevoflurane is considered safe and suitable for endotracheal intubation. It does not irritate the airway and does not significantly increase airway resistance or cough reflex, which are important factors during intubation. Additionally, its rapid onset and offset make it convenient for use during intubation procedures.
- Sevoflurane is considered safe and suitable for endotracheal intubation.
- It does not irritate the airway and does not significantly increase airway resistance or cough reflex, which are important factors during intubation.
- Additionally, its rapid onset and offset make it convenient for use during intubation procedures.
- Sevoflurane is commonly used for the induction of anesthesia in both adult and pediatric patients.
- It has a pleasant odor and a low blood-gas solubility, which means it is rapidly taken up and eliminated from the body.
- This allows for a smooth and rapid induction of anesthesia.

- Sevoflurane is also frequently used for the maintenance of anesthesia during surgical procedures.
- It provides a good depth of anesthesia and allows for precise control of the anesthetic depth by adjusting its concentration.
- Sevoflurane is well-suited for use with various anesthesia techniques, including endotracheal intubation, regional anesthesia, and balanced anesthesia.
- Sevoflurane is considered safe and suitable for endotracheal intubation.
- It does not irritate the airway and does not significantly increase airway resistance or cough reflex, which are important factors during intubation.
- Additionally, its rapid onset and offset make it convenient for use during intubation procedures.

Incorrect Options:

Option A - Halothane:

- It is primarily used for the maintenance of anesthesia and is not recommended for induction due to its potential for hepatotoxicity and its undesirable side effects.
- Halothane is also associated with a higher risk of cardiac arrhythmias than other inhalation anesthetics.

Option B - Desflurane:

- Desflurane is a volatile inhalation anesthetic that is primarily used for the maintenance of anesthesia.
- It has low blood-gas solubility and a rapid onset and offset, making it suitable for long surgical procedures.
- However, desflurane has a pungent odor that can cause airway irritation and coughing, making it less suitable for induction and endotracheal intubation.

Option D - Isoflurane:

- Isoflurane is a volatile inhalation anesthetic that is commonly used for the maintenance of anesthesia.
- It has a good safety profile and provides a smooth anesthetic depth.
- However, it is not the ideal choice for induction due to its slower onset than sevoflurane.

Solution for Question 5:

Correct Option C - Desflurane:

- Desflurane is not commonly used for the induction of anesthesia in pediatric patients.
- It is mainly used for the maintenance of anesthesia due to its pungent odor and potential airway irritation during induction.

Incorrect Options:

Options A, B, and D are used for the induction of anesthesia in pediatric patients.

Solution for Question 6:

Correct Option B - Calcium:

- The above patient presents with rigidity and hyperthermia after administration of halothane.
- This is indicative towards the diagnosis of malignant hyperthermia and the ion involved is calcium.
- It occurs due to prolonged opening of functionally altered ryanodine receptors, leading to uncontrolled release of calcium from sarcoplasmic reticulum causing sustained muscle contraction.

Incorrect Options: Options A, C, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 7:

Correct Option C: Desflurane

- Desflurane is known for its rapid onset and offset of action, making it a suitable choice for patients with cardiac conditions.
- It provides good cardiovascular stability and is associated with minimal metabolism in the body.

- It is less likely to cause myocardial depression and is suitable for patients with compromised cardiac function.

Incorrect Options:

Option A: Halothane:

- Halothane was once a commonly used anesthetic agent, but it has largely fallen out of favor due to its potential to cause arrhythmias and myocardial depression.

Option B: Isoflurane:

- Isoflurane is a widely used inhalational anesthetic agent with a relatively stable cardiovascular profile.
- It is often considered suitable for patients with cardiac conditions, but it might not be the absolute preferred choice due to newer agents with even better cardiovascular stability.

Option D: Sevoflurane:

- Sevoflurane is also a popular choice for patients with cardiac conditions.
- It offers a smooth induction and emergence, as well as good cardiovascular stability.
- It is less likely to cause myocardial depression and is also associated with minimal metabolism in the body.
- Desflurane and Sevoflurane are often the preferred inhalational agents for patients with compromised cardiac function due to their rapid onset and offset, minimal metabolism, and overall good cardiovascular stability, but the best inhalational agent is Desflurane.

Solution for Question 8:

Correct Option C - Halothane:

- Halothane is a volatile inhalation anesthetic that has been associated with hepatotoxicity in some patients.
- It can cause a rare but severe adverse reaction called halothane hepatitis, which can lead to liver damage or failure.

Incorrect Options: Options A, B, and D are not hepatotoxic.

Solution for Question 9:

Correct Option C - Both 1 and 2:

- Hypotensive anesthesia involves deliberate lowering of blood pressure during surgery to reduce bleeding and improve surgical field visibility.
- Isoflurane is an inhalation anesthetic that can be used to achieve controlled hypotension.
- Nitroglycerin, a vasodilator, can also be administered to lower blood pressure during surgery.

Incorrect Options:

Option D - Dantrolene: Dantrolene is a muscle relaxant used for the treatment of malignant hyperthermia, a rare but life-threatening reaction to certain anesthetic agents, not specifically for achieving hypotensive anesthesia.

Solution for Question 10:

Correct Option B - Desflurane:

- Blood gas coefficient indicates speed of onset and recovery, solubility of inhalational agent in the blood.
- It is inversely proportional to speed of onset High B/G has a slow onset Low B/G has a fast onset
- High B/G has a slow onset
- Low B/G has a fast onset
- Desflurane has similar B/G partition co-efficient similar to Nitrous oxide.
- Desflurane Least metabolized in liver: < 0.01% in liver and 99.9% in lungs Inhalational agent of choice in: Liver and renal transplant surgery Prolonged surgeries Elderly patients
- Least metabolized in liver: < 0.01% in liver and 99.9% in lungs
- Inhalational agent of choice in: Liver and renal transplant surgery Prolonged surgeries Elderly patients

- Liver and renal transplant surgery
- Prolonged surgeries
- Elderly patients
- High B/G has a slow onset
- Low B/G has a fast onset
- Least metabolized in liver: < 0.01% in liver and 99.9% in lungs
- Inhalational agent of choice in: Liver and renal transplant surgery Prolonged surgeries Elderly patients
- Liver and renal transplant surgery
- Prolonged surgeries
- Elderly patients
- Liver and renal transplant surgery
- Prolonged surgeries
- Elderly patients

Incorrect Options:

Option A - Isoflurane :

- Most commonly used in CVS and CNS surgeries
- Cost effective

Option C - Sevoflurane :

- Induction agent of choice in children
- Second fastest acting inhalational agent

Option D - Halothane :

- Undergoes maximum metabolism in liver (20%)
- Most hepatotoxic- can cause Halothane induced hepatitis if given repeatedly

Anaesthesia for Special Conditions and Fluid Management

1. A 65 y/o male with coronary artery disease (CAD) is scheduled for an elective laparoscopic cholecystectomy. The patient is stable on medications with no recent chest pain or dyspnea. The anesthesiologist selects anaesthetic agents for the patient. Which of the following options is FALSE regarding the choice of agents?

- A. Propofol is used in CAD patients as the intravenous anaesthetic agent of choice.
- B. Isoflurane is used as an inhalational agent for patients with good myocardial contractility.
- C. Vecuronium is used as a muscle relaxant for cardiac patients.
- D. Etomidate is used as an intravenous anaesthetic agent.

2. Which of the following is the IV induction agent of choice in a patient with decompensated Cirrhosis?

- A. Thiopentone
- B. Propofol
- C. Etomidate
- D. Ketamine

3. Which of the following anaesthetic agents should be avoided in patients undergoing neurosurgery who are at risk for developing pneumocephalus?

- A. Propofol
- B. Nitrous Oxide
- C. Fentanyl

D. Mannitol

4. A pregnant patient with pre-eclampsia is scheduled for a cesarean section. The anaesthesiologist opts for general anaesthesia. Which drug commonly used in general anaesthesia may have a prolonged effect in a patient receiving magnesium sulphate?

- A. Bupivacaine
 - B. Fentanyl
 - C. Lidocaine
 - D. Succinylcholine
-

5. A 60-year-old obese patient with a history of ischemic heart disease is scheduled for surgery. The anesthesiologist decides to use Propofol for intravenous induction and Desflurane for maintenance. What is a crucial consideration for this anaesthetic management?

- A. The need to keep the patient in a supine position during induction.
 - B. The potential for increased intraocular pressure with Desflurane.
 - C. The likelihood of respiratory depression with Propofol.
 - D. The requirement for a ramped position during induction.
-

6. A pediatric anesthesiologist is evaluating the best practice for inducing anaesthesia in a 5-year-old child with a history of asthma. Which of the following is the most appropriate choice for inhalational induction?

- A. Isoflurane
- B. Sevoflurane

- C. Nitrous oxide
 - D. Halothane
-

7. A 70-year-old patient with a history of hypertension is scheduled for cataract surgery under local anaesthesia. Which of the following is true regarding anaesthesia considerations in ocular procedures?

- A. The use of ketamine as the induction agent to prevent complications.
 - B. Atropine is preferred to treat bradycardia due to the oculocardiac reflex.
 - C. Nitrous oxide is the preferred analgesic agent during vitreoretinal surgeries.
 - D. Retrobulbar block for anaesthesia is preferred due to its least side effects.
-

8. A 35 y/o asthmatic patient is scheduled for a minor surgical procedure. Which induction agent and muscle relaxant combination is safest for this patient?

- A. Propofol and Succinylcholine
 - B. Ketamine and d-tubocurarine
 - C. Ketamine and Vecuronium
 - D. Thiopental and Rocuronium
-

9. A 50-year-old patient is scheduled for a short elective procedure in a daycare setting. The anesthesiologist needs to ensure rapid induction, short-term muscle relaxation, and quick recovery to facilitate early discharge. Which muscle relaxant would be the most suitable for this scenario?

- A. Vecuronium
- B. Succinylcholine
- C. Mivacurium

D. Pancuronium

10. A 70 kg patient is scheduled for elective open-heart surgery that is expected to last over 4 hours. According to fluid replacement guidelines, which of the following is the most appropriate additional fluid replacement rate to compensate for the third space volume loss for this patient?

- A. 140 mL/hr
 - B. 280 mL/hr
 - C. 560 mL/hr
 - D. 1000 mL/hr
-

11. A 45 y/o male undergoes abdominal surgery and requires fluid replacement. The patient has a history of chronic kidney disease. Which of the following fluids is most appropriate for maintenance during surgery to avoid the risk of hyperchloremic metabolic acidosis?

- A. 0.9% Normal Saline
 - B. Ringer's Lactate
 - C. 3% NaCl
 - D. 5% Dextrose in Water
-

12. A 6 y/o child weighs 20 kg and requires maintenance fluid based on the Holliday-Segar formula. What is the correct amount of fluid the child should receive in 24 hours?

- A. 1500 mL
- B. 1600 mL

C. 1800 mL

D. 2000 mL

13. A 50 kg patient has been fasting for 6 hours before surgery. If the maintenance fluid rate is 1.2 mL/kg/h, what is the total maintenance fluid requirement?

A. 360 mL

B. 420 mL

C. 500 mL

D. 720 mL

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	4
Question 5	4
Question 6	2
Question 7	2
Question 8	3
Question 9	3
Question 10	3
Question 11	2
Question 12	1
Question 13	1

Solution for Question 1:

Correct Answer: A) Propofol is used in CAD patients as the intravenous anaesthetic agent of choice.

Explanation:

- Propofol significantly lowers blood pressure and heart rate, making it less suitable for patients with coronary artery disease (CAD). It is not the anaesthetic agent of choice in CAD patients.
- Regional anaesthesia is preferred over general anaesthesia for cardiac patients when the procedure allows for it.

General Anesthetics for Cardiac Patients

Intravenous Anesthetics:

Narcotics:

- Preferred for cardiac stability and analgesia; newer agents like fentanyl and sufentanil also provide stability.

Inhalational Agents:

Muscle Relaxants:

Anticholinergics:

Glycopyrrolate: Preferred over atropine for less tachycardia; use only as needed.

Solution for Question 2:

Correct Answer: B) Propofol

Explanation:

- Propofol is the IV induction agent of choice in patients with hepatic disease because it has a rapid onset and shorter recovery time, making it safer for these patients.

Anesthesia for patients with hepatic diseases:

- IV induction agent of choice: Propofol
- Muscle relaxants of choice: Cis-atracurium/Atracurium due to organ-independent Hoffman degradation (i.e do not rely on renal or hepatic clearance)
- Inhalational agent: Desflurane or Isoflurane or Sevoflurane
- Avoid using halothane: Risk of halothane-induced hepatitis
- Narcotics used: Fentanyl, Sufentanil
- Sedatives used: Oxazepam, Lorazepam

- All long-acting narcotics and sedatives should be avoided in Cirrhotic patients.

Thiopentone (Option A), Etomidate (Option C), and Ketamine (Option D) are not the induction agents of choice in hepatic disease.

Solution for Question 3:

Correct answer: B) Nitrous Oxide

Explanation:

- Nitrous Oxide is known for its rapid onset and strong anaesthetic properties, but it should be avoided in patients at risk for pneumocephalus.
- Nitrous oxide can diffuse into closed-air spaces, leading to the expansion of trapped air. In neurosurgical patients, this could worsen pneumocephalus by enlarging intracranial air pockets, leading to increased intracranial pressure and potential complications.

Note: Nitrous oxide is also contraindicated in bowel obstruction because it diffuses into closed air spaces, increasing bowel distension and the risk of complications.

Propofol (Option A):

- It reduces cerebral blood flow and intracranial pressure, making it a good choice for neurosurgical patients.
- However, it does not affect air-filled spaces in the body, so there is no risk of exacerbating pneumocephalus when using propofol.

Fentanyl (Option C):

- Fentanyl is an opioid that is commonly used to manage pain during surgery.
- It does not directly affect cerebral blood flow or alter intracranial air spaces.

Mannitol (Option D):

- Mannitol is an osmotic diuretic that is frequently administered in neurosurgery to lower intracranial pressure by drawing fluid out of brain tissue.
 - While it helps reduce cerebral oedema, it does not affect air-filled spaces.
-

Solution for Question 4:

Correct answer: D) Succinylcholine

Explanation:

Magnesium sulfate is commonly used in pre-eclampsia to prevent seizures (eclampsia). It has several effects on neuromuscular transmission:

- Magnesium sulfate inhibits the release of acetylcholine at the neuromuscular junction and decreases the sensitivity of the motor endplate to acetylcholine.
- This potentiates the action of succinylcholine, a depolarizing neuromuscular blocker, which leads to a prolonged neuromuscular blockade and, thus, a prolonged duration of muscle paralysis.

Bupivacaine (Option A): Bupivacaine is a local anaesthetic used in epidural and spinal anaesthesia, and its duration of action is not significantly prolonged by magnesium sulphate.

Fentanyl (Option B): Fentanyl, an opioid, does not have a prolonged effect due to magnesium sulphate and is used for pain relief during general and regional anaesthesia.

Lidocaine (Option C): Lidocaine is a local anaesthetic used in regional blocks. It is not commonly affected by magnesium sulphate in terms of duration of action.

Solution for Question 5:

Correct Answer: D) The requirement for a ramped position during induction.

Explanation: A ramped position is commonly used in obese patients to help maintain a clear airway and reduce the risk of respiratory complications during induction



Anaesthesia for Obese Patients

- High Risk Factors: Increased risk of regurgitation and aspiration. Perioperative hypoxia; hypoxic hypoxia is most common.
- Increased risk of regurgitation and aspiration.
- Perioperative hypoxia; hypoxic hypoxia is most common.
- Induction: Rapid Sequence Induction (RSI) is utilized in the presence of any risk factors(aspiration) Inhalational Agent: Desflurane is preferred. IV Induction Agent: Propofol is commonly used. Muscle Relaxant: Any muscle relaxant can be used.
- Rapid Sequence Induction (RSI) is utilized in the presence of any risk factors(aspiration)
- Inhalational Agent: Desflurane is preferred.
- IV Induction Agent: Propofol is commonly used.
- Muscle Relaxant: Any muscle relaxant can be used.
- Posture for Induction: Ramped position is commonly used for obese patients. Pillows are placed below the shoulders to improve airway alignment.
- Ramped position is commonly used for obese patients.
- Pillows are placed below the shoulders to improve airway alignment.
- Other Complications: Conditions such as diabetes mellitus, hypertension (HTN), and ischemic heart disease (IHD) may be present.
- Conditions such as diabetes mellitus, hypertension (HTN), and ischemic heart disease (IHD) may be present.
- Premedications: Avoid sedatives like benzodiazepines to minimize respiratory depression.
- Avoid sedatives like benzodiazepines to minimize respiratory depression.

- Postoperative Considerations: Due to the increased risk of hypoxia, be prepared with CPAP or BiPAP oxygen supplementation.
- Due to the increased risk of hypoxia, be prepared with CPAP or BiPAP oxygen supplementation.

High Risk Factors:

- Increased risk of regurgitation and aspiration.
- Perioperative hypoxia; hypoxic hypoxia is most common.
- Rapid Sequence Induction (RSI) is utilized in the presence of any risk factors(aspiration)
- Inhalational Agent: Desflurane is preferred.
- IV Induction Agent: Propofol is commonly used.
- Muscle Relaxant: Any muscle relaxant can be used.
- Ramped position is commonly used for obese patients.
- Pillows are placed below the shoulders to improve airway alignment.
- Conditions such as diabetes mellitus, hypertension (HTN), and ischemic heart disease (IHD) may be present.
- Avoid sedatives like benzodiazepines to minimize respiratory depression.
- Due to the increased risk of hypoxia, be prepared with CPAP or BiPAP oxygen supplementation.

The need to keep the patient in a supine position during induction (Option A) : While the supine position is common, the ramped position is more appropriate for obese patients to optimize airway management.

The potential for increased intraocular pressure with Desflurane (Option B): Desflurane is generally associated with minimal changes in intraocular pressure; thus, this is less of a concern compared to airway management.

The likelihood of respiratory depression with Propofol (Option C): Although respiratory depression can occur with Propofol, the more critical consideration in this scenario is airway management in an obese patient, which can significantly impact safety during induction.

Solution for Question 6:

Correct Answer: B) Sevoflurane

Explanation: Sevoflurane is the preferred inhalational agent for induction in pediatric patients, especially asthmatics because it has a sweet odour, low pungency and causes minimal airway irritation.

Anaesthesia in Pediatric Surgeries

- Concerns : Placing an IV cannula can be challenging. Bradycardia is the most common response to hypoxia.
- Placing an IV cannula can be challenging.
- Bradycardia is the most common response to hypoxia.
- Induction Agents : Inhalational Agent : Sevoflurane is preferred due to its sweet odour and minimal airway irritation. Sevoflurane and halothane are less likely to irritate the airway or cause breath-holding and laryngospasm during induction. Volatile anaesthetics tend to depress ventilation more in infants than older children, with sevoflurane causing the least respiratory depression. IV Induction : Ketamine is commonly used.
- Inhalational Agent : Sevoflurane is preferred due to its sweet odour and minimal airway irritation.
- Sevoflurane and halothane are less likely to irritate the airway or cause breath-holding and laryngospasm during induction.
- Volatile anaesthetics tend to depress ventilation more in infants than older children, with sevoflurane causing the least respiratory depression.
- IV Induction : Ketamine is commonly used.
- Laryngoscope : Miller laryngoscope is utilized for intubation.
- Miller laryngoscope is utilized for intubation.
- Placing an IV cannula can be challenging.
- Bradycardia is the most common response to hypoxia.
- Inhalational Agent : Sevoflurane is preferred due to its sweet odour and minimal airway irritation.
- Sevoflurane and halothane are less likely to irritate the airway or cause breath-holding and laryngospasm during induction.
- Volatile anaesthetics tend to depress ventilation more in infants than older children, with sevoflurane causing the least respiratory depression.
- IV Induction : Ketamine is commonly used.
- Miller laryngoscope is utilized for intubation.



- Endotracheal Tube : A micro-sized cuffed endotracheal tube is employed to minimize tracheal mucosal damage and reduce aspiration risk.
- A micro-sized cuffed endotracheal tube is employed to minimize tracheal mucosal damage and reduce aspiration risk.
- A micro-sized cuffed endotracheal tube is employed to minimize tracheal mucosal damage and reduce aspiration risk.

Isoflurane (Option A): Isoflurane can irritate the airway, making it less suitable for patients with asthma compared to other agents.

Nitrous oxide (Option C): Nitrous oxide is not the best choice for induction in this scenario because it is a weak anaesthetic and is typically used as a carrier gas rather than a sole agent for anaesthesia induction.

Halothane (Option D) is associated with potential complications, such as hepatic dysfunction, bradycardia and arrhythmias, and is not preferred in pediatric patients, especially those with asthma.

Solution for Question 7:

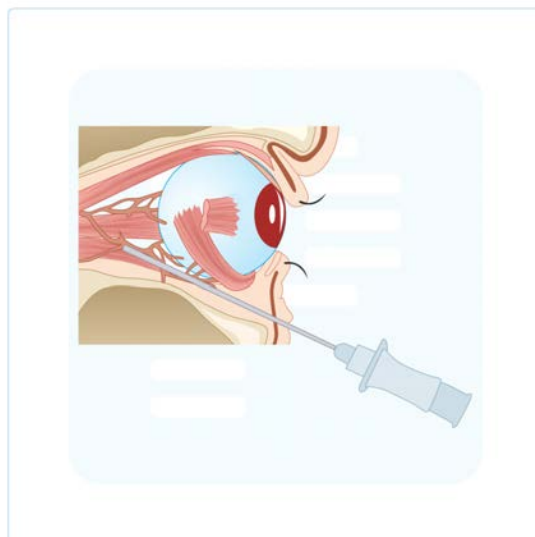
Correct Answer: B) Atropine is preferred to treat bradycardia due to the oculocardiac reflex.

Explanation:

- Atropine is preferred over glycopyrrolate, due to its fast action to treat bradycardia and other symptoms occurring during the event of an oculocardiac reflex.
- The surgery is halted immediately.

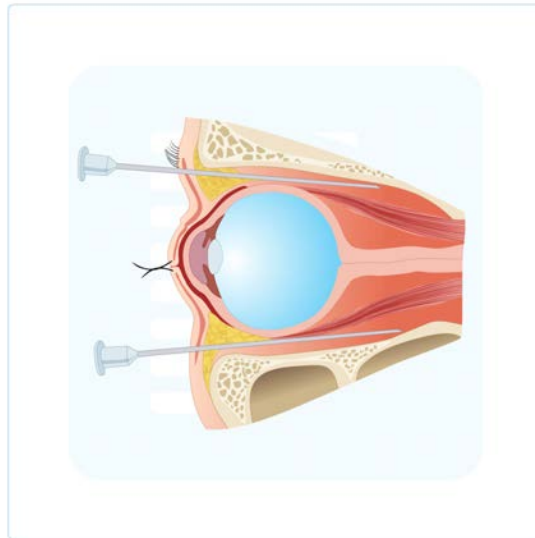
Anaesthesia in Ocular Surgeries:

- Topical : Commonly used for cataract and glaucoma surgeries. Beneficial for patients on anticoagulants as it reduces injection-related bleeding risks. Topical eye drops or gels are available. Eg. Lidocaine
- Commonly used for cataract and glaucoma surgeries.
- Beneficial for patients on anticoagulants as it reduces injection-related bleeding risks.
- Topical eye drops or gels are available. Eg. Lidocaine
- Local Anesthesia : (Option D ruled out) Retrobulbar : Generally avoided due to risks of optic nerve damage, globe perforation and haemorrhage.
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- Commonly used for cataract and glaucoma surgeries.
- Beneficial for patients on anticoagulants as it reduces injection-related bleeding risks.
- Topical eye drops or gels are available. Eg. Lidocaine
- Retrobulbar : Generally avoided due to risks of optic nerve damage, globe perforation and haemorrhage.



- Peribulbar : Most commonly used. Safer than retrobulbar block. (Has the least side effects)
- Most commonly used.

- Safer than retrobulbar block. (Has the least side effects)
- Most commonly used.
- Safer than retrobulbar block. (Has the least side effects)



- General Anesthesia : Induction agents (e.g., propofol, thiopental, etomidate) and volatile anaesthetics decrease IOP significantly. Avoid Ketamine : Increases intraocular pressure (IOP) and may cause nystagmus. (Option A) Avoid Nitrous Oxide (N₂O) : This should be avoided in surgeries involving gas bubble placement (Vitreoretinal surgery) as it can Increase the size of the bubble, leading to increased ICP. (Option C)
- Induction agents (e.g., propofol, thiopental, etomidate) and volatile anaesthetics decrease IOP significantly.
- Avoid Ketamine : Increases intraocular pressure (IOP) and may cause nystagmus. (Option A)
- Avoid Nitrous Oxide (N₂O) : This should be avoided in surgeries involving gas bubble placement (Vitreoretinal surgery) as it can Increase the size of the bubble, leading to increased ICP. (Option C)
- Neuromuscular Blockade: Preferably non-depolarizing agents are used e.g. Atracurium, Cisatracurium Sugammadex for reversal of non-depolarizing blockers to avoid IOP increases associated with anticholinergics.
- Preferably non-depolarizing agents are used e.g. Atracurium, Cisatracurium
- Sugammadex for reversal of non-depolarizing blockers to avoid IOP increases associated with anticholinergics.
- Oculocardiac reflex: Bradycardia may occur due to a stretch of the rectus muscle during surgery

- Bradycardia may occur due to a stretch of the rectus muscle during surgery
- Induction agents (e.g., propofol, thiopental, etomidate) and volatile anaesthetics decrease IOP significantly.
- Avoid Ketamine : Increases intraocular pressure (IOP) and may cause nystagmus. (Option A)
- Avoid Nitrous Oxide (N₂O) : This should be avoided in surgeries involving gas bubble placement (Vitreoretinal surgery) as it can increase the size of the bubble, leading to increased ICP. (Option C)
- Bradycardia may occur due to a stretch of the rectus muscle during surgery

Treatment : Atropine (due to its fast onset of action) is immediately given, and surgery is halted. (Option B)

Solution for Question 8:

Correct Answer: C) Ketamine and Vecuronium

Explanation:

Ketamine is beneficial in asthmatics as it provides analgesia and induces bronchodilation without significant hemodynamic instability and Vecuronium is a non-depolarizing muscle relaxant that does not release histamine, making it safe for use in asthmatic patients.

Anaesthesia in Asthmatics

- IV Induction Agent : Propofol : Preferred for stable patients, promotes bronchodilation. Ketamine : Ideal for hemodynamically unstable asthmatics, promotes bronchodilation.
- Propofol : Preferred for stable patients, promotes bronchodilation.
- Ketamine : Ideal for hemodynamically unstable asthmatics, promotes bronchodilation.
- Muscle Relaxant : Vecuronium , Rocuronium , Cis-atracurium : Safe for use in asthmatics, no significant histamine release. Pancuronium : Releases low levels of histamine.
- Vecuronium , Rocuronium , Cis-atracurium : Safe for use in asthmatics, no significant histamine release.
- Pancuronium : Releases low levels of histamine.
- Inhalational Agent : Sevoflurane , Isoflurane , Halothane : Ideal choices as they depress airway reflexes and relax bronchial smooth muscle. Desflurane : This should be avoided due

to airway irritation and increased airway resistance.

- Sevoflurane , Isoflurane , Halothane : Ideal choices as they depress airway reflexes and relax bronchial smooth muscle.
- Desflurane : This should be avoided due to airway irritation and increased airway resistance.
- Propofol : Preferred for stable patients, promotes bronchodilation.
- Ketamine : Ideal for hemodynamically unstable asthmatics, promotes bronchodilation.
- Vecuronium , Rocuronium , Cis-atracurium : Safe for use in asthmatics, no significant histamine release.
- Pancuronium : Releases low levels of histamine.
- Sevoflurane , Isoflurane , Halothane : Ideal choices as they depress airway reflexes and relax bronchial smooth muscle.
- Desflurane : This should be avoided due to airway irritation and increased airway resistance.

Propofol and Succinylcholine (Option A) : Succinylcholine may provoke bronchospasm.

Ketamine and d-tubocurarine (Option B) : d-tubocurarine releases histamine, which can trigger bronchospasm.

Thiopental and Rocuronium (Option D) : Thiopental has slower recovery times and can cause respiratory depression.

Solution for Question 9:

Correct Answer: C) Mivacurium

Explanation: Mivacurium is a short-acting, rapidly metabolized muscle relaxant, ideal for daycare surgeries where fast recovery is desired.

Duration of Action	Muscle Relaxant
Long-Acting	Pancuronium
Intermediate-Acting	Rocuronium, Vecuronium, Atracurium, Cisatracurium
Short-Acting	Mivacurium

Ultrashort-Acting	Gantacurium
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Anaesthesia in Day Care Surgery

- IV Induction Agent : Propofol : Rapid onset and recovery; commonly used for induction and maintenance.
- Propofol : Rapid onset and recovery; commonly used for induction and maintenance.
- Muscle Relaxant : Mivacurium : Short-acting; suitable for short procedures.
- Mivacurium : Short-acting; suitable for short procedures.
- Inhalational Agent : Desflurane : Best for rapid recovery but less commonly used due to cost and airway irritability. Sevoflurane : Most commonly used; provides smooth induction with minimal airway irritation.
- Desflurane : Best for rapid recovery but less commonly used due to cost and airway irritability.
- Sevoflurane : Most commonly used; provides smooth induction with minimal airway irritation.
- Propofol : Rapid onset and recovery; commonly used for induction and maintenance.
- Mivacurium : Short-acting; suitable for short procedures.
- Desflurane : Best for rapid recovery but less commonly used due to cost and airway irritability.
- Sevoflurane : Most commonly used; provides smooth induction with minimal airway irritation.

Vecuronium (Option A) is an intermediate-acting muscle relaxant, not ideal for short procedures where quick recovery is needed

Succinylcholine (Option B) : While Succinylcholine provides short-term paralysis, it has significant side effects (e.g., hyperkalemia, malignant hyperthermia) that make it less favourable for routine daycare surgeries.

Pancuronium (Option D) is long-acting and is not suitable for short, daycare procedures due to delayed recovery time.

Solution for Question 10:

Correct Answer: C) 560 mL/hr

Explanation:

- For major surgery, such as open-heart surgery, the fluid replacement guideline recommends a rate of 6-8 mL/kg/hr .
- For a 70 kg patient: Minimum rate: $70 \text{ kg} \times 8 \text{ mL/kg/hr} = 560 \text{ mL/hr}$

Fluid Replacement Based on Type of Surgery

- Minor Surgery : Eg. Herniorrhaphy Fluid replacement: 2-4 ml/kg/hr Duration: 45 minutes to 1 hour
 - Eg. Herniorrhaphy
 - Fluid replacement: 2-4 ml/kg/hr
 - Duration: 45 minutes to 1 hour
- Moderate Surgery : Eg. Cholecystectomy Fluid replacement: 4-6 ml/kg/hr Duration: Up to 2 hours
 - Eg. Cholecystectomy
 - Fluid replacement: 4-6 ml/kg/hr
 - Duration: Up to 2 hours
- Major Surgery : Eg. Bowel resection Eg. Fluid replacement: 6-8 ml/kg/hr Duration: Usually more than 4 hours
 - Eg. Bowel resection
 - Eg. Fluid replacement: 6-8 ml/kg/hr
 - Duration: Usually more than 4 hours
- Eg. Herniorrhaphy
 - Fluid replacement: 2-4 ml/kg/hr
 - Duration: 45 minutes to 1 hour
- Eg. Cholecystectomy
 - Fluid replacement: 4-6 ml/kg/hr
 - Duration: Up to 2 hours
- Eg. Bowel resection
 - Eg. Fluid replacement: 6-8 ml/kg/hr
 - Duration: Usually more than 4 hours

Types of Fluids and Volume Replacement	
Crystalloids	Colloids

Small molecules Less effective than colloids at stabilizing hemodynamic endpoints Higher volumes are needed in the ICU compared to colloids (3:1)	Larger molecules More effective at stabilizing hemodynamic endpoints
Advantages : Cost-effective Easy to use Provide immediate fluid resuscitation	Advantages : May provide a more swift volume expansion in the intravascular space (1:1)
Disadvantages : May increase edema	Disadvantages : Expensive Risk of allergic reactions. Potential for blood clotting disorders Risk of kidney failure
Examples: Normal saline Balanced electrolyte solutions (e.g., Lactated Ringer's, PlasmaLyte) 5% dextrose in water (D5W) Hypertonic 3% saline	Examples: Albumin (5%, 25%) Plasma protein fraction (5%) Dextran (Dextran 70, Dextran 40) Hetastarch (hydroxyethyl starch)

Solution for Question 11:

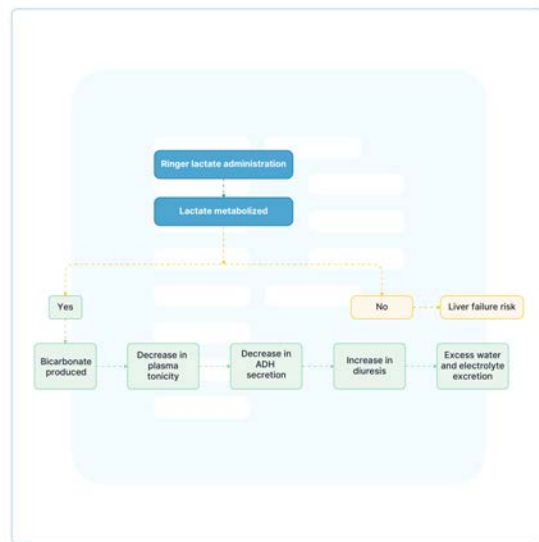
Correct Answer: B) Ringer's Lactate

Explanation:

- Ringer's Lactate is often preferred because it contains a buffer (lactate) that helps to mitigate the risk of acidosis.
- It is isotonic and provides electrolytes in a more balanced way, making it a suitable option for patients with CKD.

Ringer's Lactate (Balanced Crystalloid)

- Usage : Maintenance fluid during surgery.
- Advantage : Prevents hyperchloremic metabolic acidosis, unlike normal saline.
- Excretion : Rapid elimination due to suppression of ADH secretion, promoting diuresis.
- Rapid elimination due to suppression of ADH secretion, promoting diuresis.
- Rapid elimination due to suppression of ADH secretion, promoting diuresis.



- Buffer : Lactate metabolised to bicarbonate, maintaining or slightly increasing HCO_3^- levels, hence preventing acidosis.

$\frac{1}{2}$ NS (0.45% NaCl, Hypotonic Saline)

- Usage : Maintenance fluid in the postoperative period along with 4% dextrose.
- Risk : To minimize the risk of hyponatremia, excessive use of hypotonic solutions, particularly in the postoperative phase, is discouraged.

3% NS (3% NaCl, Hypertonic Saline)

- Usage : Treatment of symptomatic hyponatremia and cerebral oedema.
- Mechanism : Draws water from intracellular to extracellular compartments, aiding in plasma volume expansion.
- Caution : Careful electrolyte and osmolality monitoring is required to avoid complications like osmotic demyelination. There is also a potential to cause fluid overload and hypernatremia, not recommended as a maintenance fluid. (Option C)

0.9% Normal Saline (Option A) : It may lead to hyperchloremic metabolic acidosis due to its high chloride content.

5% Dextrose in Water (Option D) :

- This hypotonic solution is used primarily for free water replacement, not surgical maintenance.
- It is usually used perioperatively and must be avoided in the postoperative period as it can severely dilute plasma electrolytes and osmolality.

Note: An isotonic solution in the postoperative period could lead to potential GI dysfunction, when administered in excess, due to increased mucosal edema.

Solution for Question 12:

Correct Answer: A) 1500 mL

Explanation:

- Using the Holliday-Segar "4-2-1" rule :

To calculate maintenance fluid per hour:

- First 10kgs $\times 4 = a$
- Next 10kgs $\times 2 = b$
- Next kgs $\times 1 = c$
- Total = $a+b+c$

Hence, for this patient:

- For the first 10 kg of weight: $4 \text{ mL/kg/hour} \rightarrow 4 \text{ mL} \times 10 \text{ kg} = 40 \text{ mL/hour}$
- For the next 10 kg (10-20 kg): $2 \text{ mL/kg/hour} \rightarrow 2 \text{ mL} \times 10 \text{ kg} = 20 \text{ mL/hour}$
- Total per hour: $40 \text{ mL} + 20 \text{ mL} = 60 \text{ mL/hour}$

In 24 hours: $60 \text{ mL/hour} \times 24 \text{ hours} = 1440 \text{ mL}$ (rounded to 1500 mL)

Solution for Question 13:

Correct Answer: A) 360 mL

Explanation:

- Maintenance Rate: 1.2 mL/kg/h
- Hours of Fasting: 6 hours

- Body Weight: 50 kg

Total Maintenance Fluid = Maintenance rate (mL/kg/h) × Hours of fasting (h) × Body weight (kg)

Total Maintenance Fluid = $1.2 \times 6 \times 50 = 360$ mL

Maintenance Fluid Requirements:

- Purpose : Covers insensible losses and urine output during fasting before surgery.

Rate : Typically 1 to 1.5 mL/kg/h, adjusted for surgery type and anticipated fluid shifts.

Monitoring during anaesthesia

1. Refer to the readings depicted in the image. Which statement is inaccurate based on these readings?



- A. 76% is due to shivering
- B. 76% is due to Methemoglobinemia
- C. 76% is due to Carboxyhemoglobinemia
- D. 76% is due to Peripheral Vasoconstriction

2. Which combination of ECG leads is considered the best for diagnosing intraoperative myocardial ischemia?

- A. Leads I, II, and III
- B. Leads II, V4, and V5
- C. Leads V1, V2, and V3

D. Leads aVR, aVL, and aVF

3. Which of the following is the correct function of the shown image?



- A. Monitor Intracardiac pressures
- B. Infuse solutions
- C. Measure Temperature
- D. All of the above

4. Postoperatively, a patient reports feeling pain and recalling specific details from the operating room, during the procedure. Which of the following has no relevance to prevent this issue in the future?

- A. Patient state analyzer
- B. Visual Evoked Potentials
- C. Evan's Score

D. Anxiolytics before anesthesia induction

5. Which of the following is an incorrect statement for the image below?



- A. The higher the value lower the awareness
 - B. Elevated EEG is seen in Ketamine
 - C. Depressed EEG is seen in COPD
 - D. Nitrous oxide shows elevated EEG
-

6. Which of the following is the most common nerve-muscle combination tested?

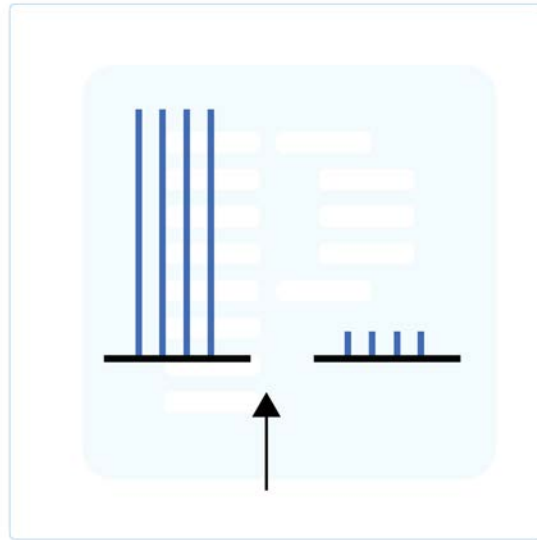


- A. Ulnar Nerve & Adductor Pollicis Muscle
 - B. Ulnar Nerve & Abductor Pollicis Muscle
 - C. Radial Nerve & Adductor Pollicis Muscle
 - D. Radial Nerve & Abductor Pollicis Muscle
-

7. A 65-year-old male patient is undergoing major abdominal surgery under general anesthesia. After induction of anesthesia, the patient's CVP is measured at 3 cm H₂O. The patient's heart rate is 110/min, blood pressure is 80/50 mmHg. Which of the following is the most appropriate initial management step based on the CVP and clinical findings?

- A. Administer vasopressors
 - B. Administer a fluid bolus
 - C. Transfuse blood
 - D. Increase the anesthetic depth
-

8. A 40-year-old female patient is undergoing a thyroidectomy under general anesthesia. A Train of Four (TOF) stimulation is performed to monitor neuromuscular function, the result of which is shown below. Which neuromuscular blocking agent was most likely administered?

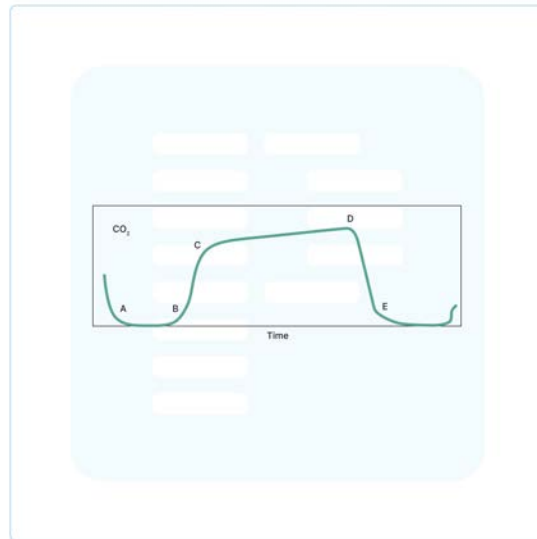


- A. Succinylcholine
- B. Rocuronium
- C. Vecuronium
- D. Cisatracurium

9. Which stimulation pattern is specifically designed to assess deep neuromuscular blockade?

- A. Train-of-Four (TOF) stimulation
- B. Single-Twitch stimulation
- C. Post Tetanic Count (PTC) stimulation
- D. Double-Burst stimulation (DBS)

10. Which of the following is incorrect about the given image?



- A. D represents the best reflection of the alveolar CO₂
- B. B to C represents CO₂ of alveoli mixing with upper airway
- C. A to B represents Dead space ventilation
- D. E is the value seen on the monitor

11. A 62-year-old male with asthma presents to the emergency department with worsening shortness of breath and wheezing. He is placed on supplemental oxygen and bronchodilators. What is the most likely Capnographic waveform for this patient?

- A.
- B.
- C.
- D.

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	4
Question 4	4
Question 5	1
Question 6	1
Question 7	2
Question 8	1
Question 9	3
Question 10	4
Question 11	3

Solution for Question 1:

Correct Answer: C) 76% is due to Carboxyhemoglobinemia

Explanation:

- The given image shows a pulse oximetry with a low SpO₂.
- Out of 4 options, only carboxyhemoglobin can show false high readings, not false low readings.
- Easily monitored using a pulse oximeter for arterial oxygenation.

- Principle: Based on Beer-Lambert's law , which relates solute concentration to light intensity.
- Offers continuous tracking of hemoglobin saturation during the perioperative period.



Principles of Pulse Oximetry

- Oxygenated hemoglobin (HbO₂) and reduced hemoglobin (Hb) absorb light differently at two wavelengths: 660 nm (Red Light): Reduced hemoglobin absorbs about 10x more light than oxyhemoglobin. 940 nm (Near Infrared Light): Oxyhemoglobin has a greater absorption coefficient than reduced hemoglobin.
- 660 nm (Red Light): Reduced hemoglobin absorbs about 10x more light than oxyhemoglobin.
- 940 nm (Near Infrared Light): Oxyhemoglobin has a greater absorption coefficient than reduced hemoglobin.
- 660 nm (Red Light): Reduced hemoglobin absorbs about 10x more light than oxyhemoglobin.
- 940 nm (Near Infrared Light): Oxyhemoglobin has a greater absorption coefficient than reduced hemoglobin.
- Absorbance varies due to changes in arterial blood volume, allowing for measurement of oxygen saturation.

False low reading:

- Methemoglobinemia (Option B).
- Peripheral vasoconstriction (shock, hypotension) (Option C).
- Anemia
- Shivering (Option A).

- Nail paint (Dark blue - maximum)

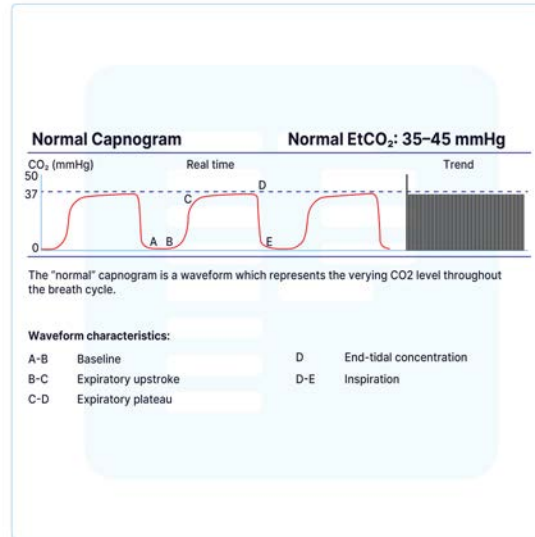
False high reading: Carboxyhaemoglobin

No change in reading:

- Skin pigmentation
- Sulph Hb
- Polycythemia
- Jaundice

Ventilation:

- Key Parameters to Monitor : Chest Movement : Observe symmetry and effectiveness during ventilation. End-Tidal CO₂ (ETCO₂) (Capnography): Normal Range: 35 to 45 mmHg
- Chest Movement : Observe symmetry and effectiveness during ventilation.
- End-Tidal CO₂ (ETCO₂) (Capnography): Normal Range: 35 to 45 mmHg
- Capnography : Measurement and numeric representation of CO₂ concentration during inspiration and expiration. Capnogram : Continuous display showing CO₂ concentration over time at the patient's airway. Provides a non-invasive , continuous, real-time reflection of ventilation status.
- Measurement and numeric representation of CO₂ concentration during inspiration and expiration.
- Capnogram : Continuous display showing CO₂ concentration over time at the patient's airway.
- Provides a non-invasive , continuous, real-time reflection of ventilation status.
- Chest Movement : Observe symmetry and effectiveness during ventilation.
- End-Tidal CO₂ (ETCO₂) (Capnography): Normal Range: 35 to 45 mmHg
- Measurement and numeric representation of CO₂ concentration during inspiration and expiration.
- Capnogram : Continuous display showing CO₂ concentration over time at the patient's airway.
- Provides a non-invasive , continuous, real-time reflection of ventilation status.



Solution for Question 2:

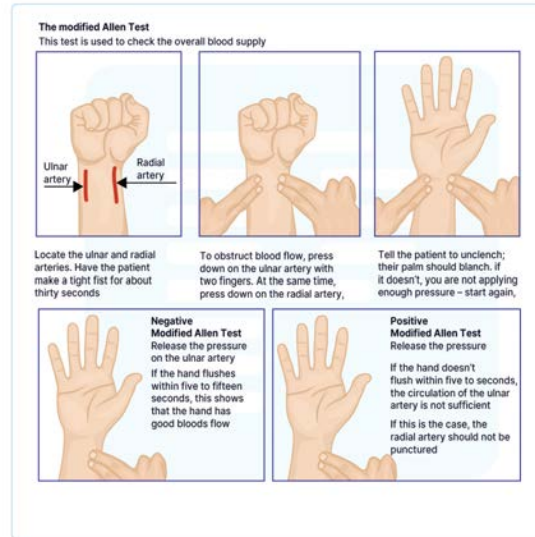
Correct Answer: B) Leads II, V4, and V5

Explanation:

Leads II, V4, and V5 are optimal for detecting intraoperative myocardial ischemia.

CARDIAC MONITORING	
Blood Pressure Monitoring	Non-invasive: Via sphygmomanometer. Invasive: Catheter placement in radial or femoral artery.
ECG Monitoring	12 Leads: Leads II and V5 are crucial. Lead II: Detects intraoperative arrhythmias. Lead V5: Detects ischemia.
Transesophageal Echocardiography (TEE)	Indication: Essential for major cardiac surgeries. Key Benefit: Most sensitive and reliable for detecting venous air embolism.

- Radial Artery: The most commonly used artery for invasive arterial monitoring.
- Allen's Test: Checks the patency of collateral circulation of the hand.



- Best ECG Lead Combination: Leads II, V4, and V5 for intraoperative myocardial ischemia monitoring.
- Central Venous Catheter Position: Ideal tip position at the junction of the superior vena cava (SVC) and right atrium.
- Pulmonary Artery (PA) Catheter: Measures Pulmonary Capillary Wedge Pressure (PCWP), the best indicator of Left Ventricular End Diastolic Pressure (LVEDP).

Solution for Question 3:

Correct Answer: D) All of the above

Explanation:

- The given image is of the Swan-Ganz catheter.
- These catheters monitor intracardiac pressures, and cardiac output, provide infusing solutions, and measure temperature.
- Importance: Essential for detecting malignant hyperthermia and quantifying hyperthermia/hypothermia.
- Common Disturbance: Inadvertent hypothermia is the most prevalent thermal issue during surgery.

- Thermoregulatory Control: General anesthetics significantly impair the body's ability to regulate temperature.
- Core Temperature Measurement: Always monitor core temperature during anesthesia.
- Pulmonary Artery (Best site) Monitored via Swan-Ganz catheter.
- Monitored via Swan-Ganz catheter.
- Monitored via Swan-Ganz catheter.



- Tympanic Membrane
- Oesophagus
- Nasopharynx

Monitoring Requirements

- Major Surgeries: Core temperature monitoring is essential during major surgical procedures.

Solution for Question 4:

Correct Answer: D) Anxiolytics before anesthesia induction

Explanation:

- This is a case of Awareness during surgery

- While anxiolytics can help with preoperative anxiety, they have no relevance in preventing intraoperative awareness.
- Other options hold relevance in measuring the depth of anesthesia, which in turn leads to less intraoperative awareness.

Awareness During Surgery (Light anesthesia):

- Patients recovering from surgery often express that they recall conversations and they were able to feel but were unable to move during the procedure.
- Insufficient inhalational agent
- No inhalational agent in vaporizer

Depth of Anesthesia Monitoring

- Purpose: To prevent: Very deep anesthesia (delayed emergence, cardiovascular complications) Light anesthesia (post-traumatic stress disorder)
- Very deep anesthesia (delayed emergence, cardiovascular complications)
- Light anesthesia (post-traumatic stress disorder)
- Very deep anesthesia (delayed emergence, cardiovascular complications)
- Light anesthesia (post-traumatic stress disorder)

Methods of Monitoring Depth of Anesthesia	
Clinical Techniques	Clinical Signs: PRST or Evan's Score (Option C): P: Systolic blood Pressure R: Heart Rate S: Sweating T: Tears
Isolated Forearm Technique: Tourniquet applied to upper arm before muscle relaxants. Movement indicates wakefulness (not necessarily explicit awareness).	
Brain Electrical Activity Monitoring	Spontaneous EEG Activity Monitors: Bispectral Index Entropy Narcotrend Patient State Analyzer (Option A) Evoked Brain Electrical Activity Monitors: Somatosensory Evoked Potentials (SSEP) Visual Evoked Potentials (VEP) (Option B) Auditory Evoked Potentials (AEP)

Solution for Question 5:

Correct Answer: A) The higher the value, the lower the awareness

Explanation:

The given image shown is of Bispectral index (BIS). Higher values on BIS indicate higher awareness

- Monitoring Setup: EEG connected to the patient's frontal area. BIS is calculated based on EEG activity.
- EEG connected to the patient's frontal area.
- BIS is calculated based on EEG activity.
- EEG connected to the patient's frontal area.
- BIS is calculated based on EEG activity.



- BIS Values: 100: Full Awareness 65 to 85: Moderate Sedation 40 to 65: Fully Anesthetized (adequate depth of anaesthesia) < 40: Cortical Depression or Deep Hypnotic State
- 100: Full Awareness
- 65 to 85: Moderate Sedation
- 40 to 65: Fully Anesthetized (adequate depth of anaesthesia)
- < 40: Cortical Depression or Deep Hypnotic State
- Depressed EEG Observations: Anesthetized Patients (IV induction and inhalational agents) Hypoxia or Hypercarbia (e.g., COPD) (Option C ruled out)
- Anesthetized Patients (IV induction and inhalational agents)
- Hypoxia or Hypercarbia (e.g., COPD) (Option C ruled out)
- Elevated EEG Observations: Ketamine (Option B ruled out) Nitrous Oxide (N₂O) - has a low MAC value (Option D ruled out)
- Ketamine (Option B ruled out)

- Nitrous Oxide (N₂O) - has a low MAC value (Option D ruled out)
- 100: Full Awareness
- 65 to 85: Moderate Sedation
- 40 to 65: Fully Anesthetized (adequate depth of anaesthesia)
- < 40: Cortical Depression or Deep Hypnotic State
- Anesthetized Patients (IV induction and inhalational agents)
- Hypoxia or Hypercarbia (e.g., COPD) (Option C ruled out)
- Ketamine (Option B ruled out)
- Nitrous Oxide (N₂O) - has a low MAC value (Option D ruled out)

Solution for Question 6:

Correct Answer: A) Ulnar Nerve & Adductor Pollicis Muscle

Explanation:

The most commonly tested pair in neuromuscular testing is Ulnar Nerve & Adductor Pollicis Muscle.

Importance of Neuromuscular Monitoring

- Pharyngeal dysfunction
- Aspiration and pneumonia
- Acute respiratory events (hypoxemia, airway obstruction)
- Need for re-intubation
- Patient and surgeon discomfort
- Prolonged PACU stay
- Prolonged infusions or use of long-acting NMBAs
- Extended surgery or anesthesia duration
- Patients with: Severe respiratory disease Morbid obesity Cardiac issues (tachyarrhythmias, heart failure) Liver or renal dysfunction (altered pharmacokinetics) Neuromuscular disorders (e.g., myasthenia gravis)
- Severe respiratory disease
- Morbid obesity

- Cardiac issues (tachyarrhythmias, heart failure)
- Liver or renal dysfunction (altered pharmacokinetics)
- Neuromuscular disorders (e.g., myasthenia gravis)
- Severe respiratory disease
- Morbid obesity
- Cardiac issues (tachyarrhythmias, heart failure)
- Liver or renal dysfunction (altered pharmacokinetics)
- Neuromuscular disorders (e.g., myasthenia gravis)

Common Nerve-Muscle Combinations:

Solution for Question 7:

Correct Answer: B) Administer a fluid bolus

Explanation:

Low CVP (3 cm H₂O) suggests hypovolemia, which is consistent with the patient's tachycardia and hypotension. Administering a fluid bolus is the first-line treatment to increase preload and improve hemodynamics.

Central Venous Pressure (CVP) Monitoring:

CVP measures the pressure in the thoracic vena cava near the right atrium, reflecting right ventricular preload and overall fluid status.

Measurement Method (Invasive):

Central venous catheter placed in large veins (e.g., subclavian or internal jugular), connected to a pressure transducer

Normal Range: 3-8 mmHg.

Low CVP Causes:

- Hypovolemia (blood loss or dehydration).
- Venodilation (e.g., in septic shock).

Elevated CVP Causes:

- Heart failure, fluid overload, valve dysfunction.

- Increased intrathoracic pressure (e.g., mechanical ventilation, tension pneumothorax).

Clinical Use:

- Guides fluid resuscitation.
- Helps assess heart function and venous return.

Vasopressors (Option A) may be required if hypotension persists after adequate fluid resuscitation, but not as the initial step.

Blood transfusion (Option C) is not indicated unless there is evidence of blood loss or anemia.

Increasing anesthetic depth (Option D) is inappropriate here, as the patient is likely hypovolemic.

Solution for Question 8:

Correct Answer: A) Succinylcholine

Explanation:

The absence of fade with all twitches equally reduced indicates the use of a depolarizing neuromuscular blocking agent, such as Succinylcholine .

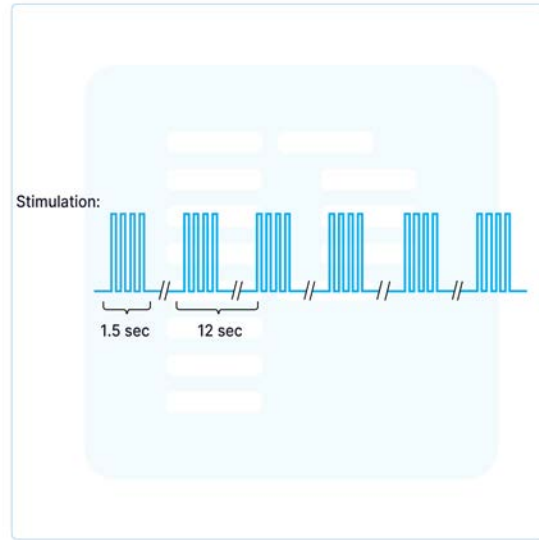
Train-of-Four (TOF) Pattern of Stimulation:

The Train-of-Four (TOF) stimulation pattern is a commonly used method to evaluate neuromuscular function, particularly during anesthesia and in the management of neuromuscular blocking agents.

Mechanism:

TOF stimulation involves the delivery of four supramaximal electrical stimuli at a frequency of 2 Hz (one stimulus every 0.5 seconds).

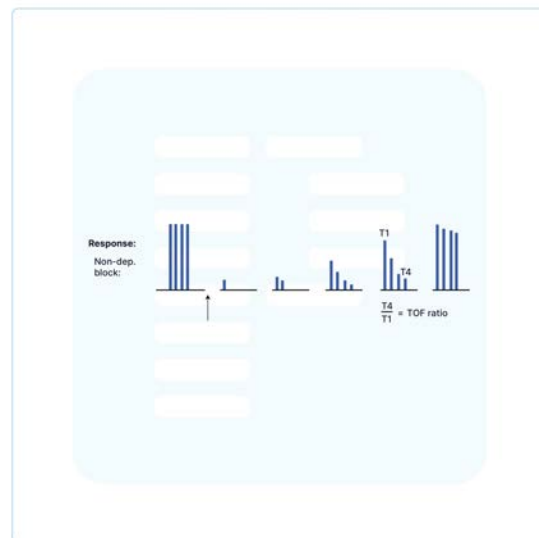
An interval of at least 10 seconds should be allowed between each set (train) of four stimuli to avoid fade during the measurement.



Muscle response:

In normal, non-paralyzed muscle, all four muscle contractions are of equal strength.

NMBA	Response
Depolarizing	All four responses are equally diminished TOF ratio = 1
Non- depolarizing	The strength of the contractions diminishes progressively, showing fade between the responses.



Interpretation:

TOF Count:

- This refers to the number of muscle contractions visible in response to the four stimuli.
- If all four responses are present, it indicates partial blockade, while fewer responses suggest a deeper blockade.

TOF Ratio:

- This is the ratio of the amplitude of the fourth twitch to the first twitch (T4/T1).
- For safe extubation: TOF ratio >0.9

Type of Block	TOF ratio	Interpretation
No block	1	All twitches are of equal strength
Nondepolarizing block (Rocuronium, Vecuronium, and Cisatracurium)	<1 (e.g., 0.7, 0.5)	The strength of the contractions diminishes progressively, showing fade between the responses. Lower ratios representing a stronger blockade.
Depolarizing block (succinylcholine)	1	All four responses are equally diminished Fade is not present until a phase II block develops.

Limitations:

Unable to Detect Deep Block: TOF stimulation cannot detect intense or deep levels of neuromuscular block, where no responses are seen. In such cases, Post Tetanic Count (PTC) stimulation is more useful.

No Monitoring of Depolarizing Phase I Block: TOF is not suitable for monitoring the depolarizing phase I block caused by succinylcholine, as all four responses in the TOF are equally diminished without fade.

Advantages:

- Simple and Common
- Non-Invasive
- Reliable for Nondepolarizing Blocks

Rocuronium (replacing succinylcholine for intubation), Vecuronium, and Cisatracurium (Options B, C and D) are all non-depolarizing agents that would typically show a fade in the TOF response.

Solution for Question 9:

Correct Answer: C) Post Tetanic Count (PTC) stimulation

Explanation:

Post Tetanic Count (PTC) stimulation is used to assess deep levels of neuromuscular blockade that are not detectable by TOF stimulation.

Patterns of Nerve Stimulation for the evaluation of the neuromuscular function:

Stimulation Pattern	Mechanism	Interpretation	Clinical Implication and Limitation
Single-Twitch Stimulation (Option B)	Single electrical stimulus applied at a frequency of 0.1 Hz to 1 Hz.	Requires comparison to pre-NMBA administration value. Muscle response to the stimulus reflects neuromuscular block onset; changes in twitch height help assess blockade.	Used to evaluate onset of neuromuscular block. Limited utility as a standalone test.
Train-of-Four (TOF) Stimulation (Option A)	Four supramaximal stimuli delivered at 0.5-second intervals (2 Hz).	Fade between responses indicates degree of block. TOF count assesses onset . TOF ratio measures recovery . Fade suggests nondepolarizing block.	Most common pattern for intraoperative monitoring. Cannot measure deep block or phase I block.
Double-Burst Stimulation (DBS) (Option D)	Two bursts of 50-Hz stimulation with a 750-ms interval.	Fade between the two bursts indicates block. Fade between bursts correlates with TOF ratio. DBS is more sensitive for detecting fade.	Better tactile detection of residual block than TOF but cannot reliably assess recovery at higher TOF ratios.
Tetanic Stimulation	High-frequency stimulation (50-100 Hz for 5 seconds) causing sustained muscle contraction .	Fade indicates nondepolarizing block; sustained response suggests normal transmission or depolarizing block.	Painful and limited use in practice. May cause lasting muscle antagonism, affecting recovery assessment.
Post Tetanic Count (PTC) Stimulation (Option C)	Tetanic stimulation (50 Hz for 5 seconds) followed by single twitches at 1 Hz after 3 seconds.	Counts discernible twitches after tetanic stimulation; assesses deep block not detectable by TOF.	Used for monitoring deep neuromuscular blockade. Requires 6-minute intervals, limiting continuous use.

Solution for Question 10:

Correct answer: D) E is the value seen on the monitor

Explanation:

From Phase D to E

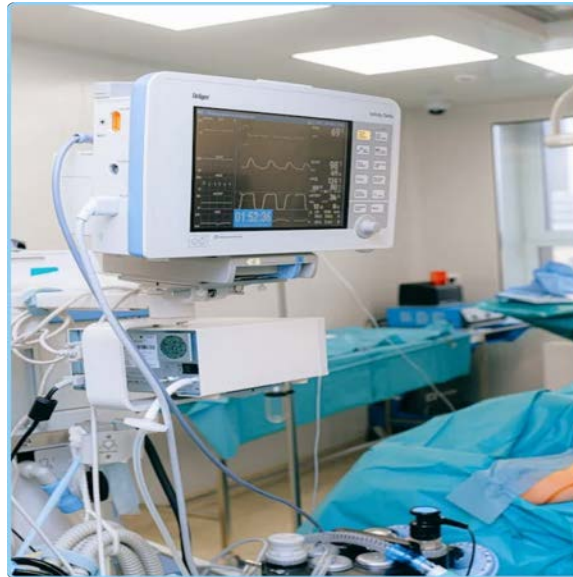
- Inhalation begins
- CO₂ level drops to zero as oxygen fills the airway
- Point E does not represent the value seen on a capnograph monitor
- It is Point D, which represents the value seen on a monitor

Capnography and Capnometry difference:

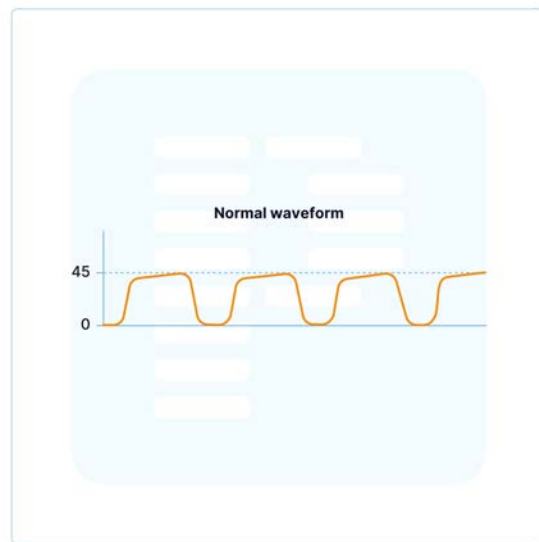


Capnography:

- Measures CO₂ in the airway
- Provides breath-to-breath status of ventilation.
- Detects CO₂ only.



Phase I	Dead Space Ventilation	Beginning of exhalation No CO ₂ present Air from trachea and pharynx No gas exchange occurs here Called dead space ventilation (Option C)
Phase II	Ascending Phase	CO ₂ from alveoli reaches upper airway, mixes with dead space air Causes rapid rise in CO ₂ concentration. (Option B)
Phase III	Alveolar Plateau	Majority of exhaled air is CO ₂ -rich Uniform CO ₂ concentration
End-Tidal	Highest CO ₂ concentration is the end-tidal value. This is the value you see on the monitor Normal EtCO ₂ is 35-45mmHg Best reflection of Alveolar CO ₂ (Option A)	
Phase IV	Descending Phase	Inhalation begins CO ₂ level drops to zero as oxygen fills the airway.



Solution for Question 11:

Correct Answer: C) Option C

Explanation:

- The capnographic waveform shows a characteristic Shark fin appearance.
- This occurs in Bronchospasms (asthma, COPD exacerbation), partially obstructed ET tube, or breathing circuit.

Different types of Capnographic Waveforms:

A normal wave in controlled mechanical ventilation	
Normal wave during spontaneous breathing	
Bronchospasm (asthma, COPD exacerbation) / partially obstructed ET tube or breathing circuit Shark fin appearance	

Spontaneous breathing effort during controlled mechanical ventilation (Curare notch) Curare cleft, or notch, is seen during spontaneous ventilation. Patients partially paralyzed with muscle relaxants may make respiratory efforts at any time during the respiratory cycle with controlled ventilation. As the muscle relaxant is reversed, the curve becomes normal in shape. (Option D ruled out)	
Role in Intubation: End tidal is used as the gold standard for confirmation of endotracheal tube position. Esophageal intubation can initially produce an EtCO ₂ due to gastric CO ₂ however this should taper after a few breaths. Right main-stem intubation may still have a normal EtCO ₂ , so get a CXR to confirm optimal tube positioning.	
Role during Cardiac Arrest: EtCO ₂ > 20 suggests good quality CPR. If it starts to drop, consider switching compressors or making sure the LUCUS is in the appropriate position. A sudden increase in EtCO ₂ greater than 10 from the patient's baseline can be an early indication of ROSC.	
Hypoventilation (Option A ruled out)	
Hyperventilation (Option B ruled out)	
Flat Capnogram: Sudden and abrupt fall of etco ₂ to zero can be seen in: Accidental extubation (most common) Disconnection of CO ₂ transducer Sudden cardiac arrest Massive pulmonary embolism	

Previous Year Questions

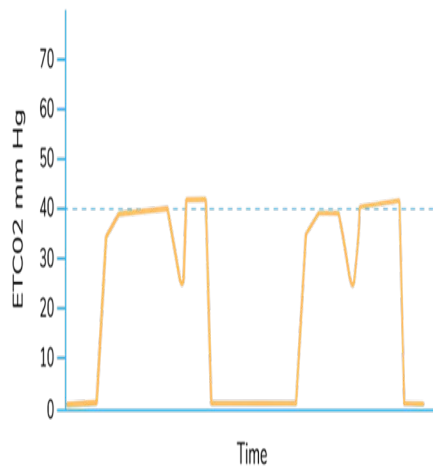
1. Which drug is most likely responsible for the occurrence of hyperthermia and muscle rigidity in a patient undergoing surgery while being administered anesthesia with halothane?

- A. D-curare
 - B. Suxamethonium
 - C. Cis-atracurium
 - D. Rocuronium
-

2. What is the most effective way to verify the placement of the endotracheal tube in a 30-year-old male patient who has been intubated for surgery by a final-year resident?

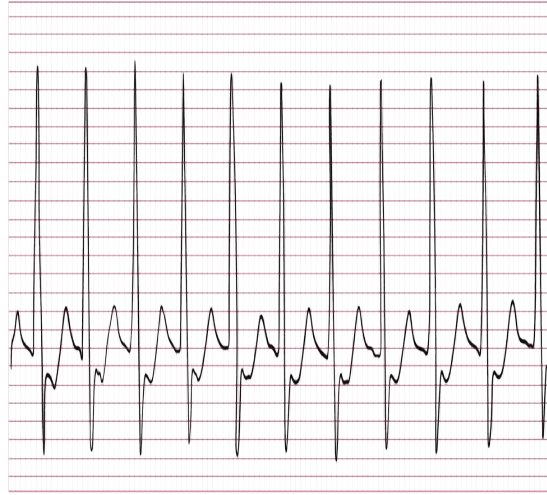
- A. X-ray chest
 - B. Auscultation
 - C. End-tidal CO₂ concentration
 - D. Chest rise
-

3. What could be the probable reason for the observed result in the capnograph of a patient who is intubated and undergoing controlled ventilation?



- A. Esophageal intubation
 - B. Inspiration with cardiac oscillations
 - C. Bronchospasm
 - D. Spontaneous respiratory effort
-

4. What should be the next course of action in managing a 50-year-old patient who has a medical history of hypertension and diabetes mellitus, and is currently presenting with a feeble pulse and a blood pressure reading of 90/40 mm Hg?



- A. IV adenosine
- B. IV diltiazem
- C. DC cardioversion
- D. Ibutilide

5. Which of the following post-bronchodilator pulmonary function test (PFT) results would necessitate a further assessment of lung function in a 50-year-old male patient with grade 3 American Society of Anesthesiology (ASA) status?

- A. $FEV_1 < 0.85$ and $FEV_1/FVC < 0.8$
- B. $FEV_1 < 0.85$ and $FEV_1/FVC < 0.75$
- C. $FEV_1 < 0.85$ and $FEV_1/FVC < 0.7$
- D. $FEV_1 < 0.8$ and $FEV_1/FVC < 0.75$

6. Which among the following is most commonly used to monitor the depth of anaesthesia?

- A. EEG
 - B. Provoked lower oesophageal contractility
 - C. Entropy
 - D. Bispectral index
-

7. Which of the following is measured by the device shown in the given image?



- A. Oxygen saturation
 - B. Oxygen content of the blood
 - C. Partial pressure of oxygen
 - D. Amount of inspired oxygen
-

8. Which of the following nerves is popularly used in clinical anesthesia to monitor neuromuscular blockade?

- A. Ulnar Nerve

- B. Median Nerve
 - C. Radial nerve
 - D. Mandibular nerve
-

9. What is the most probable cause of the patient's hypotension and cardiac arrhythmia during modified radical neck dissection (MRND) for laryngeal malignancy, when the internal jugular vein is being ligated, leading to a drop in end-tidal CO₂ (etCO₂) from 40 mm Hg to 14 mmHg?

- A. Sympathetic overactivity
 - B. Vagal stimulation
 - C. Venous air embolism
 - D. Carotid body stimulation
-

10. Which of the following methods cannot be used to determine cardiac output?

- A. Thermal dilution technique
 - B. Capnography
 - C. Esophageal doppler
 - D. Echocardiography
-

11. A 60-year-old male patient is undergoing laparoscopic cholecystectomy. During the surgery, a sudden decrease in etCO₂ to 8 mm Hg was observed. The patient's SpO₂ was 90% and peak airway pressure was 18 mm Hg. His blood pressure also dropped to 80/50 mm Hg. Which of the following is the most probable diagnosis?

- A. Pneumothorax
- B. CO₂ embolism

C. Endobronchial intubation

D. Bronchospasm

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	3
Question 5	3
Question 6	4
Question 7	1
Question 8	1
Question 9	3
Question 10	2
Question 11	2

Solution for Question 1:

Correct Option B -Suxamethonium :

- Suxamethonium is a depolarizing neuromuscular blocker that is used during surgery to induce muscle relaxation. However, in some individuals, especially those with a genetic predisposition known as malignant hyperthermia susceptibility, administration of suxamethonium can trigger a life-threatening reaction characterized by hyperthermia (dangerously elevated body temperature) and muscle rigidity. This condition is known as

malignant hyperthermia, a rare but potentially fatal reaction to certain anesthesiamedications, including suxamethonium.

Incorrect Options:

Option A: D-curare : "D-curare" is likely a typo or confusion with d-tubocurarine, a non-depolarizing neuromuscular blocker. Unlike suxamethonium, d-tubocurarine does not cause hyperthermia or muscle rigidity.

Option C: Cis-atracurium : Cis-atracurium is also a non-depolarizing neuromuscular blocker, similar to d-tubocurarine. It does not cause hyperthermia and muscle rigidity as a side effect.

Option D: Rocuronium : Rocuronium is another non-depolarizing neuromuscular blocker, and it does not cause malignant hyperthermia.

Solution for Question 2:

Correct Option C - End-tidal CO₂ concentration:

- End-tidal CO₂ concentration is the ideal method to confirm the endotracheal tube position during surgery.
- This technique is also known as capnography. This technique measures the amount of CO₂ in the exhaled breath.

Incorrect Options:

Option A - X-ray chest:

- While an X-ray chest can help confirm the position of the endotracheal tube, it is difficult to manage and arrange for the same.

Option B - Auscultation:

- Auscultation can provide initial clues for endotracheal tube placement but isn't definitive due to potential misleading breath sounds and false results.

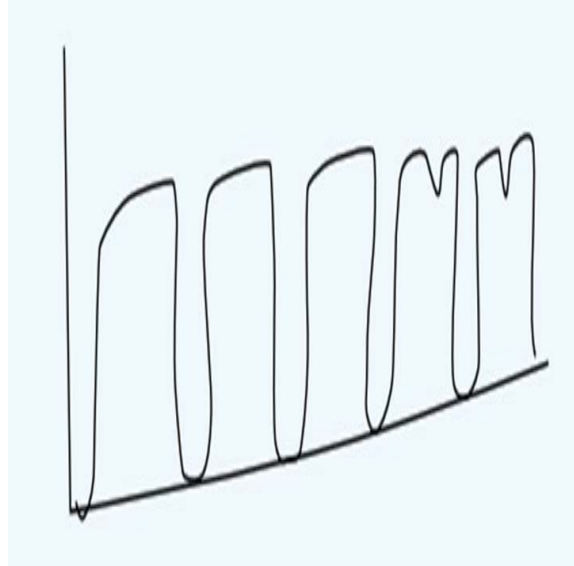
Option D - Chest rise:

- Chest rise alone isn't reliable for endotracheal tube placement as it can occur even if the tube is in the esophagus.

Solution for Question 3:

Correct Option D - Spontaneous respiratory effort:

- During controlled ventilation, spontaneous respiratory effort occurs when the patient initiates breaths alongside ventilator-delivered breaths. This can result from inadequate sedation or neuromuscular blockade.
- Spontaneous effort during the plateau phase disrupts CO₂ exhalation, leading to a distinct curare notch in the capnograph waveform, indicating patient-generated breaths.



Incorrect Options:

Option A - Esophageal intubation:

- Esophageal intubation results in an abnormal capnograph waveform with minimal CO₂ detection, but it doesn't account for the specific finding of a curare notch in the plateau phase.

Option B - Inspiration with cardiac oscillations:

- Cardiac oscillations near the endotracheal tube during inspiration can cause minor CO₂ waveform fluctuations but wouldn't specifically result in a curare notch during the plateau phase.

Option C - Bronchospasm:

- While bronchospasm can change the capnograph waveform by prolonging expiration or reducing CO₂ levels, it doesn't cause a curare notch in the plateau phase.

Solution for Question 4:

Correct Option C - DC cardioversion:

- Paroxysmal Supraventricular Tachycardia (PSVT) is likely due to the rapid heart rhythm. With the patient showing low blood pressure and a weak pulse, immediate intervention is crucial for stability.
- DC cardioversion, delivering a synchronized electrical shock, is the best course of action. It swiftly ends the tachycardia, restoring normal rhythm and improving cardiac output and blood pressure.

Incorrect Options:

Option A - IV adenosine:

- In unstable patients like this one, adenosine isn't ideal as it can worsen hemodynamic stability by causing transient asystole or severe bradycardia.

Option B - IV Diltiazem:

- In unstable patients with low blood pressure and a feeble pulse, Diltiazem might not be the best initial choice for treating SVT. It can further lower blood pressure by slowing the heart rate.

Option D - Ibutilide:

- Ibutilide has a slower onset and may not provide immediate relief in critical situations.
-

Solution for Question 5:

Correct Option C - $FEV1 < 0.85$ and $FEV1/FVC < 0.7$:

- $FEV1$ below 0.85 liters and $FEV1/FVC$ ratio less than 0.7 indicate significant airflow limitation, suggesting conditions like COPD. In patients with grade 3 ASA, further evaluation, including lung volume measurements or gas diffusion capacity assessment, is needed to

determine the severity and type of lung disease.

Incorrect Options:

Option A - $FEV_1 < 0.85$ and $FEV_1/FVC < 0.8$:

- An FEV_1/FVC ratio of 0.8 is normal, usually not needing more assessment. However, an FEV_1 under 0.85 liters suggests airflow limitation, possibly requiring further evaluation.

Option B - $FEV_1 < 0.85$ and $FEV_1/FVC < 0.75$:

An FEV_1/FVC ratio of 0.75 suggests airflow limitation, while an FEV_1 below 0.85 liters indicates reduced air volume expelled in the first second. Despite the exact FEV_1/FVC ratio, a low FEV_1 necessitates further evaluation. Option D - $FEV_1 < 0.8$ and $FEV_1/FVC < 0.75$:

- An FEV_1 below 0.8 liters suggests significant airflow limitation, warranting further evaluation. However, with an FEV_1/FVC ratio of 0.75 within the normal range, immediate additional assessment may not be necessary.

Solution for Question 6:

Correct Option D - Bispectral index (BIS):

- The BIS is a monitoring tool that combines various parameters, including EEG activity, to provide a numerical value that correlates with the depth of anesthesia.
- It is widely used in clinical practice to guide the administration of anesthesia and ensure the patient remains at an appropriate level of unconsciousness.

Incorrect Options:

Option A - EEG (Electroencephalogram):

- EEG measures brain electrical activity and can gauge anesthesia depth, but it's not the primary method for monitoring anesthesia depth.

Option B - Provoked lower esophageal contractility:

- Monitoring lower esophageal contractility isn't commonly used to assess anesthesia depth, although it might be relevant in certain situations.

Option C - Entropy:

- Entropy monitoring analyzes brainwave activity to indicate anesthesia depth, but it's less common compared to the Bispectral index (BIS).

Solution for Question 7:

Correct Option A - Oxygen saturation:

- The device shown in the image is a pulse oximeter, which is used to measure the oxygen saturation level in the blood. It provides a numerical value, typically expressed as a percentage, indicating the percentage of hemoglobin in the blood that is saturated with oxygen.

Incorrect Options:

Option B - Oxygen content of the blood:

- The pulse oximeter measures oxygen saturation, not the actual oxygen content in the blood.

Option C - Partial pressure of oxygen:

- The pulse oximeter measures oxygen saturation, not the partial pressure of oxygen (PO₂).

Option D - Amount of inspired oxygen:

- The pulse oximeter only measures blood oxygen saturation, not the amount or concentration of inspired oxygen.
-

Solution for Question 8:

Correct Option A - Ulnar Nerve:

- To monitor a neuromuscular blockade, a supramaximal electrical stimulus is given to a peripheral nerve that can be accessed quickly, and the muscle response to the stimulus is noted.
- One of the easily accessible nerves is the ulnar nerve, so it is preferred for monitoring it. Hence, option A is the correct answer to the question given.

Incorrect Options:

Option B - Median Nerve:

- It is not commonly used in clinical anesthesia to monitor neuromuscular blockade.

Option C - Radial Nerve:

- It is not commonly used in clinical anesthesia to monitor neuromuscular blockade.

Option D - Mandibular Nerve:

- It is not commonly used in clinical anesthesia to monitor neuromuscular blockade.
-

Solution for Question 9:

Correct Option C - Venous air embolism:

- Venous air embolism, common during neck surgeries like modified radical neck dissection (MRND), happens when air enters the venous system, often from internal jugular vein manipulation. This air can block blood flow to the heart's right side, leading to reduced cardiac output, systemic hypotension, and cardiac arrhythmias. It can also lower etCO₂ levels due to impaired perfusion and ventilation. Transesophageal echocardiography (TEE): The most sensitive and best monitor to detect venous air embolism

Incorrect Options:

Option A - Sympathetic overactivity:

- Sympathetic overactivity usually increases blood pressure, heart rate, and respiratory rate, but it's unlikely to cause the sudden drop in etCO₂, hypotension, and cardiac arrhythmia observed here..

Option B - Vagal stimulation:

Vagal stimulation triggers the parasympathetic nervous system, causing bradycardia and occasionally hypotension, but it's not typically associated with significant etCO₂ drops or cardiac arrhythmias.

Option D - Carotid body stimulation:

- Stimulating the carotid body increases sympathetic activity, raising blood pressure and heart rate. However, it's not linked to the sudden etCO₂ drop, hypotension, and cardiac arrhythmia described.

Solution for Question 10:

Correct Option B - Capnography:

- Capnography is not a method for directly determining cardiac output. Capnography is a monitoring technique that measures the partial pressure of carbon dioxide (CO₂) in exhaled breath. It is commonly used to assess ventilation and end-tidal CO₂ levels but does not provide direct information about cardiac output.

Incorrect Options:

Option A - Thermal dilution technique:

The thermal dilution technique measures cardiac output by injecting cold fluid into a central vein and tracking the temperature change as it mixes with blood.

Option C - Esophageal Doppler:

- Esophageal Doppler estimates cardiac output by analyzing the velocity-time profile.

Option D - Echocardiography:

- Echocardiography evaluates cardiac function, including cardiac output, by assessing parameters like left ventricular stroke volume and heart rate.
-

Solution for Question 11:

Correct Option B - CO₂ embolism:

- The most probable diagnosis in this scenario is CO₂ embolism. The sudden decrease in end-tidal carbon dioxide (etCO₂), along with a drop in oxygen saturation (SpO₂), low blood pressure, and increased peak airway pressure, is indicative of a gas embolism. In laparoscopic procedures, CO₂ is commonly used to insufflate the abdomen, and an unintentional entry of CO₂ into the vascular system can lead to a gas embolism. This condition can cause cardiovascular collapse, hypoxia, and increased airway pressure.

Incorrect Options:

Option A - Pneumothorax:

- While pneumothorax can cause decreased oxygen saturation (SpO₂) and increased peak airway pressure, it is less likely in the context of laparoscopic cholecystectomy.

Option C - Endobronchial intubation:

- While it can cause hypoxia, it is less likely to result in the sudden decrease in etCO₂ and the other cardiovascular manifestations described in the scenario.

Option D - Bronchospasm:

- Bronchospasm can raise airway pressure and lower oxygen saturation, but it's less likely to cause the sudden drop in etCO₂ and cardiovascular collapse seen in this case.
-

1. A 30-year-old man collapses and is unresponsive. After tapping his shoulders and getting no response, what is your next step?

- A. Check his pulse
 - B. Start high-quality CPR
 - C. Start rescue breaths
 - D. Shout for help
-

2. A 30-year-old man is found unconscious after a motorcycle accident. There are no signs of external bleeding, but you suspect a possible head or neck injury. You need to open his airway to check for breathing. What is the preferred method for opening the airway in this situation?

- A. Head tilt-chin lift
 - B. Jaw thrust
 - C. Head tilt-neck lift
 - D. Avoid opening the airway
-

3. A 55-year-old man collapses at a shopping mall, and bystanders begin performing CPR. An automated external defibrillator (AED) is brought to the scene. After applying the pads, the AED begins analyzing the heart rhythm. You are assisting with the resuscitation. What action should you take during the AED's analysis?

- A. Check the pulse.
- B. Continue chest compressions.

- C. Give rescue breaths only.
 - D. Stand clear of the victim.
-

4. A 60-year-old female is found unresponsive at home. Cardiopulmonary resuscitation (CPR) is initiated, and she is found to be in asystole. After several minutes of resuscitation efforts, her rhythm converts to ventricular fibrillation. Which of the following is most likely associated with improved survival in this case?

- A. Defibrillation after the conversion to a shockable rhythm
 - B. Administration of high-dose epinephrine
 - C. Immediate initiation of therapeutic hypothermia
 - D. Continuous chest compressions without defibrillation
-

5. In pediatric Basic Life Support (BLS), which of the following is the most recent modification recommended by the American Heart Association (AHA) to improve resuscitation outcomes in children?

- A. Starting with Airway, Breathing, Chest Compressions
 - B. Pulse check for 30 seconds before initiating CPR
 - C. Delivering chest compressions at a depth of 1 cm for infants
 - D. Shifting the resuscitation sequence to Chest Compressions, Airway, Breathing
-

6. A 55-year-old male is found unresponsive and without a pulse. Emergency responders are activated, and advanced cardiac life support (ACLS) protocols are initiated. Which of the following steps should be prioritized immediately after confirming cardiac arrest?

- A. Administer epinephrine (1 mg every 3-5 minutes)
 - B. Ensure proper airway management using bag-mask ventilation
 - C. Initiate high-quality chest compressions at a rate of 100-120 per minute
 - D. Deploy an Automated External Defibrillator (AED)
-

7. A healthcare provider is managing a patient with cardiac arrest due to pulseless ventricular tachycardia. Which of the following is a key advantage of using manual defibrillators in this scenario?

- A. They are easier to operate than Automated External Defibrillators (AEDs).
 - B. Manual defibrillators can allow the users to adjust energy levels based on their clinical judgement.
 - C. They can be used by bystanders.
 - D. They are used independently, without ECG monitoring.
-

8. A 60-year-old man presents to the emergency department after being resuscitated from cardiac arrest. The medical team suspects a reversible cause and starts assessing possible factors. Which of the following pairs do not represent causes to be prioritized for immediate treatment based on their roles in causing cardiac arrest?

- A. Hypovolemia and Tension Pneumothorax
 - B. Hyperkalemia and Acidosis
 - C. Drug Overdose and Cardiac Tamponade
 - D. Hyperglycemia and alkalosis
-

9. A 60-year-old male patient presents to the emergency department with palpitations and shortness of breath. Upon assessment, he is found to have narrow-complex tachycardia at a rate of 180 beats per minute. He reports no chest pain or hypotension. What is the most appropriate initial management for this stable patient?

- A. Synchronized cardioversion
- B. Administration of amiodarone
- C. Administration of adenosine
- D. Observation and reassurance

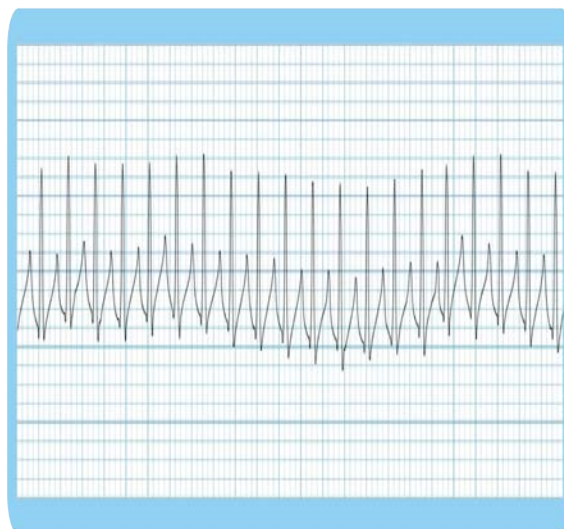
10. A 68-year-old man presents to the emergency department with palpitations, dizziness, and chest discomfort. His vital signs are blood pressure 75/50 mmHg, heart rate 190 beats per minute, and oxygen saturation 94% on room air. He appears pale and diaphoretic. ECG is shown below. What is the most appropriate next step in the management of this patient? {{caption_text}}



- A. Administer IV amiodarone 150 mg over 10 minutes
- B. Perform immediate unsynchronized defibrillation
- C. Perform immediate synchronized cardioversion

D. Administer intravenous adenosine 6 mg

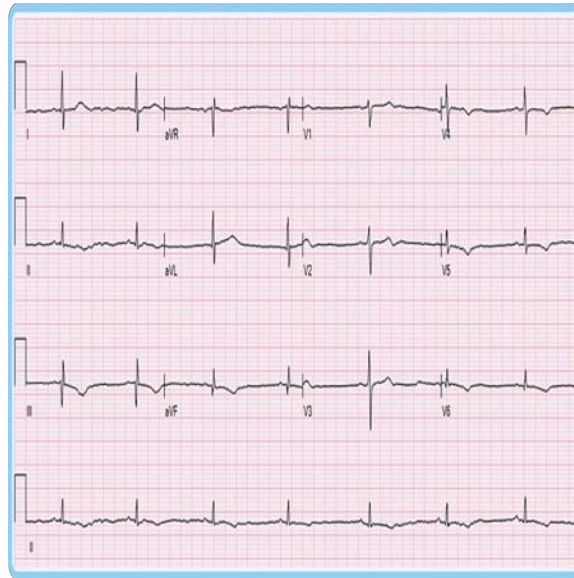
11. A 35-year-old woman presents to the emergency department with palpitations that started suddenly 30 minutes ago. She reports feeling lightheaded but denies chest pain or shortness of breath. Her vital signs show a heart rate of 180 beats per minute, blood pressure of 120/80 mmHg, and oxygen saturation of 96% on room air. ECG is shown below. Reveals a regular narrow-complex tachycardia without discernible P waves. What is the most appropriate initial management?
{{caption_text}}



- A. Immediate synchronized cardioversion
 - B. Intravenous adenosine 6 mg push
 - C. Intravenous amiodarone 150 mg over 10 minutes
 - D. Perform carotid sinus massage
-

12. A 65-year-old man with a history of coronary artery disease and heart failure presents to the emergency department after experiencing lightheadedness and

near syncope. On examination, he is alert but pale, with a heart rate of 43 beats per minute and a blood pressure of 80/50 mmHg. The ECG is shown below. He remains hypotensive and symptomatic despite the IV administration of a maximum dose of atropine. What is the most appropriate next step in the management of this patient? {{caption_text}}



- A. Epinephrine IV bolus
- B. Amiodarone IV infusion
- C. Dopamine IV infusion
- D. Digoxin IV bolus

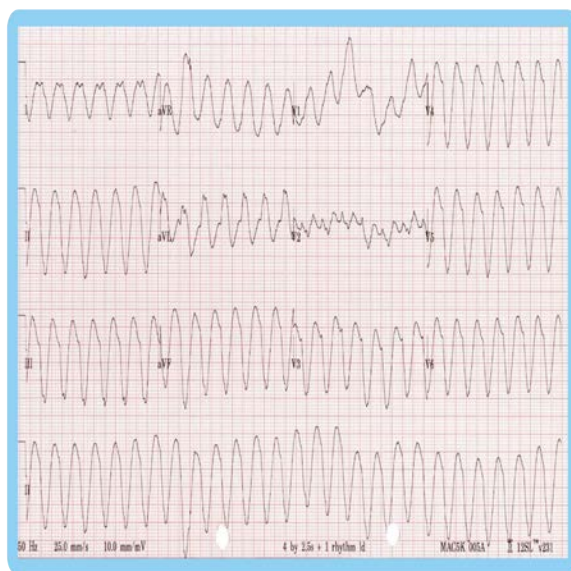
13. A 58-year-old man is found unresponsive and pulseless at home. Paramedics initiate CPR and provide bag-mask ventilation. Upon arrival at the emergency department, the patient is intubated. IV access is delayed due to difficult peripheral access, and intraosseous access is not immediately available. The decision is made to administer drugs via the endotracheal tube. Which of the following drugs is appropriate for endotracheal administration during cardiac arrest?

- A. Amiodarone
- B. Epinephrine

C. Adenosine

D. Atropine

14. A 70-year-old man with a history of ischemic cardiomyopathy (ejection fraction 25%) and a previous history of myocardial infarction presents to the emergency department with dizziness, palpitations, and mild breathlessness. On arrival, his heart rate is 130 beats per minute, and his blood pressure is 100/64 mmHg. His ECG is shown below. Which among the following options is the most appropriate in the management of this patient? {{caption_text}}



A. Intravenous amiodarone 150 mg over 10 minutes

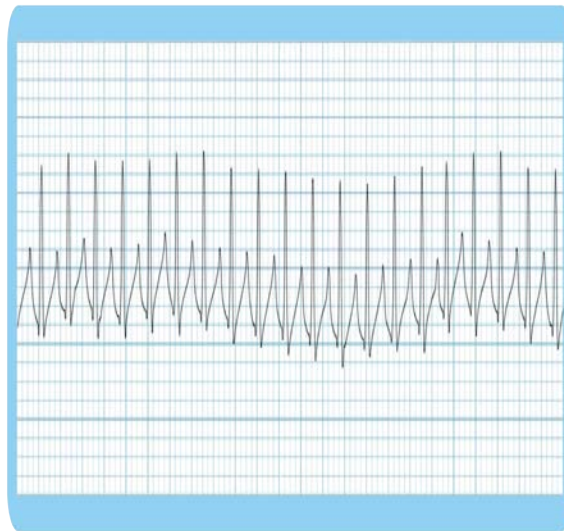
B. Adenosine 12 mg IV push

C. Intravenous lidocaine 1 mg/kg

D. Intravenous magnesium sulfate 2 g

15. A 54-year-old woman presents to the emergency department with palpitations, dizziness, and mild shortness of breath. Her vital signs are heart rate of 150 bpm, blood pressure of 124/80 mmHg, and oxygen saturation of 95% on room air. The rhythm strip of the ECG is shown below. Vagal maneuvers are attempted without success. You decide to administer adenosine. Which of the following is the most appropriate initial dose of adenosine, and what is the mechanism of its action?

{{caption_text}}



- A. 6 mg IV push, briefly depresses sinus node and AV node conduction
- B. 12 mg IV push, briefly depresses sinus node and AV node conduction
- C. 6 mg IV push, increases AV node conduction and depresses sinus node
- D. 12 mg IV push, decreases AV node conduction and increases sinus node rate

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	4
Question 4	1
Question 5	4
Question 6	3
Question 7	2
Question 8	4
Question 9	3
Question 10	3
Question 11	4
Question 12	3
Question 13	2
Question 14	1
Question 15	1

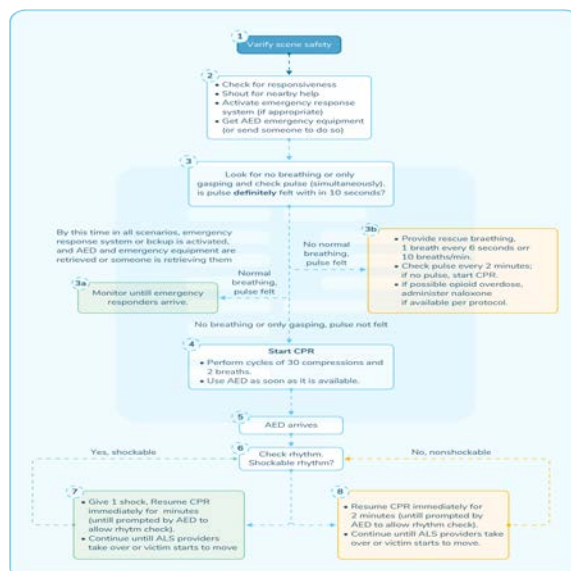
Solution for Question 1:

Correct Answer: D) Shout for help

Explanation:

The first step after confirming unresponsiveness is to shout for help and activate emergency services. This ensures that additional resources (like an AED or more rescuers) are available while you continue to assess and provide care.

Steps in Basic Life Support Skills:



Solution for Question 2:

Correct Answer: B) Jaw thrust

Explanation:

- In cases of suspected head or neck trauma, the jaw thrust manoeuvre is preferred to open the airway while minimizing neck movement, reducing the risk of exacerbating spinal injuries.
- The CAB algorithm highlights the key elements needed to ensure high-quality CPR in BLS.

Compressions (C)	Position: Place the patient in a supine position on a firm surface. Site: Lower ■ rd of sternum Rate: 100 to 120 compressions per minute. Depth: Compress 2 to 2.4 inches (5-6 cm); avoid exceeding 2.4 inches. Compression-to-Ventilation Ratio: 30 compressions to 2 breaths (30:2). Chest Recoil: Allow full recoil between compressions to optimize blood flow. Interruptions: Minimize interruptions in compressions (<10 seconds). Chest Compression Fraction (CCF): The proportion of time spent performing chest compressions during CPR: Maintain a CCF $\geq 60\%$. For pregnant women: perform manual lateral uterine displacement (LUD) to relieve pressure on large abdominal vessels.
Airway (A)	Head Tilt-Chin Lift: Used to open the airway in patients without suspected neck trauma. Jaw Thrust: Preferred method, if neck trauma is suspected. The most common cause of airway obstruction in unconscious patients is the tongue falling back.
Breathing (B)	Rescue Breaths: Give 2 breaths after every 30 compressions. Method: Mouth-to-mouth or with a bag-mask device, ensuring chest rise with each breath. Breath Duration: Each breath should last about 1 second.

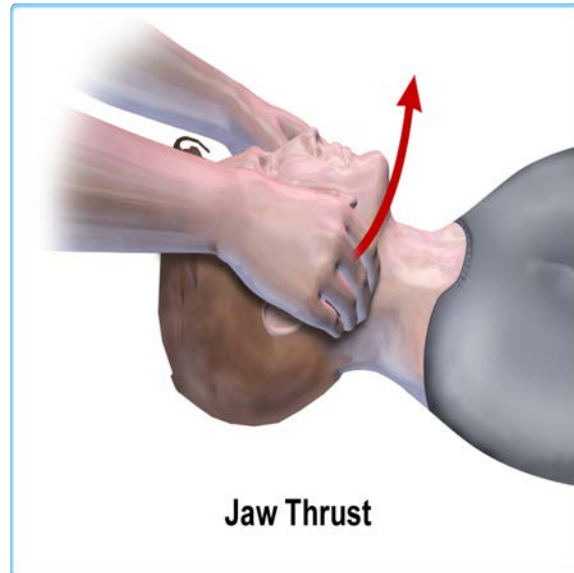
Head Tilt Chin Lift Maneuver:

- The head is tilted back while lifting the chin upwards to prevent the tongue from obstructing the airway.
- Used to open the airway in an unconscious patient without suspected spinal injury.



Jaw Thrust Maneuver:

- The lower jaw is lifted forward by placing fingers behind the angle of the mandible, moving the tongue forward and clearing the airway without moving the neck.
- Used to open the airway in patients with suspected cervical spine injury.



Solution for Question 3:

Correct Answer: D) Stand clear of the victim

Explanation:

During the AED analysis, it is crucial to avoid touching the patient to prevent interference with the heart rhythm reading and ensure the safety of those around.

Automated External Defibrillator (AED)

- An Automated External Defibrillator (AED) is a portable, computerized device designed to identify life-threatening cardiac arrhythmias and, if necessary, deliver an electrical shock to restore a normal heart rhythm.
- It is an automated, user-friendly device for layperson use, while a conventional defibrillator requires professional expertise for manual operation and rhythm interpretation.

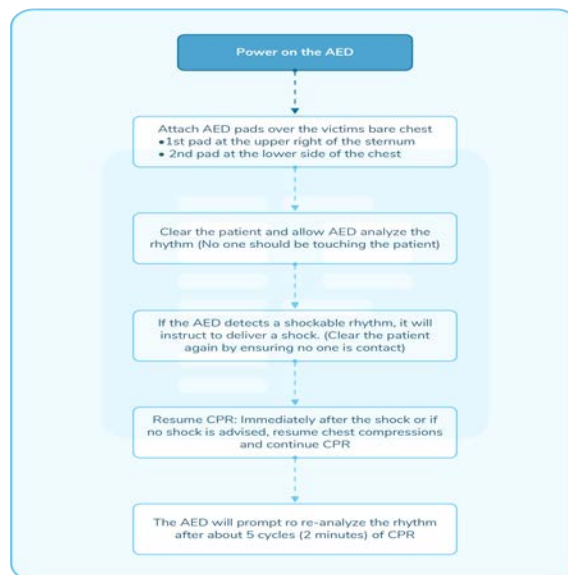


Indications for AED use:

AEDs are indicated in cardiac arrest due to shockable rhythms such as:

- Pulseless Ventricular Tachycardia (pVT)
- Ventricular Fibrillation (VF)

Steps for Operating an AED



Solution for Question 4:

Correct Answer: A) Defibrillation after the conversion to a shockable rhythm

Explanation:

- Initially, the patient is in asystole, a non-shockable rhythm, and later converts to ventricular fibrillation, a shockable rhythm. Once the patient converts from a non-shockable to a shockable rhythm, defibrillation is the most effective intervention.

Shockable vs Non-Shockable Rhythm definitions

- Shockable Rhythms : Ventricular Fibrillation (VF) Pulseless Ventricular Tachycardia (pVT) Respond to defibrillation.
- Ventricular Fibrillation (VF)
- Pulseless Ventricular Tachycardia (pVT)
- Respond to defibrillation.
- Non-Shockable Rhythms : Asystole Pulseless Electrical Activity (PEA) Do not respond to defibrillation.
- Asystole
- Pulseless Electrical Activity (PEA)
- Do not respond to defibrillation.
- Ventricular Fibrillation (VF)
- Pulseless Ventricular Tachycardia (pVT)
- Respond to defibrillation.
- Asystole
- Pulseless Electrical Activity (PEA)
- Do not respond to defibrillation.

Prognostic Implications

- Survival Rates : Shockable rhythms have better survival and neurological outcomes compared to non-shockable rhythms.
- Shockable rhythms have better survival and neurological outcomes compared to non-shockable rhythms.
- Conversion from Non-Shockable to Shockable : Early conversion improves outcomes, enhancing survival rates and neurological recovery.
- Early conversion improves outcomes, enhancing survival rates and neurological recovery.

- Shockable rhythms have better survival and neurological outcomes compared to non-shockable rhythms .
- Early conversion improves outcomes, enhancing survival rates and neurological recovery.

Factors Influencing Rhythm Type

Factor	Non-Shockable Rhythms	Shockable Rhythms
Demographics	Female, older age, unwitnessed arrests	-
Medical History	Non-cardiovascular diseases (e.g., COPD), medications (antidepressants, antipsychotics)	History of cardiovascular disease

Administration of high-dose epinephrine (Option B) is not standard and may not significantly improve survival compared to defibrillation in shockable rhythms.

Immediate initiation of therapeutic hypothermia (Option C) is used post-resuscitation for neuroprotection but does not address the immediate need for defibrillation.

Continuous chest compressions without defibrillation (Option D) would delay the critical intervention for a shockable rhythm, reducing survival chances.

Solution for Question 5:

Correct Answer: D) Shifting the resuscitation sequence to Chest Compressions, Airway, Breathing

Explanation:

The AHA now emphasizes starting with chest compressions (C) , followed by airway and breathing (A-B), to improve outcomes in pediatric cardiac arrest.

Modifications in Pediatric Basic Life Support (BLS)

- Sequence Shift: C-A-B (Option A ruled out) (Option C) Emphasis on chest compressions is first , followed by airway and breathing to improve survival in cardiac arrest.
- Emphasis on chest compressions is first , followed by airway and breathing to improve survival in cardiac arrest.

- "Look, Listen, Feel" Removed Immediate CPR initiation is recommended if unsure about a pulse; it minimizes treatment delay.
- Immediate CPR initiation is recommended if unsure about a pulse; it minimizes treatment delay.
- Pulse Check Lay rescuers: Pulse check not required. Healthcare providers: Pulse check within 10 seconds; begin CPR if uncertain. (Option B ruled out)
- Lay rescuers: Pulse check not required.
- Healthcare providers: Pulse check within 10 seconds; begin CPR if uncertain. (Option B ruled out)
- Compression Depth and Rate Depth: At least 1/3 of chest depth (4 cm for infants, 5 cm for children). (Option C ruled out) Rate: 100-120 compressions per minute.
- Depth: At least 1/3 of chest depth (4 cm for infants, 5 cm for children). (Option C ruled out)
- Rate: 100-120 compressions per minute.
- Use of AEDs: AEDs recommended for infants if no manual defibrillator is available.
- Training Adjustments: Simplified BLS training models improve knowledge and skill retention.
- High-Quality CPR Focus: Minimal interruptions and proper ventilation post-advanced airway insertion are emphasized.
- Emphasis on chest compressions is first , followed by airway and breathing to improve survival in cardiac arrest.
- Immediate CPR initiation is recommended if unsure about a pulse; it minimizes treatment delay.
- Lay rescuers: Pulse check not required.
- Healthcare providers: Pulse check within 10 seconds; begin CPR if uncertain. (Option B ruled out)
- Depth: At least 1/3 of chest depth (4 cm for infants, 5 cm for children). (Option C ruled out)
- Rate: 100-120 compressions per minute.

Solution for Question 6:

Correct Answer: C) Initiate high-quality chest compressions at a rate of 100-120 per minute

Explanation:

- Initiate high-quality chest compressions at a rate of 100-120 per minute : High-quality chest compressions should be started immediately after confirming cardiac arrest to improve chances of survival.

ACLS Steps

Advanced Cardiac Life Support (ACLS) is crucial for managing cardiac arrest and cardiovascular emergencies, following the 2016 guidelines.

Key Steps in ACLS

Step	Description
1. Early Recognition and Activation	Identify cardiac arrest signs quickly and alert emergency services.
2. High-Quality Chest Compressions (Option C)	Initiate compressions at a rate of 100-120 per minute, with a depth of at least 2 inches (5 cm) for adults, allowing full chest recoil.
3. Early Defibrillation (Option D ruled out)	Deploy an AED promptly; follow prompts for defibrillation if a shockable rhythm is present.
4. Airway Management (Option B ruled out)	Ensure airway patency using techniques like bag-mask ventilation or advanced airway placement; provide oxygen to support ventilation.
5. Drug Administration (Option A ruled out)	Administer epinephrine (1 mg every 3-5 minutes); consider antiarrhythmics like amiodarone for shock-resistant VF/pulseless VT.
6. Identify and Treat Reversible Causes	Assess for reversible causes ("H's and T's"):
Hypoxia	
Hypovolemia	
Acidosis (Hydrogen ion)	
Hyper/hypokalemia	
Hypothermia	
Tension pneumothorax	
Cardiac tamponade	
Toxins	
Thrombosis (pulmonary/coronary)	

7. Post-Resuscitation Care	Provide comprehensive care after return of spontaneous circulation (ROSC), including monitoring and supportive therapies.
8. Continuous Quality Improvement	Assess and enhance resuscitation efforts, focusing on minimizing interruptions in compressions and following protocols.

Solution for Question 7:

Correct Answer: B) Manual defibrillators can allow the users to adjust energy levels based on their clinical judgement

Explanation:

- One of the key advantages of manual defibrillators is the ability to synchronize shocks with the patient's heartbeat.

Uses of Manual Defibrillators

- Manual defibrillators are crucial for treating cardiac arrest caused by ventricular fibrillation (VF) or pulseless ventricular tachycardia (VT) by delivering controlled electric shocks to restore normal heart rhythm.

Use	Details
Termination of VF and Pulseless VT	Treats life-threatening arrhythmias, the leading causes of sudden cardiac arrest. Stops chaotic electrical activity, allowing heart reset.
Early Defibrillation in Cardiac Arrest	Critical for survival; chances decrease 7-10% each minute without defibrillation. Timely shocks significantly enhance resuscitation success.
Differentiation from AEDs	AEDs are for bystanders; manual defibrillators are used by healthcare professionals in clinical settings. Offer advanced features like energy adjustment. (Options A & C ruled out)
Integration with ACLS Protocols	Integral to Advanced Cardiac Life Support protocols for managing cardiac emergencies. Used with chest compressions, airway management, and drug administration.

Ongoing Monitoring and Rhythm Analysis	Incorporate ECG monitoring for continuous heart rhythm assessment. (Option D ruled out) Enable synchronized cardioversion for stable patients with abnormal rhythms.
--	--

Solution for Question 8:

Correct Answer: D) Hyperglycemia and alkalosis

Explanation:

While Hyperglycemia (high blood sugar) is a significant metabolic disturbance, it is not a direct reversible cause of cardiac arrest compared to the other pairs.

5 H's of Cardiac Arrest

Cause	Description	Interventions
Hypovolemia (Option A ruled out)	Significant loss of blood volume leading to inadequate circulation.	Identify blood loss; administer fluid resuscitation.
Hypoxia	Inadequate oxygen supply causing cardiac arrest.	Ensure airway is clear; provide supplemental oxygen.
Hydrogen Ion (Acidosis) (Option B ruled out)	Disruption of heart function due to metabolic or respiratory acidosis.	Monitor arterial blood gasses; administer sodium bicarbonate.
Hyper-/Hypokalemia (Option B ruled out)	Abnormal potassium levels leading to cardiac arrest.	Treat with calcium, insulin, or dialysis.
Hypoglycemia	Low blood sugar levels precipitating cardiac arrest.	Administer glucose.

5 T's of Cardiac Arrest

Cause	Description	Interventions
Toxins (Option C ruled out)	Drug overdoses or toxic substance exposure causing cardiac arrest.	Identify toxins; provide antidotes or supportive care.
Tamponade (Cardiac) (Option C ruled out)	Fluid accumulation in the pericardial sac leading to decreased cardiac output.	Perform pericardiocentesis to relieve pressure.

Tension Pneumothorax (Option A ruled out)	Air in the pleural space causing lung collapse.	Immediate needle thoracostomy for decompression.
Thrombosis (Coronary and Pulmonary)	Blockages causing cardiac arrest (heart attack or pulmonary embolism).	Thrombolytics for coronary thrombosis; anticoagulation for pulmonary embolism.
Trauma	Severe injuries leading to cardiac arrest through hemorrhage or direct heart injury.	Assess for injuries; provide appropriate interventions.

Solution for Question 9:

Correct Answer: C) Administration of adenosine

Explanation:

- Administration of adenosine because it is the first-line treatment for stable narrow-complex tachycardia. Adenosine acts quickly to potentially terminate the arrhythmia by interrupting the reentrant pathways, making it the most appropriate choice in this scenario.

Adult Tachycardia Algorithm

- Management follows the Advanced Cardiac Life Support (ACLS) guidelines.
- Treatment decisions depend on patient stability and tachycardia characteristics.

Key Steps

- Check for signs of tachycardia.
- Determine stability: Signs of instability: Chest pain Shortness of breath Altered mental status Hypotension Pulmonary edema Ischemic changes on ECG
- Signs of instability: Chest pain Shortness of breath Altered mental status Hypotension Pulmonary edema Ischemic changes on ECG
- Chest pain
- Shortness of breath
- Altered mental status
- Hypotension
- Pulmonary edema
- Ischemic changes on ECG

- Signs of instability: Chest pain Shortness of breath Altered mental status Hypotension Pulmonary edema Ischemic changes on ECG

- Chest pain
- Shortness of breath
- Altered mental status
- Hypotension
- Pulmonary edema
- Ischemic changes on ECG
- Chest pain
- Shortness of breath
- Altered mental status
- Hypotension
- Pulmonary edema
- Ischemic changes on ECG

- Classify based on QRS complex "list-style-type:square" aria-level="3"> Narrow-complex : QRS < 0.12 seconds

- Wide-complex : QRS $\geq$ 0.12 seconds

Stability	Narrow-Complex Management	Wide-Complex Management
Stable	Pharmacological options: Adenosine Beta-blockers Calcium channel blockers	Antiarrhythmics, e.g., amiodarone (Option B ruled out)
Unstable (Option C ruled out)	Synchronized cardioversion	Synchronized cardioversion

- Start at 50 joules, increase if needed.
- Ensure synchronization with R wave to prevent ventricular fibrillation.
- Monitor for complications after cardioversion or stabilization.

Observation and reassurance (Option D) :

- This is insufficient management for a tachycardic patient, even if stable.
- Immediate treatment is warranted.

Solution for Question 10:

Correct Answer: C) Perform immediate synchronized cardioversion

Explanation:

- The clinical presentation and ECG suggesting a monomorphic wide-complex tachycardia with a regular rhythm is consistent with a diagnosis of an unstable monomorphic ventricular tachycardia.
- The first line of treatment for unstable VT with a pulse is synchronized cardioversion.

Wide QRS Tachycardia (QRS ≥ 0.12 second):

- Ventricular in origin
- It includes ventricular tachycardia and ventricular fibrillation, pre-excited tachycardias (Wolff-Parkinson-White syndrome), and ventricular-paced rhythms.
- Ventricular tachycardia is classified as monomorphic or polymorphic
- Treatment is based on the type of ventricular tachycardia

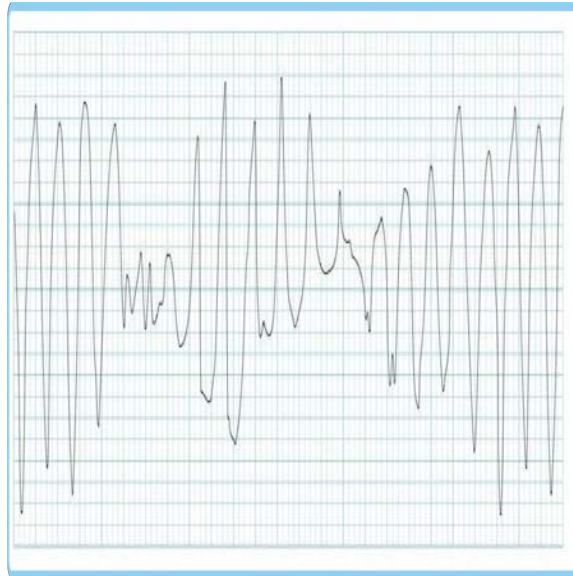
Monomorphic VT:



- Regular, stable with a pulse- DOC is Adenosine 6mg IV push followed by 12mg IV push if needed
- Unstable patient with pulse-synchronized cardioversion

- No pulse- Unsynchronized shock

Polymorphic VT:

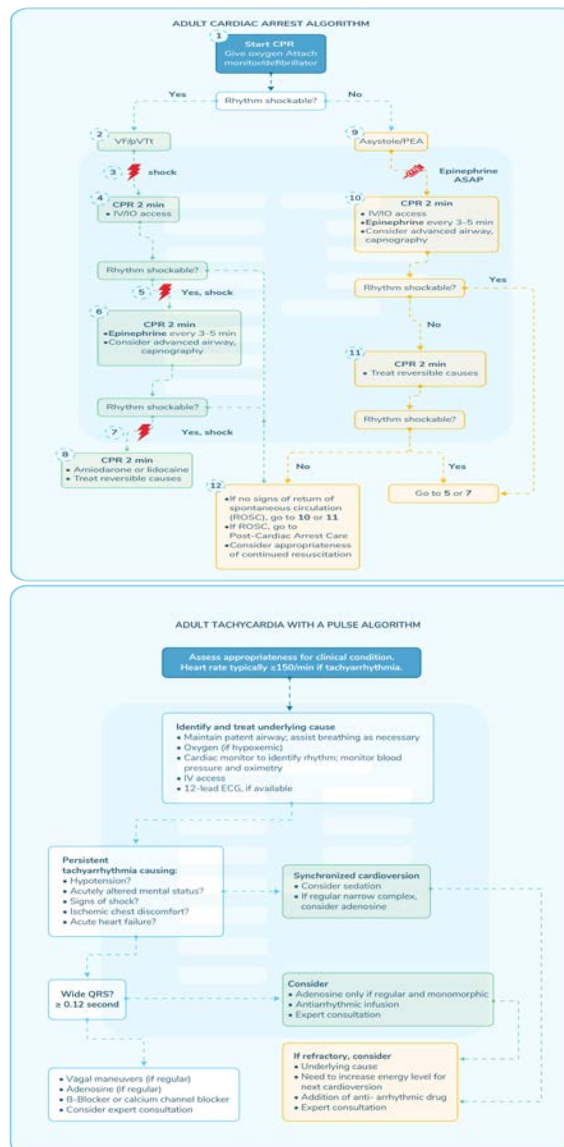


- With pulse- Amiodarone or procainamide
- Without pulse/unstable- Unsynchronized shock

Ventricular fibrillation:



Note: Almost always presents as unstable, without a pulse- Unsynchronized shock



Administer IV amiodarone 150 mg over 10 minutes (Option A):

- It is not the first-line treatment in a patient with hemodynamic instability and shock.
- Immediate cardioversion is preferred in unstable patients.

Perform immediate unsynchronized defibrillation (Option B):

- Defibrillation is indicated for pulseless VT or ventricular fibrillation (VF).
- This patient has a wide-complex tachycardia with a pulse, making synchronized cardioversion the appropriate approach.

Administer intravenous adenosine 6 mg (Option D):

- Adenosine is used to treat stable, regular, monomorphic wide complex tachycardia with a pulse.
- This patient is unstable with pulse making synchronized cardioversion the appropriate approach.

Solution for Question 11:

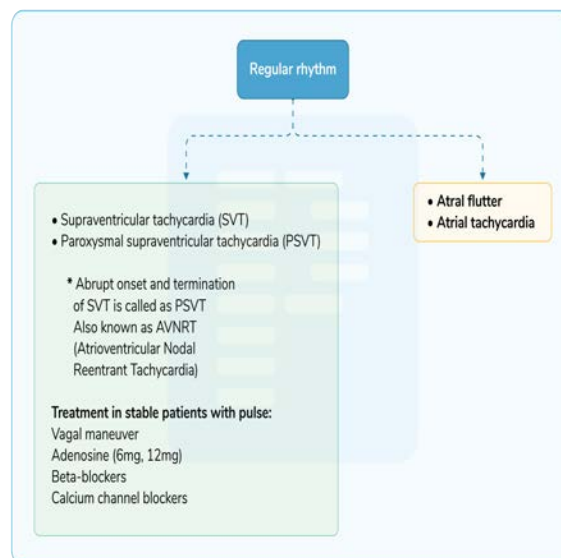
Correct Answer: D) Perform carotid sinus massage

Explanation:

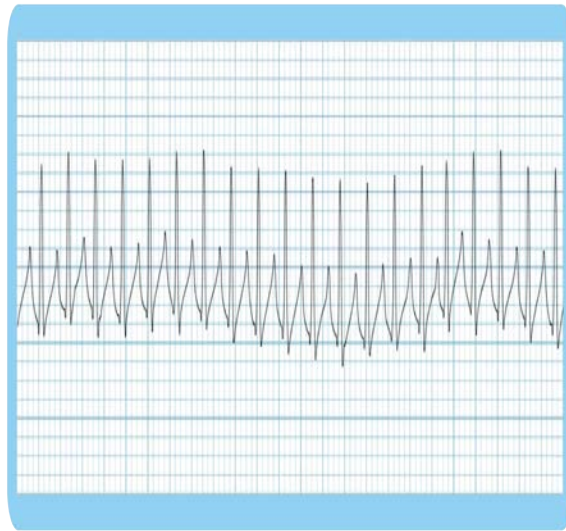
- The patient is likely experiencing stable supraventricular tachycardia (SVT). Initial management for stable SVT includes vagal maneuvers such as carotid sinus massage or the Valsalva maneuver.

Narrow complex tachycardia (QRS $\leq$ 0.12 second):

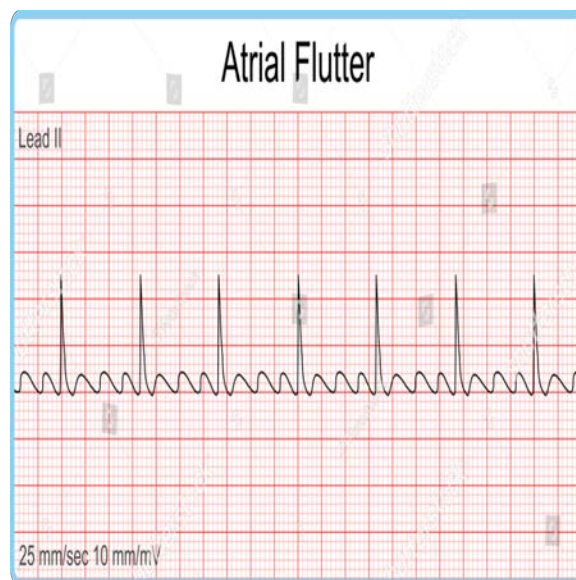
- They arise above the ventricles
- Types: Regular and irregular rhythm

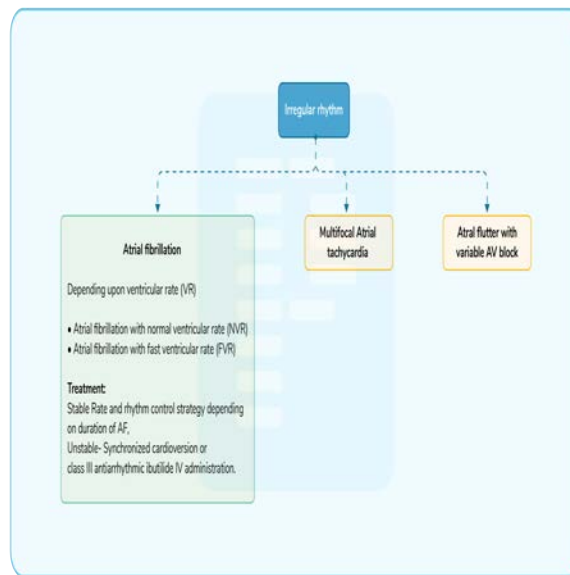


SVT:

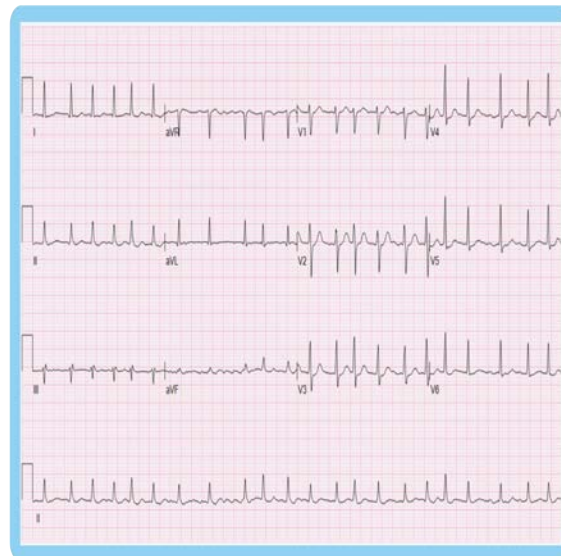


Atrial flutter:





Atrial fibrillation:



Immediate synchronized cardioversion (Option A):

- In this case, the patient is hemodynamically stable with a blood pressure of 120/80 mmHg, no chest pain, and no shortness of breath.
- Therefore, synchronized cardioversion is unnecessary as a first-line treatment in this stable scenario.

Intravenous adenosine 6 mg push (Option B): It is the next step after vagal manoeuvres fail in SVT.

Intravenous amiodarone 150 mg over 10 minutes (Option C): It is not a first-line treatment for narrow-complex tachycardia like SVT in a stable patient.

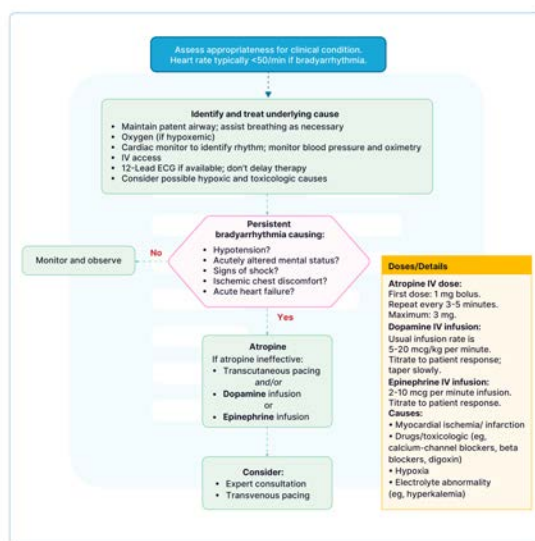
Solution for Question 12:

Correct Answer: C) Dopamine IV infusion

Explanation:

- The clinical presentation and ECG suggesting sinus bradycardia without evidence of heart block are consistent with symptomatic sinus bradycardia with hypotension.
- After atropine fails, the next step in the management is dopamine infusion.

Adult bradycardia algorithm



Epinephrine IV bolus (Option A): The recommendation is IV infusion and not as a bolus administration of the drug.

Amiodarone IV infusion (Option B) and digoxin IV bolus (Option D): They are not used in treating bradyarrhythmias.

Solution for Question 13:

Correct Answer: B) Epinephrine

Explanation:

- Epinephrine can be administered via the endotracheal route if IV or IO access is not available during cardiac arrest. The recommended endotracheal dose is generally 2 to 2.5 times the intravenous dose.

Drugs with good tracheal absorption ("NAVEL" mnemonic):

- N aloxone
- A tropine
- V asopressin
- V alium
- E pinephrine (Option B)
- L idocaine

Amiodarone (Option A) and Adenosine (Option C) : They are not suitable for the endotracheal route of administration.

Atropine (Option D): It can be administered via the endotracheal route but it is not used according to the recent ACLS guidelines in the event of a cardiac arrest.

Solution for Question 14:

Correct Answer: A) Intravenous amiodarone 150 mg over 10 minutes

Explanation:

- The clinical presentation and ECG are suggestive of a stable, monomorphic wide-complex tachycardia, consistent with ventricular tachycardia (VT).
- Amiodarone 150 mg IV over 10 minutes, followed by 1 mg/min infusion for 6 hours is the antiarrhythmic of choice for hemodynamically stable monomorphic ventricular tachycardia

(VT) with impaired cardiac function (EF<45%).

Amiodarone

Mechanism of action:

- Multichannel blocker (sodium, potassium, calcium channel, and non-competitive α/β -blocker)

Indications:

- Hemodynamically stable monomorphic VT (wide complex)
- Stable irregular narrow complex tachycardia (atrial fibrillation)
- Stable regular narrow complex tachycardia
- Unstable VT and VF
- To control rapid ventricular rate due to accessory pathway conduction in pre-excited atrial arrhythmias
- Polymorphic VT with normal QT interval
- Antiarrhythmic of choice in cardiac function impairment (EF<40%) (Option A)

Dosage:

- 150 mg given over 10 min and repeated if necessary
- Followed by a 1 mg/min infusion for six hours
- Followed by 0.5 mg/min
- The total dose over 24 h should not exceed 2.2 g

Side effects:

- Bradycardia
- Hypotension
- Phlebitis

Adenosine 12 mg IV push (Option B): It is used in stable, regular, monomorphic, wide complex tachycardia and narrow complex tachycardias (stable and unstable) and the initial dose is 6mg IV push, followed by 12mg IV push. Since the option is 12mg IV push as an initial dose, the correct answer is amiodarone.

Intravenous lidocaine 1 mg/kg (Option C): It can also be used to treat VT, especially in ischemic heart disease, but amiodarone is generally preferred due to its broader antiarrhythmic effects and better safety profile.

Intravenous magnesium sulfate 2 g (Option D): It is used to treat polymorphic VT associated with QT prolongation (torsades de pointes).

Solution for Question 15:

Correct Answer: A) 6 mg IV push, briefly depresses sinus node and AV node conduction

Explanation:

- The clinical presentation and ECG are suggestive of stable, narrow-complex regular tachycardia consistent with supraventricular tachycardia (SVT).
- Adenosine is an endogenous purine nucleoside that temporarily depresses the sinus node and AV node function, essential in treating regular, narrow-complex tachycardias like SVT. The initial dose of 6 mg IV should be administered rapidly, followed by a 20 mL saline flush.

Adenosine

Mechanism of action:

- Endogenous purine nucleoside
- Briefly depresses sinus node rate and AV node conduction
- Vasodilator

Half-life: 10-30s

Indication:

- Narrow-complex tachycardia: Stable, regular tachycardias Unstable, regular tachycardias while preparing for electrical cardioversion
- Stable, regular tachycardias
- Unstable, regular tachycardias while preparing for electrical cardioversion
- Wide-complex tachycardia: Stable, regular, monomorphic, as a therapeutic and diagnostic maneuver
- Stable, regular, monomorphic, as a therapeutic and diagnostic maneuver
- Stable, regular tachycardias
- Unstable, regular tachycardias while preparing for electrical cardioversion
- Stable, regular, monomorphic, as a therapeutic and diagnostic maneuver

Dosage:

- 6 mg IV as a rapid push followed by a 20 mL saline flush
- Repeat if required as 12 mg IV push

Side effects:

- Hypotension
 - Bronchospasm
 - Chest discomfort
-

Previous Year Questions

1. According to the American Heart Association (AHA), what is the 6th link added to the chain of survival?

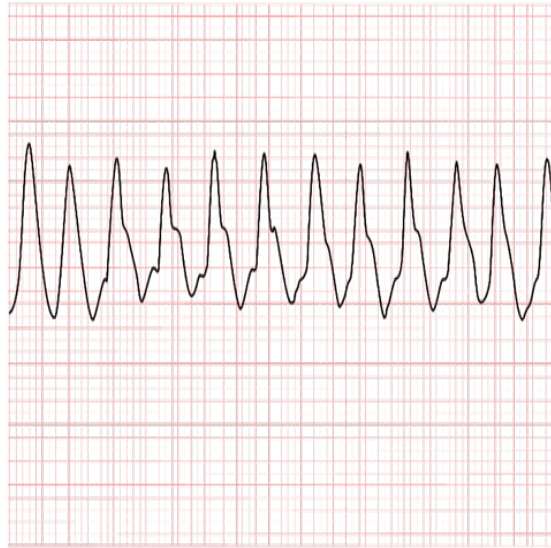
- A. Rapid defibrillation
 - B. Recovery and rehabilitation
 - C. High-quality CPR
 - D. Advanced airway management
-

2. When do you put a victim in the following position?



- A. Conscious patient with pulse & breathing present.
- B. Unconscious patient with pulse & breathing present.
- C. Unconscious patient with pulse present but no breathing.
- D. Unconscious patient without pulse but breathing present.

3. A patient presents with no pulse, and the ECG shows the following rhythm. What is the next appropriate step?



- A. Defibrillate, check pulse and then perform CPR
- B. Cardioversion
- C. Defibrillate, perform CPR, and then check the pulse
- D. Administer epinephrine

4. In a cardiac arrest victim, when there is more than one rescuer present, what is the recommended compression-to-ventilation ratio during CPR?

- A. 15:2
- B. 30:2
- C. 15:1
- D. 30:1

5. Which of the following is not an ideal step for effective CPR in adults?

- A. 100 -120 compressions per minute
 - B. Depth of compression 2-3 cm
 - C. Compression to ventilation ratio - 30:2
 - D. Allow complete recoil in between compressions
-

6. In a 60-year-old patient who has just experienced a cardiac arrest, which of the following interventions is not a part of the Advanced Cardiovascular Life Support (ACLS) protocol?

- A. Amiodarone
 - B. Defibrillation
 - C. Adrenaline
 - D. Soda Bicarbonate
-

7. Steps of intubation sequence are: a. Flexion of lower cervical spine and extension of Atlanto-occipital joint b. Inflate cuff of endotracheal tube c. Introduce a laryngoscope from the right side of mouth d. Confirm the position of tube

- A. a, b, c, d
 - B. a, b, d, c
 - C. b, a, c, d
 - D. a, c, b, d
-

8. Identify the device shown in the image below



- A. Automated External Desynchronizer
 - B. Automated external defibrillator
 - C. Automated electronic defibrillator
 - D. Automated external decompressor
-

9. You are working as an intern in ICU. Which of the following statements is incorrect about ABG report?

- A. Normal pH cannot rule out acid base disorder
 - B. Flush syringe with 0.5 ml heparin before taking the sample
 - C. If the modified Allen test is positive use an alternative site for sampling
 - D. Radial artery is preferred site for taking sample
-

10. Which of the following is not a risk factor for postoperative pulmonary complications?

- A. BMI >30
 - B. Well controlled asthma
 - C. Upper abdominal surgery
 - D. Ag>70 years
-

11. What should be done after immediate defibrillation is performed on a young male patient who was brought to the emergency department unconscious after collapsing at a railway station, with no pulse felt and a monitor showing ventricular fibrillation?

- A. Check for a pulse
 - B. Resume chest compressions
 - C. Epinephrine 1 mg IV
 - D. Amiodarone 300 mg IV bolus
-

12. While taking your morning stroll in the park, you come across an individual who is unconscious and lying on the ground. What would be the most appropriate course of action to take next from the given choices?

- A. Check carotid pulse, start CPR, and call for help; Defibrillate if no pulse
 - B. Call for help, check pulse, do CPR, defibrillate if no pulse
 - C. Observe
 - D. Ignore and continue to walk
-

13. How many chest compressions per minute are performed during CPR in adults?

- A. 30-50 per minute
 - B. 50-72 per minute
 - C. 100-120 per minute
 - D. 120-200 per minute
-

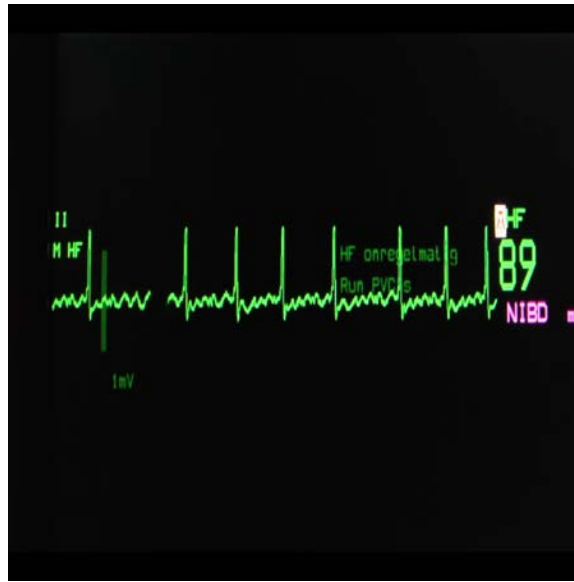
14. Which of the following can be considered effective CPR? Respiratory rate: 20-24 breaths per minute Chest compressions: 100-120/min Allow complete chest recoil 5-6cm deep compression

- A. 1,2,3 and 4
 - B. 1,2 and 4
 - C. 2,3 and 4
 - D. 1 and 2 only
-

15. If epinephrine fails to be effective, what other medication would you consider using for cardiac arrest?

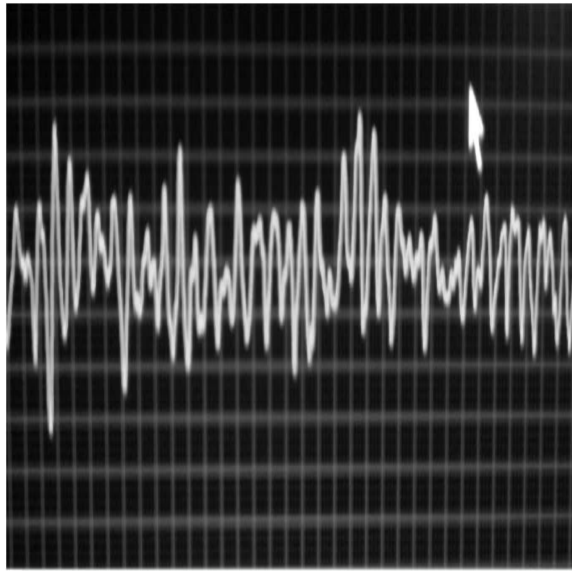
- A. Vasopressin
 - B. Atropine
 - C. Amiodarone
 - D. Adenosine
-

16. What is the diagnosis for a 23-year-old female who has been experiencing fever for the past week and displays the following rhythm on the monitor?



- A. Atrial fibrillation
- B. Atrial flutter
- C. Supra ventricular tachyarrhythmia
- D. Ventricular tachyarrhythmia

17. An unresponsive patient in ambulance on attachment of monitors showed following rhythm. What should be done immediately?



- A. Check for pulse and breathing
 - B. Chest compressions and defibrillation
 - C. Put the patient in left lateral position
 - D. Cardioversion
-

18. Which of the following statements given below is incorrect regarding CPR in adults?

- A. Depth of chest compression up to 5-6 cm
 - B. Allow adequate chest recoil
 - C. Ventilation: 22-25 breaths/ min
 - D. Chest compression rate 100-120/min
-

19. When should the indicated position be used during resuscitation?



- A. Unconsciousness with pulse absent and breathing absent
 - B. Unconsciousness with pulse present and breathing absent
 - C. Unconsciousness with pulse and breathing present
 - D. Unconsciousness with pulse absent and breathing present
-

20. Please identify the maneuver being executed in the image provided.

(or)

Identify the maneuver being performed in the given image.



- A. Head tilt, chin lift
 - B. Jaw thrust
 - C. Head extension
 - D. In-line manual stabilization
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	2
Question 5	2
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	2
Question 11	2
Question 12	2
Question 13	3
Question 14	3
Question 15	3
Question 16	1
Question 17	2
Question 18	3

Question 19	3
Question 20	1

Solution for Question 1:

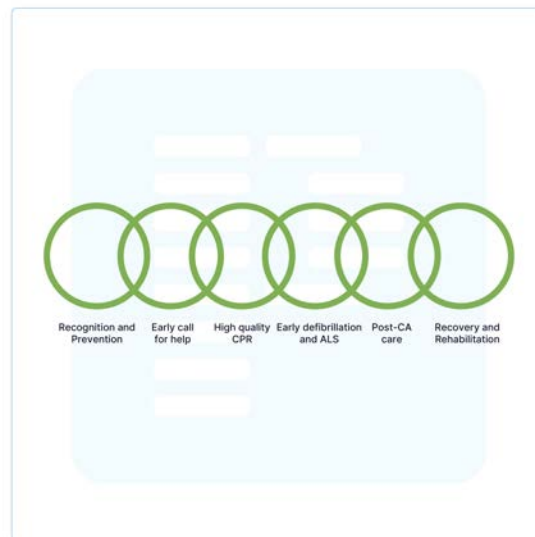
Correct Answer: B) Recovery and rehabilitation

Explanation:

The American Heart Association (AHA) has expanded the Chain of Survival concept by adding a sixth link: recovery and rehabilitation . This addition reflects the growing recognition that survival extends beyond the immediate resuscitation period .

The complete AHA Chain of Survival now includes:

- Early recognition and activation of emergency response
- Early CPR with an emphasis on chest compressions
- Rapid defibrillation
- Advanced life support
- Post-cardiac arrest care
- Recovery/rehabilitation



Rapid defibrillation (Option A) is already included as the third link. Rapid defibrillation remains a critical component for improving survival in cardiac arrest, particularly for shockable rhythms like ventricular fibrillation and pulseless ventricular tachycardia.

High-quality CPR (Option C) constitutes the second link. The AHA emphasizes the importance of proper chest compression depth, rate, allowing full chest recoil, and minimizing interruptions in CPR to optimize outcomes.

Advanced airway management (Option D) is part of the fourth link (advanced life support) in the Chain of Survival. While airway management is an important component of resuscitation, the AHA guidelines emphasize that it should not interfere with or delay other critical interventions such as chest compressions and defibrillation.

Solution for Question 2:

Correct Option B - Unconscious patient with pulse and breathing present

- The given image shows the recovery position and is used for an unconscious patient who is breathing and has a pulse. This position helps keep the airway open and allows any fluids to drain, reducing the risk of airway obstruction.

Incorrect Options:

- Options A, C and D are not the indications for recovery position.

Solution for Question 3:

Correct Option C - Defibrillate, perform CPR, and then check the pulse

- According to Advanced Cardiovascular Life Support (ACLS) guidelines, for a shockable rhythm like ventricular tachycardia, the first step is to defibrillate the patient. After delivering the shock, you immediately start CPR (Cardiopulmonary Resuscitation) to maintain blood circulation. Checking the pulse is done after completing a cycle of CPR to assess the effectiveness of the resuscitation efforts.

Incorrect Options:

Options A, B, and D are incorrect, as explained above.

Solution for Question 4:

Correct Option B - 30:2:

- The recommended compression-to-ventilation ratio for CPR with more than one rescuer is 30:2. This ratio involves providing 30 chest compressions followed by 2 rescue breaths.

Incorrect Options:

- Options A, C and D are not the recommended compression-to-ventilation ratio during CPR.
-

Solution for Question 5:

Correct Option B - Depth of compression 2 to 3 cm:

- The depth of chest compression for effective CPR in adults is 5 to 6 cm (2 to 2.4 inches). A depth of 2 to 3 cm is insufficient for effective compressions.

Incorrect Options:

Option A - 100-120 compressions per minute: This is the recommended compression rate for effective CPR.

Option C - Compression to ventilation ratio is 30:2: This signifies the correct ratio for adult CPR when performed by a single rescuer.

Option D - Allow complete recoil in between compressions: Allowing complete chest recoil is essential to facilitate adequate venous return during CPR.

Solution for Question 6:

Correct Option D- Soda Bicarbonate:

- While sodium bicarbonate used to be a part of ACLS protocols in the past, its use has become more limited in recent guidelines.
- It was commonly used to correct acidosis (increased acidity) during cardiac arrest.
- However, current evidence suggests that routine use of sodium bicarbonate in cardiac arrest may not provide significant benefits and could potentially cause harm.
- Therefore, it is no longer routinely recommended in ACLS protocols, especially in cases of cardiac arrest due to non-hypoxic causes.

Incorrect Options:

Option A- Amiodarone: Amiodarone is included in the ACLS protocol and is used for the treatment of certain types of irregular heart rhythms, particularly ventricular fibrillation and ventricular tachycardia that are resistant to other interventions. It helps stabilize the heart's electrical activity.

Option B- Defibrillation: Defibrillation is a critical component of the ACLS protocol. It involves delivering an electric shock to the heart to reset its electrical rhythm in cases of ventricular fibrillation or pulseless ventricular tachycardia, which are life-threatening arrhythmias that can lead to cardiac arrest.

Option C- Adrenaline: Adrenaline (also known as epinephrine) is an essential medication in the ACLS protocol. It is used to stimulate the heart and blood vessels, increasing heart rate and blood pressure. It is administered during cardiac arrest to improve blood flow to vital organs.

Solution for Question 7:

Correct Option D- a, c, b, d: Is the correct sequence -

Steps of intubation:

- Flexion of the lower cervical spine and extension of the atlanto-occipital joint - It is an optimal and ideal position for intubation it is also called the Morning sniffing position.

- If you are a right-handed person then you will hold the laryngoscope in your left hand and start entering from the right side of your mouth till you see the epiglottis, once the epiglottis is visible, push the epiglottis anteriorly in adults then vocal cords will be visible
- After this place the endotracheal tube, then inflate it.
- Cuff pressure should not exceed 25

Incorrect Options:

Option A- a, b, c, d: This option is incorrect.

Option B- a, b, d, c: This option is incorrect.

Option C- b, a, c, d: This option is incorrect.

Solution for Question 8:

Correct Option B - Automated external defibrillator:

- An automated external defibrillator (AED) is a portable electronic device used to diagnose and treat life-threatening cardiac arrhythmias, specifically ventricular fibrillation and ventricular tachycardia, through the application of an electric shock to the heart. It is designed to be user-friendly and can be used by individuals with minimal medical training.

Incorrect Options:

Option A - Automated External Desynchronizer: This is not a recognized device in the context of medical emergency management. It appears to be an incorrect or fictional term.

Option C - Automated Electronic Defibrillator: This is not a recognized term for a specific medical device. It may be a combination of terms, but it does not accurately represent the device shown in the image.

Option D - Automated External Decompressor: This is not a recognized term for a medical device. It does not relate to the purpose or function of the device shown in the image.

Solution for Question 9:

Correct Option C: If the modified Allen test is positive use an alternative site for sampling

A positive Allen test means that the patient has an adequate dual blood supply to the hand. A negative Allen test shows the patient may not have adequate dual blood supply, which would be a contraindication for catheterization, radial forearm flap harvest, or any procedure that may result in the occlusion of the vessel. Therefore this statement is incorrect.

Incorrect options:

Option A. Normal pH can't rule out an acid-base disorder: This statement is correct. Normal pH values in an ABG analysis do not rule out the presence of an acid-base disorder. Acid-base disorders involve disturbances in the pH, partial pressure of carbon dioxide (PaCO₂), and bicarbonate (HCO₃⁻) levels. As the patients having a combination of metabolic acidosis and metabolic alkalosis will have normal pH.

Option B. Add 0.5 ml Heparin to the syringe before taking blood: This statement is correct. Heparin is typically added to the syringe before obtaining the arterial blood sample to prevent clotting of the blood in the syringe. Flushing the syringe will prevent clotting but leaving heparin behind will cause falsely low pH and bicarbonate.

Option D. Radial artery is the preferred site: This statement is correct. The radial artery is preferred as it usually has good collateral circulation with the ulnar artery.

Solution for Question 10:

Correct Option B - Well controlled asthma:

- Well controlled asthma is not considered a significant risk factor for postoperative pulmonary complications. When asthma is well controlled, the risk of complications is generally lower compared to uncontrolled or severe asthma.

Incorrect Options:

Option A - BMI >30:

- Obesity, indicated by a BMI greater than 30, is a known risk factor for postoperative pulmonary complications. It can impair lung function and increase the risk of respiratory complications after surgery.

Option C - Upper abdominal surgery:

- Upper abdominal surgeries, such as surgeries involving the chest, abdomen, or diaphragm, can impact respiratory function due to the proximity to the respiratory muscles and potential effects on lung expansion. This increases the risk of postoperative pulmonary complications.

Option D - Age >70 years:

- Advanced age is associated with decreased respiratory reserve and decreased lung function, making older individuals more susceptible to postoperative pulmonary complications.

Solution for Question 11:

Correct Option B - Resume chest compressions:

- Ventricular fibrillation (V-fib) is a dangerous heart rhythm condition that can cause cardiac arrest. Hence, it is necessary to stop V-fib immediately to restore pulse and blood pressure. The second focus should be on reducing the development of V-fib in the future.
- The treatment for ventricular fibrillation(V-fib) includes cardiopulmonary resuscitation, i.e., (CPR). Recent consensus guidelines for CPR advocate that chest compression immediately resumes after the defibrillation process.
- Pulse checks and rhythm checks can be postponed until the completion of five compressions, minimally for two minutes.

Incorrect Options:

Option A - Check for a pulse: The pulse check can be postponed for 2 minutes until the five-chest compression is completed.

Option C - Epinephrine 1 mg IV: Epinephrine, 1 mg, is recommended every 3-5 minutes once IV or IO access is established until the completion of five compressions.

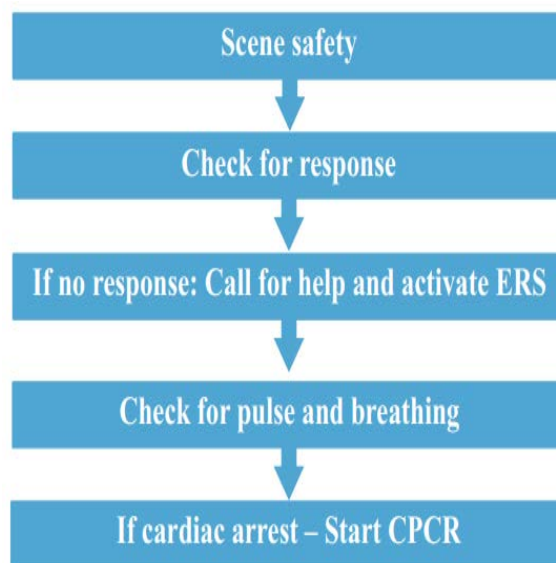
Option D - Amiodarone 300 mg IV bolus: Amiodarone can be provided for ventricular tachycardia or ventricular fibrillation (VT/VF) cardiac arrest via intraosseous (IO)/intravenous (IV) infusion as a 300mg rapid bolus. A 150 mg IO/IV additional bolus is provided if VT persists.

Solution for Question 12:

Correct Option B - Call for help, check pulse, do CPR, defibrillate if no pulse:

Encountering an unconscious person requires prompt action to assess their condition and provide necessary assistance. The correct sequence of steps to take in this situation is as follows:

- Call for help: Immediately alert emergency medical services (EMS) or call for assistance to ensure that professional medical help is on the way. This ensures that trained personnel can provide advanced life support as needed.
- Check pulse: Assess the person's carotid pulse, which is located on the side of the neck, to determine if they have a heartbeat. This helps determine if immediate CPR is required.
- Perform CPR: If the person does not have a pulse, begin cardiopulmonary resuscitation (CPR) immediately. This involves providing chest compressions to help circulate blood and maintain vital organ perfusion.
- Defibrillate if no pulse: If an automated external defibrillator (AED) is available and the person remains unresponsive with no pulse, follow the AED's instructions to deliver an electrical shock (defibrillation) if appropriate. AEDs are specifically designed for individuals who are in cardiac arrest due to certain heart rhythm abnormalities, such as ventricular fibrillation or ventricular tachycardia.



Incorrect Options:

Option A - Check carotid pulse, start CPR, and call for help; Defibrillate if no pulse: The correct order of actions is to call for help first, then assess the pulse, perform CPR if necessary, and

consider defibrillation if appropriate. The given option has the correct steps but in an incorrect sequence.

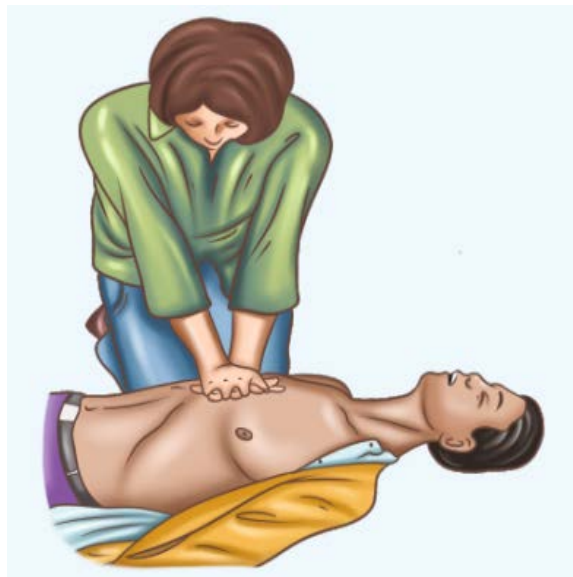
Option C - Observe: Observing an unconscious person without taking any immediate action can be dangerous. Unconsciousness may be a sign of a serious medical emergency, and timely intervention is crucial for the person's well-being.

Option D - Ignore and continue to walk: This option is highly inappropriate and potentially life-threatening. Encountering an unconscious person requires immediate attention and appropriate action to ensure their safety and potentially save their life.

Solution for Question 13:

Correct Option C - 100-120 per minute:

- Chest compressions are critical to providing circulation to the heart and brain during CPR (cardiopulmonary resuscitation). The recommended rate for chest compressions in adult CPR is 100-120 compressions per minute, which ensures adequate blood flow to vital organs.



Incorrect Options:

Option A - 30-50 per minute: This range is incorrect as it is too low for effective CPR.

Option B - 50-72 per minute: This range is also incorrect as it is below the recommended rate for effective chest compressions.

Option D - 120-200 per minute: This range is too high and not recommended for effective CPR. An excessive compression rate may compromise the quality of compressions.

Solution for Question 14:

Correct Option C - 2,3 and 4:

Effective CPR (Cardiopulmonary Resuscitation) is crucial in providing life-saving measures during cardiac arrest. It includes:

Incorrect Options:

Options A, B, and D are incorrect, as statement 1 is incorrect.

Solution for Question 15:

Correct Option C - Amiodarone:

- Amiodarone is an antiarrhythmic medication that is commonly used in the management of cardiac arrest. It is used when initial treatments, such as epinephrine, fail to restore a stable heart rhythm. Amiodarone works by stabilizing the electrical activity of the heart and is effective in treating various types of life-threatening arrhythmias.

Incorrect Options:

Option A - Vasopressin:

- It has been used in the past as an alternative to epinephrine, current guidelines recommend epinephrine as the primary vasopressor in cardiac arrest. Amiodarone is the preferred alternative to epinephrine if it proves ineffective.

Option B - Atropine:

- Atropine is a medication used to increase heart rate and treat certain types of bradycardia (slow heart rate). It is not typically effective in treating cardiac arrest, particularly in cases of ventricular fibrillation or pulseless ventricular tachycardia. Therefore, it is not the preferred alternative to epinephrine.

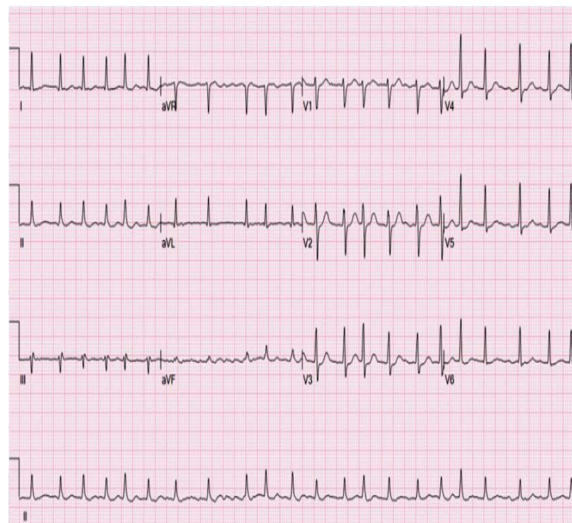
Option D - Adenosine:

- Adenosine is primarily used in the treatment of supraventricular tachycardia (SVT), which is a different type of arrhythmia compared to ventricular fibrillation or pulseless ventricular tachycardia seen in cardiac arrest. Adenosine works by briefly slowing down the heart rate and interrupting the re-entry pathway responsible for SVT. It is not indicated for use during cardiac arrest and is not an alternative to epinephrine in that context.

Solution for Question 16:

Correct Option A - Atrial fibrillation:

- Atrial fibrillation is a cardiac rhythm disorder characterized by rapid, irregular electrical impulses in the atria (upper chambers of the heart). This results in a rapid and irregular ventricular response, leading to an irregular heart rate on the monitor.



Atrial Fibrillation

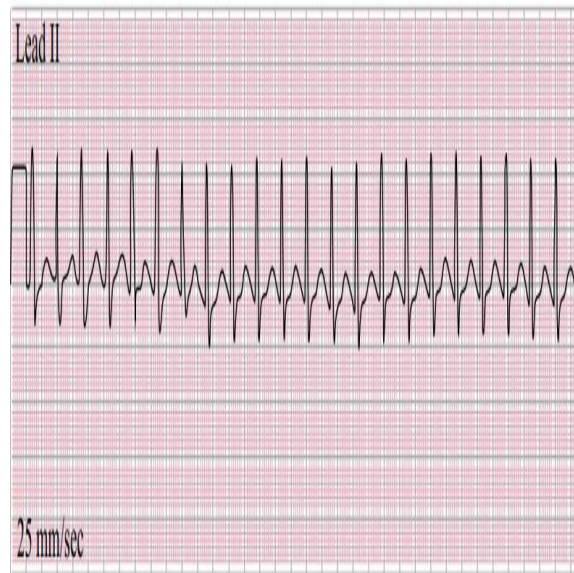
Incorrect Options:

Option B - Atrial flutter:

- Atrial flutter is another type of abnormal heart rhythm characterized by rapid, regular contractions of the atria. Unlike atrial fibrillation, atrial flutter typically appears as a regular, sawtooth pattern on the monitor, rather than the irregular pattern seen in this case.

Option C - Supraventricular tachyarrhythmia:

- Supraventricular tachyarrhythmia is a broad term that encompasses various rapid heart rhythms originating from above the ventricles. It includes conditions like atrial fibrillation, atrial flutter, and other supraventricular tachycardias. While atrial fibrillation is a type of supraventricular tachyarrhythmia, it is more specific than the general term used in this option.



Option D - Ventricular tachyarrhythmia:

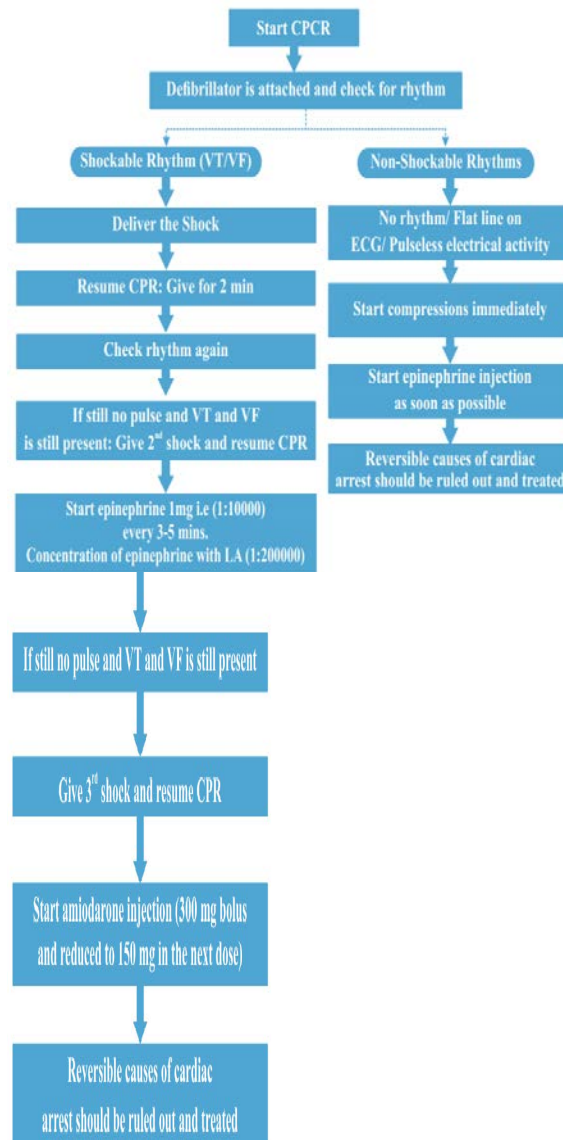
- Ventricular tachyarrhythmia refers to abnormal rapid rhythms that originate in the ventricles (lower chambers of the heart). This rhythm disorder is characterized by a wide QRS complex on the monitor, unlike the narrow QRS complex typically seen in atrial fibrillation.

Solution for Question 17:

Correct Option B - Chest compressions and defibrillation:

- Chest compressions and defibrillation are the appropriate immediate actions to take when a patient in cardiac arrest is unresponsive and shows an abnormal rhythm on the attached monitors. Chest compressions help maintain blood circulation to vital organs, while

defibrillation delivers an electric shock to the heart to restore its normal rhythm.



Incorrect Options:

Option A - Check for pulse and breathing:

- This option is not sufficient when dealing with a patient in cardiac arrest. In cardiac arrest, the heart has stopped pumping effectively, and the patient will not have a palpable pulse. Therefore, checking for a pulse is not reliable in this situation. Instead, immediate initiation of cardiopulmonary resuscitation (CPR) is required.

Option C - Put the patient in the left lateral position:

- Placing the patient in the left lateral position (recovery position) is typically done for conscious individuals who are breathing adequately. It is not appropriate for an unresponsive patient in cardiac arrest since it does not address the underlying issue of cardiac dysfunction and lack of circulation.

Option D - Cardioversion:

- Cardioversion is a procedure used to convert an abnormal heart rhythm back to normal by delivering a synchronized electric shock. However, cardioversion is not the first-line treatment for patients in cardiac arrest. Defibrillation, which delivers an unsynchronized shock, is the appropriate initial intervention in cardiac arrest cases.

Solution for Question 18:

Correct Option C - Ventilation: 22-25 breaths/ min

- Ventilation rate should be 10 - 12 breaths/min.
- Clear the airway before provide rescue breaths.
- CPR is carried out at a ratio of 30:2, i.e. give 2 breaths after 30 chest compressions if one person is performing CPR.
- If two persons are performing CPR, give two breaths after every 15 chest compressions i.e, a ratio of 15:2.

Incorrect Options:

Option A - Depth of chest compression up to 5-6 cm:

- The depth of the chest compression has to be up to 5-6 cm. The rule nowadays is to “push hard and fast in the center of the chest” in adults.

Option B - Allow adequate chest recoil:

- Chest recoil should be complete between two compressions. As the compression depth is 5 cm, the chest must recoil before the next compression.

Option D - Chest compression rate 100-120/min:

- The rescuers should deliver at least 100-120 compressions/minute.
-

Solution for Question 19:

Correct Option C - Unconsciousness with pulse and breathing present:

- Resuscitation is restoring life or consciousness to an individual who has stopped breathing or whose heart has stopped beating. It involves a series of interventions designed to support the body's vital functions, such as delivering oxygen to the tissues and removing carbon dioxide.
- Unconsciousness with pulse and breathing present: This can be used in this situation to help maintain an open airway and improve ventilation. The given position involves placing the patient on their side with the lower arm extended and the upper leg flexed at the hip and knee. This helps prevent the tongue from blocking the airway and allows secretions to drain from the mouth. It can help to maintain an open airway and improve ventilation.
- The given position is particularly useful in cases where the unconsciousness has been caused by alcohol or drug use, as these substances can increase the risk of aspiration (inhaling vomit or other liquids) and airway obstruction. By maintaining an open airway, the given position can help to prevent these complications and improve oxygenation of the tissues.

Incorrect Options:

Option A - Unconsciousness with pulse absent and breathing absent: In this situation, the position is not indicated. The given position is used for patients who are unconscious but still have a pulse and are breathing.

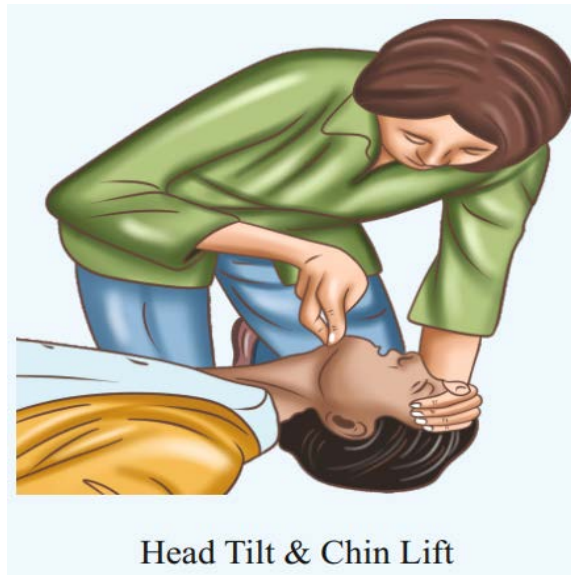
Option B - Unconsciousness with pulse present and breathing absent: The given position is not indicated in this situation either. This medical emergency requires immediate initiation of cardiopulmonary resuscitation (CPR).

Option D - Unconsciousness with pulse absent and breathing present: This is another medical emergency requiring immediate CPR initiation. The given position is not indicated in this situation.

Solution for Question 20:

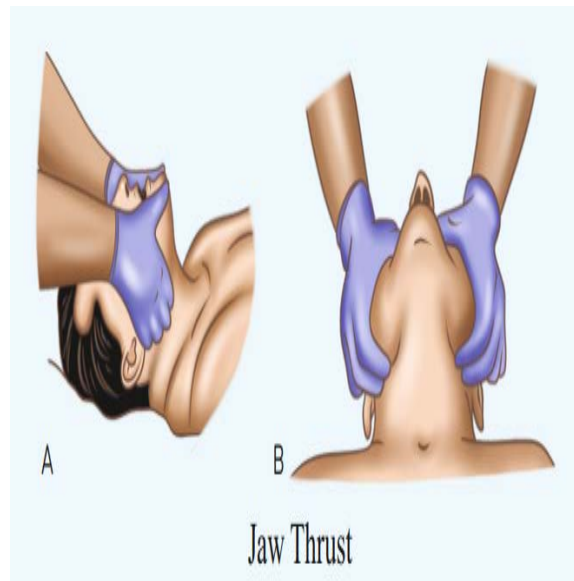
Correct Option A - Head tilt, chin lift:

- The given maneuver is Head tilt and chin lift. It is done in any patient except trauma patients. Backward pressure on the head (head is extended) and chin are lifted forward, pulling the tongue away from the posterior pharyngeal wall and epiglottis away from the laryngeal inlet. Neck muscles are stretched.
- It is done in any patient except trauma patients.
- Backward pressure on the head (head is extended) and chin are lifted forward, pulling the tongue away from the posterior pharyngeal wall and epiglottis away from the laryngeal inlet.
- Neck muscles are stretched.
- It is done in any patient except trauma patients.
- Backward pressure on the head (head is extended) and chin are lifted forward, pulling the tongue away from the posterior pharyngeal wall and epiglottis away from the laryngeal inlet.
- Neck muscles are stretched.



Incorrect Options:

Option B - Jaw thrust: Similar to the head tilt and chin lift, this technique is used to open the airway in unconscious or unresponsive individuals who may have a suspected neck injury. Instead of tilting the head, the rescuer will use both hands to lift the angle of the jaw forward, which raises the tongue away from the back of the throat.



Option C - Head extension: This technique opens the airway in individuals with suspected neck or spinal injuries. Rather than tilting or lifting the head, the rescuer gently lifts the chin and extends the neck while keeping the head neutral.

Option D - In-line manual stabilization: This technique stabilizes the head and neck to prevent further damage when there is a suspected spinal injury. The rescuer will place their hands on either side of the patient's head and hold it in a neutral position while avoiding twisting or neck rotation.

Anaesthesia Machine

1. Schimmelbusch mask is an equipment used in the past to deliver liquid volatile anesthetics. What kind of breathing system is this?

- A. Open
 - B. Closed
 - C. semi -open
 - D. Semi-closed
-

2. While performing panretinal photocoagulation for diabetic retinopathy in an elderly male patient, the patient's airway caught sudden fire. All of the following steps can be done in the immediate management of this airway fire except

- A. Continue ventilation
 - B. Flood area with 0.9% saline
 - C. Stop using laser
 - D. Remove the tube and extinguish in water
-

3. All of the following are components of Baralyme except?

- A. Ca(OH)_2
 - B. NaOH
 - C. KOH
 - D. Ba(OH)_2
-

4. A 40 year male was undergoing laparoscopic cholecystectomy, in the middle of surgery, he developed tachypnoea with increased sweating, HR - 120 bpm, spO₂ - 92%. Anesthetist observed the change and promptly responded by adding a CO₂ absorber substance as shown in the image below, following which the signs and symptoms reverted back to normal limits. Which among the following inhalational agents is completely safe to be administered with the added substance?

{{caption_text}}



- A. Desflurane
- B. Halothane
- C. Sevoflurane
- D. Trilene

5. Which of the following is the CO₂ absorption capacity of soda lime?

- A. 50L CO₂ per 100 gram
- B. 10L CO₂ per 100 gram

- C. 50L CO₂ per 100 gram
 - D. 23L CO₂ per 100 gram
-

6. How is a circle system made into a fully closed system?

- A. APL valve is opened
 - B. APL valve is closed
 - C. APL valve is closed and the fresh gas flow is set equal to the uptake by the patient
 - D. None of the above
-

7. Which of the following is a standard breathing system for pediatric patients up to 25 kg?

- A. Mapleson A
 - B. Mapleson B
 - C. Mapleson C
 - D. Mapleson F
-

8. Which of the following is not a feature of the Bain system?

- A. Fresh gas flow at patient end
 - B. Valve at machine end
 - C. Bag at the machine end
 - D. None of the above
-

9. Which of the following statements is false regarding Magill breathing system

- A. Efficient for spontaneous ventilation
 - B. Inefficient for controlled ventilation
 - C. Semi-open system
 - D. Suitable for pediatrics
-

10. Which of the following safety features of Boyle's anesthesia machine reduces the supply of other gases in response to a drop in oxygen pressure?

- A. Second stage pressure regulator
 - B. Oxygen supply failure alarm
 - C. Oxygen failure safety device
 - D. All of the above
-

11. You notice the following vaporizer being used in OT. What is the likely anesthetic agent that could be used here? {{caption_text}}



- A. Desflurane
 - B. Isoflurane
 - C. Sevoflurane
 - D. Halothane
-

12. A 60 year old female patient posted for some surgery under general anesthesia, was maintaining saturation but suddenly got desaturated. Now the clinician decided to supply oxygen via a device shown below. What is the ideal pressure to be maintained for delivering O₂ in this scenario?



- A. 2000 psi
 - B. 400 psi
 - C. 40 psi
 - D. 20 psi
-

13. Choose the correct set of true statements regarding the central supply line and Diameter index safety system (DISS) 1) DISS is used for preventing wrong attachment of cylinder to machine 2) White color hoses are used for Oxygen 3) Part of low pressure system 4) Ideal pressure of gasses is fixed at 45psi

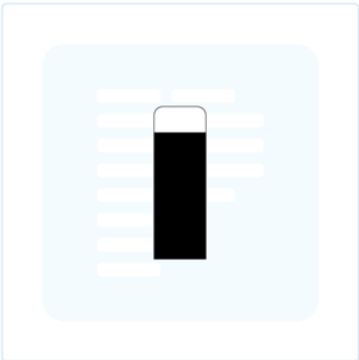
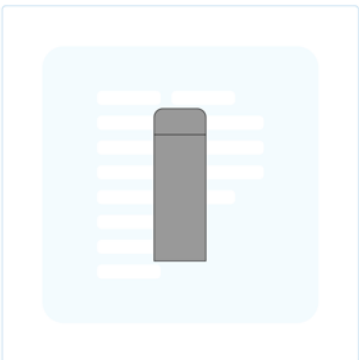
- A. 1, 2
 - B. 2, 4
 - C. 1, 3
 - D. 2, 3
-

14. Mark the correct option regarding the scale reading of circular and cylindrical bobbin in the following device {{caption_text}}



- A. Circular - from top ; cylindrical - parallel with the equator
 - B. Circular - parallel with the equator ; cylindrical - from top
 - C. Both to be read from top
 - D. Both to be read from side, parallel with the equator
-

15. Match Column A with Column B appropriately: Column A (Cylinder color)
Column B (Pin index) 1) a) 1, 5 2) b) 2, 5 3) c) 3, 5 4) d) 1, 6 e) 7

Column A (Cylinder color)	Column B (Pin index)	
1)		a) 1, 5
2)		b) 2, 5
3)		c) 3, 5

4)		d) 1, 6
	e) 7	

- A. 1-c; 2-b; 3-d; 4-e
- B. 1-c; 2-a; 3-d; 4-e
- C. 1-e; 2-b; 3-d; 4-c
- D. 1-e; 2-a; 3-d; 4-c

16. Label the following image appropriately



- A. 1 - Hanger yoke assembly; 2 - DISS
- B. 1 - DISS; 2 - Hanger yoke assembly
- C. 1 - PIN Index system; 2 - Hanger yoke assembly
- D. 1 - Hanger yoke assembly; 2 - PIN Index system

17. Match the gas to its respective cylinder body and shoulder colors. Column A (Gas) Column B (Body Color) Column C (Shoulder Color)

1. Oxygen a) Black i) White

2. N₂O b) Blue ii) Blue

3. Nitrogen c) Black iii) Black

4. Carbon dioxide d) Gray iv) Gray

5. Entonox e) Blue v) White & blue (ITS ONLY WHITE SHOULDER)

6. Air f) Black vi) Black & white

Column A (Gas)	Column B (Body Color)	Column C (Shoulder Color)
1. Oxygen	a) Black	i) White
2. N ₂ O	b) Blue	ii) Blue
3. Nitrogen	c) Black	iii) Black
4. Carbon dioxide	d) Gray	iv) Gray
5. Entonox	e) Blue	v) White & blue (ITS ONLY WHITE SHOULDER)
6. Air	f) Black	vi) Black & white

- A. 1-a-i ; 2-b-ii ; 3-c-iii; 4-d-iv ; 5-e-v ; 6-f-vi
- B. 1-a-vi ; 2-b-ii ; 3-c-iii; 4-d-iv ; 5-d-v ; 6-f-i
- C. 1-a-i ; 2-b-v ; 3-c-iii; 4-d-iv ; 5-d-ii ; 6-f-vi
- D. 1-a-i ; 2-b-ii ; 3-d-iv; 4-c-iii ; 5-d-v ; 6-f-vi

18. Mark the cylinder used in MRI rooms and its corresponding wall thickness.

- A. Steel cylinders; 3mm

B. Aluminum cylinders; 3mm

C. Steel cylinders; 6mm

D. Aluminum cylinders; 6mm

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	2
Question 4	2
Question 5	4
Question 6	3
Question 7	4
Question 8	4
Question 9	4
Question 10	3
Question 11	1
Question 12	3
Question 13	2
Question 14	2
Question 15	3
Question 16	4
Question 17	1
Question 18	4

Solution for Question 1:

Correct Answer: A) Open

Explanation:

- Open breathing systems have no reservoir breathing bag, functional rebreathing of exhaled gases, tubing, or valves.
- Examples: Use of chloroform dripped onto gauze (Eg- Schimmelbusch mask) Delivery of the fresh gas into an anaesthetist's cupped hands for inhalational induction in pediatrics
- Use of chloroform dripped onto gauze (Eg- Schimmelbusch mask)
- Delivery of the fresh gas into an anaesthetist's cupped hands for inhalational induction in pediatrics
- Use of chloroform dripped onto gauze (Eg- Schimmelbusch mask)
- Delivery of the fresh gas into an anaesthetist's cupped hands for inhalational induction in pediatrics



Solution for Question 2:

Correct answer: A) Continue ventilation

Explanation:

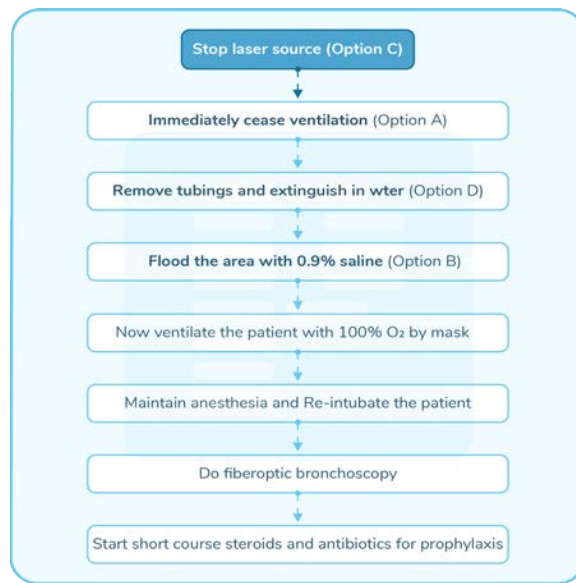
- Panretinal photocoagulation done for Diabetic retinopathy is a type of photocoagulation laser . Laser beams used in surgery may cause ignition of tissues and equipment in the presence of combustible gasses such as oxygen and nitrous oxide.
- During such airway fires, ventilation should be stopped immediately and the tubings should be removed and extinguished with water .



Prophylactic measures:

- Use of nitrous oxide in laser surgery is contraindicated.
- Preferred alternative: Heliox.
- Endotracheal tubes made up of stainless steel or aluminum are preferred during laser surgery.
- Proper training given to theater staff regarding the airway fire drill.

Management of airway fire:



Solution for Question 3:

Correct answer: B) NaOH

Explanation:

- Baralyme, a CO₂ absorber does not contain NaOH and rather contains Ba(OH)₂ and hence called Baralyme.

Baralyme composition:

- 80% Ca(OH)₂
- 11% Ba(OH)₂
- 2% KOH
- 11% H₂O

Soda Lime and Baralyme:

	Soda Lime	Baralyme
Composition	80% Ca(OH) ₂ 3% NaOH 2% KOH 16% H ₂ O	80% Ca(OH) ₂ 11% Ba(OH) ₂ 2% KOH 11% H ₂ O
Absorptive capacity	20-26 L of CO ₂ /100g/min	9-18 L of CO ₂ /100g/min

Indicator	Ethyl violet (changes color upon exhaustion)	
Exhaustion	Indicated by change of color Results in rebreathing of CO ₂ (tachypnoea, tachycardia, hypoxia)	
Interactions	Soda Lime + Trilene ■ Phosgene (cause ARDS) + Dichloroacetylene (neurotoxic) Soda Lime + Sevoflurane ■ COMPOUND - A (nephrotoxic) Minor interaction: with Desflurane, Enflurane and Isoflurane, produces carbon monoxide (more with Baralyme)	Carbon monoxide formation (major interaction): leading to carboxyhemoglobinemia. CO production is max with Baralyme >> Soda Lime and with Desflurane > Enflurane > Isoflurane > Halothane = Sevoflurane.

Solution for Question 4:

Correct answer: B) Halothane

Explanation:

Patient's symptoms and signs are attributed to rebreathing CO₂ which causes tachypnoea, hypoxia and tachycardia.



- The CO₂ absorber substance most commonly used (and shown in the image) is Soda lime , which can interact with the inhalational anesthetic agents.
- Among the options, the single best answer would be Halothane as Soda lime least interacts with it.
- Soda lime interactions: Trilene (trichlorethylene): dichloroacetaldehyde (neurotoxic) (Option D) Sevoflurane: Compound - A (nephrotoxic) (Option C) Carbon monoxide production: Desflurane (Option A) > Enflurane > Isoflurane > Halothane = Sevoflurane ; (although significantly more carbon monoxide production is seen with Baralyme)
- Trilene (trichlorethylene): dichloroacetaldehyde (neurotoxic) (Option D)
- Sevoflurane: Compound - A (nephrotoxic) (Option C)
- Carbon monoxide production: Desflurane (Option A) > Enflurane > Isoflurane > Halothane = Sevoflurane ; (although significantly more carbon monoxide production is seen with Baralyme)
- Trilene (trichlorethylene): dichloroacetaldehyde (neurotoxic) (Option D)
- Sevoflurane: Compound - A (nephrotoxic) (Option C)
- Carbon monoxide production: Desflurane (Option A) > Enflurane > Isoflurane > Halothane = Sevoflurane ; (although significantly more carbon monoxide production is seen with Baralyme)

Soda Lime:

Composition	80% Ca(OH) ₂ 3% NaOH 2% KOH 16% H ₂ O
Absorptive capacity	20-26 L of CO ₂ /100g/min
Indicator	Ethyl violet (changes color upon exhaustion)
Exhaustion	Indicated by change of color Results in rebreathing of CO ₂ (tachypnoea, tachycardia, hypoxia)
Interactions	Soda Lime + Trilene ■ Phosgene (cause ARDS) + Dichloroacetylene (neurotoxic) Soda Lime + Sevoflurane ■ COMPOUND - A (nephrotoxic) Minor interaction: with Desflurane, Enflurane and Isoflurane, produces carbon monoxide (more with Baralyme)

Solution for Question 5:

Correct Answer: D) 23L CO₂ per 100 gram

Explanation:

Soda lime absorbs up to 23 L of CO₂ per 100 g of absorbent

Soda lime:

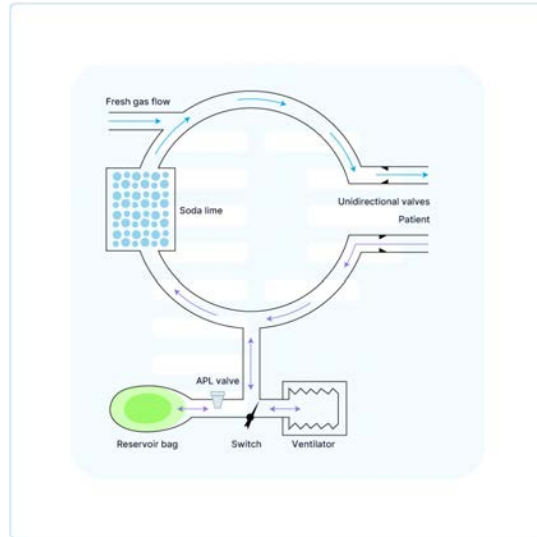
Function	Removes carbon dioxide through an exothermic reaction in a closed breathing circuit
Composition	Sodium hydroxide Calcium hydroxide Potassium hydroxide Water
Granule size	4 to 8 mesh size
Absorption capacity	Soda lime absorbs up to 23 L of CO ₂ per 100 g of absorbent.
Chemical Process	Sodium and potassium hydroxide neutralize and trap carbon dioxide. Gaseous carbon dioxide is converted into solid calcium carbonate (chalk).
By-products	Heat and water are produced during the reaction. Helps maintain the temperature and humidity of the anesthesia circuit.
Indicator	A dye (e.g., ethyl violet) changes color from white to purple with increasing hydrogen ion concentration and absorbent exhaustion.
Reaction with anesthetics	Dry soda lime can degrade volatile anesthetics (e.g., desflurane, sevoflurane) and produce carbon monoxide. Carbon monoxide formation is greatest with desflurane.

Solution for Question 6:

Correct Answer: C) APL valve is closed and the fresh gas flow is set equal to the uptake by the patient

Explanation:

Circle system - closed breathing system



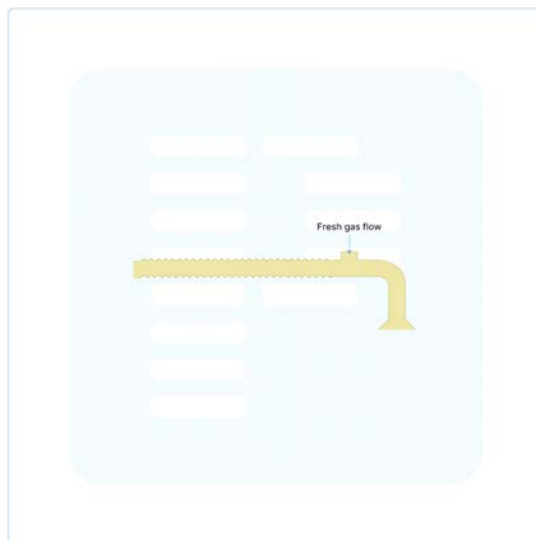
Components	Fresh gas inlet Reservoir bag Two one-way valves (inspiratory and expiratory limbs) Y-piece connector to the patient APL valve Soda lime canister for CO ₂ absorption Kink-resistant corrugated tubing connecting components and patient
Fully closed system	To make the circle system a fully closed system - the APL valve is closed and the fresh gas flow is set equal to the uptake by the patient so that the overall volume of gas in the system is constant. It is difficult to estimate the patient's oxygen uptake accurately and therefore fully closed systems are rarely used in practice.
Advantages	Conserves anesthetic gases, heat, and moisture. Allows low-flow anesthesia with monitoring. Low dead space; Y-piece tubing creates mechanical dead space comparable to non-rebreathing circuits. Soda lime canister is distant from the patient's airway, reducing the inhalation risk of soda lime dust. Reduces atmospheric pollution by recycling anesthetic gases.

Solution for Question 7:

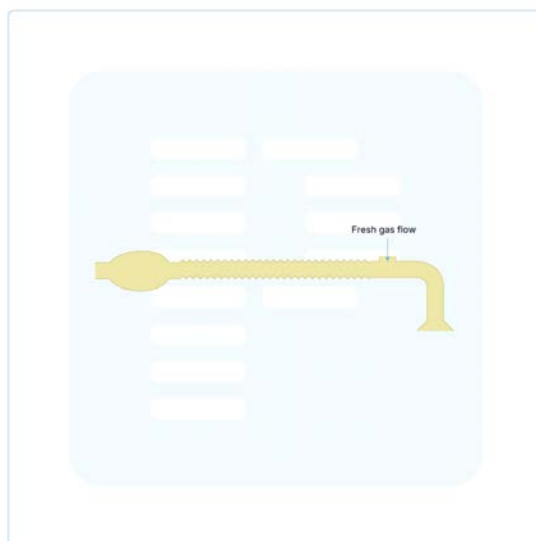
Correct Answer: D) Mapleson F

Explanation:

Ayre's T-piece=Mapleson E - Valveless, bagless breathing system consisting of a T-shaped rigid tube with connections for Fresh Gas Flow (FGF), patient, and reservoir tubing. Used for premature babies (cannot handle dead space).



Jackson-Rees modification (Mapleson F):



Structure	Modification of Ayre's T-piece with an open-ended bag attached to the reservoir tubing.
Advantages	Minimal dead space (as for Mapleson E). Allows positive pressure ventilation and PEEP. More suitable for inhalational induction than a circle system. Standard breathing system for pediatric patients (up to 25 kg).

Disadvantages	Scavenging exhaled gases is difficult. Requires high FGFs, making the system inefficient. Technique of partially occluding the bag tail while squeezing it requires moderate skill.

Solution for Question 8:

Correct Answer: D) None of the above

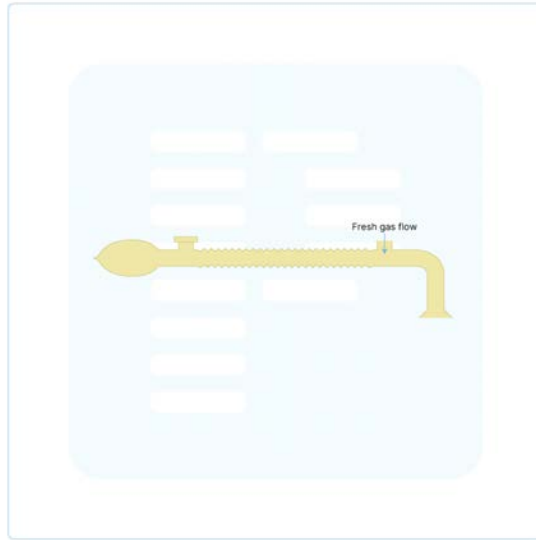
Explanation:

- All are features of the Bain system

Mapleson D system:

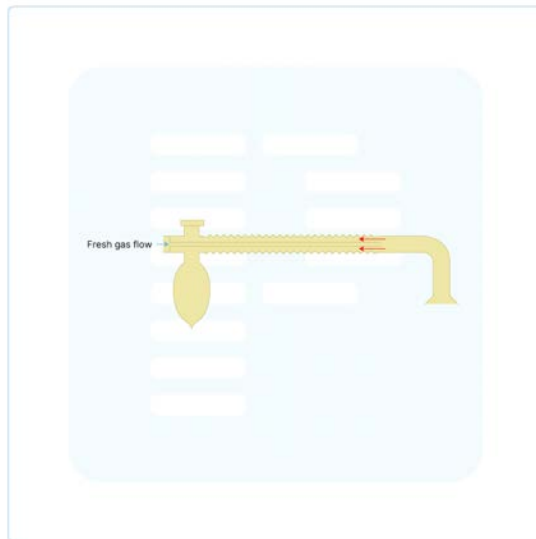
- Fresh Gas Flow (FGF) enters at the patient end
- Adjustable Pressure-Limiting (APL) valve and bag are located at the machine end

Daniel Aston Equipment in Anaesthesia and Critical Care, page



Bain system (Coaxial Mapleson D):

- Modification of Mapleson D system
- Fresh gas flows down an inner tube and exhaled gas passes down an outer tube



Advantages	Compact system, components located at the machine end, facilitating scavenging Low dead space due to the APL valve at the machine end Compatible with Penlon Nuffield 200 ventilator

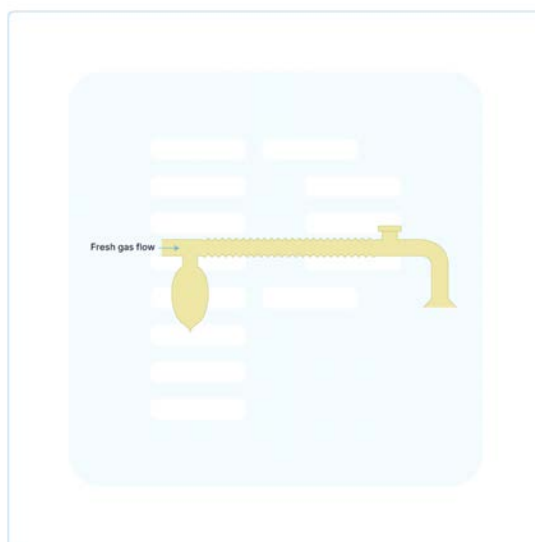
Disadvantages	Inefficient for spontaneous ventilation If the inner tube disconnects or breaks, the entire system becomes dead space.

Solution for Question 9:

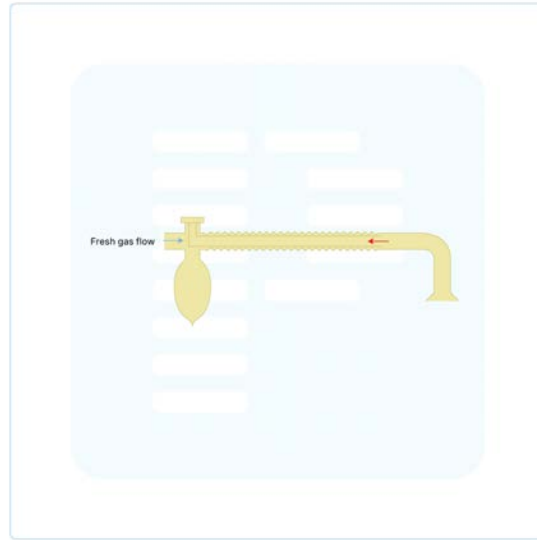
Correct Answer: D) Suitable for pediatrics

Explanation:

- Magill breathing system is not suitable for pediatrics
- Mapleson A system is also called the Magill breathing system
- Mapleson systems are semi-open



Advantages	Efficient for spontaneous ventilation
Disadvantages	Inefficient for controlled ventilation APL valve at patient end adds bulk and drags on circuit connections Not suitable for pediatrics (Option D is false)



- Note - Lack system or coaxial Mapleson A is a modification of the Magill breathing system with an outer tube for inspiration and an inner tube for expiration

Ventilation rates in Mapleson circuits:

Circuit type	Spontaneous	Controlled
Type A	= mV	3 x mV
Type B	2 x mV	2.25 x mV
Type C	2 x mV	2.25 x mV
Type D	2.5 x mV	1.6 x mV
Type E	2.5 x mV	3 x mV
Type F	2.5 x mV	1.5-2 x mV

Solution for Question 10:

Correct Answer: C) Oxygen failure safety device

Explanation:

The oxygen failure safety device in Boyle's machine reduces or cuts off other gases when oxygen pressure drops.

Boyle's machine

- Pressurized gases are supplied by cylinders/pipelines and controlled by the machine. It passes through a vaporizer and is then delivered to the patient via the breathing circuit.



- Key components: High-pressure system: Cylinders, yoke assemblies, pressure indicators, regulators Pin index safety system: Prevents incorrect cylinder attachment NIST connection: Unique hose connection for each gas service Low-pressure system: Flow meters, hypoxia prevention, vaporizers, unidirectional valves, pressure relief
- High-pressure system: Cylinders, yoke assemblies, pressure indicators, regulators
- Pin index safety system: Prevents incorrect cylinder attachment
- NIST connection: Unique hose connection for each gas service
- Low-pressure system: Flow meters, hypoxia prevention, vaporizers, unidirectional valves, pressure relief
- High-pressure system: Cylinders, yoke assemblies, pressure indicators, regulators
- Pin index safety system: Prevents incorrect cylinder attachment
- NIST connection: Unique hose connection for each gas service
- Low-pressure system: Flow meters, hypoxia prevention, vaporizers, unidirectional valves, pressure relief
- Safety Features Oxygen failure safety device: Reduces or cuts off other gases when oxygen pressure drops. (Option C) Oxygen supply failure alarm: Detects oxygen supply failure based on pressure. (Option B ruled out)

- Oxygen failure safety device: Reduces or cuts off other gases when oxygen pressure drops. (Option C)
- Oxygen supply failure alarm: Detects oxygen supply failure based on pressure. (Option B ruled out)
- Oxygen failure safety device: Reduces or cuts off other gases when oxygen pressure drops. (Option C)
- Oxygen supply failure alarm: Detects oxygen supply failure based on pressure. (Option B ruled out)

Second stage pressure regulator: Maintains constant flow despite pipeline pressure changes. (Option A ruled out)

Solution for Question 11:

Correct answer: A) Desflurane

Explanation:

The image in the question shows a Tec-6 Vaporizer with blue color coding, which is used for safe administration of Desflurane .



Vaporizers:

Description	The most commonly used type is variable bypass vaporizers (except for Desflurane). Made up of copper.	
Principle	Based on the vapor pressure of the inhaled anesthetic agent	
Color coding	Desflurane	Blue
Isoflurane	Purple (Option B)	
Sevoflurane	Yellow (Option C)	
Halothane	Red (Option D)	
Modification	Tec-6 vaporizer is used for Desflurane (Option A) (Reason: low boiling point of Desflurane)	



Solution for Question 12:

Correct answer: C) 40 psi

Explanation:

Emergency oxygen flush, pipeline inlets are part of the intermediate pressure system of gas delivery and the ideal pressure maintained for the same is 40-50 psi .

Emergency Oxygen flush:

- Used in cases of sudden desaturation.
- Can be used bedside near the breathing circuit.
- Delivers emergency O₂ at a rate of 35-70 L/min.

Pressure regulators and pressure reducing valves:

Component	Function	Pressure Settings
Pressure regulators:		
Primary Regulator	Reduces high cylinder pressure to a consistent level	400 kPa
Secondary Regulator	Stabilizes fluctuations in pipeline pressure	-
Pressure reducing/relief valve:		
Pressure Relief Valve	Provides a backup in case of regulator failure	40-50 psi (Option C)
Non-Return Pressure Relief	Protects flowmeters and vaporizers from downstream pressure, prevents back pressure effects	35 kPa

Solution for Question 13:

Correct answer: B) 2, 4

Explanation:

Diameter index safety system (DISS) is used for prevention of wrong attachment of pipeline to machine , and the gas supply devices are a part of intermediate pressure system (not low).

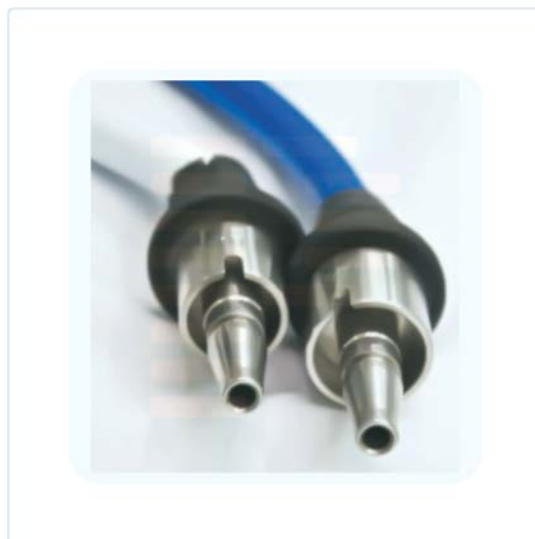
Central supply line and DISS:

- Medical gas supply includes pipelines linking VIEs, cylinder banks and air compressors to wall outlets in wards and theater suites.

- Part of the intermediate pressure delivery system. (Statement-3)
- Indexed connectors (DISS) prevent cross-connection.
- Schrader probes are used for connection.
- DISS (Diameter Index safety system): prevents wrong attachment of pipeline to machine . (Statement-1)
- Hoses are made up of copper and are color coded:

O2	White (Statement-2)
N2O	Blue
Air	Black
Suction	Yellow

- Ideal pressures: Gasses: 45-55 psi (Statement-4) Air: 60 psi
- Gasses: 45-55 psi (Statement-4)
- Air: 60 psi
- Gasses: 45-55 psi (Statement-4)
- Air: 60 psi

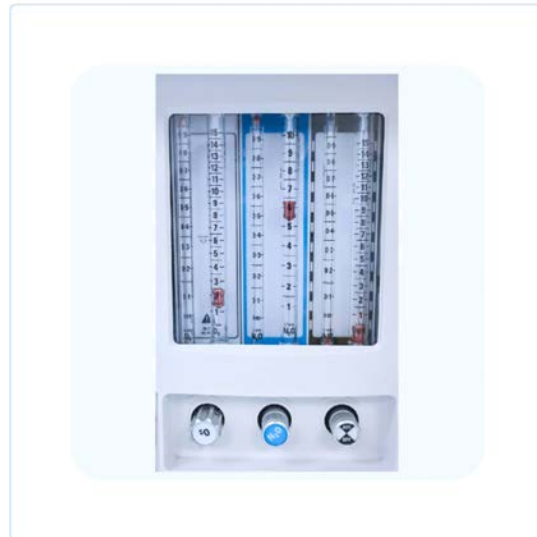


Solution for Question 14:

Correct answer: B) Circular - parallel with the equator ; cylindrical - from top

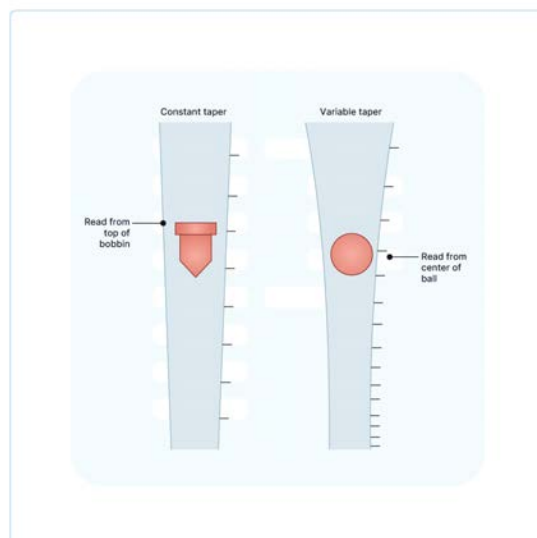
Explanation:

The device shown is a flowmeter/rotameter anesthesia equipment, used to control the flow of gasses.



Two shapes of Bobbin:

- Cylindrical: to be read from top
- Spherical: to be read from side, parallel with the equator



Flowmeter / Rotameter:

Overview	Part of low pressure system (14-28psi)
Principle	Constant pressure, variable orifice principle is used. Gas flow is controlled via a needle valve, and flow is measured by the position of a bobbin inside the cone.
Parts	Thorpe's tube: made up of Glass ; Conical in shape Bobbin/float: made up of aluminum
Bobbin	Made up of aluminum The bobbin inside the conical flow meter moves upwards with gas flow, creating a variable orifice as the gap between the bobbin and the flowmeter wall increases. The bobbin's position reaches equilibrium when the upward force of the gas equals the downward force of gravity. Bobbin types: Cylindrical bobbin: to be read from top (upper part) Spherical bobbin: to be read from side, parallel with the equator (middle part) (Option B)
Disadvantages	Bobbin can stick due to static or dirt, leading to inaccurate readings
O2 position	O2 (oxygen) should be kept most downstream of all gasses

Solution for Question 15:

Correct answer: C) 1-e; 2-b; 3-d; 4-c

Explanation:

Column A (Cylinder color)	Column B (Pin index)
1)	e) 7
2)	b) 2, 5
3)	d) 1, 6
4)	c) 3, 5

Gasses and their PIN Indices:

Gas	Cylinder color	PIN Index system
-----	----------------	------------------

Air	Black body with black & white shoulder	1, 5
Oxygen	Black body with white shoulder	2, 5
N2O	Blue	3, 5
Carbon dioxide	Grey	1, 6 (> 7.5%); 2, 6 (< 7.5%)
Entonox	Blue body with white shoulders	7
Cyclopropane	Orange	3, 6

Entonox:



Composition:	50% N2O + 50% O2
PIN Index:	7 (Option D)
Uses:	Only provides analgesia Used for short-term pain relief such as labor analgesia/dental procedures
Disadvantage:	No anesthesia effect

Solution for Question 16:

Correct answer: D) 1 - Hanger yoke assembly; 2 - PIN Index system

Explanation:

Shown image is that of a gas delivery device that connects the anesthesia machine with the gas cylinder via an interlink PIN Index safety system .



Hanger-Yoke assembly:

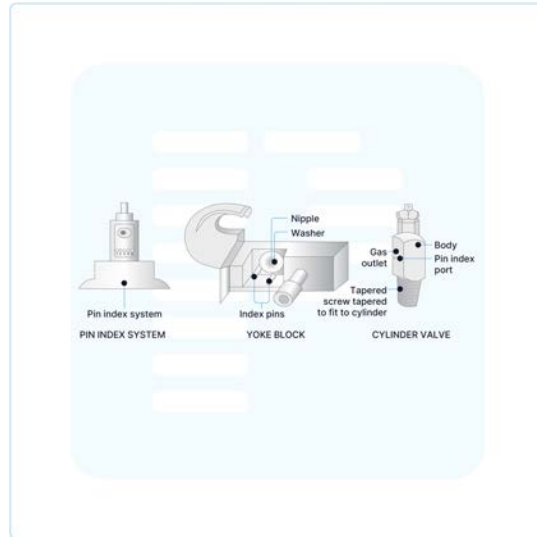
- Steel valve assembly that helps in attachment of cylinders to the anesthesia machine.
- A type of high-pressure gas delivery system
- Ideal pressures: O₂ - 2000 psi; N₂O - 760 psi

Gas Cylinders:

- Materials: molybdenum steels; Aluminium- MRI compatible
- Sizes: A (smallest) to H (largest); most commonly used - E (emergency gas supply)

PIN Index Safety system:

- Prevents wrong attachment of the cylinder to the machine.
- Pins are present on the machine.
- Ports are present on the cylinder.



DISS:

- Diameter index safety system
- Prevents wrong attachment of pipeline to the machine

Gasses and their PIN Indices:

Gas	Cylinder color	PIN Index system
Air	Black body with black & white shoulder	1,5
Oxygen	Black body with white shoulder	2,5
N2O	Blue	3,5
Carbon dioxide	Grey	1,6 (> 7.5%) ; 2,6 (< 7.5%)
Entonox	Blue body with white shoulders	7
Cyclopropane	Orange	3,6

Solution for Question 17:

Correct Answer: A) 1-a-i ; 2-b-ii ; 3-c-iii; 4-d-iv ; 5-e-v ; 6-f-vi

Explanation:

Gas	Cylinder color	
Body	Shoulder	
Air	Black	Black & white
Oxygen	Black	White
N ₂ O	Blue	Blue
Nitrogen	Black	Black
Carbon dioxide	Grey	Grey
Entonox	Blue	White & blue (ITS ONLY WHITE SHOULDER)
Helium	Brown	Brown

Cylinder contents	Body colour	Shoulder colour
Oxygen	Black	White
Nitrous oxide	Blue	Blue
Air	Black	Black & white
Entonox	Blue	White & blue
Carbon dioxide	Grey	Grey
Helium	Brown	Brown
Heliox	White	White & brown

Solution for Question 18:

Correct Answer: D) Aluminum cylinders; 6mm

Explanation:

Aluminum cylinders are MRI compatible. It is made with a cylinder wall thickness of 6 mm.

Properties of cylinder

- Constructed from lightweight materials like chrome molybdenum steel, aluminum, or composites (e.g., aluminum wrapped in carbon fiber).
 - MRI compatibility : Special aluminum cylinders are safe for use in MRI rooms. (Option D) (Option A & C ruled out)
 - Wall thickness : Typical thickness of 3 mm (Steel cylinders), while aluminum cylinders are 6 mm thick. (Option D) (Option B ruled out)
 - Composite cylinders : Made from steel or aluminum liners encased in high-density polyethylene or wrapped in carbon fiber, Kevlar, Twaron, or glass fiber. Advantages : Ultra-lightweight and durable Can hold 30% more gas than aluminum cylinders 70% lighter than steel cylinders Can be pressurized up to 4000 kPa
 - Advantages : Ultra-lightweight and durable Can hold 30% more gas than aluminum cylinders 70% lighter than steel cylinders Can be pressurized up to 4000 kPa
 - Ultra-lightweight and durable
 - Can hold 30% more gas than aluminum cylinders
 - 70% lighter than steel cylinders
 - Can be pressurized up to 4000 kPa
 - Advantages : Ultra-lightweight and durable Can hold 30% more gas than aluminum cylinders 70% lighter than steel cylinders Can be pressurized up to 4000 kPa
 - Ultra-lightweight and durable
 - Can hold 30% more gas than aluminum cylinders
 - 70% lighter than steel cylinders
 - Can be pressurized up to 4000 kPa
 - Ultra-lightweight and durable
 - Can hold 30% more gas than aluminum cylinders
 - 70% lighter than steel cylinders
 - Can be pressurized up to 4000 kPa
-

Previous Year Questions

1. A 42-year-old male who had tested positive for COVID-19 and was isolating at home presents to the emergency room with difficulty breathing. The patient is conscious and responsive but has an increased breathing rate. An oxygen saturation (SpO₂) measurement reveals a reading of 86%. The patient is given oxygen through a face mask equipped with a reservoir bag at a flow rate of 10 L/min. Following an hour-long trial, the SpO₂ level improves to 88%. What is the subsequent course of action in managing this patient?

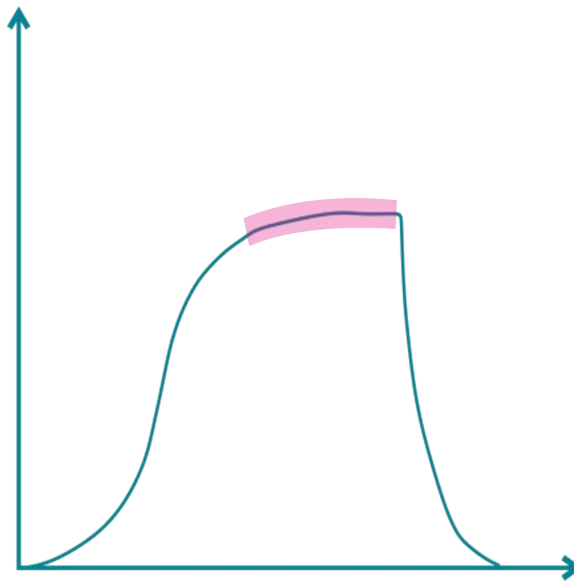
- A. Increase flow rate to 15 L/min
- B. High – flow nasal cannula oxygen therapy
- C. Endotracheal intubation
- D. No specific intervention necessary

2. In which of the following is the given position used?



- A. Neurosurgery
 - B. Intubation
 - C. NG tube insertion
 - D. Bougie insertion
-

3. In the given capnograph, what does the marked area denote?



- A. Expiratory upstroke
 - B. Alveolar Plateau
 - C. Inspiratory downstroke
 - D. Inspiratory baseline
-

4. Which of these is a second-generation Laryngeal Mask Airway (LMA)?

- A. LMA Proseal
- B. Ambu

- C. Intubating LMA
 - D. Classic LMA
-

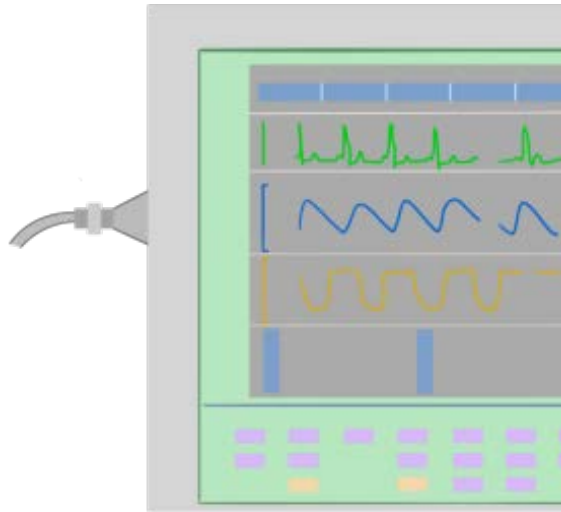
5. Where can the devices in Murphy's eye be observed?

- A. Nasopharyngeal airway
 - B. Laryngoscope
 - C. Bronchoscope
 - D. Endotracheal tube
-

6. The color of the nitrous oxide cylinder is

- A. Completely Blue
 - B. Blue body with shoulder
 - C. Black body & white shoulder
 - D. Grey body & black white shoulder
-

7. What does the yellow-colored tracing on this device in the operating room signify?



- A. Capnography
- B. Airway pressure
- C. O₂ pressure in inhaled air
- D. O₂ pressure in exhaled air

8. A 60 year old male patient was undergoing laparoscopic cholecystectomy. During the surgery, a sudden drop in EtCO₂ to 8 mmHg was noted. His SpO₂ became 90%, his blood pressure dropped to 80/50 mmHg, and peak airway pressure was 18 cm of H₂O. What is the diagnosis? CO₂ embolism Severe bronchospasm Pneumothorax Single bronchus intubation EtCO₂- End-tidal carbon dioxide

- A. 4 only
- B. Both 3 and 4
- C. 1 only
- D. 2 only

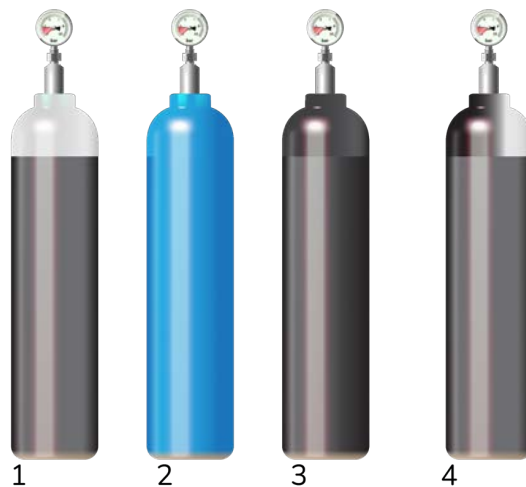
9. Which device in conventional oxygen therapy will provide the highest FiO₂?

- A. Venturi mask
 - B. Hudson mask
 - C. Non rebreathing mask
 - D. High Flow Nasal Cannula
-

10. An 18-year-old boy is rushed to the ER after having choked on a bone while eating. He is responsive but unable to speak. What is to be done next?

- A. Bag and mask ventilation
 - B. Endoscopic removal
 - C. Deliver 5 back slaps and 5 chest compressions
 - D. Perform Heimlich maneuver
-

11. Which type of cylinder should be utilized to administer oxygen therapy to a COVID patient?



- A. 1
 - B. 2
 - C. 3
 - D. 4
-

12. What is the maximum rate of oxygen flow that can be maintained using a traditional oxygen mask?

- A. 5 L/min
 - B. 15 L/min
 - C. 10 L/min
 - D. 20 L/min
-

13. Which of the following airway devices helps maintain Fio₂ of 24 - 60%, irrespective of the patient's breathing effort?

- A. Nasal cannula
 - B. Simple face mask
 - C. Venturi mask
 - D. Non-rebreathing mask
-

14. A child was being delivered oxygen by nasal cannula at 3L/min. What is the FiO₂ being delivered?

- A. 24%
- B. 30%

C. 44%

D. 40%

15. A known case of asthmatic bronchitis was reported to the hospital with breathlessness, palpitations, and oxygen saturation of 88%. The hospital does not have a ventilator. All the following can be given to the patient, except?

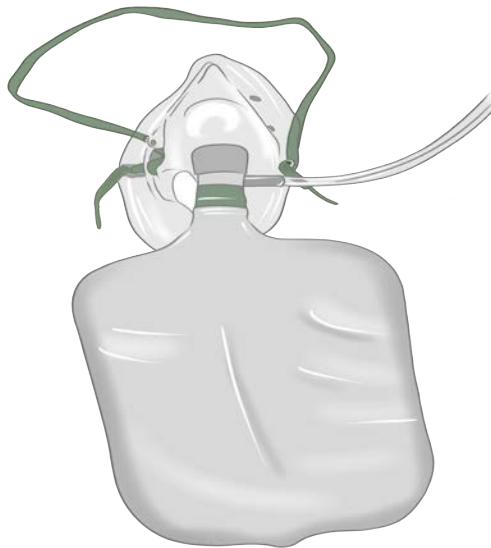
A. Hydrocortisone

B. Theophylline

C. Oxygen with oxygen mask, and reservoir

D. Leukotriene antagonist

16. Identify the mask in the image used for patients with COVID-19 infection.



A. Venturi mask

B. Hudson mask

- C. Nebuliser
 - D. Non-rebreathing mask
-

17. Which of the following is not a clear benefit of using high-pressure oxygen via nasal cannula?

- A. Decreases the need of intubation
 - B. Reduces nasopharyngeal dead space
 - C. Heating and humidification
 - D. PEEP
-

18. These instruments are utilized for:



- A. Central line insertion
- B. Arterial line insertion
- C. Airway management

D. Umbilical catheter insertion

19. Identify the equipment shown in the image below



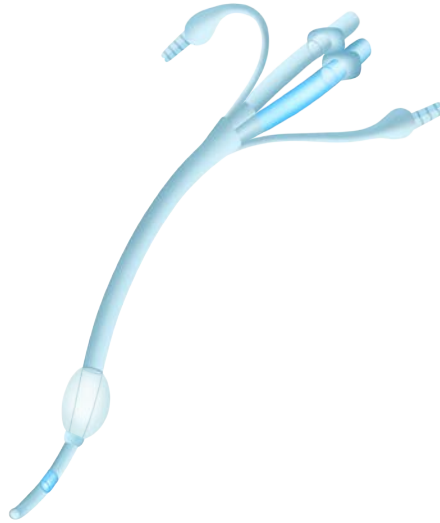
- A. Mechanical ventilator
 - B. Infusion pump
 - C. CPAP machine
 - D. Nasal cannula
-

20. Identify the given instrument?



- A. Guedel airway
- B. Laryngeal mask airway
- C. Endotracheal tube
- D. Nasogastric tube

21. What is this device called?



- A. Armored endotracheal tube
- B. Double-lumen endotracheal tube
- C. Ring-adair-elwin type
- D. Endotracheal tubes equipped with bronchial blockers

22. The mask given below works on which of the following principles?



- A. Bohr effect
 - B. Boyle's principle
 - C. Bernoulli principle
 - D. Mapleson's circuit
-

23. Which is the best way to recognize the cylinder?

- A. Color coding
 - B. Label
 - C. Pin index system
 - D. All of the above
-

24. What is the cylinder associated with a black body and white shoulder?

- A. Oxygen
- B. Nitric oxide

- C. Helium
 - D. Entonox
-

25. Which of the following conditions is a contraindication for the use of bi-level positive airway pressure?

- A. Obstructive sleep apnea
 - B. Unconscious/ sedation
 - C. High oxygen levels
 - D. Low partial pressure of oxygen
-

26. In a patient with severe maxillofacial trauma following a road traffic accident, which airway management option is most appropriate initially?

- A. Tracheostomy
 - B. Orotracheal intubation
 - C. Nasotracheal intubation
 - D. Oxygen therapy
-

27. What is the specific action being demonstrated in the provided image?



- A. Head tilt, chin lift
 - B. Jaw thrust
 - C. Head extension
 - D. In line manual stabilisation
-

28. Which statement in rapid sequence intubation is incorrect?

- A. Preoxygenation with 100% oxygen for 3 min is done
 - B. High-volume, low-pressure cuff is used
 - C. It is the first choice of treatment in cardiac arrest patients
 - D. The induction agent and muscle relaxant are given simultaneously
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	1
Question 5	4
Question 6	1
Question 7	1
Question 8	3
Question 9	4
Question 10	4
Question 11	1
Question 12	3
Question 13	3
Question 14	2
Question 15	3
Question 16	4
Question 17	1
Question 18	3

Question 19	4
Question 20	2
Question 21	2
Question 22	3
Question 23	2
Question 24	1
Question 25	2
Question 26	1
Question 27	1
Question 28	3

Solution for Question 1:

Correct Option B - High-flow nasal cannula oxygen therapy:

- In this scenario, the patient has developed breathing difficulty and has a low oxygen saturation (SpO₂) of 86%. Oxygen therapy is initiated using a face mask with a reservoir bag at a flow rate of 10 L/min. However, after one hour, the SpO₂ only improved to 88%. This suggests the patient is not adequately responding to the current oxygen delivery method.
- The next step in management would be to transition the patient to high-flow nasal cannula oxygen therapy. High-flow nasal cannula delivers a higher flow rate of oxygen and a higher fraction of inspired oxygen (FiO₂) than traditional nasal cannulas. It provides warmed and humidified oxygen at a flow rate of 30-60 L/min, which helps to improve oxygenation, reduce work of breathing, and provide positive end-expiratory pressure (PEEP).
- High-flow nasal cannula oxygen therapy is often the preferred choice in patients with respiratory distress who have not responded adequately to conventional oxygen delivery methods. It offers several advantages, including improved oxygenation, better patient comfort, and the ability to provide PEEP to recruit collapsed alveoli and improve lung

function.

Incorrect Options:

Option A - Increase flow rate to 15 L/min:

- Increasing the flow rate to 15 L/min may provide a small improvement in oxygenation. However, given that the patient's SpO₂ only improved from 86% to 88% after one hour of oxygen therapy at 10 L/min, a further increase in flow rate alone is unlikely to be sufficient.
- Switching to high-flow nasal cannula therapy would be a more appropriate next step.

Option C - Endotracheal intubation:

- Endotracheal intubation is a more invasive procedure typically reserved for patients with severe respiratory distress who cannot maintain adequate oxygenation or ventilation despite non-invasive interventions.
- In this case, the patient is alert and responsive, suggesting that their mental status is preserved and may not require immediate intubation. Therefore, endotracheal intubation is not the next step in management.

Option D - No specific intervention necessary:

- After an hour of oxygen therapy, the patient's SpO₂ is still below the target range, indicating that a specific intervention is required to improve oxygenation.
- Continuing without modifying the current treatment would not be appropriate and may lead to worsening respiratory distress and hypoxia.

Solution for Question 2:

Correct Option B – Intubation:

- The given image shows the sniffing position. It is used during intubation to align the oral, pharyngeal, and laryngeal axes, making it easier to visualize the vocal cords and insert the endotracheal tube.

Incorrect Options:

Options A, C, and D: The sniffing position is not typically used for these.

Solution for Question 3:

Correct Option B – Alveolar Plateau:

- In a normal capnograph, the marked area represents the alveolar plateau phase. This is the flat part of the capnograph, indicating constant exhalation of CO₂.

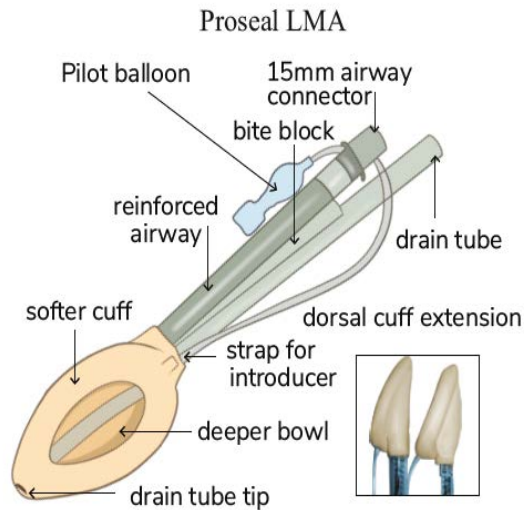
Incorrect Options:

Options A, C, and D are not marked in the given capnograph.

Solution for Question 4:

Correct Option A - LMA Proseal:

- LMA Proseal is a second-generation LMA designed with two ports: Tracheal port
Esophageal port
- Tracheal port
- Esophageal port
- Tracheal port
- Esophageal port



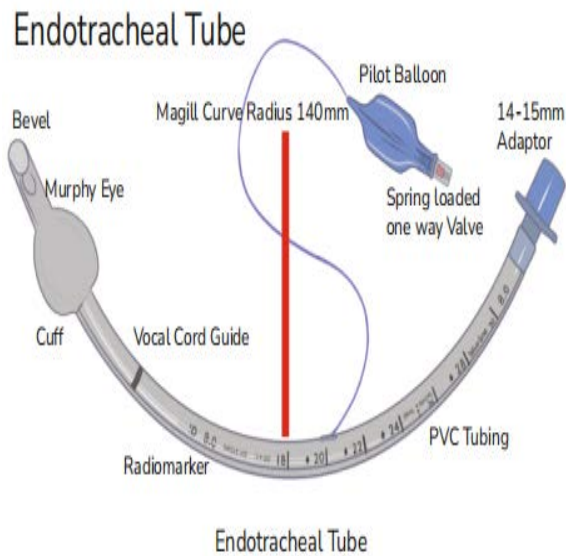
Incorrect Options:

Options B, C, and D are not considered second-generation LMAs as they have only one port.

Solution for Question 5:

Correct Option D - Endotracheal tube:

- The Murphy's eye is a feature found in certain medical devices, particularly in endotracheal tubes (ETT) and tracheostomy tubes.
- It consists of an additional opening or eyelet located on the side of the tube near the distal end, opposite to the main opening through which the tube is inserted into the airway.
- The purpose of the Murphy's eye is to provide an alternative pathway for airflow in case the main opening becomes obstructed. This can happen due to factors such as mucosal swelling, secretions, or misplacement of the tube against the tracheal wall.
- When the tube is correctly positioned in the airway, the Murphy's eye remains outside the tracheal wall. However, if the tube tip is pressed against the tracheal wall, the Murphy's eye allows air to flow into the lungs, preventing complete obstruction.



Incorrect Options:

Option A - Nasopharyngeal airway: The nasopharyngeal airway is used in conscious patients with a compromised airway. It is made of soft silicon and has a single opening.

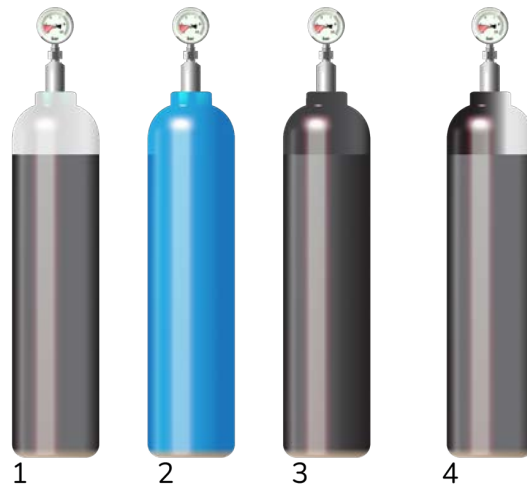
Option B - Laryngoscope: The laryngoscope is an instrument used to view the larynx or voice box.

Option C - Bronchoscope: The bronchoscope is a tube-like instrument with a lens to check for abnormalities in the lungs and bronchioles.

Solution for Question 6:

Correct Option A - Completely Blue:

- The color code for nitrous oxide (liquid) cylinders is complete blue (french blue).
- Anesthetic gasses are supplied in cylinders. The cylinders are made up of molybdenum steel which is an alloy.
- Nitrous oxide is stored under high pressure in cylinders. It cannot be directly given to patients as it can lead to the danger of barotrauma, and the flow rates at high pressure are difficult to control.



Cylinder	Colour
Oxygen (gas)	Black body and white shoulder.
Nitrous Oxide (gas)	Blue
Nitrogen (gas)	Black
Air (gas)	Gray/black body with black/white shoulder
Entonox (gas)	Blue body and white/blue shoulder

Incorrect Options:

Option B - Blue body with shoulder: Blue body with the blue shoulder is the color code for Entonox.

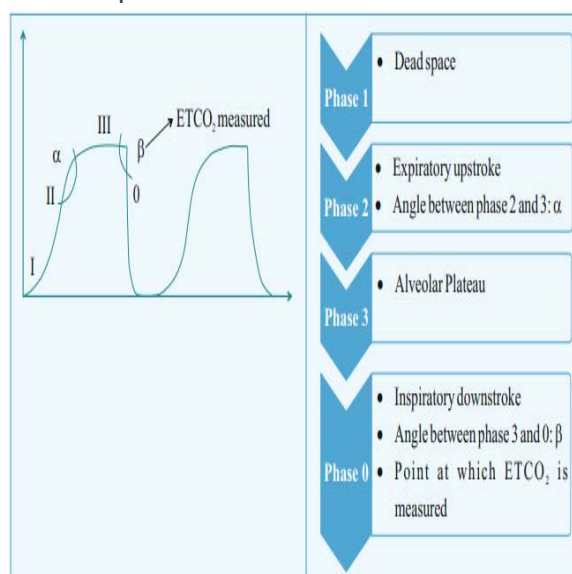
Option C - Black body & white shoulder: Black body with a white shoulder is the color code for oxygen (gas).

Option D - Grey body & black white sholuder: Grey body and black-white shoulder is the color code for nitrogen.

Solution for Question 7:

Correct Option A - Capnography:

- Capnography is a monitoring technique that measures the concentration or partial pressure of carbon dioxide (CO₂) in respiratory gases, typically in exhaled air. The tracing in yellow color on the device indicates capnography monitoring.
- Capnography provides real-time information about the patient's respiratory status and ventilation effectiveness. It is commonly used during anesthesia and in critical care settings to assess the patient's ventilation, detect changes in the end-tidal CO₂ (EtCO₂) levels, and monitor the integrity of the airway and respiratory system.
- Capnography measures the amount of CO₂ present at the end of exhalation (EtCO₂) and displays it as a waveform and numerical value. The waveform typically appears as a continuous waveform with peaks and valleys representing the inhalation and exhalation phases of the respiratory cycle. The numerical value represents the EtCO₂ level, which indicates the patient's ventilation and CO₂ elimination.
- The best test to confirm the placement of the endotracheal tube.



Incorrect Options:

Option B - Airway pressure:

- Airway pressure monitoring typically involves measuring the pressure within the airway and respiratory system, such as through the use of an esophageal pressure monitor or ventilator pressure monitoring. It does not directly relate to the yellow tracing on the device, which indicates capnography.

Option C - O₂ pressure in inhaled air:

- Monitoring the oxygen (O₂) pressure in inhaled air is typically not depicted by a yellow tracing on a device. O₂ pressure in inhaled air can be measured using various methods,

such as oxygen analyzers or oxygen concentration sensors, but it does not pertain to the yellow tracing seen on the device.

Option D - O₂ pressure in exhaled air:

- Monitoring the O₂ pressure in exhaled air is not typically indicated by a yellow tracing on a device. The O₂ concentration in exhaled air can be measured using specific devices.

Solution for Question 8:

Correct Option C - 1 Only:

- The sudden drop in EtCO₂ to 8 mmHg, along with the accompanying signs and symptoms, suggests a diagnosis of CO₂ embolism. Let's break down the information:
- CO₂ embolism: CO₂ embolism occurs when carbon dioxide gas enters the bloodstream, leading to a blockage of blood flow in the vessels. In this scenario, the sudden drop in EtCO₂ to 8 mmHg indicates a significant decrease in carbon dioxide elimination. CO₂ embolism can result from the inadvertent entry of carbon dioxide gas into the bloodstream during laparoscopic surgery, usually due to a breach in the abdominal cavity. It can cause a range of symptoms, including hypotension (low blood pressure), hypoxemia (low oxygen saturation), and increased airway pressures.

Incorrect Options:

Option A - 4 only: Single bronchus intubation: Single bronchus intubation occurs when the endotracheal tube is mistakenly placed into one of the mainstem bronchi, leading to inadequate ventilation of the other lung. While this can result in decreased EtCO₂ in the affected lung, it would not cause a sudden drop in EtCO₂ to 8 mmHg. Additionally, single bronchus intubation alone would not explain the hypotension. Therefore, option B is incorrect.

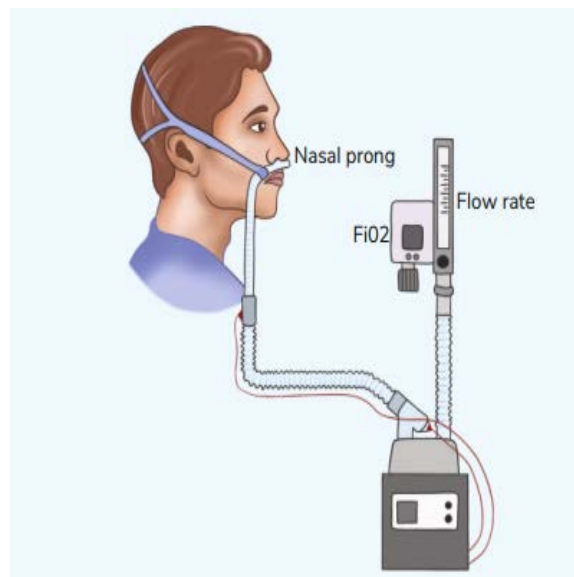
Option B - Both 3 and 4: Pneumothorax: Pneumothorax refers to the presence of air in the pleural space, causing lung collapse. It can lead to decreased oxygenation, hypotension, and increased airway pressures. However, pneumothorax alone would not explain the sudden drop in EtCO₂. Therefore, option B (both 3 and 4) is incorrect.

Option D - 2 only: Severe bronchospasm: While severe bronchospasm can cause a decrease in EtCO₂ and hypoxemia, it typically does not cause a sudden drop in EtCO₂ to such a low value (8 mmHg). Additionally, severe bronchospasm is not associated with significant hypotension or increased airway pressures. Therefore, option D (2 only) is incorrect.

Solution for Question 9:

Correct Option D - High Flow Nasal Cannula:

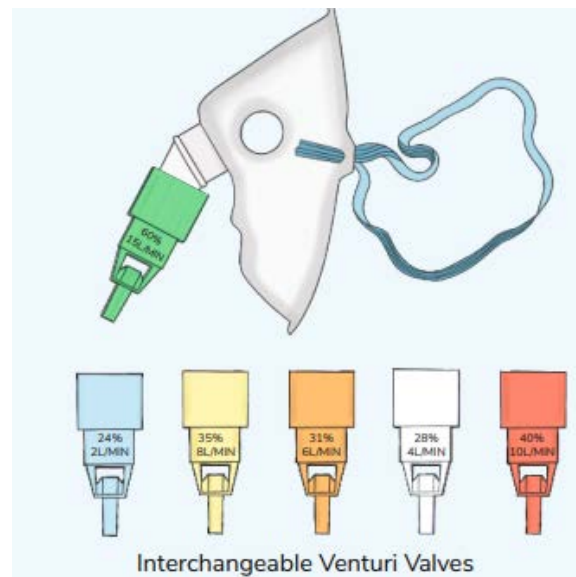
- High-Flow Nasal Cannula (HFNC) therapy is a mode of respiratory support that delivers a high flow of heated and humidified oxygen-enriched air through nasal prongs to patients with respiratory distress.
- Components of HFNC: Flow generator: Provides gas flow rates up to 60 L/min. Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate. Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).
- Flow generator: Provides gas flow rates up to 60 L/min.
- Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate.
- Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).
- Flow generator: Provides gas flow rates up to 60 L/min.
- Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate.
- Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).



Incorrect Options:

Option A - Venturi mask:

- A Venturi mask is a device that delivers a precise and fixed concentration of oxygen.
- It works by mixing a specific amount of oxygen with room air to achieve the desired FiO₂.
- The Venturi mask delivers a range of FiO₂ levels, depending on the specific Venturi adaptor used.
- While it can provide accurate oxygen concentrations, it typically delivers lower FiO₂ levels compared to a HFNC.



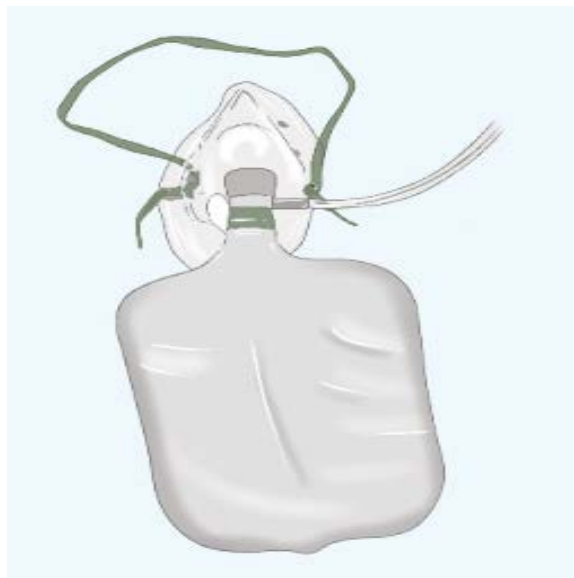
Option B - Hudson mask:

- The Hudson mask is a simple mask that covers the patient's nose and mouth.
- It is commonly used in conventional oxygen therapy.
- The FiO₂ delivered by a Hudson mask is influenced by the patient's breathing pattern, the fit of the mask, and the oxygen flow rate.
- It generally delivers a lower FiO₂ than a HFNC.



Option C- Non-rebreathing mask:

- Non-rebreather (NRB) masks deliver high concentrations of oxygen via a reservoir bag (600-1000 mL) and a one-way valve. The reservoir bag must be filled before use.
- Oxygen flow rates of 10-15 L/min achieve FiO_2 levels of 65%- 95%. NRB masks are used in conditions such as smoke inhalation, carbon monoxide poisoning, and severe hypoxia.
- Contraindications include facial trauma and claustrophobia, with caution advised in COPD patients. NRB masks are for short-term use only and limit patients' ability to eat, drink, or self-expectorate.



Solution for Question 10:

Correct Option D - Perform Heimlich maneuver:

- The patient is an adult who has choked on a bone while eating. The next best step is to perform the Heimlich maneuver by giving 5 abdominal thrusts.
- Heimlich Maneuver: The rescuer should stand or kneel behind the patient and wrap his arms around the patient's waist. With one hand, he makes a fist, and using the other hand, he drives the fist into the patient's abdomen with a rapid, forceful upward thrust to dislodge the foreign body. This is continued until the foreign body is expelled or if the patient becomes unresponsive.
- If the patient becomes unresponsive, immediately start CPR.
- In the case of an infant, 5 back slaps between the shoulder blades and 5 downward thrusts on the chest are given.



Incorrect Options:

Option A - Bag and mask ventilation: This intervention may be necessary if the patient becomes unresponsive and stops breathing, it is not the initial step in managing a conscious patient with an airway obstruction due to choking.

Option B - Endoscopic removal: Endoscopic removal may be necessary if other methods fail or if the obstruction is not cleared by basic interventions, it may not be the initial step in managing a conscious patient with a clear airway obstruction.

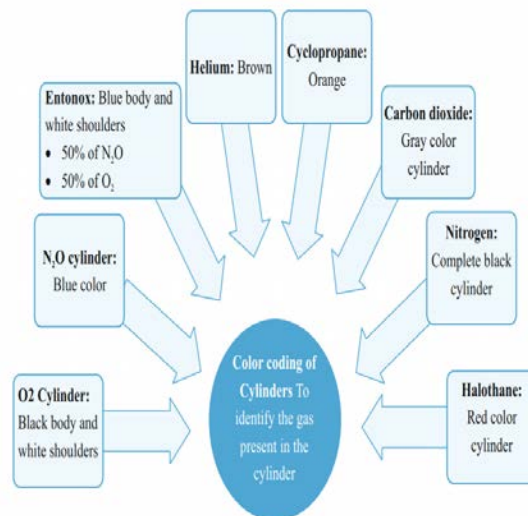
Option C - Deliver 5 back slaps and 5 chest compressions: This is done in the case of an infant and not an adult.

Solution for Question 11:

Correct Option A - 1:

- The oxygen cylinder has a white shoulder and a black body.

Cylinder Color Coding



Incorrect Options:

Option B, C, and D: Refer to the explanation of option A.

Solution for Question 12:

Correct Option C - 10 L/min:

- A conventional oxygen mask, also known as a simple face mask, is a device used for oxygen delivery to patients who require supplemental oxygen. The mask covers the nose and mouth, allowing the patient to breathe in a mixture of oxygen and room air.
- The maximum oxygen flow rate that can be delivered with a conventional oxygen mask is typically around 10 L/min. This flow rate ensures that a sufficient amount of oxygen is delivered to the patient, maintaining an adequate oxygen concentration or fraction of inspired oxygen (FiO₂). Higher flow rates may not significantly increase the delivered oxygen concentration and may lead to discomfort for the patient.

Incorrect Options:

Option A - 5 L/min: A flow rate of 5 L/min is commonly used for nasal cannula oxygen therapy, where the oxygen is delivered through small nasal prongs. However, this flow rate is generally considered insufficient for a conventional oxygen mask, which requires a higher flow rate to deliver an appropriate FiO₂.

Option B - 15 L/min: A flow rate of 15 L/min is higher than what a conventional oxygen mask can effectively deliver. While it may be possible to achieve such a flow rate with certain masks or adaptations, it is not the standard maximum flow rate for a conventional oxygen mask.

Option D - 20 L/min: A flow rate of 20 L/min is higher than what a conventional oxygen mask can effectively deliver. Such a high flow rate would exceed the capacity of the mask and could lead to discomfort and difficulty for the patient.

Solution for Question 13:

Correct Option C - Venturi mask:

- The venturi mask, or air entrainment mask, is a medical device that delivers a known oxygen concentration to patients on controlled oxygen therapy.
- A venturi flow meter is a type of differential pressure that generates a flow measurement by measuring the pressure difference at two locations in a pipe.
- Venturi mask used in maintaining FiO₂ of 24 - 60%.

Incorrect Options:

Option A - Nasal cannula: It can effectively provide only up to 4-6 L/min of supplemental oxygen, which equates to FiO₂ of approximately 24-44%.

Option B - Simple face mask: It can deliver 35-60% oxygen with an appropriate flow rate of 6-10 L/min.

Option D - NRBM (Non-rebreathing mask): It has a flow rate of 15 L/min and can deliver up to 85% oxygen.

Solution for Question 14:

Correct option B: 30%

- For each liter per minute (L/min) of oxygen flow, you can estimate an increase in FiO₂ of about 4%. 1L/min of O₂ corresponds to an FiO₂ of 24% 1 l/min – 24% 2 l/min -28% 3 l/min – 32% 4 l/min – 36% 5 l/min – 40% 6 l/min – 44%

- 1 l/min – 24%

- 2 l/min -28%

- 3 l/min – 32%

- 4 l/min – 36%

- 5 l/min – 40%

- 6 l/min – 44%

- In this case, with an oxygen flow rate of 3 L/min, the FiO₂ delivered is approximately 32%

- 1 l/min – 24%

- 2 l/min -28%

- 3 l/min – 32%

- 4 l/min – 36%

- 5 l/min – 40%

- 6 l/min – 44%

Other options:

Option A - 24%: An FiO₂ of 24% corresponds to 1L/min of oxygen.

Option C - 44%: An FiO₂ of 44% corresponds to 6L/min of oxygen.

Option D - 40%: An FiO₂ of 40% corresponds to 5L/min of oxygen.

Solution for Question 15:

Correct Option C - Oxygen with oxygen mask, and reservoir:

- Bronchial asthma is an inflammatory chronic disease condition of airways that is characterized by paroxysmal wheezing, cough, and chest tightness.
- Asthma causes tightening of muscles around the airways and causes airways to narrow. Hence, when both acute bronchitis and asthma occur together, this situation is called asthmatic bronchitis.
- Currently, nasal cannula, non-breathing bag face mask reservoir, venturi masks, are used in the clinical settings as conventional oxygen therapies. However, non-invasive ventilation (NIV) in airways causes trauma, need for sedation, infections.
- Conventional oxygen therapy (COT) involves a simple face mask that delivers oxygen.
- COT is not able to provide sufficient respiratory support to the patients that are suffering from bronchial asthma. As it needs an extra attempt to breath and work for the breathing process to maintain oxygenation.

Incorrect Options:

Option A - Hydrocortisone: In case of bronchial asthma and hay fever, oral hydrocortisone is an effective therapy for symptomatic relief.

Option B - Theophylline: Theophylline is applied for the treatment of shortness of breath, wheezing, chest tightness caused by asthma, emphysema, chronic bronchitis and other lung diseases. Lungs passages are opened and relaxed to make breathing easier.

Option D - Leukotriene antagonist: Chronic asthma/bronchial asthma is treated with Leukotriene-receptor antagonists such as Montelukast and Zafirlukast.

Solution for Question 16:

Correct Option: D - Non-Rebreathing Mask:

- The Non-Rebreathing Mask (NRB) is a type of oxygen mask frequently used in medical settings to provide high oxygen concentration to the patient.
- This mask provides the highest concentration of oxygen at a flow rate of 6 to 15 liters/min.
- It is sometimes called a demand oxygen mask or a partial non-rebreather mask. The NRB mask consists of a face mask that covers the patient's mouth and nose, connected to an oxygen reservoir bag.
- The patient can breathe in pure oxygen because of the reservoir bag, which boosts the amount of oxygen in the bloodstream and facilitates better breathing. Therefore, the mask used for patients with COVID-19 infection is the Non-rebreathing mask.



Incorrect Options:

Option A - Venturi mask:

- Oxygen is delivered at a concentration of 24 to 60% with the help of a Venturi Mask. The flow rate is 4 to 15 liters per minute.
- A constant amount is delivered through this mask which is not dependent on the respiratory rate and the tidal volume of the patient.
- Color coding is done on these masks, according to which They deliver a specific percentage of oxygen to the patient.

Colour Flowrate	Oxygen concentration
Blue	2L-24%
White	4L-28%
Yellow	6L-35%

Red	8L-40%
Green	12L-60%



Option B - Hudson mask:

- Advantages An increased delivery of oxygen can be made but for a shorter duration due to damage to the skin surface from the mask.
- An increased delivery of oxygen can be made but for a shorter duration due to damage to the skin surface from the mask.
- Disadvantages Tight seals are required, which can make many patients feel claustrophobic. There lies a potential for skin breaching. It can be uncomfortable for the patient while eating or talking.
- Tight seals are required, which can make many patients feel claustrophobic.
- There lies a potential for skin breaching.
- It can be uncomfortable for the patient while eating or talking.
- An increased delivery of oxygen can be made but for a shorter duration due to damage to the skin surface from the mask.
- Tight seals are required, which can make many patients feel claustrophobic.
- There lies a potential for skin breaching.
- It can be uncomfortable for the patient while eating or talking.

Option C - Nebulizer:

- It is a tool used to spray an aerosol of medication into the patient's airways.
- Patients with respiratory diseases like asthma frequently use it.

- A risk factor for COVID-19 transmission is the creation of a fine mist by the nebulizer, which could cause the virus to circulate in the air.
-

Solution for Question 17:

Correct Option A. Decreases the need for intubation

- High-flow oxygen therapy delivered via nasal cannula, also known as high-flow nasal cannula (HFNC) therapy, has several benefits, but it does not necessarily decrease the need for intubation in all cases.
- HFNC therapy can improve oxygenation, provide consistent and comfortable humidification, and reduce nasopharyngeal dead space, making it an effective therapy for patients with respiratory distress. However, the decision to intubate a patient is complex and depends on various factors, including the underlying condition, the severity of respiratory distress, and the patient's response to other therapies.
- HFNC therapy may delay the need for intubation in some cases, but it does not eliminate the need entirely.

Incorrect Options

Option B. Reduces nasopharyngeal dead space : HFNC therapy delivers a high flow of oxygen that washes out the nasopharyngeal dead space, allowing for more efficient ventilation and oxygenation.

Option C. Heating and humidification : HFNC systems effectively heat and humidify the oxygen, preventing drying and irritation of the airways. This is particularly beneficial for patients requiring high-flow oxygen for extended periods.

Option D. PEEP (Positive End-Expiratory Pressure) : HFNC can provide a degree of positive end-expiratory pressure, which helps keep the airways open, improve lung recruitment, and enhance oxygenation. This is especially useful in patients with atelectasis or other conditions where maintaining lung volume is important.

Solution for Question 18:

Correct Option C - Airway management:

- The image showing a tray in an airway cart with instruments strongly suggests that the instruments are used for airway management.
- Airway management involves various techniques and tools to establish and maintain a patient's airway, ensuring adequate oxygenation and ventilation.
- The instruments on the tray are likely to include items such as endotracheal tubes, laryngoscopes, stylets, oral airways, suction catheters, and other devices used for airway assessment and intervention.

Incorrect Options:

Options A - Central line insertion:

- Central line insertion refers to the placement of a catheter into a large central vein, such as the subclavian vein or internal jugular vein.
- The instruments used for this procedure typically include a needle, guidewire, dilators, and catheterization kit. The image provided does not correspond to instruments commonly used for central line insertion.

Options B - Arterial line insertion:

- Arterial line insertion involves the placement of an arterial catheter to monitor blood pressure continuously and obtain arterial blood samples.
- The instruments used for arterial line insertion include a needle, guidewire, dilators, and arterial catheterization kit. The image does not match the instruments commonly used for arterial line insertion.

Options D - Umbilical catheter insertion:

- Umbilical catheter insertion refers to the placement of a catheter into the umbilical artery or vein in newborns for medical purposes.
- The instruments used for this procedure typically include umbilical catheters, catheterization tray, and securing devices. The image provided does not correspond to the instruments commonly used for umbilical catheter insertion.

Solution for Question 19:

Correct Option D: Nasal cannula

- A nasal cannula is a medical device used to deliver supplemental oxygen or increased airflow to a patient or person in need of respiratory help.
- It consists of a lightweight tube that splits into two prongs that are placed in the nostrils. The other end of the tube is connected to an oxygen supply.
- It is commonly used in conditions where patients need oxygen support but are still able to breathe on their own, such as in cases of chronic obstructive pulmonary disease (COPD), post-surgical recovery, or during oxygen therapy.

A nasal cannula is a medical device used to deliver supplemental oxygen or increased airflow to a patient or person in need of respiratory help.

It consists of a lightweight tube that splits into two prongs that are placed in the nostrils. The other end of the tube is connected to an oxygen supply.

It is commonly used in conditions where patients need oxygen support but are still able to breathe on their own, such as in cases of chronic obstructive pulmonary disease (COPD), post-surgical recovery, or during oxygen therapy.

Incorrect options

Option A: Mechanical ventilator

- A mechanical ventilator is a device used to assist or control a patient's breathing. It delivers a mixture of air and oxygen, controls the respiratory rate and tidal volume, and provides positive pressure to support ventilation.
- It is more complex than a high-flow nasal cannula and is used in cases where more advanced respiratory support is needed.

Option B: Infusion pump

- An infusion pump is a device used to deliver fluids, such as medications or nutrients, into a patient's body in a controlled manner. It is not related to respiratory support and is used for intravenous administration of fluids or medications.

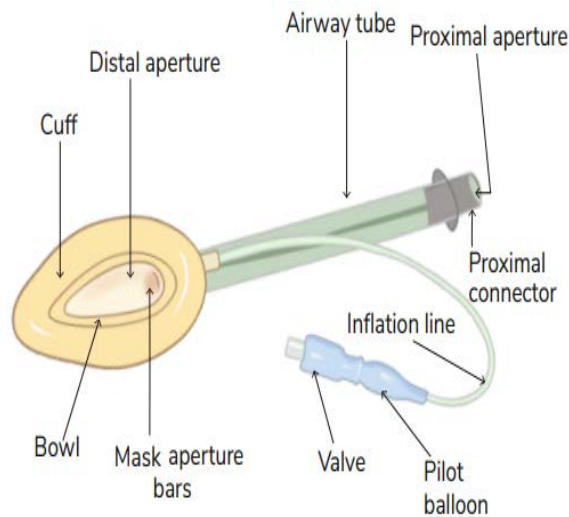
Option C: CPAP machine

- CPAP (Continuous Positive Airway Pressure) is a respiratory therapy used to treat sleep apnea. It involves providing a continuous flow of air pressure to keep the airways open during sleep. A CPAP machine is different from a high-flow nasal cannula, which delivers a high flow rate of oxygen.

Solution for Question 20:

Correct Option B: Laryngeal Mask Airway.

- A laryngeal mask airway (LMA) is a device that is inserted into the patient's throat to establish an airway during anesthesia or in cases where ventilation support is required. It consists of an inflatable mask that sits in the back of the throat and a tube that connects to a ventilation device. The mask forms a seal around the larynx, allowing for the delivery of oxygen and anesthesia gases directly into the patient's airway.



Classic LMA

Incorrect Options

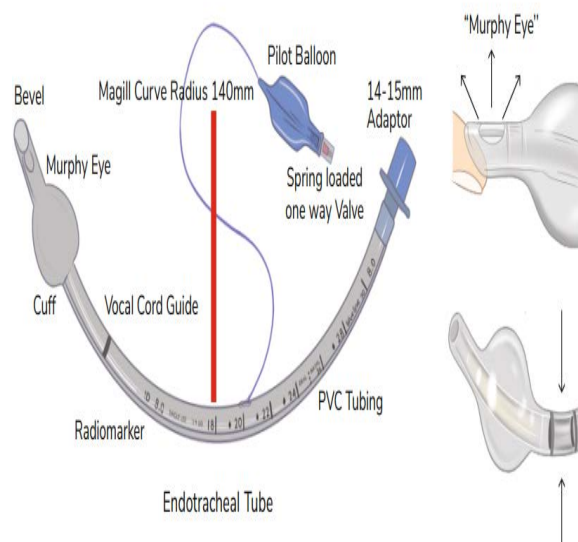
Option A: Guedel airway:

- A Guedel airway, also known as an oropharyngeal airway, is a curved plastic tube that is inserted into the patient's mouth to maintain an open airway. It is commonly used during anesthesia or in cases of unconsciousness to prevent the tongue from obstructing the airway. The Guedel airway is designed to fit over the tongue and extend into the back of the throat, allowing for easier passage of air.



Option A: Endotracheal tube

- An endotracheal tube (ETT) is a flexible tube that is inserted into the patient's trachea (windpipe) to establish and secure the airway. It is commonly used during general anesthesia, respiratory distress, or in cases where mechanical ventilation is required. The ETT is inserted through the mouth or nose and extends into the trachea, allowing for the delivery of oxygen and the removal of carbon dioxide. It is typically connected to a mechanical ventilator to provide controlled breathing support.



Option D: Nasogastric tube

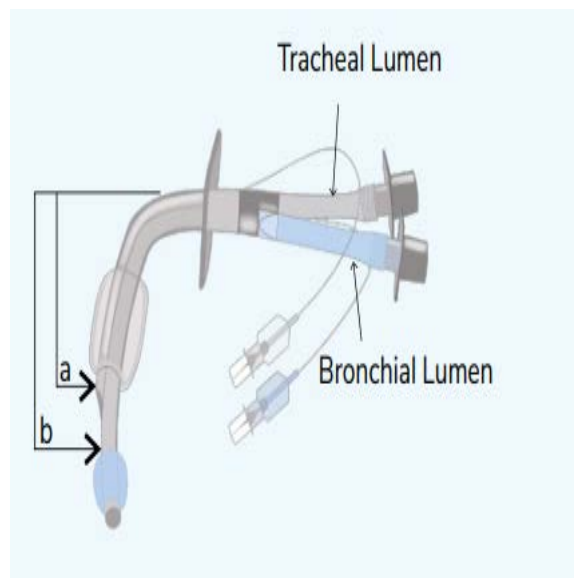
- A nasogastric tube (NG tube) is a thin tube that is inserted through the nose and down into the stomach. It is used for various purposes, such as feeding, medication administration, or

drainage of stomach contents. The NG tube is typically made of flexible plastic and can be temporarily or long-term placed depending on the patient's needs.

Solution for Question 21:

Correct Option B - Double lumen endotracheal tube:

- A double-lumen endotracheal tube is designed with two separate lumens or channels. It allows independent ventilation of each lung during thoracic surgeries or procedures requiring lung isolation. The tube has one lumen that enters the trachea and provides ventilation to one lung, while the other lumen enters the bronchus of the opposite lung, allowing selective lung collapse and ventilation control.



Incorrect Options:

Option A - Armored endotracheal tube:

- An armored endotracheal tube is a specialized type of endotracheal tube that has an internal wire coil or reinforcement. This provides rigidity and prevents kinking or collapse of the tube, ensuring a patent airway. Armored tubes are typically used in situations where there is a risk of tube compression or obstruction, such as during head and neck surgery.



Option C- Ring-Adair-Elwin (RAE) tube:

- The Ring-Adair-Elwin tube is a type of cuffed endotracheal tube that has a preformed curvature near the distal end. This curved design helps to facilitate placement and positioning of the tube within the trachea. RAE tubes are commonly used in anesthesia and critical care settings.



Option D - Endotracheal tubes equipped with bronchial blockers:

- These are endotracheal tubes that have an additional component called a bronchial blocker. A bronchial blocker is a small balloon-tipped catheter that can be inserted through the endotracheal tube and into one of the main bronchi. It is used to selectively block the airflow to one lung while allowing ventilation of the other lung. This technique is employed for

lung isolation during thoracic surgeries or procedures.

Solution for Question 22:

Correct Option C - Bernoulli's principle:

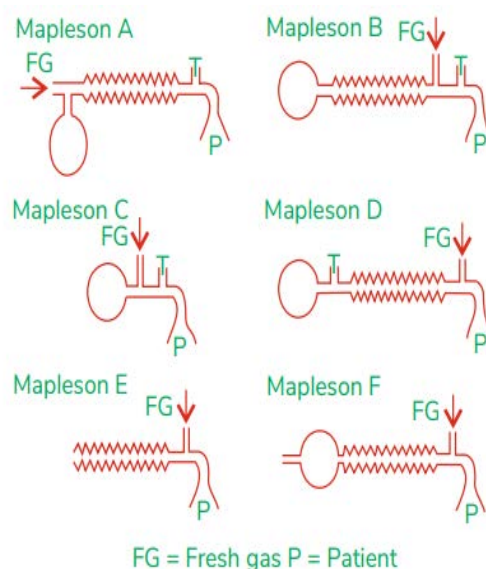
- The picture depicts a High Flow Venturi mask that uses Bernoulli's principle to deliver a predetermined and fixed oxygen concentration to the patient.

Incorrect Options:

Option A - Bohr effect: The Bohr effect describes the ability of the RBCs to adapt to the changes in the biochemical changes taking place in the body by maximizing the ability of hemoglobin to bind with O₂ in the Lungs and simultaneously delivering optimal O₂ to the tissues with the greatest demand.

Option B - Boyle's Principle: This principle defines the relationship between Pressure and Volume at a given Mass and Temperature. When the lungs are in a resting state with normal Tidal Volume, when the alveoli are not collapsed, and when they are not at a maximum capacity, they comply with Boyle's principle and follow proportional changes of Pressure and Volume.

Option D - Mapleson's Circuits: Mapleson's Circuits or Mapleson's Breathing systems are Semi-closed types of Breathing Circuits used during anesthesia to deliver oxygen and anesthetic agents and effectively washout CO₂.



Solution for Question 23:

Correct Option B - Label:

- The best way to recognize a cylinder is by its label. The color also helps to identify the cylinders.
- The name of the gas is usually written on the neck of the cylinder.
- The labels on the cylinder also had details about the weight of the gas, the pressure, and volume, the manufacturer, etc.
- There are also safety instructions given on the label (including danger signs and exposure to air warnings).

Incorrect Options:

Option A - Color coding:

- Color coding is a way to identify a cylinder, but it's not the best way, as some people may not be aware of the color coding given for a cylinder.

Option C - Pin index:

- A pin index system is also not the best way to recognize a cylinder, as the person must understand the arrangement of the pins.

Option D - All of the above:

- All of the above are methods used to identify a cylinder but only the label is the best method.
-

Solution for Question 24:

Correct Option A - Oxygen:

- Oxygen cylinders have a black body with white shoulders. This is the color coding of cylinders. Anesthetic gasses are usually supplied in cylinders that are made of molybdenum

steel.

- This allows the cylinders to be made thinner and lighter for comparable working pressures. Oxygen is stored under high pressure in cylinders and they cannot be directly administered to a patient.

Incorrect Options:

Option B - Nitric oxide: Nitric oxide cylinders have a blue color code.

Option C - Helium: Helium cylinders have a brown color code.

Option D - Entonox: Entonox cylinders have a blue body with a white shoulder color code.

Solution for Question 25:

Correct Option B - Unconscious/ sedation:

- BiPAP (bilevel positive airway pressure) therapy is best for short term use and in patients who have the ability to cough and protect their airway.

Incorrect Options

- Indications for using BiPAP machine COPD Obstructive sleep apnea Patients with conditions causing low partial pressure of oxygen
- COPD
- Obstructive sleep apnea
- Patients with conditions causing low partial pressure of oxygen
- Contraindications
- COPD
- Obstructive sleep apnea
- Patients with conditions causing low partial pressure of oxygen

Absolute contraindication	Relative contraindication
Facial trauma/ burn	Altered mental status/ unconscious
Fixed upper airway obstruction	Inability to protect the airway
Active vomiting	Life threatening hypoxemia

Solution for Question 26:

Correct Option A - Tracheostomy :

- In any emergency situation, the first priority is to maintain a patent airway.
- In this case, the way to secure airway is through the tracheostomy among the given options (given that there is maxillofacial trauma).
- Main function of tracheostomy is to relieve upper airway obstruction
- Indications for tracheostomy Difficult airway Significant maxillofacial trauma Angioedema Upper airway tumors
- Difficult airway
- Significant maxillofacial trauma
- Angioedema
- Upper airway tumors
- Difficult airway
- Significant maxillofacial trauma
- Angioedema
- Upper airway tumors

Incorrect Options:

Options B and C - Orotracheal intubation and Nasotracheal intubation:

- These two procedures are contraindicated in the maxillofacial traumas, as it is very hard to intubate.

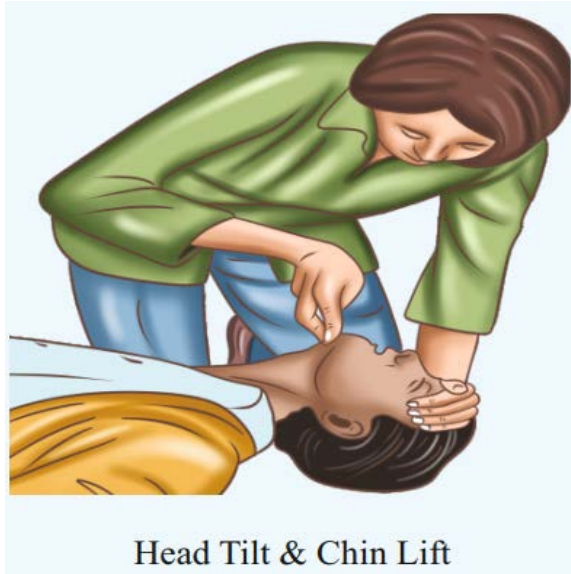
Option D - Oxygen therapy:

- This is possible only when there is a patent airway.
 - Hence, maintaining a patent airway (tracheostomy) is prior to oxygen therapy
-

Solution for Question 27:

Correct Option A - Head tilt, chin lift:

- In this image, the maneuvers being performed are head tilt and chin lift, where the patient's head is inclined backward, and the chin is tilted upwards.



Incorrect Options

Option B - Jaw thrust:

- In a jaw thrust, the patient's lower jaw is pulled in the forward direction. This is done in cases where there is a possibility of damage to the cervical spine region.

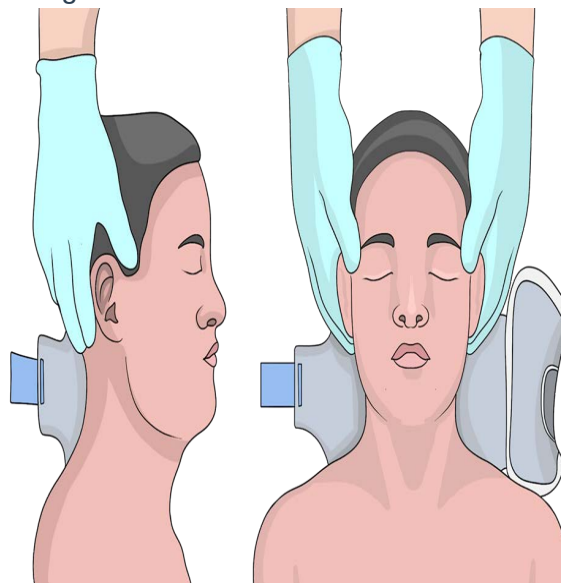


Option C - Head extension:

- The head extension is quite similar to the head and chin tilt maneuver. However, here, only the head is moved in the backward direction without moving the chin. This technique is used to open the airway in case of limited neck mobility.

Option D - In-line manual stabilization:

- When there is a severe head and neck injury, the in-line manual stabilization technique is used. Here, the neck and chin are aligned and brought in one line to minimize movement and prevent further damage.



Solution for Question 28:

Correct Option C - It is the first choice of treatment in cardiac arrest patients:

- Rapid Sequence Intubation (RSI) is a technique used to achieve prompt and safe tracheal intubation in critically ill patients. The procedure involves administering a potent induction agent and a muscle relaxant to facilitate tracheal intubation.
- RSI is not the first choice of treatment in cardiac arrest patients. In these patients, immediate defibrillation and cardiopulmonary resuscitation take priority over airway management.
- Preoxygenation with 100% oxygen for 3-5 minutes is a critical step in RSI, as it increases oxygen reserve and delays desaturation during the apneic period that occurs after administering the muscle relaxant.
- A high-volume low-pressure cuff is used to reduce the incidence of tracheal trauma and to provide a seal to prevent air leaks and aspiration.
- Induction agents and muscle relaxants are given simultaneously to ensure the rapid onset of sedation and paralysis, allowing for successful tracheal intubation.

Incorrect Options:

Options A, B, and D are components of Rapid Sequence Intubation (RSI).

Difficult airway management and Ventilators

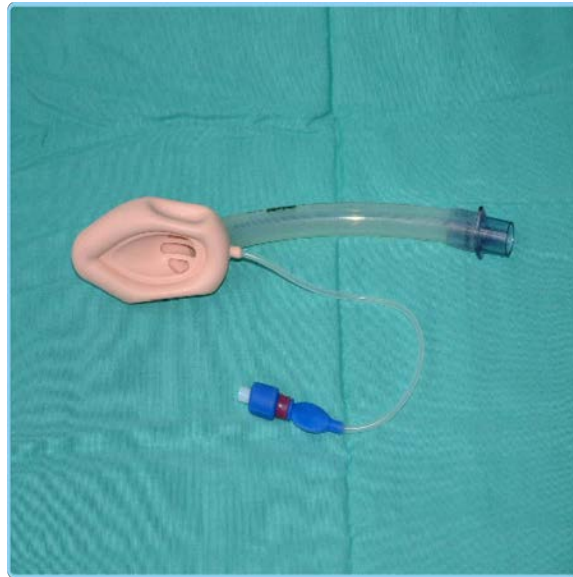
1. A male patient presents for an elective total knee replacement. During the preoperative assessment, the anesthesiologist performs a thorough evaluation and finds: Mallampati Score: Class III Thyromental Distance: 6 cm Sterno-mental distance 13 cm with head in maximum extension Neck circumference of 16 inch Which of the following features in a patient would prompt you to anticipate a difficult intubation?

- A. Mallampati Score: Class III
- B. Thyromental Distance: 6cm
- C. Sterno-mental distance 13cm with head in maximum extension
- D. Neck circumference of 16 inch

2. A 54-year-old woman is scheduled for an elective hysterectomy. During the preoperative assessment, the anesthesiologist evaluates her airway using Lemons' Law, which suggests that certain anatomical features are key predictors of difficult intubation. Which of the following features in a patient would prompt you to anticipate a difficult intubation?

- A. Mallampati Classification: Class II
- B. Thyroid to mouth distance - 2 finger breadth
- C. BMI of 20
- D. Range of neck movement is good

3. According to DAS guidelines, how will you manage a patient who is undergoing an elective cholecystectomy, if the below shown device fails to ventilate the patient?



- A. Bag and mask ventilation with 2-person technique
 - B. Continue the surgery with LMA
 - C. Invasive airway management
 - D. None of the above
-

4. Which of the following is NOT an indication for invasive mechanical ventilation?

- A. Respiratory arrest
 - B. Severe hypoxemia unresponsive to oxygen therapy
 - C. Inability to protect the airway
 - D. Cardiogenic pulmonary oedema
-

5. A 65-year-old male patient is undergoing general anesthesia for an open cholecystectomy. The anesthesiologist decides to use Continuous Mandatory Ventilation (CMV). Which of the following is NOT a characteristic of CMV?

- A. It typically requires deeper sedation than spontaneous breathing modes
 - B. It's useful for patients with severe respiratory failure
 - C. It allows for patient-triggered breaths between mandatory breaths
 - D. It may lead to respiratory muscle atrophy with prolonged use
-

6. A 60-year-old patient with acute respiratory distress syndrome (ARDS) is being mechanically ventilated. The physician is considering switching from volume-controlled assist control ventilation (VC-ACV) to pressure-controlled assist control ventilation (PC-ACV). Which of the following statements is true regarding these two modes?

- A. PC-ACV always delivers a higher tidal volume than VC-ACV
 - B. VC-ACV ensures lower peak inspiratory pressure than PC-ACV
 - C. PC-ACV provides a decelerating flow pattern which may improve gas distribution
 - D. VC-ACV is more likely to cause barotrauma than PC-ACV
-

7. A 55-year-old patient with acute respiratory failure is being weaned from mechanical ventilation using Synchronized Intermittent Mandatory Ventilation (SIMV). Which of the following statements about SIMV is correct?

- A. SIMV always delivers a fixed number of breaths per minute
 - B. SIMV prevents spontaneous breathing between mandatory breaths
 - C. SIMV is considered the most efficient technique for ventilator weaning
 - D. SIMV allows for both mandatory and spontaneous breaths
-

8. A 68-year-old patient with ARD is being mechanically ventilated. The medical team is considering the application of PEEP to improve the patient's oxygenation.

Which of the following statements about PEEP is true regarding its use in this patient?

- A. PEEP is contraindicated in patients with tension pneumothorax.
 - B. PEEP decreases FRC and lung compliance.
 - C. PEEP is indicated in Hypovolemic shock and asthma patients.
 - D. Auto-PEEP is the pressure set on the ventilator to maintain airway patency.
-

9. A 50-year-old patient with a history of emphysema presents with a large pneumothorax and is unable to maintain adequate oxygenation on conventional mechanical ventilation. The medical team decides to initiate high-frequency oscillation ventilation (HFOV) as part of the management strategy. What is the typical respiratory rate range for HFOV?

- A. 60 to 100 breaths/min
 - B. 100 to 150 breaths/min
 - C. 300 to 900 breaths/min
 - D. 180 to 300 breaths/min
-

10. A 62-year-old patient with severe COVID-19 pneumonia presents with acute respiratory distress syndrome (ARDS). After a few days of mechanical ventilation, the patient's arterial blood gas shows a PaO₂ of 58 mmHg and a P/F ratio of 120. Given the patient's condition, which of the following interventions should the team prioritize to improve oxygenation?

- A. Increase tidal volume to 8 mL/kg of ideal body weight
 - B. Implement prone positioning
 - C. Apply higher levels of PEEP to recruit collapsed alveoli
 - D. Start aggressive fluid resuscitation
-

11. A 68-year-old patient is recovering from pneumonia and has been on mechanical ventilation for the past week. The medical team is preparing to assess the patient for readiness to wean off the ventilator. Which of the following criteria should the medical team confirm before initiating the weaning process?

- A. Vital capacity < 10 mL/kg
 - B. Minute ventilation > 10 L/min
 - C. Tidal volume (VT) > 5 mL/kg
 - D. Rapid shallow breathing index (RSBI) > 100
-

12. A 60-year-old patient on a ventilator for respiratory distress develops pneumonia 48 hours after starting mechanical ventilation. The physician notes fever, leukocytosis, and patchy infiltrates on the chest X-ray. Which statement is true regarding the patient's condition?

- A. The patient likely has VAP and should be treated with ceftriaxone and azithromycin.
 - B. The patient likely has HAP and is appropriately treated with ceftriaxone and azithromycin.
 - C. Procalcitonin is not relevant in this case.
 - D. Early-onset VAP should be treated with piperacillin and carbapenem.
-

13. A 45-year-old patient is scheduled for surgery under general anesthesia. During the airway assessment using Modified Mallampati scoring, the following image shows the patient's oropharyngeal structures during assessment. Which class does this patient fall into?



- A. Class IV
 - B. Class I
 - C. Class II
 - D. Class III
-

Correct Answers

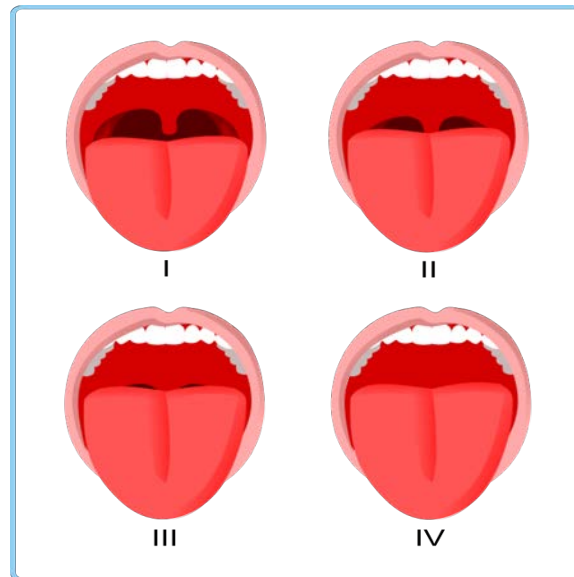
Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	4
Question 5	3
Question 6	3
Question 7	4
Question 8	1
Question 9	3
Question 10	2
Question 11	3
Question 12	1
Question 13	4

Solution for Question 1:

Correct Answer: A) Mallampati Score: Class III

Explanation :

The Mallampati score III suggests a potentially challenging airway, as visibility is reduced.



Mallampati Airway Classification

Predictors of difficult airway: Specific tests for assessment

- Mallampati score (III and IV)
- Thyromental distance less than 6 cm (Option B)
- Sterno-mental distance less than 12.5 cm with head in maximum extension (Option C)
- Increased neck circumference of more than 17 inches in men or more than 16 inches in women (Option D)
- Cormack and Lehane (laryngoscopic view)- Grade III and IV
- Upper lip bite test: Inability to protrude the mandible or lower teeth in front of the upper teeth
- Finger breadth for TMJ mobility- <3 fingers.

Risk factors of difficult airway:

- Obstructive sleep apnoea
- History of snoring
- Obesity
- Facial and neck deformities from previous surgery
- Previous head and neck radiation
- Head and neck trauma
- Congenital abnormalities of the head and neck
- Rheumatoid arthritis

- Down syndrome
- Scleroderma
- Cervical spine disease

Solution for Question 2:

Correct Answer: B) Thyroid to mouth distance - 2 finger breadth

Explanation:

Thyroid to mouth distance of 2 finger breadths is one of the key predictors of a difficult airway according to Lemon's Law.

The Lemon's Law	
Evaluation Criteria	Points
L = Look externally	
Facial trauma	1
Large incisors	1
Beard or moustache	1
Large tongue	1
E = Evaluate the 3-3-2 rule	
Incisor distance - 3 finger breadths	1
Hyoid mental distance - 3 finger breadths	1
Thyroid-to-mouth distance - 2 finger breadths	1
M = Mallampati (Mallampati score >3)	1
O = Obstruction (presence of any condition such as epiglottitis, peritonsillar abscess, trauma)	1
N = Neck mobility (limited neck mobility)	1
Total	10

- Lemon's law helps to assess the risk of a difficult airway
- Total score: 10 points
- Higher the lemon score- Higher the risk for airway management

A Mallampati score of more than III is a predictor of a difficult airway. (Option A)

A BMI of 20 means the person is not obese so, it's not a risk factor. (Option C)

Range of neck movement suggests the absence of limited neck mobility. (Option D)

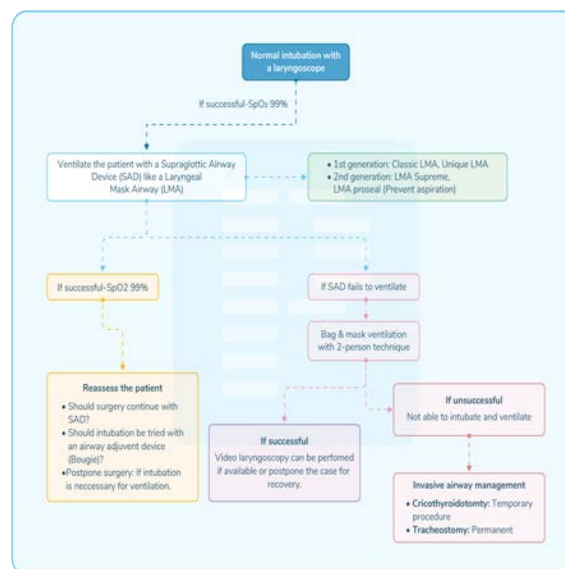
Solution for Question 3:

Correct Answer: A) Bag and mask ventilation with 2-person technique

Explanation:

The image is a Laryngeal Mask Airway (LMA) - a supraglottic airway device (SAD).

Bag and mask ventilation with a 2-person technique is performed if SAD fails to ventilate



- Continue the surgery with LMA practiced when normal intubation failed with a laryngoscope (Option B)
- Invasive airway management is performed when the Bag and mask ventilation is unsuccessful (Option C)

Solution for Question 4:

Correct Answer: D) Cardiogenic pulmonary oedema

Explanation: Cardiogenic pulmonary oedema is not typically an indication for invasive mechanical ventilation. Instead, it is often managed with non-invasive ventilation (NIV) or medical therapy alone.

Indications of invasive and non-invasive ventilators

Invasive Ventilation	Non-Invasive Ventilation
- Respiratory arrest (Option A) - Severe hypoxemia unresponsive to oxygen therapy (Option B) - Inability to protect airway (Option C) - Severe acidosis with hypercapnia - Decreased level of consciousness - Hemodynamic instability with respiratory compromise - Failure of non-invasive ventilation - Copious secretions	- Mild to moderate COPD exacerbations - Cardiogenic pulmonary edema - Immunocompromised patients with acute respiratory failure - Postoperative respiratory failure - Weaning from invasive ventilation - Mild to moderate hypoxemic respiratory failure - Palliative care in DNI patients

Solution for Question 5:

Correct Answer: C) It allows for patient-triggered breaths between mandatory breaths

Explanation:

- Continuous Mandatory Ventilation (CMV) is a mode of mechanical ventilation where the ventilator delivers a set number of breaths per minute, regardless of the patient's own respiratory efforts.
- It does not allow for patient-triggered breaths between the mandatory breaths, which is a key feature that distinguishes it from other modes like Assist-Control ventilation.

Continuous Mandatory Ventilation (CMV)

Mechanism	- Delivers a set number of breaths per minute - Ventilator controls both rate and tidal volume - Patient's respiratory efforts are not considered
Sedation/Paralysis	- Often requires deeper sedation (Option A) - Common drugs: Propofol, Midazolam, Fentanyl - May require neuromuscular blockade (e.g., Rocuronium)
Uses	- Severe respiratory failure (Option B) - ARDS - Post-operative ventilation - Neuromuscular disorders
Drawbacks	- Potential for respiratory muscle atrophy (Option D) - Increased risk of barotrauma - Patient-ventilator asynchrony - Increased sedation requirements

Solution for Question 6:

Correct Answer: C) PC-ACV provides a decelerating flow pattern which may improve gas distribution

Explanation: Pressure-controlled assist control ventilation (PC-ACV) typically delivers a decelerating flow pattern, which may improve gas distribution in the lungs, especially in patients with heterogeneous lung disease like ARDS.

Assist control ventilation- Pressure and volume control mode

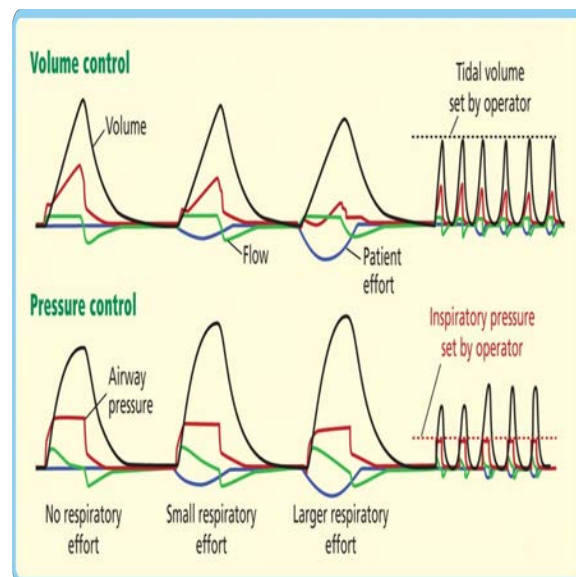
Feature	Volume-Controlled ACV (VC-ACV)	Pressure-Controlled ACV (PC-ACV)
Set parameter	Tidal volume	Inspiratory pressure
Flow pattern	Usually constant (square wave)	Decelerating
Tidal volume	Fixed (may vary slightly due to compliance changes)	Variable (depends on lung mechanics)
Peak airway pressure	Variable	Fixed
Response to changes in lung mechanics	Pressure increases if compliance decreases	Volume decreases if compliance decreases
Gas distribution	May be less uniform	May be more uniform due to decelerating flow

Risk of barotrauma	May be higher if not properly limited	May be lower due to pressure limitation
Risk of volutrauma	May be lower due to fixed tidal volume	May be higher if not properly monitored

PC-ACV does not always deliver a higher tidal volume than VC-ACV. The delivered volume in PC-ACV depends on lung mechanics and can vary (Option A)

VC-ACV can lead to higher peak inspiratory pressures, especially when lung compliance decreases, whereas PC-ACV limits peak pressure (Option B)

The risk of barotrauma is not necessarily higher with VC-ACV. Both modes can cause barotrauma if not properly managed (Option D)



Solution for Question 7:

Correct Answer: D) SIMV allows for both mandatory and spontaneous breaths

Explanation: Synchronized Intermittent Mandatory Ventilation (SIMV) is a mode of mechanical ventilation that allows for both mandatory (set by the ventilator) and spontaneous (initiated by the patient) breaths .

Synchronized Intermittent Mandatory Ventilation (SIMV)

Feature	Description
Breath Types	Mandatory (set) and spontaneous (patient-initiated)
Synchronization	Attempts to synchronize mandatory breaths with patient effort
Mandatory Breaths	Set number with fixed tidal volume or pressure
Spontaneous Breaths	Allowed between mandatory breaths, volume determined by patient effort
Pressure Support	Can be added to enhance spontaneous breath volumes
Primary use	To allow weaning of ventilatory support

SIMV does not always deliver a fixed number of breaths per minute. While it has set mandatory breaths, patients can take additional spontaneous breaths (Option A)

SIMV does not prevent spontaneous breathing between mandatory breaths. In fact, it allows for spontaneous breaths (Option B)

SIMV is not considered the most efficient technique for ventilator weaning. Studies have shown it to be less efficient than pressure support ventilation or T-piece trials (Option C)

Solution for Question 8:

Correct Answer: A) PEEP is contraindicated in patients with tension pneumothorax.

Explanation:

Applying PEEP in patients with tension pneumothorax can exacerbate the condition by increasing intrathoracic pressure and not allowing adequate ventilation.

Positive End-Expiratory Pressure (PEEP)

- PEEP is the positive pressure that remains in the alveoli at the end of exhalation, exceeding atmospheric pressure in mechanically ventilated patients. It can be set on the ventilator (extrinsic PEEP) or may occur as a complication (auto-PEEP). (Option D)
- Benefits of PEEP: Reduces alveolar trauma. Enhances oxygenation by recruiting closed alveoli, increasing gas exchange surface area. Increases functional residual capacity, improving oxygenation between breaths. (Option B) Improves ventilation/perfusion (V/Q)

matching. Increases lung compliance. (Option B)

- Reduces alveolar trauma.
- Enhances oxygenation by recruiting closed alveoli, increasing gas exchange surface area.
- Increases functional residual capacity, improving oxygenation between breaths. (Option B)
- Improves ventilation/perfusion (V/Q) matching.
- Increases lung compliance. (Option B)
- Reduces alveolar trauma.
- Enhances oxygenation by recruiting closed alveoli, increasing gas exchange surface area.
- Increases functional residual capacity, improving oxygenation between breaths. (Option B)
- Improves ventilation/perfusion (V/Q) matching.
- Increases lung compliance. (Option B)

PEEP	
Indications	Contraindications
Acute Respiratory Distress Syndrome (ARDS): Increases FRC and improves oxygenation. Chronic Obstructive Pulmonary Disease (COPD): Reduces work of breathing by splinting airways open. Pulmonary Edema: Improves lung compliance and oxygenation, especially in cardiogenic pulmonary edema. Auto-PEEP: Mitigates effects of intrinsic PEEP in patients with expiratory airflow limitation.	Tension Pneumothorax: Can exacerbate intrathoracic pressure. Hypovolemic Shock: May decrease venous return and worsen hypotension. (Option C) High Intracranial Pressure: Increased intrathoracic pressure can impair cerebral perfusion. Bronchopleural Fistula: May hinder healing due to increased airway pressures. Acute Bronchospasm (Asthma): Can lead to hyperinflation and increased gas trapping. (Option C)

Solution for Question 9:

Correct Answer: C) 300 to 900 breaths/min

Explanation:

The typical respiratory rate for high-frequency oscillation ventilation (HFOV) in adults is between 300 to 900 breaths per minute, which allows for effective gas exchange and oxygenation.

Modes used in Emergency: High-frequency ventilation

- Indications of HFV: Adults It is mainly used in severe ARDS to prevent ventilator-induced lung injury. Large air leak syndromes with the inability to keep lungs open like bronchopleural fistula, pneumothorax, pulmonary interstitial emphysema. Failure of conventional mechanical

ventilation. Refractory hypoxemia as a rescue therapy. Neonates Persistent pulmonaryhypertension. ARDS. Pulmonary interstitial emphysema. Meconium aspiration. Pulmonaryhypoplasia.

- Adults It is mainly used in severe ARDS to prevent ventilator-induced lung injury. Large air leak syndromes with the inability to keep lungs open like bronchopleural fistula, pneumothorax, pulmonary interstitial emphysema. Failure of conventional mechanical ventilation. Refractory hypoxemia as a rescue therapy.
- It is mainly used in severe ARDS to prevent ventilator-induced lung injury.
- Large air leak syndromes with the inability to keep lungs open like bronchopleural fistula, pneumothorax, pulmonary interstitial emphysema.
- Failure of conventional mechanical ventilation.
- Refractory hypoxemia as a rescue therapy.
- Neonates Persistent pulmonary hypertension. ARDS. Pulmonary interstitial emphysema. Meconium aspiration. Pulmonary hypoplasia.
- Persistent pulmonary hypertension.
- ARDS.
- Pulmonary interstitial emphysema.
- Meconium aspiration.
- Pulmonary hypoplasia.
- 4 types High-frequency jet ventilation (HFJV): Rate is 100 to 150/min (Option B)
High-frequency oscillation (HFOV): Rate is 300 to 900/min High-frequency positive pressure ventilation (HFPPV): Delivered using a conventional ventilator where respiratory rates are set at maximum limits. Rarely used. High-frequency percussive ventilation (HFPV) : Combination of HFV and conventional ventilation (pressure control mode)
- High-frequency jet ventilation (HFJV): Rate is 100 to 150/min (Option B)
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- Large air leak syndromes with the inability to keep lungs open like bronchopleural fistula, pneumothorax, pulmonary interstitial emphysema.
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- Persistent pulmonary hypertension.
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- High-frequency percussive ventilation (HFPV): Combination of HFV and conventional ventilation (pressure control mode)

60 to 100 breaths/min (Option A) and 180 to 300 breaths/min (Option D) are incorrect as they are too low for HFOV and do not meet the requirements for effective HFOV.

Solution for Question 10:

Correct Answer: B) Implement prone positioning

Explanation:

Prone positioning is beneficial for patients with severe ARDS (P/F ratio ≤ 150) as it has been shown to improve oxygenation and survival outcomes. This intervention aligns with current management strategies for ARDS.

- Lung Protective Ventilation (LPV): Tidal volumes : ≤ 6 mL/kg of ideal body weight to prevent alveolar overdistension. PEEP : Use higher levels to avoid alveolar collapse. Plateau pressure : Limit to ≤ 30 cm H₂O to minimize barotrauma. Permissive hypercapnia : Allow mild-to-moderate respiratory acidosis to avoid lung injury. FiO₂ : Minimize while targeting PaO₂ > 55 mmHg or SpO₂ $> 88\%$.
- Tidal volumes : ≤ 6 mL/kg of ideal body weight to prevent alveolar overdistension.
- PEEP : Use higher levels to avoid alveolar collapse.
- Plateau pressure : Limit to ≤ 30 cm H₂O to minimize barotrauma.
- Permissive hypercapnia : Allow mild-to-moderate respiratory acidosis to avoid lung injury.
- FiO₂ : Minimize while targeting PaO₂ > 55 mmHg or SpO₂ $> 88\%$.
- Recruitment Strategies: Use of PEEP to recruit collapsed alveoli and improve oxygenation. Prone positioning : Shown to improve survival in severe ARDS (P/F ratio ≤ 150).
- Use of PEEP to recruit collapsed alveoli and improve oxygenation.
- Prone positioning : Shown to improve survival in severe ARDS (P/F ratio ≤ 150).
- Advanced Techniques: Neuromuscular blockade : Early use may benefit patients with severe ARDS (P/F ratio < 150). ECMO : Considered for severe, refractory ARDS in specialized centers.
- Neuromuscular blockade : Early use may benefit patients with severe ARDS (P/F ratio < 150).
- ECMO : Considered for severe, refractory ARDS in specialized centers.
- Rescue Therapies (Limited evidence for survival benefit): High-frequency oscillatory ventilation : No significant outcome improvements. Inhaled vasodilators (iNO, prostaglandins): Transient oxygenation improvement but no long-term benefit.
- High-frequency oscillatory ventilation : No significant outcome improvements.
- Inhaled vasodilators (iNO, prostaglandins): Transient oxygenation improvement but no long-term benefit.
- Fluid Management: Conservative fluid management : Improves oxygenation and ventilator-free days.
- Conservative fluid management : Improves oxygenation and ventilator-free days.

- Corticosteroids : Mixed results; may benefit ARDS from bacterial sepsis but not viral pneumonia (except SARS-CoV2).
- Tidal volumes : ≤ 6 mL/kg of ideal body weight to prevent alveolar overdistension.
- PEEP : Use higher levels to avoid alveolar collapse.
- Plateau pressure : Limit to ≤ 30 cm H₂O to minimize barotrauma.
- Permissive hypercapnia : Allow mild-to-moderate respiratory acidosis to avoid lung injury.
- FiO₂ : Minimize while targeting PaO₂ > 55 mmHg or SpO₂ $> 88\%$.
- Use of PEEP to recruit collapsed alveoli and improve oxygenation.
- Prone positioning : Shown to improve survival in severe ARDS (P/F ratio ≤ 150).
- Neuromuscular blockade : Early use may benefit patients with severe ARDS (P/F ratio < 150).
- ECMO : Considered for severe, refractory ARDS in specialized centers.
- High-frequency oscillatory ventilation : No significant outcome improvements.
- Inhaled vasodilators (iNO, prostaglandins): Transient oxygenation improvement but no long-term benefit.
- Conservative fluid management : Improves oxygenation and ventilator-free days.

Increase tidal volume to 8 mL/kg of ideal body weight (Option A): Increasing tidal volume above the recommended ≤ 6 mL/kg can lead to alveolar overdistension and further lung injury, which is contrary to lung protective ventilation strategies.

Apply higher levels of PEEP to recruit collapsed alveoli (Option C): While using PEEP can help recruit collapsed alveoli, in patients with severe ARDS, this intervention must be balanced carefully. Excessive PEEP can lead to hemodynamic instability or worsen ventilation-perfusion mismatch. Prone positioning is generally prioritized for improving oxygenation in severe cases.

Start aggressive fluid resuscitation (Option D): In ARDS management, conservative fluid management is preferred to avoid fluid overload, which can exacerbate pulmonary edema and impair oxygenation. Aggressive fluid resuscitation may lead to increased extravascular lung water and worsen respiratory function.

Solution for Question 11:

Correct Answer: C) Tidal volume (VT) > 5 mL/kg

Explanation:

A tidal volume greater than 5 mL/kg supports the decision to initiate weaning, indicating adequate respiratory function.

Discontinuing mechanical ventilation involves two phases: Readiness testing and Weaning.

- Readiness Testing: Assess if the underlying cause for ventilation is controlled or reversed. Manage complicating factors such as bronchospasm, heart failure, infection, malnutrition, increased CO₂ production, acid-base imbalances, anemia, altered mental status, and sleep deprivation. Weaning is considered when the patient meets the following criteria: pH > 7.25, Adequate oxygenation (FiO₂ < 0.5, PaO₂ saturation > 90%), Hemodynamic stability, and no signs of myocardial ischemia. Adequate cough reflex present.
- Assess if the underlying cause for ventilation is controlled or reversed.
- Manage complicating factors such as bronchospasm, heart failure, infection, malnutrition, increased CO₂ production, acid-base imbalances, anemia, altered mental status, and sleep deprivation.
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- Adequate oxygenation (FiO₂ < 0.5, PaO₂ saturation > 90%),
- Hemodynamic stability, and no signs of myocardial ischemia.
- Adequate cough reflex present.
- Weaning or Liberation: Key measurements for weaning criteria: Inspiratory pressure : < -25 cm H₂O Tidal volume (VT) : > 5 mL/kg Vital capacity : > 10 mL/kg (Option A) Minute ventilation : < 10 L/min (Option B) Rapid shallow breathing index (RSBI) : < 100 (with < 105 indicating potential for extubation). (Option D) Weaning methods: SIMV : Gradually decrease mechanical breaths and pressure support to 5–8 cm H₂O while maintaining acceptable arterial CO₂ tension and respiratory rate. PSV : Gradually reduce pressure support to maintain a VT of 4–6 mL/kg and respiratory rate < 30. T-Piece/CPAP : Observe spontaneous breathing; use sequential trials if needed. CPAP (5 cm H₂O) may prevent atelectasis during prolonged T-piece trials by maintaining functional residual capacity (FRC).
- Key measurements for weaning criteria: Inspiratory pressure : < -25 cm H₂O Tidal volume (VT) : > 5 mL/kg Vital capacity : > 10 mL/kg (Option A) Minute ventilation : < 10 L/min (Option B) Rapid shallow breathing index (RSBI) : < 100 (with < 105 indicating potential for extubation). (Option D)
- Inspiratory pressure : < -25 cm H₂O
- Tidal volume (VT) : > 5 mL/kg

- Vital capacity : > 10 mL/kg (Option A)
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- Assess if the underlying cause for ventilation is controlled or reversed.
- Manage complicating factors such as bronchospasm, heart failure, infection, malnutrition, increased CO₂ production, acid-base imbalances, anemia, altered mental status, and sleep deprivation.
- Weaning is considered when the patient meets the following criteria: pH > 7.25, Adequate oxygenation (FiO₂ < 0.5, PaO₂ saturation > 90%), Hemodynamic stability, and no signs of myocardial ischemia. Adequate cough reflex present.
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- Hemodynamic stability, and no signs of myocardial ischemia.
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- pH > 7.25,
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- Inspiratory pressure : < -25 cm H₂O
 - Tidal volume (VT) : > 5 mL/kg
 - Vital capacity : > 10 mL/kg (Option A)
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 - Inspiratory pressure : < -25 cm H₂O
 - Tidal volume (VT) : > 5 mL/kg
 - Vital capacity : > 10 mL/kg (Option A)
 - Minute ventilation : < 10 L/min (Option B)
 - Rapid shallow breathing index (RSBI) : < 100 (with < 105 indicating potential for extubation). (Option D)
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 - PSV : Gradually reduce pressure support to maintain a VT of 4–6 mL/kg and respiratory rate < 30.
 - T-Piece/CPAP : Observe spontaneous breathing; use sequential trials if needed. CPAP (5 cm H₂O) may prevent atelectasis during prolonged T-piece trials by maintaining functional residual capacity (FRC).
-

Solution for Question 12:

Correct Answer: A) The patient likely has VAP and should be treated with ceftriaxone and azithromycin.

Explanation:

- The patient developed pneumonia 48 hours after starting mechanical ventilation, fitting the criteria for VAP. Symptoms such as fever, leukocytosis, and patchy X-ray infiltrates support this diagnosis.
- The prescribed treatment of ceftriaxone and azithromycin is appropriate for early-onset VAP, effectively targeting common pathogens while adhering to clinical guidelines for managing hospital-acquired infections.

Criteria	Hospital-Acquired Pneumonia (HAP)	Ventilator-Associated Pneumonia (VAP)
Timing	Develops after 48 hours of admission (Option B)	Occurs after 48 hours of intubation Early: First 48 to 72 hours Late: Occurs after 72 hours
Pneumonia Features	Fever Leukocytosis Patchy infiltration on X-ray	Fever Leukocytosis Patchy infiltration on X-ray
Treatment	Ceftriaxone and Azithromycin	Early-Onset: Ceftriaxone plus azithromycin for 8 days (Option D) Late-Onset: Vancomycin or linezolid plus cefepime for 8 days
Common Organisms	MRSA, Pseudomonas aeruginosa and other aerobic Gram-negative bacilli, such as Klebsiella pneumoniae, Escherichia coli and Acinetobacter.	Haemophilus influenzae, Streptococcus pneumoniae, MRSA, and other antibiotic-sensitive oral flora and viruses and fungi
Infection Marker	Procalcitonin (Option C)	

Solution for Question 13:

Correct Answer: D) Class III

Explanation:

Class III means that only the soft palate and the base of the uvula are visible. This classification is significant as it predicts a moderate risk of difficult intubation.

Modified Mallampati scoring		
Class 0	Any part of the epiglottis is visible	
Class I	Faucial pillars, uvula, and soft palate are visualized. (Option B)	
Class II	Uvula and soft palate are visualized. Tip of uvula is not visualized. (Option C)	
Class III	Only Soft palate and base of the uvula are visible.	Predict difficult intubation
Class IV	Only hard palate is visible. (Option A)	

Previous Year Questions

1. Which of the following is the use of the Mallampatti classification?

- A. Endotracheal intubation
 - B. To evaluate the risk of surgery
 - C. To evaluate the pros and cons of surgery
 - D. To evaluate the fitness of the patient
-

2. What is the primary purpose of PEEP (Positive End-Expiratory Pressure) in mechanical ventilation?

- A. To increase respiratory rate
 - B. To reduce carbon dioxide levels
 - C. To increase the VR
 - D. To open alveoli and prevent atelectasis
-

3. A patient undergoing general anesthesia develops left lung collapse following intubation. On auscultation, breath sounds are heard only on the right side. What is the most likely cause of this condition?

- A. Right endobronchial intubation
 - B. Mucus secretions obstructing the endotracheal tube
 - C. Pneumothorax on the left side due to positive pressure ventilation
 - D. Bronchospasm
-

4. A patient in the ICU with an endotracheal tube now needs a tracheostomy tube. Which type of tube will you use?

- A. Uncuffed tracheostomy tube
 - B. Cuffed tracheostomy tube
 - C. Metallic tracheostomy tube.
 - D. Endotracheal tube
-

5. Resistance of the tube shown below is primarily because of ____.



- A. Curvature
 - B. Circumference
 - C. Length
 - D. Diameter
-

6. What should be the next course of action in managing a patient with pneumonia who is on a ventilator, weighs 50 kg, has a respiratory rate of 14 breaths per minute, a bicarbonate level of 18, a pH of 7.33, a paCO_2 level of 48 mmHg, a paO_2 level of 110 mmHg, a PEEP of 12 cm H₂O, a tidal volume of 420 mL, and a SpO_2 level of 100% with an FiO_2 of 90%?

- A. Increase PEEP
 - B. Decrease fiO_2
 - C. Decrease RR
 - D. Increase tidal volume
-

7. What is the most likely cause of a young patient experiencing desaturation and bilateral basal crepitations shortly after undergoing surgery with general anesthesia using O₂, Nitrous oxide, and Desflurane?

- A. ARDS
 - B. Pleural effusion
 - C. Negative pressure pulmonary oedema
 - D. Atelectasis
-

8. Regarding the initial setting for a patient with blunt chest trauma who has been initiated on mechanical ventilation, which of the following statements is incorrect?

- A. Respiratory rate – 12 breaths/ min
 - B. PEEP - 5 cm H₂O
 - C. FiO_2 -100%
 - D. Tidal volume – 12ml/kg of actual BW
-

9. Which of the following is associated with respiratory alkalosis ?

- A. Assisted control mode ventilation
 - B. Controlled mandatory
 - C. Pressure controlled
 - D. SIMV
-

10. A neonate is brought to the pediatric ER in acute respiratory distress. On examination, the abdomen is scaphoid. Which of the following should not be performed in this patient?

- A. PPV
 - B. Bag and mask ventilation
 - C. Oxygen therapy
 - D. Intubation
-

11. Ventilatory support is required if SpO₂ becomes less than 90% because

- A. Tissues become vulnerable for hypoxia below 90%
 - B. Peripheral chemoreceptors becomes insensitive to hypoxia below 90%
 - C. Central chemoreceptors becomes insensitive to hypoxia below 90%
 - D. Patient capacity to generate adequate tidal volume decreases below 90%
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	1
Question 4	2
Question 5	4
Question 6	2
Question 7	3
Question 8	4
Question 9	1
Question 10	2
Question 11	1

Solution for Question 1:

Correct Option A - Endotracheal Intubation:

- The Mallampati classification is primarily used to assess the difficulty of endotracheal intubation. It evaluates the visibility of anatomical structures in the oral cavity to predict the ease or difficulty of securing the airway.

Incorrect Options :

- Options B, C, and D: The Mallampati classification is not used to evaluate the overall risk associated with surgery and does not provide insights into the pros and cons of surgical procedures. It does not assess a patient's general fitness for surgery.
-

Solution for Question 2:

Correct Option D – To open alveoli and prevent atelectasis

- PEEP is used in mechanical ventilation to keep alveoli open at the end of exhalation, preventing them from collapsing (atelectasis). This helps improve oxygenation and maintain lung function, especially in patients with acute respiratory distress syndrome (ARDS) or other conditions leading to alveolar collapse.

Incorrect Options:

Options A, B, and C are not the purpose of PEEP.

Solution for Question 3:

Correct Answer: A) Right endobronchial intubation

Explanation:

The physical finding of breath sounds heard only on the right side with absence on the left strongly supports this diagnosis of right endobronchial intubation (right mainstem intubation). When an endotracheal tube is advanced too far, it typically enters the right mainstem bronchus due to its more vertical orientation compared to the left bronchus. This results in ventilation of only the right lung while the left lung collapses due to lack of ventilation.

Mucus secretions obstructing the endotracheal tube (Option B) would typically cause bilateral decreased air entry or generalized poor air movement rather than unilateral findings. If the tube were completely obstructed, there would be no or diminished breath sounds on both sides.

Pneumothorax on the left side due to positive pressure ventilation (Option C) could present with diminished breath sounds on the left but would typically have additional signs such as

hypoxemia, hypotension, tracheal deviation toward the right, and hyperresonance to percussion on the left side.

Bronchospasm (Option D) typically presents with bilateral wheezing, prolonged expiration, and increased airway pressures. It would not cause a complete absence of breath sounds on one side with normal findings on the other.

Solution for Question 4:

Correct Option B - Cuffed tracheostomy tube:

- When a patient in the ICU requires a tracheostomy tube, the preferred choice is a cuffed tracheostomy tube.
- A cuffed tracheostomy tube has an inflatable cuff that helps to create a seal between the tube and the tracheal wall.
- This seal prevents the leakage of air and secretions around the tube, reducing the risk of aspiration and improving ventilation efficiency.

Incorrect Options:

Option A - Uncuffed tracheostomy tube:

- An uncuffed tracheostomy tube has no inflatable cup.
- These are most commonly used in the neonatal population or for patients who can manage their own secretions.
- These allow airway clearance but don't protect from aspiration.

Option C - Metallic tracheostomy tube:

- A 2-3 inch long curved metal or plastic tube placed in a surgically created opening in the trachea to keep it open.
- It is indicated in case of obstruction in the upper airway, establishing and maintaining a secure airway when it is unsafe, searches in the face, head and neck injuries, bronchial toilet and secretion removal, and airway production in case of neuromuscular disorders.

Option D - Endotracheal tube:

- It is made of polyvinyl chloride.

- The main indication is to secure a definitive airway when there is the inability to maintain airway patency, inability to protect the air against aspiration, failure to ventilate, failure to oxygenate, and anticipation of a deteriorating course leading to respiratory failure.
-

Solution for Question 5:

Correct Option D - Diameter:

- Airway resistance (R) in the endotracheal tube is because of diameter.
- Resistance is directly proportional to viscosity and length and inversely proportional to the radius (r^4) according to the Hagen Poiseuille law.

Incorrect Options:

Option A , B and C are incorrect.

Solution for Question 6:

Correct Option B - Decrease FiO₂:

Respiratory Parameters Given	Normal Values
pH – 7.33	7.35 – 7.45
PaO ₂ 110 mmHg	80-100
PaCO ₂ – 48 mmHg	35 – 45
Bicarbonate – 18	22-26
Ventilatory settings Given	Normal Value
RR – 14/min	12-18
FiO ₂ – 90%	

PEEP- 12 cm H ₂ O	
TV - 420 mL	

- Lung Protective Ventilation is the current standard in mechanical ventilation which includes the following: Low TV to prevent barotrauma Increase RR – RR can be increased up to 30-35/min to maintain minute ventilation Increase PEEP – Can increase PEEP from 4-12 even up to 16 cm H₂O if needed to prevent collapse of alveoli especially in ARDS FiO₂ - < 0.6 or 60% - If FiO₂ > 60%, oxygen radical formation can occur causing worsening of symptoms
- Low TV to prevent barotrauma
- Increase RR – RR can be increased up to 30-35/min to maintain minute ventilation
- Increase PEEP – Can increase PEEP from 4-12 even up to 16 cm H₂O if needed to prevent collapse of alveoli especially in ARDS
- FiO₂ - < 0.6 or 60% - If FiO₂ > 60%, oxygen radical formation can occur causing worsening of symptoms
- FiO₂ has to be decreased in this patient to prevent free radical formation and since paO₂ is higher than normal.
- Low TV to prevent barotrauma
- Increase RR – RR can be increased up to 30-35/min to maintain minute ventilation
- Increase PEEP – Can increase PEEP from 4-12 even up to 16 cm H₂O if needed to prevent collapse of alveoli especially in ARDS
- FiO₂ - < 0.6 or 60% - If FiO₂ > 60%, oxygen radical formation can occur causing worsening of symptoms

Incorrect Options:

Option A - Increase PEEP:

- The patient shows an SpO₂ of 100% demonstrating adequate oxygen saturation.
- Thus, it is not needed to increase PEEP, which is done to prevent the alveoli from collapsing and SpO₂ is low.

Option C - Decrease RR:

- The patient is in acidosis and we do not want to decrease RR which will cause CO₂ retention leading to further respiratory acidosis.

Option D - Increase tidal volume:

- Low tidal volume – 6-8wt/kg of body weight is followed in lung protective ventilation to prevent barotrauma to the lungs.

Solution for Question 7:

Correct Option C - Negative pressure pulmonary oedema:

- In the above given case, the patient desaturates right after extubation and upon auscultation presents with bilateral basal crepitations.
- This is due to negative pressure pulmonary edema and is associated with extubation.

Incorrect Options:

Option A - ARDS:

- ARDS presents with crackling sounds on auscultation.

Option B - Pleural effusion:

- This is not the case of pleural effusion it presents with decreased breath sounds and vocal fremitus.

Option D - Atelectasis:

- Atelectasis is asymptomatic and upon auscultation presents with decreased or absent breath sounds
-

Solution for Question 8:

Correct Option D - Tidal volume – 12 ml/kg of actual BW:

- In patients with blunt chest trauma, lung-protective ventilation strategies are crucial to minimize ventilator-induced lung injury (VILI).
- One of the key components of lung-protective ventilation is limiting tidal volume.
- The generally recommended tidal volume for patients with acute respiratory distress syndrome (ARDS), which includes many cases of severe blunt chest trauma, is 6 ml/kg of predicted body weight (not actual body weight) to reduce the risk of barotrauma and volutrauma.

Incorrect Options:

Option A - Respiratory rate – 12 breaths/min : A respiratory rate of 12 breaths/minute is within the normal range for an adult and can be an appropriate initial setting, although the specific rate may need adjustment based on the patient's condition and blood gas values.

Option B - PEEP - 5 cm H₂O :

- Positive end-expiratory pressure (PEEP) is used to prevent alveolar collapse and improve oxygenation.
- A setting of 5 cm H₂O is a common starting point and can be adjusted based on the patient's oxygenation status.

Option C - FiO₂-100% : In the initial stages, a high fraction of inspired oxygen (FiO₂) might be necessary to ensure adequate oxygenation, but it should be titrated down as soon as possible to the lowest level necessary to maintain adequate oxygen saturation, in order to minimize the risk of oxygen toxicity.

Solution for Question 9:

Correct Option A - Assisted control mode ventilation:

- Respiratory alkalosis is a condition characterized by a decrease in arterial carbon dioxide levels, leading to an increase in blood pH.
- It is caused by hyperventilation or excessive elimination of CO₂ from the body.
- In assisted control mode ventilation, the ventilator delivers breaths at a preset rate and tidal volume, and the patient can initiate additional breaths on their own.
- This mode of ventilation can lead to increased minute ventilation and respiratory alkalosis due to excessive elimination of CO₂.

Incorrect Options:

Option B - Controlled mandatory ventilation (CMV):

- In this mode, the ventilator delivers breaths at a preset rate and tidal volume, and the patient has no control over initiating breaths.
- It does not typically cause respiratory alkalosis as the ventilator controls the entire breathing pattern.

Option C - Pressure-controlled ventilation:

- This mode of ventilation delivers breaths with a set inspiratory pressure and allows the patient to determine the respiratory rate and tidal volume.
- While it can affect blood gas parameters, it does not specifically lead to respiratory alkalosis.

Option D - SIMV (Synchronized Intermittent Mandatory Ventilation):

- SIMV is a mode of ventilation that combines mandatory breaths delivered by the ventilator with the patient's spontaneous breaths.
 - It allows the patient to breathe spontaneously between the mandatory breaths.
 - SIMV does not directly cause respiratory alkalosis.
-

Solution for Question 10:

Correct Option B - Bag and Mask Ventilation:

- Bag and mask ventilation is a technique used to manually assist ventilation by delivering positive pressure breaths using a self-expanding bag.
- A neonate with acute respiratory distress with a scaphoid could potentially be experiencing complications of a diaphragmatic hernia.
- Performing bag and mask ventilation in a child with a diaphragmatic hernia can result in further distension of the stomach and intestines, leading to increased pressure in the thoracic cavity and potentially worsening respiratory distress.

Incorrect Options: Options A, C, and D are incorrect.

- A. PPV (Positive Pressure Ventilation): This can be performed but should be done carefully to avoid overinflating the stomach, which can worsen respiratory distress. Intubation with PPV is preferred.
 - C. Oxygen therapy: This is safe and can help improve oxygenation but is usually insufficient alone for severe respiratory distress in CDH.
 - D. Intubation: This is the preferred method to secure the airway and provide effective ventilation while minimizing the risk of inflating the stomach.
-

Solution for Question 11:

Correct Option A: Tissues become vulnerable for hypoxia below 90%:

- When the peripheral oxygen saturation (SpO₂) falls below 90%, it indicates a significant reduction in the amount of oxygen the blood carries.
- At this level, the tissues in the body become vulnerable to hypoxia.
- Ventilatory support, such as supplemental oxygen or mechanical ventilation, is necessary to ensure sufficient oxygen is delivered to the tissues, preventing the onset of hypoxia and its associated complications.

Incorrect Options:

Option B: Peripheral chemoreceptors become insensitive to hypoxia below 90%:

- Peripheral chemoreceptors are specialized sensors located in the carotid bodies and aortic bodies that detect changes in the oxygen levels in the blood.
- They are primarily responsible for triggering the respiratory response to low oxygen levels.
- These chemoreceptors remain sensitive to hypoxia even at levels below 90% and continue to stimulate the respiratory centers in the brain to increase ventilation.

Option C: Central chemoreceptors become insensitive to hypoxia below 90%:

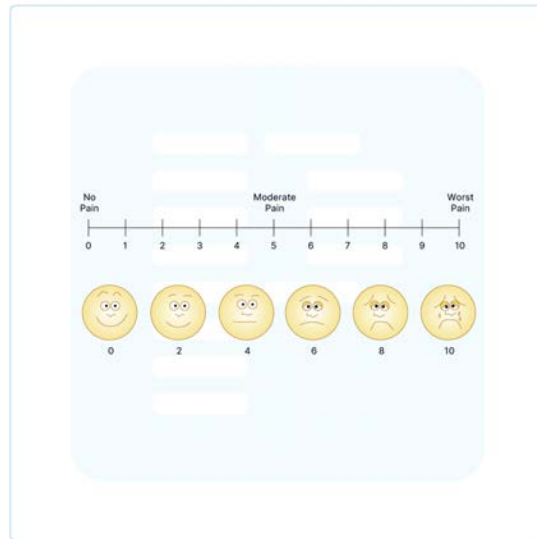
- Central chemoreceptors are located in the brain and respond to changes in the carbon dioxide (CO₂) levels in the cerebrospinal fluid.
- They play a vital role in regulating breathing based on the body's need to remove excess CO₂.
- They respond to changes in CO₂ levels, which indirectly affect the regulation of respiration.

Option D: Patient capacity to generate adequate tidal volume decreases below 90%:

- Tidal volume refers to the amount of air that is inspired or expired during a normal breath.
 - The percentage value of SpO₂ does not directly correlate with the patient's ability to generate tidal volume.
-

Pain management

1. Identify the method of assessment of Pain as depicted in the image.



- A. Wong-Baker Scale
- B. Visual Analog Scale
- C. CHEOPS
- D. McGill Pain Scale

2. A 45-year-old female presents to the clinic with a 6-month history of persistent lower back pain. She describes the pain as a dull ache that worsens throughout the day is associated with difficulty sleeping, and has a burning sensation radiating down her left leg. Over-the-counter ibuprofen has provided minimal relief. What is the most appropriate step in managing her pain?

- A. Increase the dosage of ibuprofen.
- B. Amitriptyline

- C. Tramadol
 - D. Exercise therapy
-

3. You are caring for a 20-year-old woman with metastatic breast carcinoma and renal failure. While she is admitted to the oncology ward, she develops a pain of 8/10 score. What is the best medication to treat her pain?

- A. IV Paracetamol
 - B. PO morphine
 - C. PO Tramadol
 - D. IV Pethidine
-

4. A 65-year-old woman is recovering in the postoperative unit after hip replacement surgery. She is receiving strong analgesics. The doctor checked her vital signs and found that her heart rate was 52 beats per minute. Which other adverse effect should the medical team be alert for in this patient due to strong analgesics?

- A. Rash
 - B. Mydriasis
 - C. Diarrhea
 - D. Hypertension
-

5. A 45-year-old man is in the postanesthesia care unit following a routine elective surgery. You are asked to monitor the patient for morphine overdose. Which of the following sets of symptoms best represents the classic triad associated with morphine overdose?

- A. Mydriasis, hypertension, and tachycardia
 - B. Hypertension, respiratory depression, and miosis
 - C. Respiratory depression, miosis, and altered mental status
 - D. Agitation, hyperreflexia, and diaphoresis
-

6. A 45-year-old male in the post-anesthesia care unit (PACU) following surgery is found unresponsive with a respiratory rate of 6 breaths/min and pinpoint pupils. His oxygen saturation is 85%, and his blood pressure is 90/60 mmHg. He had received morphine for intraoperative analgesia. What is the most appropriate next step?

- A. Administer Naltrexone.
 - B. Administer Naloxone.
 - C. Administer Methadone.
 - D. Administer Buprenorphine.
-

7. Which of the following statements accurately describes a key difference between sufentanil and remifentanil?

- A. Remifentanil is more potent than Sufentanil
 - B. Remifentanil is metabolized by non-specific plasma esterases, while sufentanil is metabolized by the liver.
 - C. Sufentanil acts on μ -opioid receptors whereas Remifentanil acts on κ -opioid receptors.
 - D. Remifentanil has a longer duration of action compared to sufentanil.
-

8. A 50-year-old patient is taking phenelzine, a monoamine oxidase inhibitor (MAOI), for depression and is scheduled for elective surgery. When should

phenelzine be discontinued before the surgery to minimize the risk of interactions with pethidine?

- A. 24 hours before surgery
 - B. 48 hours before surgery
 - C. 1 week before surgery
 - D. 3 weeks before surgery
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	1
Question 5	3
Question 6	2
Question 7	2
Question 8	4

Solution for Question 1:

Correct Answer: A) Wong-Baker Scale

Explanation:

- The given image is of Wong Baker's Faces Scale
- Facial expressions are used to analyze the severity of pain.

Pain assessment methods

Visual Analog Scale	Commonly used scale in hospitals. Readings of scale range from 0-10 0 - no pain 10 – worst possible pain (Option B ruled out)
Wong Baker's Faces	Facial expressions are used to analyze the severity of pain. Readings of scale range from 0-10 0 – no pain 10 – worst pain

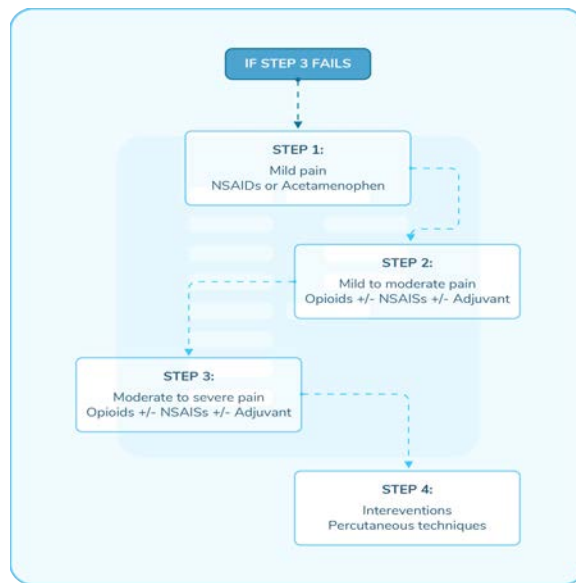
Children's Hospital of Eastern Ontario Pain Scale (CHEOPS)	Introduced by Canadian hospitals to assess pain. Observational measure of postoperative pain in children (Option C ruled out)
McGill Pain Questionnaire	Questionnaires are given to patients to assess the severity of pain. (Option D ruled out)

Solution for Question 2:

- The patient's lower back pain with a burning sensation radiating down her left leg suggests a neuropathic component.
- Since ibuprofen provided minimal relief
- Therefore an adjuvant therapy(Amitriptyline) is added.

WHO Step Ladder Management of Pain : It is WHO-given criteria for the management of severe chronic pain which is not subsiding even after conventional therapies. Management depends on the severity of pain in a 'ladder ' manner.

Step 1	Mild Pain	Non-Opioid Analgesics: Acetaminophen, NSAIDs Adjuvant therapies (e.g., corticosteroids, anticonvulsants, antidepressants)
Step 2	Moderate Pain	Fixed-Dose Combination Products: (e.g., acetaminophen + hydrocodone) Weak Opioids: (e.g., codeine, tramadol) Adjuvant agents as needed
Step 3	Severe Pain	Strong Opioids: (e.g., morphine, oxycodone) Individualized dosing Appropriate adjuvant therapies
Step 4	Complex and Uncontrolled Pain	Interventional techniques: (e.g., nerve blocks, neurolysis) Multimodal approaches for complex pain syndromes



Increasing ibuprofen (Option A): may not address the burning sensation.

Tramadol (Option C): isn't first-line for neuropathic features when amitriptyline is available.

Exercise therapy (Option D): is beneficial for general back pain, the specific symptoms make amitriptyline a more targeted approach in this case.

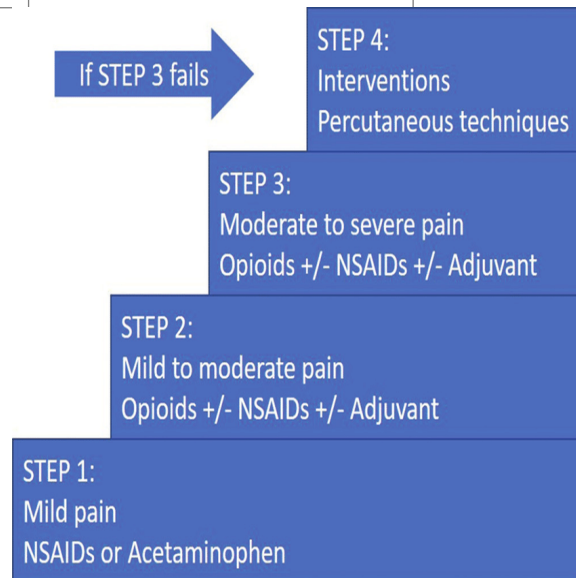
Solution for Question 3:

- In the case of a 20-year-old woman with metastatic breast carcinoma who is experiencing severe pain (8/10), the best medication to treat her pain would be PO morphine.
- Morphine is an opioid analgesic that is effective for moderate to severe pain

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IV paracetamol (Option A): less effective for severe pain compared to opioids.

PO Tramadol (Option C): less effective for severe pain compared to opioids.

IV Pethidine (Option D): Contraindicated in renal failure

Solution for Question 4:

- The patient's low heart rate (bradycardia) may indicate central nervous system effects from opioid analgesics, particularly morphine.
- Given that the patient is receiving Morphine, one of the potential adverse effects to monitor for is a rash

Opioids in Analgesia:

Major Opioids : Morphine, Fentanyl, Meperidine/ pethidine, Methadone.

Other Opioids: Codeine (Morphine derivatives with less affinity for μ receptors), Tramadol, Fentanyl, Buprenorphine,

Some Important Opioids:

Opioids	Effects
Morphine	Produces respiratory depression and cough suppression. May produce bradycardia and hypotension (Option D ruled out) Nausea and vomiting are common side effects Histamine release may lead to rash, itching, and bronchospasm (in susceptible patients) (Option A) Meiosis is common (Option B ruled out) Constipation (Option C ruled out)
Fentanyl	More potent than opioid Chest wall rigidity (fast and high dose administration) Fast onset and short duration
Sufentanil	It is 10 times more potent than fentanyl Fast onset and short duration
Remifentanyl	It is an ultra-short-acting opioid among all opioids. Fast onset and short duration Rapidly broken down by non-specific RBC and tissue esterase
Meperidine/ pethidine	Synthetic opioids acting on mu and kappa receptors having fewer side effects than other opioids Antimuscarinic property Accumulation in renal failure (seizures) Histamine release (bronchospasm) urticaria Contraindicated with Mao inhibitors causes cheese reaction Doc for prevention & treatment of postoperative shivering
Naloxone	Drug of choice for the treatment of opioid-induced respiratory depression Caution must be used in opioid addicts as giving naloxone may cause an acute withdrawal state with hypertension, pulmonary edema and cardiac arrhythmias
Tramadol	Inhibits neuronal reuptake of norepinephrine. Potentiates release of serotonin and causes descending inhibition of nociception

Solution for Question 5:

The classic morphine triad consists of respiratory depression, miosis (pinpoint pupils), and altered mental status (decreased level of consciousness). This triad is a hallmark of opioid overdose.

Morphine:

Mechanism of Action	Mu(μ) opioid receptor agonism: Primary action. Kappa(κ) and Delta(δ) receptor agonism : Lesser extent. Inhibition of pain transmission: Reduces the release of neurotransmitters like substance P and glutamate. Activation of descending inhibitory pathways: Increases pain modulation via descending inhibitory pathways.
Pharmacological Effects	Central Nervous System: Analgesia, sedation, and euphoria. Respiratory System: Suppression of respiratory centers, leading to respiratory depression. Gastrointestinal System: Decreases peristalsis, causing constipation. Reduces secretion of digestive juices. Cardiovascular System: Vasodilation and histamine release, leading to hypotension. Pupils: Causes miosis (pinpoint pupils).
Uses	Severe pain management: For chronic or acute severe pain, including post-operative and cancer pain. Myocardial infarction (MI): Relieves pain and anxiety; reduces cardiac workload. Pulmonary edema: Used to alleviate the sensation of dyspnea in acute cases. Cough suppressant: Morphine derivatives (like codeine) are used in some cough formulations.
Adverse Effects	Respiratory depression: Major life-threatening side effect. Constipation: Due to reduced GI motility. Nausea and vomiting: Activates the brain's chemoreceptor trigger zone (CTZ). Sedation and drowsiness: Common with therapeutic doses, worsened with higher doses. Tolerance and dependence: Long-term use leads to tolerance, requiring higher doses for the same effect.
Morphine Triad (Option C)	Respiratory depression: Most critical sign in overdose. Miosis : Pinpoint pupils are a hallmark feature of opioid overdose. Coma: Altered consciousness, leading to coma in severe overdose. Treatment: Naloxone(antidote)

Mydriasis, hypertension, tachycardia, agitation, hyperreflexia, and diaphoresis (Options A, B, and D) are not associated with morphine overdose.

Solution for Question 6:

The patient shows signs of morphine overdose (respiratory depression, pinpoint pupils, hypotension). Naloxone is the specific opioid antagonist that reverses these effects quickly.

Morphine (Opioid) toxicity:

For a non-tolerant adult, doses of 50 mg i.m. can produce serious toxicity.

The lethal dose of morphine is estimated at around 250 mg.

Clinical Manifestations:

- Stupor or coma
- Flaccid muscles
- Shallow and infrequent breathing
- Cyanosis
- Pinpoint pupils (miosis)
- Low blood pressure (hypotension)

Treatment:

Primary goal: Support respiration (positive pressure ventilation) and Maintain blood pressure (Intravenous fluids and vasoconstrictors).

Gastric lavage: With potassium permanganate can help remove any unabsorbed drug, even if morphine was injected (due to partitioning in the stomach)

Naloxone is the specific antidote:

- Administer 0.4-0.8 mg i.v., repeated every 2-3 minutes until respiration improves.
- Naloxone is effective because it acts rapidly and does not have agonistic actions or depress respiration.
- Due to its short duration of action, naloxone needs to be repeated every 1-4 hours depending on the patient's response.

Administer Naltrexone (Option A) : Is for the prevention of relapse once detoxified.

Administer Methadone (Option C) : Used in maintenance programs for opiate use disorder, not during acute poisoning. Also used in neonatal abstinence syndrome.

Administer Buprenorphine (Option D): Used in maintenance programs for opiate use disorder, not during acute poisoning.

Solution for Question 7:

- Remifentanyl is metabolized by non-specific plasma esterases, while sufentanil is metabolized by the liver.
- It is less potent than sufentanil, has a longer duration of action, and acts on μ -opioid receptors.

	Sufentanil	Remifentanyl
Mechanism of Action	μ -opioid receptor agonist	μ -opioid receptor agonist (Option C ruled out)
Onset of Action	Rapid onset (within 1-3 minutes)	Very rapid onset (within 1 minute)
Duration of Action	Intermediate duration (30-45 minutes)	Ultra-short duration (5-10 minutes) (Option D ruled out)
Metabolism (Option B)	Hepatic metabolism	Rapid hydrolysis by plasma esterases
Elimination	Renal excretion	Independent of renal or hepatic function
Potency	Approximately 5-10 times more potent than fentanyl (Option A ruled out)	Similar potency to fentanyl
Clinical Uses	Analgesia in major surgeries, anesthesia adjunct	Short procedures, anesthesia, ICU sedation
Adverse Effects	Respiratory depression, bradycardia, hypotension	Respiratory depression, bradycardia, hypotension
Special Considerations	Requires careful monitoring due to prolonged effects	Requires rapid titration due to short duration

Solution for Question 8:

MAOI (Mono Amine Oxidase Inhibitors) are stopped 3 weeks before surgery, to prevent the risk of developing serotonin syndrome.

MAO inhibitors inhibit the breakdown of serotonin, while pethidine may also have serotonergic effects, leading to excessive serotonin accumulation causing serotonin syndrome, and may present with symptoms such as agitation, confusion, muscle rigidity, fever, and autonomic instability.

Pethidine (Meperidine)	
Mechanism of Action	Interacts with μ -opioid receptors Metabolite normeperidine also has analgesic activity.
Pharmacological Effects	Analgesic potency : 1/10th of morphine but nearly equally effective Rapid onset after i.m. injection, shorter duration (2-4 hours) Less spasmogenic action (less miosis, constipation, urinary retention)
Uses	Occasionally used to control shivering post-anesthesia Shivering control during IV infusions Pain relief, but clinical use has declined
Adverse Effects	Dry mouth, blurred vision, tachycardia Less constipation, urinary retention, and miosis compared to morphine Tachycardia due to antimuscarinic action Tremors, mydriasis, hyperreflexia, delirium, myoclonus, convulsions (due to overdose) Prone to excitatory effects in renal failure patients
Interactions	Excitatory effects are exacerbated when combined with MAO inhibitors or SSRIs (leading to serotonin syndrome), Hence stopped 3 weeks before surgery. Drugs causing Serotonin syndrome: 1. Psychiatric drugs: MAO Inhibitors SSRIs SNRIs TCAs Vilazodone Vortioxetine Buspirone 2. Nonpsychiatric drugs: Tramadol, Ondansetron Triptans, Linezolid, MDMA, Dextromethorphan, Meperidine, St. John's wort

Previous Year Questions

1. Which of the following airway devices helps maintain Fio₂ of 24 - 60%, irrespective of the patient's breathing effort?

- A. Nasal cannula
 - B. Simple face mask
 - C. Venturi mask
 - D. Non-rebreathing mask
-

2. The color of the nitrous oxide cylinder is

- A. Completely Blue
 - B. Blue body with shoulder
 - C. Black body & white shoulder
 - D. Grey body & black white shoulder
-

3. Which device in conventional oxygen therapy will provide the highest FiO₂?

- A. Venturi mask
 - B. Hudson mask
 - C. Non rebreathing mask
 - D. High Flow Nasal Cannula
-

4. A 42-year-old male who had tested positive for COVID-19 and was isolating at home presents to the emergency room with difficulty breathing. The patient is conscious and responsive but has an increased breathing rate. An oxygen saturation (SpO₂) measurement reveals a reading of 86%. The patient is given oxygen through a face mask equipped with a reservoir bag at a flow rate of 10 L/min. Following an hour-long trial, the SpO₂ level improves to 88%. What is the subsequent course of action in managing this patient?

- A. Increase flow rate to 15 L/min
 - B. High – flow nasal cannula oxygen therapy
 - C. Endotracheal intubation
 - D. No specific intervention necessary
-

5. A 60 year old male patient was undergoing laparoscopic cholecystectomy. During the surgery, a sudden drop in EtCO₂ to 8 mmHg was noted. His SpO₂ became 90%, his blood pressure dropped to 80/50 mmHg, and peak airway pressure was 18 cm of H₂O. What is the diagnosis? CO₂ embolism Severe bronchospasm Pneumothorax Single bronchus intubation EtCO₂- End-tidal carbon dioxide

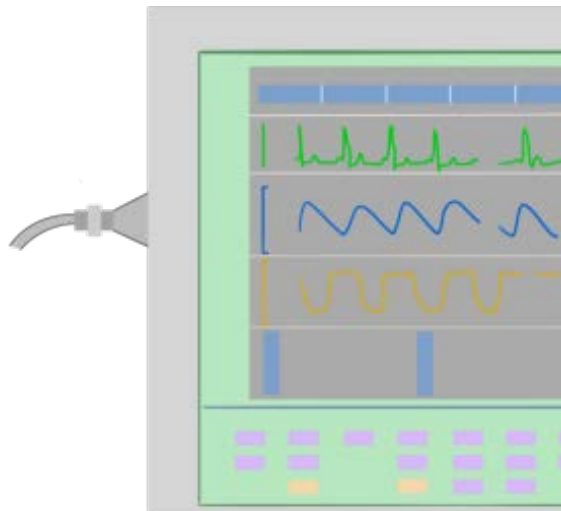
- A. 4 only
 - B. Both 3 and 4
 - C. 1 only
 - D. 2 only
-

6. What is the maximum rate of oxygen flow that can be maintained using a traditional oxygen mask?

- A. 5 L/min
- B. 15 L/min
- C. 10 L/min

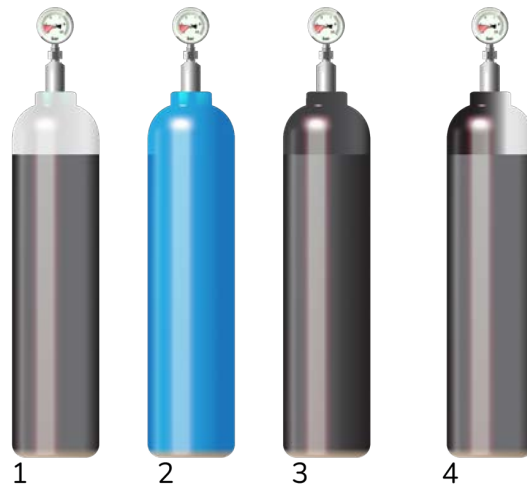
D. 20 L/min

7. What does the yellow-colored tracing on this device in the operating room signify?



- A. Capnography
 - B. Airway pressure
 - C. O₂ pressure in inhaled air
 - D. O₂ pressure in exhaled air
-

8. Which type of cylinder should be utilized to administer oxygen therapy to a COVID patient?



- A. 1
- B. 2
- C. 3
- D. 4

9. An 18-year-old boy is rushed to the ER after having choked on a bone while eating. He is responsive but unable to speak. What is to be done next?

- A. Bag and mask ventilation
- B. Endoscopic removal
- C. Deliver 5 back slaps and 5 chest compressions
- D. Perform Heimlich maneuver

10. Where can the devices in Murphy's eye be observed?

- A. Nasopharyngeal airway

- B. Laryngoscope
 - C. Bronchoscope
 - D. Endotracheal tube
-

11. A known case of asthmatic bronchitis was reported to the hospital with breathlessness, palpitations, and oxygen saturation of 88%. The hospital does not have a ventilator. All the following can be given to the patient, except?

- A. Hydrocortisone
 - B. Theophylline
 - C. Oxygen with oxygen mask, and reservoir
 - D. Leukotriene antagonist
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	2
Question 5	3
Question 6	3
Question 7	1
Question 8	1
Question 9	4
Question 10	4
Question 11	3

Solution for Question 1:

Correct Option C - Venturi mask:

- The venturi mask, or air entrainment mask, is a medical device that delivers a known oxygen concentration to patients on controlled oxygen therapy.
- A venturi flow meter is a type of differential pressure that generates a flow measurement by measuring the pressure difference at two locations in a pipe.
- Venturi mask used in maintaining FiO₂ of 24 - 60%.

Incorrect Options:

Option A - Nasal cannula: It can effectively provide only up to 4-6 L/min of supplemental oxygen, which equates to FiO_2 of approximately 24-44%.

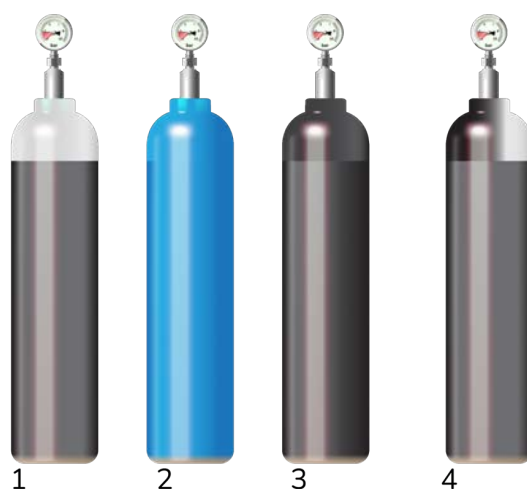
Option B - Simple face mask: It can deliver 35-60% oxygen with an appropriate flow rate of 6-10 L/min.

Option D - NRBM (Non-rebreathing mask): It has a flow rate of 15 L/min and can deliver up to 85% oxygen.

Solution for Question 2:

Correct Option A - Completely Blue:

- The color code for nitrous oxide (liquid) cylinders is complete blue (french blue).
- Anesthetic gasses are supplied in cylinders. The cylinders are made up of molybdenum steel which is an alloy.
- Nitrous oxide is stored under high pressure in cylinders. It cannot be directly given to patients as it can lead to the danger of barotrauma, and the flow rates at high pressure are difficult to control.



Cylinder	Colour
----------	--------

Oxygen (gas)	Black body and white shoulder.
Nitrous Oxide (gas)	Blue
Nitrogen (gas)	Black
Air (gas)	Gray/black body with black/white shoulder
Entonox (gas)	Blue body and white/blue shoulder

Incorrect Options:

Option B - Blue body with shoulder: Blue body with the blue shoulder is the color code for Entonox.

Option C - Black body & white shoulder: Black body with a white shoulder is the color code for oxygen (gas).

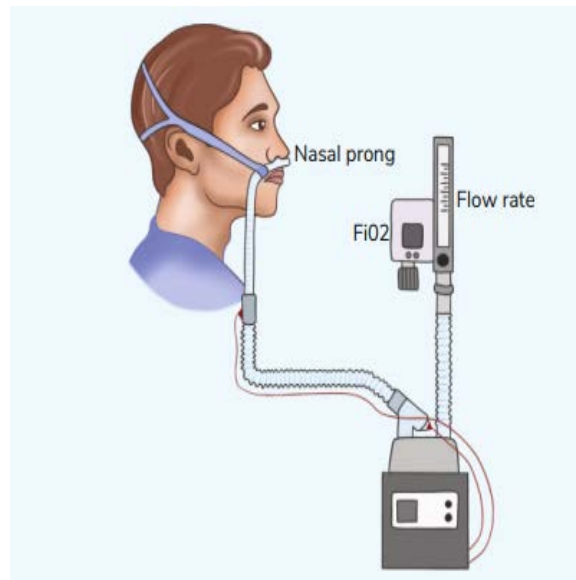
Option D - Grey body & black white sholuder: Grey body and black-white shoulder is the color code for nitrogen.

Solution for Question 3:

Correct Option D - High Flow Nasal Cannula:

- High-Flow Nasal Cannula (HFNC) therapy is a mode of respiratory support that delivers a high flow of heated and humidified oxygen-enriched air through nasal prongs to patients with respiratory distress.
- Components of HFNC: Flow generator: Provides gas flow rates up to 60 L/min. Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate. Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).
- Flow generator: Provides gas flow rates up to 60 L/min.
- Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate.
- Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).
- Flow generator: Provides gas flow rates up to 60 L/min.
- Air-oxygen blender: Achieves FiO₂ escalation from 21% to 100%, irrespective of flow rate.

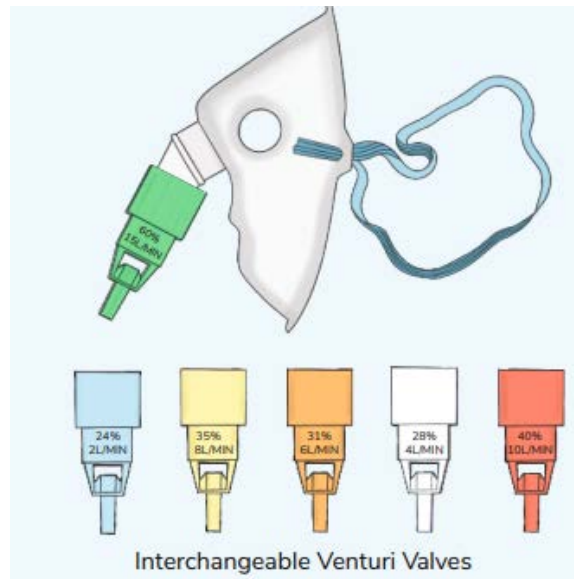
- Heated humidifier: Saturates the gas mixture at temperatures between 87.8°F to 98.6°F (31°C to 37°C).



Incorrect Options:

Option A - Venturi mask:

- A Venturi mask is a device that delivers a precise and fixed concentration of oxygen.
- It works by mixing a specific amount of oxygen with room air to achieve the desired FiO₂.
- The Venturi mask delivers a range of FiO₂ levels, depending on the specific Venturi adaptor used.
- While it can provide accurate oxygen concentrations, it typically delivers lower FiO₂ levels compared to a HFNC.



Option B - Hudson mask:

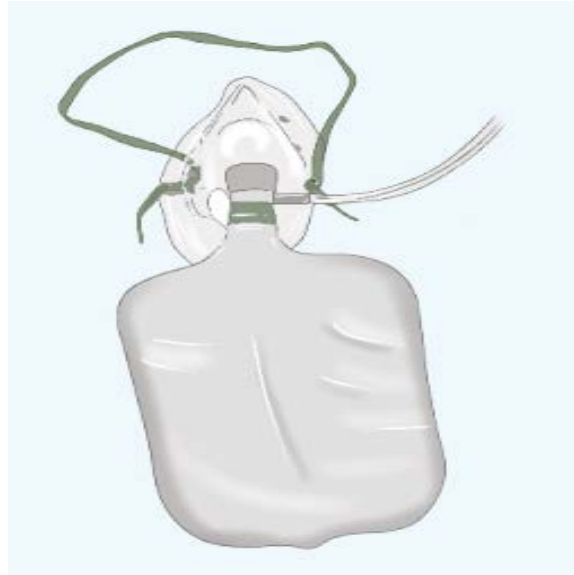
- The Hudson mask is a simple mask that covers the patient's nose and mouth.
- It is commonly used in conventional oxygen therapy.
- The FiO₂ delivered by a Hudson mask is influenced by the patient's breathing pattern, the fit of the mask, and the oxygen flow rate.
- It generally delivers a lower FiO₂ than a HFNC.



Option C- Non-rebreathing mask:

- Non-rebreather (NRB) masks deliver high concentrations of oxygen via a reservoir bag (600-1000 mL) and a one-way valve. The reservoir bag must be filled before use.

- Oxygen flow rates of 10-15 L/min achieve FiO₂ levels of 65%- 95%. NRB masks are used in conditions such as smoke inhalation, carbon monoxide poisoning, and severe hypoxia.
- Contraindications include facial trauma and claustrophobia, with caution advised in COPD patients. NRB masks are for short-term use only and limit patients' ability to eat, drink, or self-expectorate.



Solution for Question 4:

Correct Option B - High-flow nasal cannula oxygen therapy:

- In this scenario, the patient has developed breathing difficulty and has a low oxygen saturation (SpO₂) of 86%. Oxygen therapy is initiated using a face mask with a reservoir bag at a flow rate of 10 L/min. However, after one hour, the SpO₂ only improved to 88%. This suggests the patient is not adequately responding to the current oxygen delivery method.
- The next step in management would be to transition the patient to high-flow nasal cannula oxygen therapy. High-flow nasal cannula delivers a higher flow rate of oxygen and a higher fraction of inspired oxygen (FiO₂) than traditional nasal cannulas. It provides warmed and humidified oxygen at a flow rate of 30-60 L/min, which helps to improve oxygenation, reduce work of breathing, and provide positive end-expiratory pressure (PEEP).
- High-flow nasal cannula oxygen therapy is often the preferred choice in patients with respiratory distress who have not responded adequately to conventional oxygen delivery

methods. It offers several advantages, including improved oxygenation, better patient comfort, and the ability to provide PEEP to recruit collapsed alveoli and improve lung function.

Incorrect Options:

Option A - Increase flow rate to 15 L/min:

- Increasing the flow rate to 15 L/min may provide a small improvement in oxygenation. However, given that the patient's SpO₂ only improved from 86% to 88% after one hour of oxygen therapy at 10 L/min, a further increase in flow rate alone is unlikely to be sufficient.
- Switching to high-flow nasal cannula therapy would be a more appropriate next step.

Option C - Endotracheal intubation:

- Endotracheal intubation is a more invasive procedure typically reserved for patients with severe respiratory distress who cannot maintain adequate oxygenation or ventilation despite non-invasive interventions.
- In this case, the patient is alert and responsive, suggesting that their mental status is preserved and may not require immediate intubation. Therefore, endotracheal intubation is not the next step in management.

Option D - No specific intervention necessary:

- After an hour of oxygen therapy, the patient's SpO₂ is still below the target range, indicating that a specific intervention is required to improve oxygenation.
- Continuing without modifying the current treatment would not be appropriate and may lead to worsening respiratory distress and hypoxia.

Solution for Question 5:

Correct Option C - 1 Only:

- The sudden drop in EtCO₂ to 8 mmHg, along with the accompanying signs and symptoms, suggests a diagnosis of CO₂ embolism. Let's break down the information:
- CO₂ embolism: CO₂ embolism occurs when carbon dioxide gas enters the bloodstream, leading to a blockage of blood flow in the vessels. In this scenario, the sudden drop in EtCO₂ to 8 mmHg indicates a significant decrease in carbon dioxide elimination. CO₂ embolism can result from the inadvertent entry of carbon dioxide gas into the bloodstream during

laparoscopic surgery, usually due to a breach in the abdominal cavity. It can cause a range of symptoms, including hypotension (low blood pressure), hypoxemia (low oxygen saturation), and increased airway pressures.

Incorrect Options:

Option A - 4 only: Single bronchus intubation: Single bronchus intubation occurs when the endotracheal tube is mistakenly placed into one of the mainstem bronchi, leading to inadequate ventilation of the other lung. While this can result in decreased EtCO₂ in the affected lung, it would not cause a sudden drop in EtCO₂ to 8 mmHg. Additionally, single bronchus intubation alone would not explain the hypotension. Therefore, option B is incorrect.

Option B - Both 3 and 4: Pneumothorax: Pneumothorax refers to the presence of air in the pleural space, causing lung collapse. It can lead to decreased oxygenation, hypotension, and increased airway pressures. However, pneumothorax alone would not explain the sudden drop in EtCO₂. Therefore, option B (both 3 and 4) is incorrect.

Option D - 2 only: Severe bronchospasm: While severe bronchospasm can cause a decrease in EtCO₂ and hypoxemia, it typically does not cause a sudden drop in EtCO₂ to such a low value (8 mmHg). Additionally, severe bronchospasm is not associated with significant hypotension or increased airway pressures. Therefore, option D (2 only) is incorrect.

Solution for Question 6:

Correct Option C - 10 L/min:

- A conventional oxygen mask, also known as a simple face mask, is a device used for oxygen delivery to patients who require supplemental oxygen. The mask covers the nose and mouth, allowing the patient to breathe in a mixture of oxygen and room air.
- The maximum oxygen flow rate that can be delivered with a conventional oxygen mask is typically around 10 L/min. This flow rate ensures that a sufficient amount of oxygen is delivered to the patient, maintaining an adequate oxygen concentration or fraction of inspired oxygen (FiO₂). Higher flow rates may not significantly increase the delivered oxygen concentration and may lead to discomfort for the patient.

Incorrect Options:

Option A - 5 L/min: A flow rate of 5 L/min is commonly used for nasal cannula oxygen therapy, where the oxygen is delivered through small nasal prongs. However, this flow rate is generally

considered insufficient for a conventional oxygen mask, which requires a higher flow rate to deliver an appropriate FiO_2 .

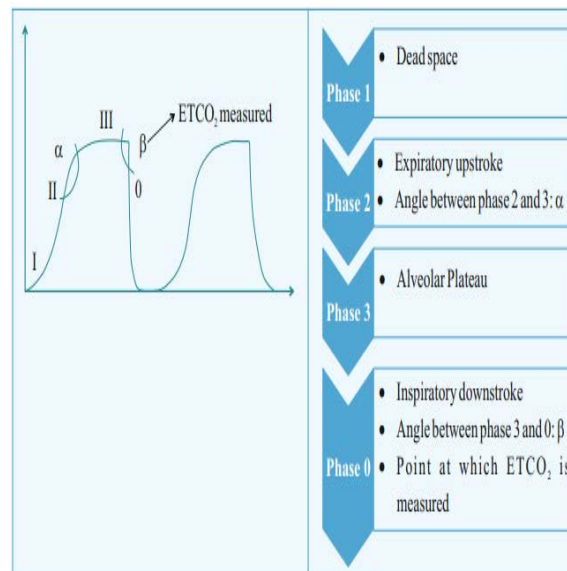
Option B - 15 L/min: A flow rate of 15 L/min is higher than what a conventional oxygen mask can effectively deliver. While it may be possible to achieve such a flow rate with certain masks or adaptations, it is not the standard maximum flow rate for a conventional oxygen mask.

Option D - 20 L/min: A flow rate of 20 L/min is higher than what a conventional oxygen mask can effectively deliver. Such a high flow rate would exceed the capacity of the mask and could lead to discomfort and difficulty for the patient.

Solution for Question 7:

Correct Option A - Capnography:

- Capnography is a monitoring technique that measures the concentration or partial pressure of carbon dioxide (CO_2) in respiratory gases, typically in exhaled air. The tracing in yellow color on the device indicates capnography monitoring.
- Capnography provides real-time information about the patient's respiratory status and ventilation effectiveness. It is commonly used during anesthesia and in critical care settings to assess the patient's ventilation, detect changes in the end-tidal CO_2 (EtCO_2) levels, and monitor the integrity of the airway and respiratory system.
- Capnography measures the amount of CO_2 present at the end of exhalation (EtCO_2) and displays it as a waveform and numerical value. The waveform typically appears as a continuous waveform with peaks and valleys representing the inhalation and exhalation phases of the respiratory cycle. The numerical value represents the EtCO_2 level, which indicates the patient's ventilation and CO_2 elimination.
- The best test to confirm the placement of the endotracheal tube.



Incorrect Options:

Option B - Airway pressure:

- Airway pressure monitoring typically involves measuring the pressure within the airway and respiratory system, such as through the use of an esophageal pressure monitor or ventilator pressure monitoring. It does not directly relate to the yellow tracing on the device, which indicates capnography.

Option C - O₂ pressure in inhaled air:

- Monitoring the oxygen (O₂) pressure in inhaled air is typically not depicted by a yellow tracing on a device. O₂ pressure in inhaled air can be measured using various methods, such as oxygen analyzers or oxygen concentration sensors, but it does not pertain to the yellow tracing seen on the device.

Option D - O₂ pressure in exhaled air:

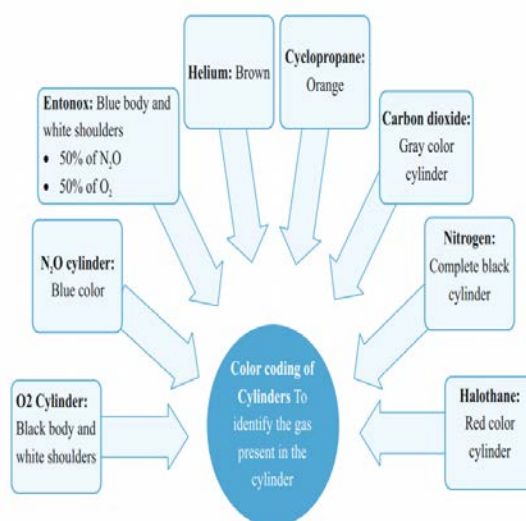
- Monitoring the O₂ pressure in exhaled air is not typically indicated by a yellow tracing on a device. The O₂ concentration in exhaled air can be measured using specific devices.

Solution for Question 8:

Correct Option A - 1:

- The oxygen cylinder has a white shoulder and a black body.

Cylinder Color Coding



Incorrect Options:

Option B, C, and D: Refer to the explanation of option A.

Solution for Question 9:

Correct Option D - Perform Heimlich maneuver:

- The patient is an adult who has choked on a bone while eating. The next best step is to perform the Heimlich maneuver by giving 5 abdominal thrusts.
- Heimlich Maneuver: The rescuer should stand or kneel behind the patient and wrap his arms around the patient's waist. With one hand, he makes a fist, and using the other hand, he drives the fist into the patient's abdomen with a rapid, forceful upward thrust to dislodge the foreign body. This is continued until the foreign body is expelled or if the patient becomes unresponsive.
- If the patient becomes unresponsive, immediately start CPR.
- In the case of an infant, 5 back slaps between the shoulder blades and 5 downward thrusts on the chest are given.



Incorrect Options:

Option A - Bag and mask ventilation: This intervention may be necessary if the patient becomes unresponsive and stops breathing, it is not the initial step in managing a conscious patient with an airway obstruction due to choking.

Option B - Endoscopic removal: Endoscopic removal may be necessary if other methods fail or if the obstruction is not cleared by basic interventions, it may not be the initial step in managing a conscious patient with a clear airway obstruction.

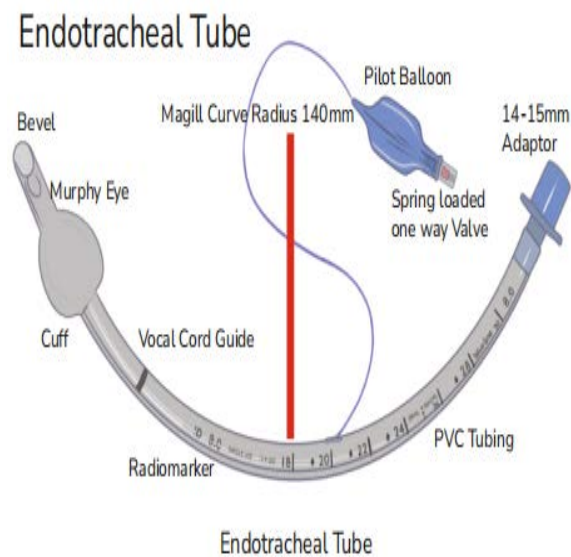
Option C - Deliver 5 back slaps and 5 chest compressions: This is done in the case of an infant and not an adult.

Solution for Question 10:

Correct Option D - Endotracheal tube:

- The Murphy's eye is a feature found in certain medical devices, particularly in endotracheal tubes (ETT) and tracheostomy tubes.
- It consists of an additional opening or eyelet located on the side of the tube near the distal end, opposite to the main opening through which the tube is inserted into the airway.

- The purpose of the Murphy's eye is to provide an alternative pathway for airflow in case the main opening becomes obstructed. This can happen due to factors such as mucosal swelling, secretions, or misplacement of the tube against the tracheal wall.
- When the tube is correctly positioned in the airway, the Murphy's eye remains outside the tracheal wall. However, if the tube tip is pressed against the tracheal wall, the Murphy's eye allows air to flow into the lungs, preventing complete obstruction.



Incorrect Options:

Option A - Nasopharyngeal airway: The nasopharyngeal airway is used in conscious patients with a compromised airway. It is made of soft silicon and has a single opening.

Option B - Laryngoscope: The laryngoscope is an instrument used to view the larynx or voice box.

Option C - Bronchoscope: The bronchoscope is a tube-like instrument with a lens to check for abnormalities in the lungs and bronchioles.

Solution for Question 11:

Correct Option C - Oxygen with oxygen mask, and reservoir:

- Bronchial asthma is an inflammatory chronic disease condition of airways that is characterized by paroxysmal wheezing, cough, and chest tightness.
- Asthma causes tightening of muscles around the airways and causes airways to narrow. Hence, when both acute bronchitis and asthma occur together, this situation is called asthmatic bronchitis.
- Currently, nasal cannula, non-breathing bag face mask reservoir, venturi masks, are used in the clinical settings as conventional oxygen therapies. However, non-invasive ventilation (NIV) in airways causes trauma, need for sedation, infections.
- Conventional oxygen therapy (COT) involves a simple face mask that delivers oxygen.
- COT is not able to provide sufficient respiratory support to the patients that are suffering from bronchial asthma. As it needs an extra attempt to breath and work for the breathing process to maintain oxygenation.

Incorrect Options:

Option A - Hydrocortisone: In case of bronchial asthma and hay fever, oral hydrocortisone is an effective therapy for symptomatic relief.

Option B - Theophylline: Theophylline is applied for the treatment of shortness of breath, wheezing, chest tightness caused by asthma, emphysema, chronic bronchitis and other lung diseases. Lungs passages are opened and relaxed to make breathing easier.

Option D - Leukotriene antagonist: Chronic asthma/bronchial asthma is treated with Leukotriene-receptor antagonists such as Montelukast and Zafirlukast.

Brain death

1. Which of the following is true about brain death?

- A. Complete loss of cerebral cortical function but preservation of brainstem reflexes.
 - B. Irreversible loss of all brain function, including the brainstem, with inability to breathe independently.
 - C. Persistent vegetative state where higher brain functions are lost, but spontaneous respiratory effort is present.
 - D. Reversible coma with absence of brainstem reflexes.
-

2. A 58-year-old woman is admitted with a traumatic brain injury and is being evaluated for brain death. Her core body temperature is 36.5°C, and her systolic blood pressure is 95 mm Hg despite volume resuscitation. Neuromuscular blocking agents were recently administered. Which of the following should be done before proceeding with the brain death diagnosis?

- A. Administer vasopressors to increase systolic blood pressure above 100 mm Hg and wait for drug clearance before proceeding.
 - B. Perform brainstem reflex testing immediately since her core temperature is adequate.
 - C. Reduce the temperature to less than 36°C to continue with the exam.
 - D. Start administering antidotes for neuromuscular blockade and proceed with the diagnosis.
-

3. A 30-year-old patient is unconscious following a massive intracerebral hemorrhage. Upon neurological examination, the pupils are fixed and dilated at 5 mm, and there is no response to cold caloric testing or corneal stimulation. However, when tracheal suctioning is performed, the patient exhibits a slight

cough. Which of the following conclusions is accurate?

- A. Brain death cannot be confirmed due to the presence of the cough reflex.
 - B. Brain death can be confirmed as the pupils are fixed and non-reactive.
 - C. Brain death can be confirmed, and the cough reflex is a false-positive result.
 - D. Cold caloric testing is unreliable in this setting.
-

4. A 40-year-old male is unconscious following a traumatic brain injury and is being evaluated for brain death. His blood pressure is stabilized with noradrenaline infusion to maintain a systolic BP of 110 mm Hg. The patient is preoxygenated for 10 minutes with 100% FiO₂, achieving a PaO₂ of 230 mm Hg. The ventilator is disconnected, and 100% FiO₂ is administered via insufflation tubing at 6 L/min near the carina. After 10 minutes, no respiratory movements are observed. Arterial blood gas analysis reveals a PaCO₂ of 65 mm Hg, and the SpO₂ remains at 98% throughout the test. What is the most appropriate conclusion regarding this apnea test?

- A. The apnea test is negative since the patient needs vasopressors.
 - B. SpO₂ should drop below 90% for a valid apnea test.
 - C. The apnea test is positive, indicating brain death.
 - D. The apnea test should be extended for another 5 minutes to confirm brain death.
-

5. A 45-year-old patient is suspected of having brain death after a severe head injury. The clinical evaluation is incomplete due to confounding factors. Which of the following ancillary tests is considered the gold standard for assessing cerebral blood flow in determining brain death?

- A. Computed Tomography (CT) Angiography
- B. Four-vessel angiography
- C. Magnetic Resonance Angiography (MRA)

D. Electroencephalogram (EEG)

6. A 30-year-old patient requires conscious sedation for a dental procedure. The anesthesiologist considers various agents for this purpose. Which of the following drugs is commonly preferred for inducing moderate sedation due to its favorable safety profile and effectiveness in achieving anxiolysis without significant cardiopulmonary complications?

- A. Propofol
 - B. Dexmedetomidine
 - C. Midazolam
 - D. Methohexital
-

7. Which of the following statements about non-operating room anesthesia (NORA) is false?

- A. NORA can be safely performed in outpatient clinics with trained personnel.
 - B. Airway management complications are less common in NORA than in operating rooms.
 - C. NORA requires specific sedation techniques based on procedure complexity.
 - D. Midazolam is the commonly used agent, owing to its safety profile.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	3
Question 5	2
Question 6	3
Question 7	2

Solution for Question 1:

Correct Answer: B) Irreversible loss of all brain function, including the brainstem, with inability to breathe independently.

Explanation:

Brain death is a permanent, irreversible, and complete loss of brain function (both cerebral and brain-stem functions).

Definition of brain death

- Brain death is a state of irreversible cessation of all cerebral and brain-stem function with preservation of cardiac activity and maintenance of respiratory and somatic function by artificial means. (Option D ruled out)
- It is the only type of brain damage recognized as morally, ethically, and legally equivalent to death.

Complete loss of cerebral cortical function but preservation of brainstem reflexes (Option A): In brain death, there is a total and irreversible loss of all brain functions, including both the cerebral cortex and the brainstem. Hence, this option is incorrect.

Persistent vegetative state where higher brain functions are lost, but spontaneous respiratory effort is present (Option C): This statement refers to a persistent vegetative state which is not the same as brain death. Cerebral cortical function is lost, but the brainstem remains functional, allowing for basic reflexes and the ability to breathe spontaneously. Thus, this is true but describes a different condition from brain death. Hence, this option is incorrect for this question.

Solution for Question 2:

- Before deciding to proceed with the diagnosis of brain death, several conditions must be evaluated and met.
- In this scenario, the patient's systolic blood pressure should be greater than 100 mm Hg to meet the criteria, and administering vasopressors would help stabilize the blood pressure to meet the required threshold.
- The team must wait for five half-lives of drug clearance (adjusted for the patient's renal and hepatic function) to ensure the drugs (neuromuscular agent) have cleared, as they can interfere with the neurologic exam by paralyzing the patient, falsely mimicking brain death.

Pre-evaluation for brain death:

Several conditions must be evaluated and met before deciding to proceed with the diagnosis of brain death.

- The cause of the coma must be identified.
- Any confounding factors, such as severe metabolic, endocrine, or acid-base abnormalities, should be ruled out and corrected.
- If drug intoxication or recent use of neuromuscular blocking agents is suspected, it is necessary to wait for five half-lives of drug clearance, adjusted for the patient's kidney and liver function. (Option B & D ruled out)
- The patient's core body temperature should be above 36°C. (Option C ruled out)
- The systolic blood pressure must be higher than 100 mm Hg, with vasopressors administered if required.

- A neurologic examination suggests widespread cortical destruction that is reflected by deep coma and unresponsiveness to all forms of simulations and should include testing for brainstem reflexes.

Solution for Question 3:

Brain death diagnosis requires the absence of all brainstem reflexes, including the cough reflex (mediated by the vagus nerve, CN X). Since the patient exhibits a cough response, this suggests that some brainstem function is still intact. Therefore, brain death cannot be confirmed.

Brainstem reflexes in a setting of brain death

In the clinical examination of a patient undergoing brain death evaluation, all brainstem reflexes must be tested and must be absent to confirm a diagnosis of brain death. (Option A)

Brainstem reflex	Cranial nerves	Findings in brain death patient.
Pupillary light reflex	II and III	Pupils should be fixed, mid-size/dilated (4-9 mm), and not reactive to light.
Oculovestibular reflex	III, VI, VIII	The oculovestibular reflex can be tested using the oculocephalic reflex (Doll's eyes) or caloric testing. In the absent oculocephalic reflex, the eyes move with the head. This test should only be done if the cervical spine is stable. Cold caloric testing involves irrigating one ear with ice water, and an absent reflex is indicated by no eye movement toward the irrigated side within 1 minute. The other ear is tested after 5 minutes.
Corneal reflex	V and VII	Loss of corneal reflex Performed using a cotton swab or drops of water/normal saline.
Gag reflex	IX	Loss of gag reflex Performed after stimulation of bilateral posterior pharyngeal membranes without elevation of the palate.

Cough reflex	X	Loss of cough reflex Confirmed after tracheal suctioning without reaction.
--------------	---	--

Brain death can be confirmed as the pupils are fixed and non-reactive (Option B): The diagnosis of brain death requires the absence of all brainstem reflexes, including the cough reflex, which is still present in this case. Therefore, brain death cannot be confirmed solely based on fixed, non-reactive pupils.

Brain death can be confirmed, and the cough reflex is a false-positive result:

This statement is false (Option C): The cough reflex is not a false-positive result but an actual brainstem reflex that indicates some remaining brainstem activity. The presence of this reflex means that brain death cannot be confirmed. This option is incorrect.

Cold caloric testing is unreliable in this setting: Cold caloric testing is a reliable method for assessing brainstem function. In this case, the absence of response to cold caloric testing suggests brainstem dysfunction. However, since the cough reflex is present, brain death cannot be confirmed. Hence, this option is incorrect.

Solution for Question 4:

Correct Answer: C) The apnea test is positive, indicating brain death.

Explanation:

In this case, all criteria for a positive apnea test are met, confirming brain death.

Apnea test for brain death

- If clinical evaluation and brain stem reflexes are inconclusive: Stimulate the brain stem with an apnea test.
- This test analyses whether the patient is brain dead or not by temporarily terminating the ventilator support or breathing.

Preparation of Apnea Test:

- Pre-evaluation for brain death should be done and corrected if any abnormalities are detected.
- Vasopressors should be adjusted to maintain a SBP ≥ 100 mmHg. (Option A ruled out)

Procedure of Apnea Test:

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Note: If PaCO₂ rise is <20 mmHg after 10 mins of providing apnea: Redo the apnea test or do ancillary tests. (Option D ruled out)

The apnea test has to be discontinued if:

- Presence of any respiratory movement (abdominal or chest excursions or brief gasp).
- SBP decreases to <90 mm Hg.
- Oxygen saturation is less than 85% for more than 30 seconds. (Option B ruled out)

Solution for Question 5:

Four-vessel angiography is considered the gold standard for assessing brain blood flow in the determination of brain death.

Types of Ancillary Tests

Category	Tests	Purpose
Traditional Tests	Four-vessel angiography	Gold standard to assess brain blood flow
	Radionuclide imaging	Evaluates cerebral blood flow
Emerging Tests	CT Angiography	Visualizes brain blood vessels
	CT Perfusion	Measures dynamic blood flow in the brain
	Magnetic Resonance Angiography (MRA)	Creates images of blood vessels using magnetic fields
	Xenon CT	Assesses cerebral blood flow with xenon gas
Functional & Electrodiagnostic Tests	Transcranial Doppler Testing	Measures blood flow velocity in cerebral arteries
	Electroencephalogram (EEG)	Records brain's electrical activity

	Evoked Potentials	Measures brain's electrical response to stimuli
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Objectives of Ancillary Testing

- Validate test accuracy compared to: Clinical diagnosis Four-vessel angiography Radionuclide imaging
- Clinical diagnosis
- Four-vessel angiography
- Radionuclide imaging
- Assess prevalence of: False negatives: Incorrectly not declaring brain death False positives: Incorrectly declaring brain death
- False negatives: Incorrectly not declaring brain death
- False positives: Incorrectly declaring brain death
- Evaluate the safety profiles of these tests.
- Clinical diagnosis
- Four-vessel angiography
- Radionuclide imaging
- False negatives: Incorrectly not declaring brain death
- False positives: Incorrectly declaring brain death

CT Angiography (Option A) is an emerging test used to visualize brain blood vessels but is not the gold standard.

MRA (Option C) uses magnetic fields to image blood vessels but is not the gold standard.

EEG (Option D) measures brain electrical activity and is used to evaluate brain function, but it is not a test for blood flow and not the gold standard for determining brain death.

Solution for Question 6:

Midazolam is commonly used for moderate sedation in dental and ambulatory procedures, with a favorable safety profile and minimal cardiopulmonary effects.

Conscious sedation: known as procedural sedation and/or moderate sedation, is a procedure to relieve anxiety and depress the level of consciousness in patients, before minor procedures.

Drug of Choice for Conscious Sedation:

Midazolam :

- Commonly used for moderate sedation in dental and ambulatory procedures
- Provides anxiolysis and sedation with minimal cardiopulmonary complications
- Has a strong safety profile

Other drugs:

Drugs	Disadvantages
Propofol	May cause respiratory depression Decreases BP/CVS depression (Option A ruled)
Etomidate E–Endocrine gland	Suppression of the adrenal gland (Endocrine) Myoclonus
Ketamine Kap on his Head	Emergence delirium (because increases cerebral blood flow) Excessive bronchial secretions and salivation Hypertension Increase ICP
Dexmedetomidine	Hypotension Bradycardia (Option C ruled out)
Methohexital	Suppression of respiration (Option D ruled out)

Solution for Question 7:

Airway management complications are more common in NORA than in operating rooms.

Non-Operating Room Anesthesia (NORA)

NORA involves sedation and anesthesia for procedures outside traditional operating rooms, increasingly utilized in settings like radiology suites and outpatient clinics. (Option A ruled out)

- Growing Demand Increased patient visits Complex procedures Greater emphasis on pain management Anesthesia providers managing sedation in diverse environments
- Increased patient visits
- Complex procedures
- Greater emphasis on pain management
- Anesthesia providers managing sedation in diverse environments

- Challenges Inadequate Support : Limited staff and resources compared to operating rooms
- Airway Management Risks : Higher incidence of airway complications requiring vigilant monitoring (Option B)
- Inadequate Support : Limited staff and resources compared to operating rooms
- Airway Management Risks : Higher incidence of airway complications requiring vigilant monitoring (Option B)
- Sedation Techniques Selection based on: (Option C ruled out) Procedure type Expected pain levels Patient characteristics (age, comorbidities) Required sedation depth
- Selection based on: (Option C ruled out) Procedure type Expected pain levels Patient characteristics (age, comorbidities) Required sedation depth
- Procedure type
- Expected pain levels
- Patient characteristics (age, comorbidities)
- Required sedation depth
- Common Agents Used Midazolam : Rapid onset, safety profile (Option D ruled out)
- Propofol : Quick recovery, necessitates careful airway monitoring
- Midazolam : Rapid onset, safety profile (Option D ruled out)
- Propofol : Quick recovery, necessitates careful airway monitoring
- Safety Considerations Major complications are rare when managed by trained personnel
- Ongoing evaluation of complication rates and mortality as procedures grow more complex
- Major complications are rare when managed by trained personnel
- Ongoing evaluation of complication rates and mortality as procedures grow more complex
- Future Trends Continued expansion of NORA practices Development of standards to improve safety and efficacy
- Continued expansion of NORA practices
- Development of standards to improve safety and efficacy
- Increased patient visits
- Complex procedures
- Greater emphasis on pain management
- Anesthesia providers managing sedation in diverse environments
- Inadequate Support : Limited staff and resources compared to operating rooms
- Airway Management Risks : Higher incidence of airway complications requiring vigilant monitoring (Option B)
- Selection based on: (Option C ruled out) Procedure type Expected pain levels Patient characteristics (age, comorbidities) Required sedation depth

- Procedure type
 - Expected pain levels
 - Patient characteristics (age, comorbidities)
 - Required sedation depth
 - Procedure type
 - Expected pain levels
 - Patient characteristics (age, comorbidities)
 - Required sedation depth
 - Midazolam : Rapid onset, safety profile (Option D ruled out)
 - Propofol : Quick recovery, necessitates careful airway monitoring
 - Major complications are rare when managed by trained personnel
 - Ongoing evaluation of complication rates and mortality as procedures grow more complex
 - Continued expansion of NORA practices
 - Development of standards to improve safety and efficacy
-

Nerve blocks

1. A 30-year-old male presents to the clinic with trauma to his lower jaw. He needs stitches on his lower lip near the incisors. Which nerve block will provide adequate anesthesia for suturing the lower lip and nearby gingiva?

- A. Infraorbital nerve block
 - B. Mental nerve block
 - C. Inferior alveolar nerve block
 - D. Supraorbital nerve block
-

2. In preparing for a superficial cervical plexus block, where should the anesthesiologist insert the needle?

- A. Along the anterior border of the sternocleidomastoid
 - B. Along the posterior border of the sternocleidomastoid, halfway between the mastoid and clavicle
 - C. At the midpoint of the clavicle
 - D. Lateral to the trapezius muscle
-

3. During the procedure as given in the image, the anesthesiologist uses a 25-gauge needle to inject the local anesthetic. If the needle inadvertently penetrates too deeply and damages nearby structures, which of the following complications might arise?



- A. Carotid artery puncture
 - B. Hemorrhage from the internal jugular vein
 - C. Mandibular nerve blockade leading to loss of mastication
 - D. Damage to the facial nerve causing unilateral weakness
-

4. A 35-year-old male presents with a laceration to his pinna after a sports injury. He expresses significant pain and anxiety about the repair procedure. The physician decides to perform a nerve block to ensure adequate analgesia during the repair. Which of the following is the preferred nerve block for pinna repair?

- A. Auriculotemporal nerve block
 - B. Greater auricular nerve block
 - C. Trigeminal nerve block
 - D. Facial nerve block
-

5. A 40-year-old female is scheduled for endoscopic surgery involving the middle turbinate and frontal recess due to chronic sinusitis. Which of the following nerve blocks is preferred for providing analgesia during this surgery?

- A. Maxillary nerve block
 - B. Anterior ethmoidal block
 - C. Infraorbital nerve block
 - D. Sphenopalatine ganglion block
-

6. Match the following: 1. Supraclavicular block A. Blocks the superior and middle trunk of the brachial plexus. 2. Interscalene block B. Block terminal nerves. 3. Infraclavicular block C. Blocks distal trunk-proximal division of the brachial plexus. 4. Axillary block D. Blocks at the level of cords of the brachial plexus.

1. Supraclavicular block	A. Blocks the superior and middle trunk
2. Interscalene block	B. Block terminal nerves.
3. Infraclavicular block	C. Blocks distal trunk-proximal division of the brachial plexus.
4. Axillary block	D. Blocks at the level of cords of the brachial plexus.

- A. 1A,2D,3C,4B
 - B. 1C,2A,3D,4B
 - C. 1C,2A,3B,4D
 - D. 1B,2C,3A,4D
-

7. A 60-year-old patient is scheduled for forearm surgery. The anesthesiologist seeks to provide an effective brachial plexus block approach with the least risk of pneumothorax and uses the block as shown below. Identify the type of brachial plexus block approach depicted in the image?



- A. Interscalene block
- B. Infraclavicular block
- C. Supraclavicular block
- D. Axillary block

8. A 45-year-old woman suffers from severe chronic pain in the upper extremities that has not been relieved with pain medications. She doesn't want to keep taking medications and wants a permanent solution. A stellate ganglion block is advised as the treatment modality. Which of the following is false about this method?

- A. The procedure carries a risk of pneumothorax
- B. Used for the treatment of PTSD and severe menopausal vasomotor symptoms
- C. A successful procedure is noted by the onset of Horner's syndrome
- D. C7 vertebra is preferred over C6 vertebra as the injection site

9. Which of the following statements is true regarding nerve blocks done at the elbow and wrist?

- A. The median nerve is accessed lateral to the brachial artery
 - B. The ulnar artery is seen at the medial side of the ulnar nerve in the wrist
 - C. Radial nerve is approached from a posterior aspect of the elbow
 - D. Ulnar nerve can be blocked at the antecubital fossa
-

10. A patient is being evaluated and prepared to undergo a mastectomy. Which regional nerve block would be the best choice?

- A. Rectus sheath block
 - B. Paravertebral block
 - C. Transversus abdominis plane block
 - D. Intercostal block
-

11. A 34-year-old male patient presents with dyspnea and pleuritic chest pain. A chest X-ray confirms bilateral pleural effusion. The physician decides to perform a thoracentesis to drain the fluid. Which of the following nerve blocks is preferred to provide analgesia during the thoracentesis procedure?

- A. Suprascapular nerve block
 - B. Intercostal nerve block
 - C. Interscalene block
 - D. Thoracic sympathetic block
-

12. A 45-year-old male is scheduled for an elective inguinal hernia repair. The anesthesiologist decides to perform a Transversus Abdominis Plane (TAP) block to

provide effective analgesia during and after the procedure. Which of the following nerves is not blocked by the TAP block?

- A. Subcostal nerve
 - B. Ilioinguinal nerve
 - C. Genitofemoral nerve
 - D. Iliohypogastric nerve
-

13. A 60-year-old patient with pancreatic cancer is in severe pain and hence is advised for celiac plexus block. Which of the following complications be rarely seen in this procedure?

- A. Postural hypotension
 - B. Back pain
 - C. Systemic local toxicity
 - D. Sexual dysfunction
-

14. A 65-year-old patient is scheduled for a knee arthroscopy. The anesthesiologist plans to use regional anesthesia for postoperative analgesia. Which block should the anesthesiologist consider for effective postoperative pain relief in this procedure?

- A. Popliteal nerve block
 - B. Femoral nerve block
 - C. Obturator nerve block
 - D. Lateral femoral cutaneous nerve block
-

15. A 70-year-old patient is scheduled for hip replacement surgery. Which nerve block should the anesthesiologist select to provide effective anesthesia for this procedure?

- A. Lumbar plexus block
 - B. Sacral plexus block
 - C. Sciatic nerve block
 - D. Ilioinguinal nerve block
-

16. A 50-year-old woman is scheduled for foot surgery to remove a neuroma. The anesthesiologist performs an ankle block and aims to provide sensation to the dorsum of the foot, excluding the first interdigital cleft. Which nerve should be blocked?

- A. Sural nerve
 - B. Tibial nerve
 - C. Deep peroneal nerve
 - D. Superficial peroneal nerve
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	2
Question 5	2
Question 6	2
Question 7	4
Question 8	4
Question 9	3
Question 10	2
Question 11	2
Question 12	3
Question 13	3
Question 14	2
Question 15	1
Question 16	4

Solution for Question 1:

Correct answer: B) Mental nerve block

Explanation:

The mental nerve provides sensation to the lower lip, labial mucosa, and gingiva near the incisors and canines , making it the correct choice for anesthesia in this case.

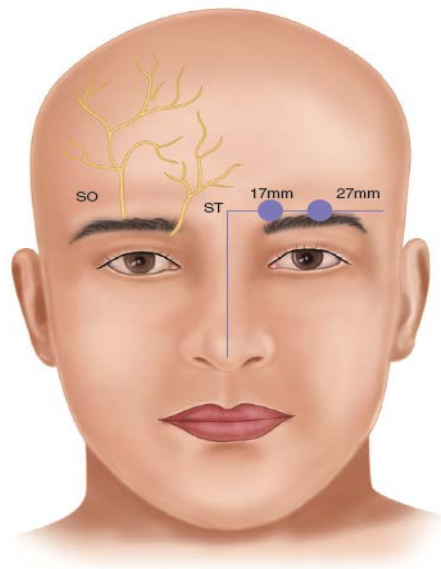
Face blocks

Most of the facial region is supplied by the Trigeminal nerve and its branches.

Anatomy:

- Cranial nerve V arises from the brainstem and forms the Gasserian (sensory) ganglion in Meckel's cave .
- The three nerve divisions (ophthalmic, maxillary, mandibular) exit the skull separately.

Nerve Block	Anatomy	Anesthesia Distribution	Technique
Supraorbital & Supratrochlear (Ophthalmic Nerve) (Option D)	Supraorbital nerve exits the supraorbital foramen on the superior surface of the orbital rim. Supratrochlear exits medially at the orbital rim .	Anesthetizes the entire forehead up to the scalp's vertex.	Inject anesthetic subcutaneously above the eyebrow, from midline to the lateral orbit.
Infraorbital (Maxillary Nerve) (Option A)	Exits infraorbital foramen below the optic rim near the first maxillary premolar.	Provides sensation loss to lower eyelid, medial cheek, side of nose, upper lip.	Retract mucosa, insert needle above the first premolar, and inject 3-5 ml between cheek and gingiva.
Mental (Mandibular Nerve) (Option B)	Mental nerve exits the mental foramen, a branch of the inferior alveolar nerve.	Anesthetizes lower lip, labial mucosa, and gingiva near incisors and canines.	Retract lip, insert needle near first/second premolars , and inject 3-5 ml in the mucosal fold
Inferior Alveolar (Mandibular Nerve) (Option C)	Arises from mandibular division of trigeminal (V3); landmarks: Mandibular sulcus, coronoid notch, pterygomandibular raphe.	Anesthetizes all lower teeth, gums, anterior labial and lingual areas.	Insert needle into pterygomandibular raphe , tap bone, withdraw slightly, and inject 2-3 ml.



Complications

- Risks include intravascular or subarachnoid injection, Horner syndrome, and motor block of jaw muscles.
- Maxillary nerve blocks carry a higher risk of hemorrhage.
- Facial nerve may be unintentionally blocked during mandibular nerve blocks.

Solution for Question 2:

Correct answer: B) Along the posterior border of the sternocleidomastoid, halfway between the mastoid and clavicle.

Explanation:

The correct insertion point for a superficial cervical plexus block is along the posterior border of the sternocleidomastoid muscle, halfway between the mastoid process and the clavicle. This approach ensures effective sensory nerve blockade while minimizing complications.

Neck block		
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Anatomy	The cervical plexus arises from the anterior rami of C1–C4 , located lateral to the vertebral transverse processes. It has four sensory branches (lesser occipital, greater auricular, transverse cervical, supraclavicular) and three motor branches (phrenic, ansa cervicalis, and posterior neck muscles).	
Indications	Used for U/L neck surgeries (e.g., carotid endarterectomy) Supplement for clavicle/shoulder anesthesia.	
Types	Superficial Block	Targets cutaneous branches, providing sensory anesthesia to the neck, jaw, and shoulder. A needle is inserted along the posterior border of the sternocleidomastoid , halfway between the mastoid and clavicle.
Deep Block	Targets nerve roots at the vertebral foramina for deeper neck anesthesia. Ultrasound-guided, targeting the nerve roots (C2–C4) with 5 mL of anesthetic at each level.	
Risks	Both types can cause hemidiaphragmatic paralysis . The deep block poses additional risks, including accidental vertebral artery or spinal canal injection.	

Along the Anterior Border of the Sternocleidomastoid (Option A) : This location is inappropriate as it does not effectively target the cervical plexus branches and may increase the risk of vascular injury.

At the Midpoint of the Clavicle (Option C) : This area is too inferior and does not align with the cervical plexus branches, making it ineffective for the intended anesthesia.

Lateral to the Trapezius Muscle (Option D) : This location is irrelevant for the cervical plexus block and may lead to inadequate anesthesia.

Solution for Question 3:

Correct answer: D) Damage to the facial nerve causing unilateral weakness

Explanation:

The procedure shown in the image is auriculotemporal nerve block , facial nerve runs superficially in the parotid gland and innervates the muscles of facial expression. If the needle penetrates too deeply, it may damage the facial nerve which leads to U/L LMN facial paralysis/weakness.

Scalp blocks	
Occipital nerve block	Greater Occipital Nerve: Arises from the dorsal primary rami of the C2 and C3 spinal nerves. Blocked approximately 3 cm lateral to the occipital prominence at the level of the superior nuchal line . The nerve is located medial to the occipital artery. Lesser Occipital Nerve: Originates from the ventral rami of the same spinal roots. Blocked 2 to 3 cm more laterally along the nuchal ridge.
Zygomaticotemporal Nerve Block	Insert the needle at outer edge of supraorbital margin at a 90° angle until bone contact is made, then deposit the LA for deep injection. Redirect the needle laterally to Zygomatic Arch and deposit LA along this trajectory.
Auriculotemporal Nerve Block	Given at 1 to 1.5 cm anterior to the ear at the tragus, above the temporomandibular joint. Palpate the superficial temporal artery to avoid intra-arterial injection. Keep the injection superficial to prevent facial nerve block.

Carotid Artery Puncture (Option A) : Less likely as the procedure is performed anterior to the ear, away from the carotid artery.

Hemorrhage from the internal Jugular Vein (Option B) : Less likely as the procedure is performed anterior to the ear, away from the internal jugular vein.

Mandibular Nerve Blockade Leading to Loss of Mastication (Option C) : This is not a direct result of an auriculotemporal nerve block, as it primarily provides sensation rather than motor function to the muscles of mastication.

Solution for Question 4:

Correct Answer: B) Greater auricular nerve block

Explanation: Greater auricular nerve block is preferred for pinna repair because it provides effective analgesia to the ear and surrounding areas, reducing the need for perioperative opioids.

Greater Auricular Nerve Block:

- Greater auricular nerve is a sensory nerve branch of the superficial cervical plexus (C3).
- Course at the level of the cricoid cartilage and follows the posterior border of the clavicular head of the sternocleidomastoid muscle.

Auricular Nerve Block" data-author="A. Thallaj, P. Marhofer, B. Moriggl, B. M. Delvi, S. C. Kettner, M. Almajed" data-hash="420" data-license="NA" data-source="https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/full/10.1111/j.1365-2044.2010.06443.x" data-tags="December2024" height="240" src="https://image.prepladder.com/notes/CRnQawYMctvHhRABeQhL1734949891.png" width="500" />

- Supplies sensory innervation to: Inferior part of the ear. Skin over the mastoid process. Angle of the mandible. Segments of the external ear: Tail of the helix, antitragus, lobule of the auricle, with variable supply to the spine of the helix, tragus, and concha.
- Inferior part of the ear.
- Skin over the mastoid process.
- Angle of the mandible.
- Segments of the external ear: Tail of the helix, antitragus, lobule of the auricle, with variable supply to the spine of the helix, tragus, and concha.
- Clinical Use: Provides analgesia for procedures: Otoplasty Tympanomastoidectomy Reduces the need for perioperative opioid use.
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- Otoplasty
- Tympanomastoidectomy
- Reduces the need for perioperative opioid use.
- Technique: Location for Block: At the level of the cricoid cartilage (C6). Local anesthetic: Injected superficially along the belly of the muscle, approximately 5-6 cm below the ear.
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- Local anesthetic: Injected superficially along the belly of the muscle, approximately 5-6 cm below the ear.
- Inferior part of the ear.
- Skin over the mastoid process.
- Angle of the mandible.

- Segments of the external ear: Tail of the helix, antitragus, lobule of the auricle, with variable supply to the spine of the helix, tragus, and concha.
- Provides analgesia for procedures: Otoplasty Tympanomastoidectomy
- Otoplasty
- Tympanomastoidectomy
- Reduces the need for perioperative opioid use.
- Otoplasty
- Tympanomastoidectomy
- Location for Block: At the level of the cricoid cartilage (C6).
- Local anesthetic: Injected superficially along the belly of the muscle, approximately 5-6 cm below the ear.

Auricular Nerve Block location" data-author=".Ritchie MK, Wilson CA, Grose BW, Ranganathan P, Howell SM, Ellison MB " data-hash="421" data-license="CC BY NC 3.0" data-source="https://openi.nlm.nih.gov/detailedresult?img=PMC4943106_cp-2016-2-856-g002&req;=4" data-tags="December2024" height="340" src="https://image.prepladder.com/notes/prRrfq QZgaMvjnkqrE6l1734950090.png" width="500" />

- Complications: Intravascular Injection due to proximity to carotid artery and jugular veins. Deep needle placement may result in: Phrenic nerve block. Cervical plexus block. Horner's syndrome.
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- Deep needle placement may result in: Phrenic nerve block. Cervical plexus block. Horner's syndrome.
- Phrenic nerve block.
- Cervical plexus block.
- Horner's syndrome.
- Intravascular Injection due to proximity to carotid artery and jugular veins.
- Deep needle placement may result in: Phrenic nerve block. Cervical plexus block. Horner's syndrome.
- Phrenic nerve block.
- Cervical plexus block.
- Horner's syndrome.
- Phrenic nerve block.
- Cervical plexus block.
- Horner's syndrome.

Auriculotemporal nerve block (Option A) is not preferred for pinna repair because it primarily targets the temporomandibular joint (TMJ) and surrounding structures.

Trigeminal nerve block (Option C) is not indicated for pinna repair because it specifically targets the treatment of trigeminal neuralgia rather than providing analgesia for ear injuries.

Facial nerve block (Option D) is not preferred for pinna repair because it primarily targets analgesia for facial procedures.

Solution for Question 5:

Correct answer: Anterior ethmoidal block

Explanation: Anterior ethmoidal block is specifically indicated for providing analgesia during surgeries involving the middle turbinate and frontal recess.

- Anterior ethmoidal nerve is a branch of the ophthalmic nerve .
- It has internal and external nasal branches that innervates: Anterior part of the septum. Lateral wall of the nasal cavity. Nasal bone. Skin to the tip of the nose.
- Anterior part of the septum.
- Lateral wall of the nasal cavity.
- Nasal bone.
- Skin to the tip of the nose.
- Anterior part of the septum.
- Lateral wall of the nasal cavity.
- Nasal bone.
- Skin to the tip of the nose.

Ethmoidal Nerve Block" data-author="Ali Kavyani Ali Manafi" data-hash="422" data-license="CC BY 4.0" data-source="https://www.researchgate.net/figure/Anterior-ethmoidal-nerve-from-Frank-H-Netter-Atlas-of-human-anatomy-6-th-edition_fig2_324510514" data-tags="December2024" height="500" src="https://image.prepladder.com/notes/hDAIYvOIXViEcIW6Z5B1734950581.png" width="500" />

- Located medially from the halfway point between the posterior palpebral fold and the eyebrow, at a depth of 1.5 cm.
- Technique

- Injection Site: Lateral nasal wall, anterior to the root of the middle turbinate. Needle insertion 9 mm superior to the superior border of the medial canthal tendon to prevent injury to the lacrimal sac.
- Lateral nasal wall, anterior to the root of the middle turbinate.
- Needle insertion 9 mm superior to the superior border of the medial canthal tendon to prevent injury to the lacrimal sac.
- Lateral nasal wall, anterior to the root of the middle turbinate.
- Needle insertion 9 mm superior to the superior border of the medial canthal tendon to prevent injury to the lacrimal sac.

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- Local anesthetic: Approximately 0.5 ml of 1% lidocaine or 0.5% ropivacaine with adrenaline.
- Administration: Injected submucosally.

ethmoidal block, Endoscopic view" data-author="Nagalingeswaran Ahilasamy,1 Veerasigamani Narendrakumar,2 Rajendran Dinesh Kumar,corresponding author3 and N Sai Sarath3" data-hash="424" data-license="NA" data-source="https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9741691/" data-tags="December2024" height="485" src="https://image.prepladder.com/notes/UnXTLjECd0e8GlvGZDye1734951272.png" width="500" />

- Indication: Used for surgeries involving the middle turbinate and frontal recess. Reduces postoperative agitation during emergence in nasal surgeries with nasal packs.
- Used for surgeries involving the middle turbinate and frontal recess.
- Reduces postoperative agitation during emergence in nasal surgeries with nasal packs.
- Maxillary nerve block (Option A) provides anesthesia primarily to the hemi-maxilla and is suited for dental procedures, not nasal cavity interventions. Infraorbital nerve block (Option C) primarily anesthetizes the anterior part of the maxilla, making it suitable for dental and cosmetic procedures but not effectively targeting the nasal cavity or surrounding structures. Sphenopalatine ganglion block (Option D) is used in the treatment of acute migraine headaches, acute cluster headaches, and facial neuralgias, rather than providing analgesia for nasal structures.
- Used for surgeries involving the middle turbinate and frontal recess.
- Reduces postoperative agitation during emergence in nasal surgeries with nasal packs.

Maxillary nerve block (Option A) provides anesthesia primarily to the hemi-maxilla and is suited for dental procedures, not nasal cavity interventions.

Infraorbital nerve block (Option C) primarily anesthetizes the anterior part of the maxilla, making it suitable for dental and cosmetic procedures but not effectively targeting the nasal cavity or surrounding structures.

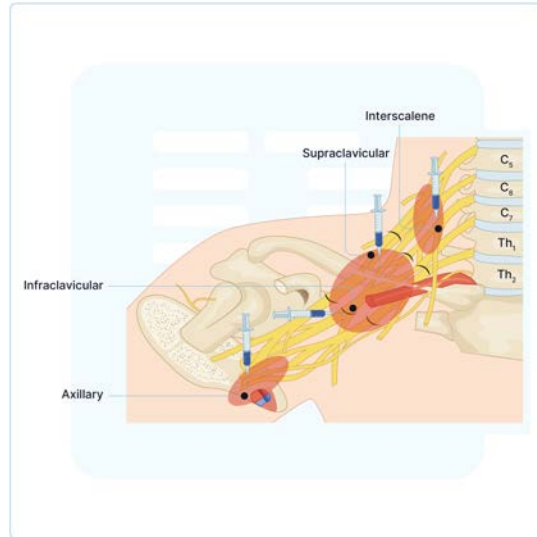
Sphenopalatine ganglion block (Option D) is used in the treatment of acute migraine headaches, acute cluster headaches, and facial neuralgias, rather than providing analgesia for nasal structures.

Solution for Question 6:

Correct Answer: B) 1C,2A,3D,4B

Explanation:

Brachial Blocks	
Supraclavicular block	Blocks distal trunk-proximal division of the brachial plexus. For surgeries distal to the mid-humerus.
Interscalene block	Blocks the superior and middle trunk of the brachial plexus. For shoulder and proximal humerus surgeries. Contraindicated in patients with respiratory insufficiency due to the high risk of ipsilateral phrenic nerve block and diaphragmatic hemiparesis. Interscalene block covers most of the brachial plexus, sparing the ulnar (C8-T1) nerve.
Infraclavicular block	Blocks at the level of cords of the brachial plexus. For surgeries distal to the mid-humerus.
Axillary block	Block terminal nerves. For surgeries distal to the mid-humerus. Axillary, musculocutaneous and medial brachial cutaneous nerve branches of brachial plexus are usually spared from blockade. No risk blockade of the phrenic nerve, nor have the potential to cause pneumothorax.



Solution for Question 7:

Correct Answer: D) Axillary block

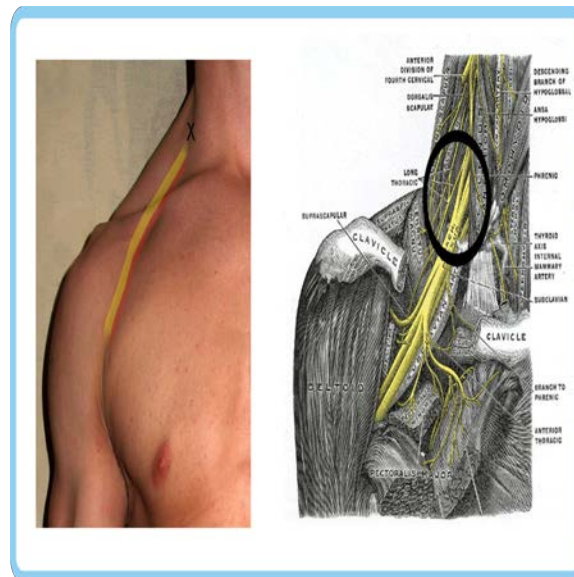
Explanation: The anesthesiologist chooses the axillary block as it primarily targets the terminal nerves and has a lower risk of serious complications compared to other approaches. This method provides greater safety, especially when using ultrasound techniques to guide the placement.

Approaches of Brachial Plexus Block:

1) Interscalene Blocks: (Option A)

- Anatomical Target: Blocks the superior and middle trunks of the brachial plexus (C5-C7).
- Indication: Shoulder surgery Upper arm surgery
- Shoulder surgery
- Upper arm surgery
- Complications: Phrenic nerve block Horner syndrome Recurrent laryngeal nerve block Vertebral artery injection Spinal/Epidural injection Pneumothorax
- Phrenic nerve block
- Horner syndrome

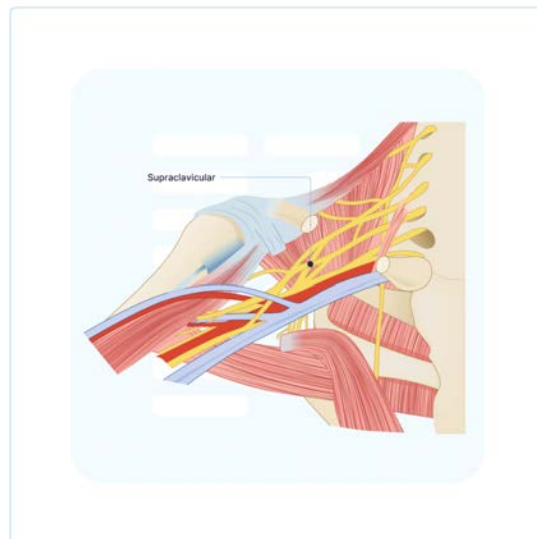
- Recurrent laryngeal nerve block
- Vertebral artery injection
- Spinal/Epidural injection
- Pneumothorax
- Shoulder surgery
- Upper arm surgery
- Phrenic nerve block
- Horner syndrome
- Recurrent laryngeal nerve block
- Vertebral artery injection
- Spinal/Epidural injection
- Pneumothorax



2) Supraclavicular Block: (Option C)

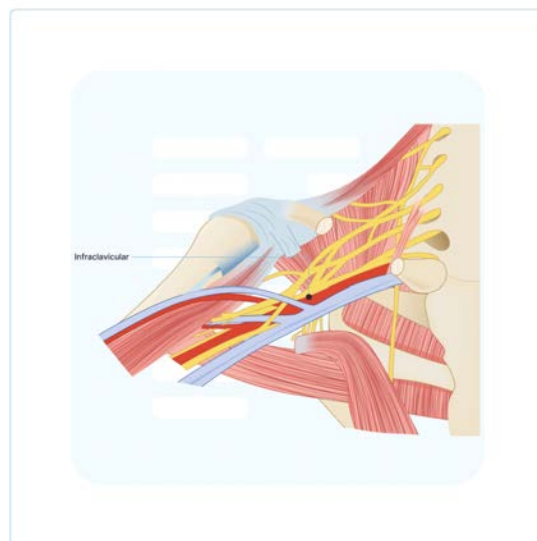
- Anatomical Target: Blocks distal trunk-proximal division of the brachial plexus.
- Indications: Operations on the elbow, forearm, and hand.
- Complications: Pneumothorax (prevalence is 0.5% to 6%) Phrenic nerve block (40%-60% of cases) Horner's syndrome Neuropathy Recurrent laryngeal nerve palsy Subclavian artery puncture
- Pneumothorax (prevalence is 0.5% to 6%)
- Phrenic nerve block (40%-60% of cases)
- Horner's syndrome

- Neuropathy
- Recurrent laryngeal nerve palsy
- Subclavian artery puncture
- Pneumothorax (prevalence is 0.5% to 6%)
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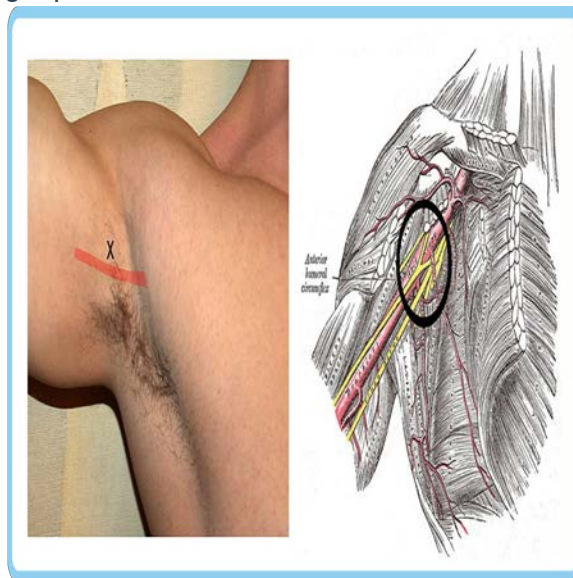
3) Infraclavicular Block: (Option B)

- Anatomical Target: Blocks at the level of cords of brachial plexus.
- Indication: Surgeries at or distal to the elbow
- Complications: Lower risk of Pneumothorax: Compared to supraclavicular blocks. Vascular puncture
- Lower risk of Pneumothorax: Compared to supraclavicular blocks.
- Vascular puncture
- Lower risk of Pneumothorax: Compared to supraclavicular blocks.
- Vascular puncture



4) Axillary Block: (Option D)

- Anatomical Target: Block terminal nerves.
- Indications: Upper extremity procedures.
- Complication: Local infection Neuropathy Bleeding risk Least risk of causing a pneumothorax.
- Local infection
- Neuropathy
- Bleeding risk
- Least risk of causing a pneumothorax.
- Local infection
- Neuropathy
- Bleeding risk
- Least risk of causing a pneumothorax.



Solution for Question 8:

Correct Answer: D) C7 vertebra is preferred over C6 vertebra as the injection site

Explanation:

In a stellate ganglion block, the C6 vertebra is the more preferred location than the C7 because, at the former location, there is a reduced possibility of injuring the vertebral artery than at the latter.

Cervicothoracic (Stellate) Block Blocks the upper thoracic and all cervical ganglia.

Indications:

- Pain management for head, neck, arm, and upper chest.
- Complex regional pain syndrome (chronic pain, sensory, motor and trophic issues of the affected extremity or region usually after a trauma or surgery).
- Used for: (Option B) Posttraumatic stress disorder (PTSD) Severe menopausal vasomotor symptoms Vasospastic disorders of the upper extremity Refractory ventricular tachycardia Herpes zoster ophthalmicus Ventricular tachyarrhythmia resulting from congenital long-QT syndrome (a left stellate block is recommended) Severe ipsilateral circulatory disorders of the upper extremity.
- Posttraumatic stress disorder (PTSD)
- Severe menopausal vasomotor symptoms
- Vasospastic disorders of the upper extremity
- Refractory ventricular tachycardia
- Herpes zoster ophthalmicus
- Ventricular tachyarrhythmia resulting from congenital long-QT syndrome (a left stellate block is recommended)
- Severe ipsilateral circulatory disorders of the upper extremity.
- Posttraumatic stress disorder (PTSD)
- Severe menopausal vasomotor symptoms
- Vasospastic disorders of the upper extremity
- Refractory ventricular tachycardia
- Herpes zoster ophthalmicus
- Ventricular tachyarrhythmia resulting from congenital long-QT syndrome (a left stellate block is recommended)
- Severe ipsilateral circulatory disorders of the upper extremity.

Location: Stellate ganglion is located at the level of C6 transverse processes (Chassaignac tubercle) or C7 (3-5 cm above the clavicle)

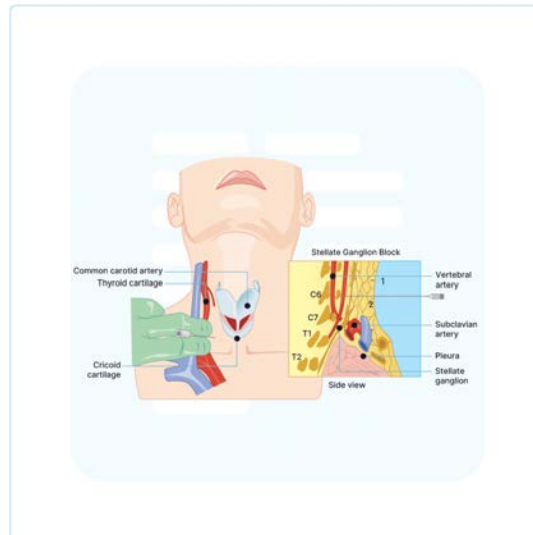
Signs of Successful Block:

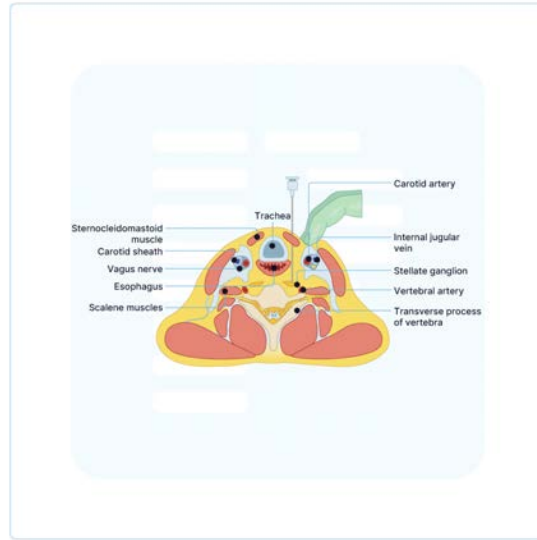
- Increased skin temperature in the ipsilateral arm.

- Onset of Horner syndrome (Option C) : ipsilateral ptosis, miosis, enophthalmos, loss of ciliospinal reflex, and anhydrosis of the neck and face.
- Guttman's sign: Nasal stuffiness after block due to vasodilatation.
- Muller's sign: vasodilation near the tympanic membrane.

Complications:

- Intravascular and subarachnoid injection
- Hematoma
- Pneumothorax (higher incidence with posterior approach) (Option A)
- Epidural anesthesia
- Brachial plexus block
- Hoarseness from recurrent laryngeal nerve blockade





Solution for Question 9:

Correct Answer: C) Radial nerve is approached from a posterior aspect of the elbow

Explanation: The radial nerve is approached from a posterior aspect of the elbow to achieve nerve block.

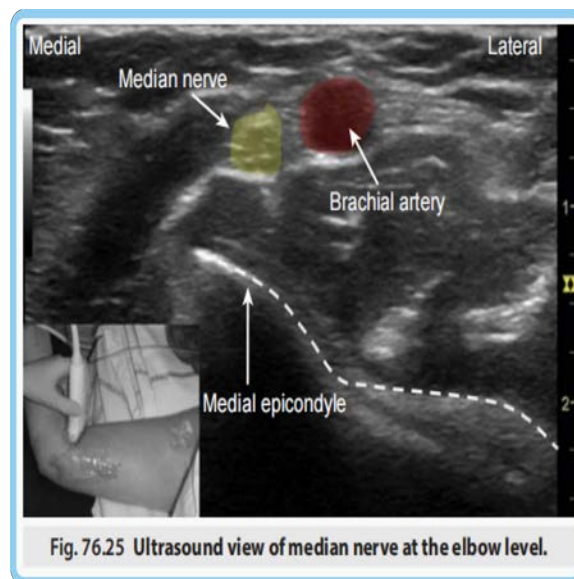
Distal conduction blocks:

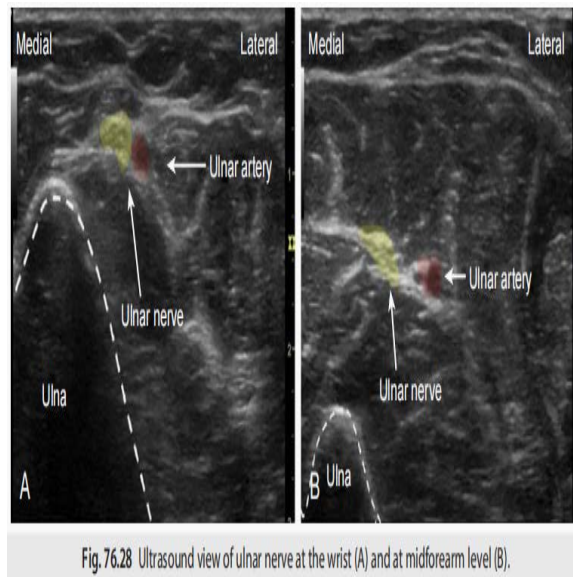
- It is indicated to complement partially failed brachial blocks.
- A very small amount of 0.05 ml/kg local anesthetic up to 1-2mL is enough for complete nerve blockade.
- Caution should be taken at the wrist as the nerves and tendons appear similar.

Nerve	Location	Ultrasound Visualization	Blocking Techniques
Median Nerve	Antecubital fossa: medial to the brachial artery (Option A) Mid-forearm: medial to radial artery	Between palmaris longus & flexor carpi radialis Difficult to distinguish from tendons at the wrist	Block at antecubital fossa (vs Option D) , mid-forearm, or wrist Caution is required at the wrist due to tendon similarity

Ulnar Nerve	Above elbow: several cm proximal At wrist: near medial side of ulnar artery (Option B)	Small hyperechoic triangle near ulnar artery Prefer tracking proximally to separate from the artery	Avoid cubital tunnel (Option D) Block above or below the elbow
Radial Nerve	Posterior to the humerus, lateral side of the elbow	Visible below the brachial muscle near the humerus	Can block at the posterior aspect of the elbow

ulnar nerve at elbow level" data-author="Courtney Aird,Kerry Thoires, Sandhya Maranna, Nicola Massy-Westropp" data-hash="435" data-license="NA" data-source="https://journals.sagepub.com/doi/10.1177/8756479319870467?icid=int.sj-full-text.similar-articles.5" data-tags="December2024" height="353" src="https://image.prepladder.com/notes/x6zC4IAgXz1WFg3gpqWJ1734957085.png" width="440" />





Solution for Question 10:

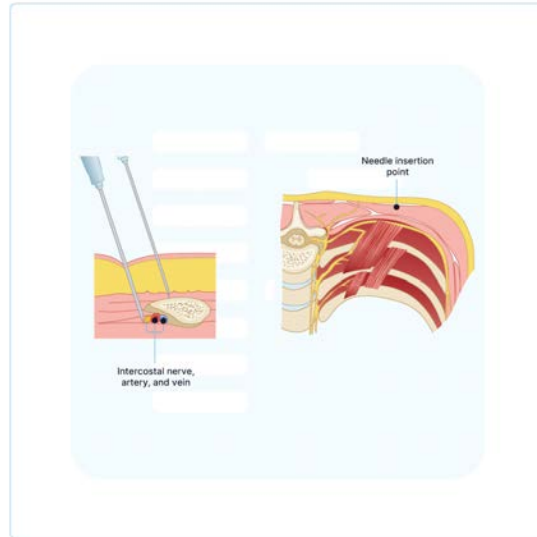
Correct Answer: B) Paravertebral block

Explanation:

The paravertebral block is preferred in surgeries like mastectomy and nephrectomy that need a great amount of analgesia.

Block Type	Indications	Technique and Nerves Targeted	Complications
Intercostal Block (Option D)	Thoracic/upper abdominal surgery, rib fractures	Individual injections at each intercostal nerve (T1-T11) at the inferior edge of the rib, with aspiration before injection.	Pneumothorax, intravascular injection, toxicity, bleeding, infection
Paravertebral Block	Thoracic/abdominal wall surgeries Mastectomy Open nephrectomy (unilateral) Cholecystectomy	Injections at T levels near transverse processes Targets thoracic spinal nerves (T2-T12) using ultrasound or landmark techniques	Pneumothorax, hypotension, retroperitoneal injury, hematoma, infection

Erector Spinae Plane Block	Thoracoabdominal surgery, rib fractures Chest wall neuropathic pain	Local anesthetic is deposited deep into erector spinae muscles, usually at a midline or lateral approach. Targets dorsal rami of thoracic spinal nerves	Pneumothorax, local anesthetic toxicity, bleeding, infection
Pectoralis I/II Block	Chest wall surgery	Injection between the pectoralis major and minor (Pecs I) and targeting intercostobrachial and intercostal nerves (3-6) for Pecs II.	Rare, but may include pneumothorax, bleeding, infection
Serratus Anterior Plane Block	Unilateral chest wall anesthesia	Injection between serratus anterior and latissimus dorsi Targets lateral cutaneous branches of intercostal nerves (T2-T6)	Risk of pneumothorax, bleeding, infection
Transversus Abdominis Plane Block (Option C)	Lower abdominal wall surgery	Injection between internal oblique and transversus abdominis muscles Targets subcostal (T12), ilioinguinal (L1), and iliohypogastric (L1) nerves	Peritoneal violation, bowel perforation, bleeding, infection
Rectus Sheath Block (Option A)	Superficial abdominal procedures	Local anesthetic injected deep into the rectus abdominis muscle Targets anterior cutaneous branches of intercostal nerves (T7-T12)	Limited visceral analgesia, bleeding, infection
Quadratus Lumborum Blocks	Lower thoracic/ lumbar anesthesia	Techniques vary: lateral (QL1), posterior (QL2), or transmuscular. Local anesthetic at quadratus lumborum Potential lumbar plexus involvement	Varies by technique; potential for visceral injury, bleeding, infection



Solution for Question 11:

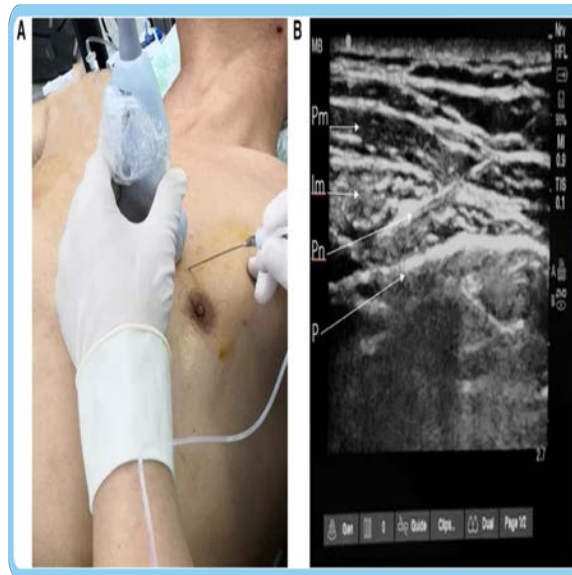
Correct answer: B) Intercostal nerve block

Explanation: Intercostal nerve block targets the intercostal nerves, which supply sensation to the chest wall, making it ideal for procedures involving the pleura or thoracic cavity.

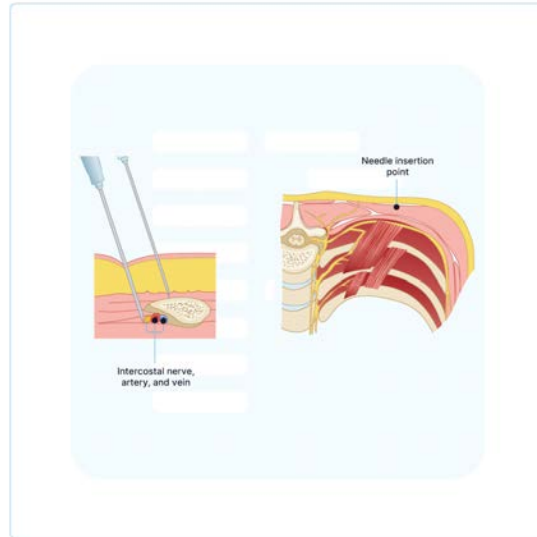
Intercostal Nerve Block:

- Intercostal nerves arise from the dorsal and ventral rami of the thoracic spinal nerves.
- Run along the underside of the corresponding rib, positioned in a neurovascular bundle with the artery and vein.
- Indications: Provides analgesia following thoracic and upper abdominal surgery. Relieves pain associated with: Rib fractures Herpes zoster Cancer
- Provides analgesia following thoracic and upper abdominal surgery.
- Relieves pain associated with: Rib fractures Herpes zoster Cancer
- Rib fractures
- Herpes zoster
- Cancer
- Provides analgesia following thoracic and upper abdominal surgery.

- Relieves pain associated with: Rib fractures Herpes zoster Cancer
- Rib fractures
- Herpes zoster
- Cancer
- Rib fractures
- Herpes zoster
- Cancer
- Procedure:



- Requires individual injections for each intercostal nerve innervating the dermatomes to be anesthetized.
- Ultrasound guidance allows for multiple levels to be reached from a single skin entry point.
- A small-gauge needle is inserted at the inferior edge of the rib.
- Aspiration to check for blood or air, then 3 to 5 mL of local anesthetic is injected at each desired level.



- Contraindication: Allergy to local anesthetics, prior nerve injury or damage, inability of the patient to consent to the procedure, active infection over the site of injection, anticoagulation, or coagulopathy.
- Complications: Has highest complication rates among peripheral nerve blocks due to proximity to the: Intercostal artery and vein: Risk of intravascular injection Pleura: Risk of pneumothorax
- Intercostal artery and vein: Risk of intravascular injection
- Pleura: Risk of pneumothorax
- Intercostal artery and vein: Risk of intravascular injection
- Pleura: Risk of pneumothorax

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Suprascapular nerve block (Option A) primarily targets shoulder-related pain rather than providing the necessary analgesia for procedures involving the thoracic wall and pleura, which the intercostal nerve block addresses more directly.

Interscalene block (Option C) is not suitable for thoracentesis because it primarily targets the upper extremity, providing analgesia for shoulder and arm surgeries rather than addressing the chest wall and pleura directly.

Thoracic sympathetic block (Option D) is not ideal for thoracentesis as it typically offers only temporary relief and is more suited for neuropathic pain management rather than procedural analgesia.

Solution for Question 12:

Correct answer: C) Genitofemoral nerve

Explanation: Transversus abdominis plane block (TAP) does not target the genitofemoral nerve, which supplies the genital area and upper thigh.

Transversus Abdominis Plane Block (TAP):

- Nerves Targeted: Subcostal nerve (T12) (Option A) Ilioinguinal nerve (L1) (Option B) Iliohypogastric nerve (L1) (Option D)
- Subcostal nerve (T12) (Option A)
- Ilioinguinal nerve (L1) (Option B)
- Iliohypogastric nerve (L1) (Option D)
- Nerve Course: These nerves travel between the internal oblique and transversus abdominis muscles.
- Area of Anesthesia: Provides anesthesia to the ipsilateral lower abdomen, below the umbilicus.
- Indication: Used for surgical anesthesia during minor, superficial procedures on the lower abdominal wall. Provides postoperative analgesia for procedures below the umbilicus. For inguinal hernia surgeries, intravenous or local supplementation may be needed for anesthesia during peritoneal traction.
- Used for surgical anesthesia during minor, superficial procedures on the lower abdominal wall.
- Provides postoperative analgesia for procedures below the umbilicus.
- For inguinal hernia surgeries, intravenous or local supplementation may be needed for anesthesia during peritoneal traction.
- Subcostal nerve (T12) (Option A)
- Ilioinguinal nerve (L1) (Option B)
- Iliohypogastric nerve (L1) (Option D)
- Used for surgical anesthesia during minor, superficial procedures on the lower abdominal wall.
- Provides postoperative analgesia for procedures below the umbilicus.

- For inguinal hernia surgeries, intravenous or local supplementation may be needed for anesthesia during peritoneal traction.

Anatomic Location for Block: The desired location for the block is between the transversus abdominis and internal oblique muscles.

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- Procedure: Patient Positioning: Ideally, the patient is positioned in lateral decubitus. Under Ultrasound Guidance: External oblique, internal oblique, and transversus abdominis muscles are identified just superior to the anterior superior iliac spine (ASIS). Needle Placement: Needle is inserted between the fascial layers of the internal oblique and transversus abdominis muscles. Local anesthetic is injected into the transversus abdominis plane.
- Patient Positioning: Ideally, the patient is positioned in lateral decubitus.
- Under Ultrasound Guidance: External oblique, internal oblique, and transversus abdominis muscles are identified just superior to the anterior superior iliac spine (ASIS).
- Needle Placement: Needle is inserted between the fascial layers of the internal oblique and transversus abdominis muscles.
- Local anesthetic is injected into the transversus abdominis plane.
- Patient Positioning: Ideally, the patient is positioned in lateral decubitus.
- Under Ultrasound Guidance: External oblique, internal oblique, and transversus abdominis muscles are identified just superior to the anterior superior iliac spine (ASIS).
- Needle Placement: Needle is inserted between the fascial layers of the internal oblique and transversus abdominis muscles.
- Local anesthetic is injected into the transversus abdominis plane.

Transversus Abdominis Plane Block 1" data-author="Vishal Uppal Sushil Francis Sancheti Hari Kalagara" data-hash="440" data-license="CC BY 4.0" data-source="https://www.researchgate.net/figure/Posterior-transversus-abdominis-plane-TAP-block-The-left-image-shows-the-ultrasound_fig2_336816293" data-tags="December2024" height="278" src="https://image.prepladder.com/notes/AwhirsxLR4RffDb9HBGq1734958925.png" width="717" />

Transversus Abdominis Plane Block 2" data-author="Nagata, Jun & Watanabe, Jun & Sawatsubashi, Yusuke & Akiyama, Masaki & Arase, Koichi & Minagawa, Noritaka & Torigoe, Takayuki & Hamada, Kotaro & Nakayama, Yoshifumi & Hirata, Keiji." data-hash="441" data-license="CC BY-NC ND" data-source="https://www.researchgate.net/figure/Transversus-abdominis-plane-block-A-Laparoscopic-view-showing-the-tip-of-a-needle_fig2_315833431"

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- Complications: Violation of the peritoneum with or without bowel perforation.

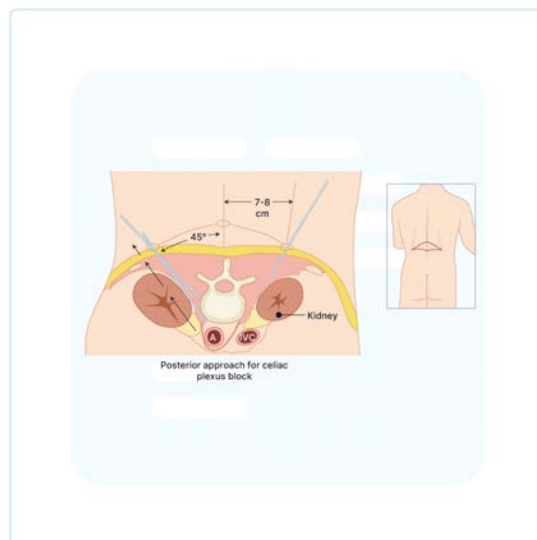
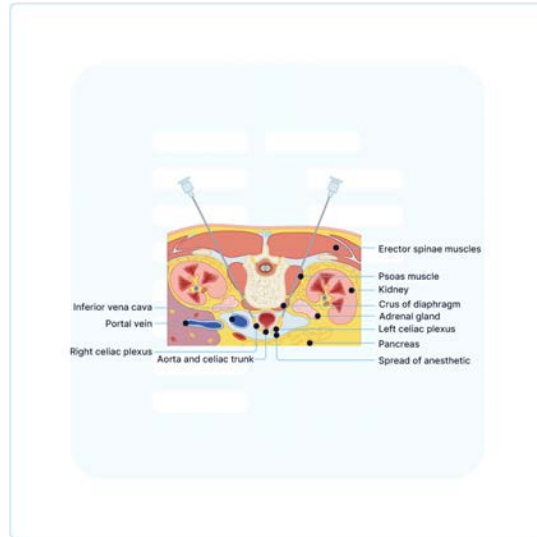
Solution for Question 13:

Correct Answer: C) Systemic local toxicity

Explanation:

Systemic local toxicity is not seen as a complication of celiac plexus block as it is seen as a complication in intercostal nerve block that is done for postoperative analgesia and as an alternative to epidural and paravertebral blocks .

Coeliac plexus block	
Indication	Patients with abdominal visceral pain, like in cancers.
Anatomy	The celiac ganglia can vary in number from one to five and differ in form and position. Typically, they are grouped around the body of the L1 vertebra , located posterior to the vena cava on the right side, just lateral to the aorta on the left, and behind the pancreas.
Approaches	Posterior and Trans-Crural Approach Anterior Approach Endoscopic Ultrasound-Guided Celiac Plexus Block
Complications	Postural hypotension (Option A) Accidental intravascular injection (risk of severe systemic reaction) Pneumothorax Retroperitoneal hemorrhage Injury to kidneys or pancreas Sexual dysfunction (Option D) Rarely, paraplegia (due to lumbar artery of Adamkiewicz injury) Increased gastrointestinal motility and diarrhea (due to unopposed parasympathetic activity) Back pain (Option B)



Solution for Question 14:

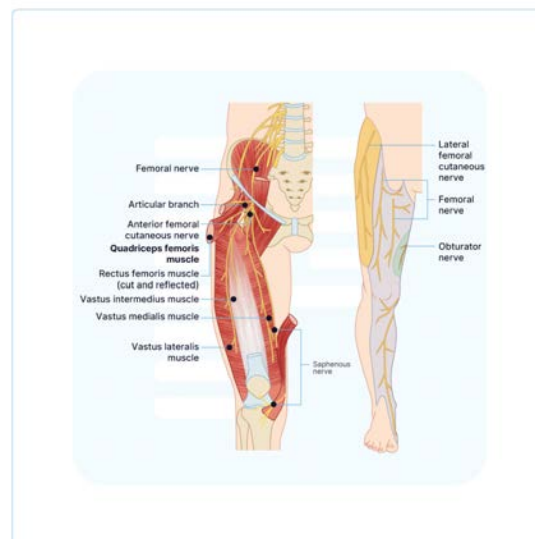
Correct Answer: B) Femoral nerve block

Explanation:

Femoral nerve block is highly effective for postoperative analgesia in knee arthroscopy, as it provides pain relief for procedures involving the hip, thigh, knee, and ankle, even though it is insufficient for surgical anesthesia on its own.

Femoral Nerve Block:

- Femoral nerve innervates the main hip flexors and knee extensors and provides sensory innervation of the hip and thigh.
- Use in Surgery: Femoral nerve block alone is insufficient for surgical anesthesia but highly effective for postoperative analgesia. (Option B) Commonly used for procedures involving the hip, thigh, knee, and ankle (via saphenous nerve).
- Femoral nerve block alone is insufficient for surgical anesthesia but highly effective for postoperative analgesia. (Option B)
- Commonly used for procedures involving the hip, thigh, knee, and ankle (via saphenous nerve).
- Femoral nerve block alone is insufficient for surgical anesthesia but highly effective for postoperative analgesia. (Option B)
- Commonly used for procedures involving the hip, thigh, knee, and ankle (via saphenous nerve).
- Anatomical Landmark: Lies lateral to the femoral artery and deep to the fascia iliaca.
- Low complication rate.
- Block causes quadriceps muscle weakness, leading to knee buckling when weight-bearing is attempted. Therefore, patients must be informed that weight-bearing on the affected leg is not possible until the block is resolved.

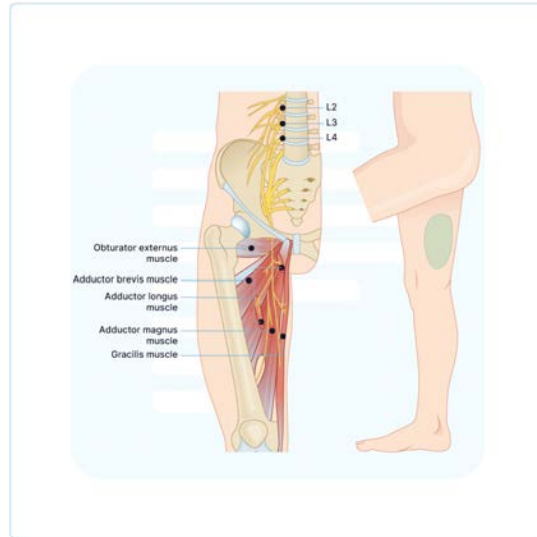


Lateral Femoral Cutaneous Nerve Block: (Option D)

- Lateral femoral cutaneous nerve (L2–L3) provides sensory innervation to the lateral thigh.
- Anatomical Location: Inferior and medial to the anterior superior iliac spine (ASIS)
- Use: Can be used as a supplement to a femoral nerve block or as an isolated block for anesthesia of the lateral thigh. Excellent for providing analgesia to a split-thickness skin graft donor site.
- Can be used as a supplement to a femoral nerve block or as an isolated block for anesthesia of the lateral thigh.
- Excellent for providing analgesia to a split-thickness skin graft donor site.
- Can be used as a supplement to a femoral nerve block or as an isolated block for anesthesia of the lateral thigh.
- Excellent for providing analgesia to a split-thickness skin graft donor site.

Obturator Nerve Block: (Option C)

- Obturator nerve provides sensory branches to the hip and knee joints, variable sensation to the medial thigh and motor innervation to the adductors of the hip.
- Anatomical Location: 2 cm lateral and distal to the pubic tubercle.
- Use: Required for complete knee anesthesia, often combined with femoral and sciatic nerve blocks. Used to prevent adductor muscle spasms during transurethral bladder resection when muscle relaxants are not employed.
- Required for complete knee anesthesia, often combined with femoral and sciatic nerve blocks.
- Used to prevent adductor muscle spasms during transurethral bladder resection when muscle relaxants are not employed.
- Required for complete knee anesthesia, often combined with femoral and sciatic nerve blocks.
- Used to prevent adductor muscle spasms during transurethral bladder resection when muscle relaxants are not employed.



Popliteal nerve block (Option A) : While popliteal nerve blocks provide excellent coverage for foot and ankle surgery and spare much of the hamstring muscles, they do not adequately target the knee joint itself, making them less effective for analgesia during knee arthroscopy compared to the femoral nerve block.

Solution for Question 15:

Correct Answer: A) Lumbar plexus block

Explanation:

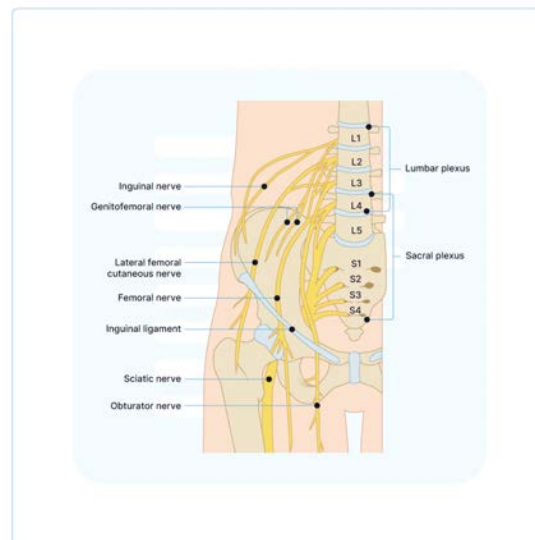
Lumbar plexus block is effective for providing analgesia in procedures involving the hip.

Lumbar Plexus Block:

- Lumbar plexus is formed by the ventral rami of L1 to L4, with occasional contribution from T12.
- Located between the psoas muscle and the quadratus lumborum muscle.
- 4 nerves of the lumbar plexus: Femoral nerve Lateral femoral cutaneous nerve Obturator nerve Genitofemoral nerve
- Femoral nerve
- Lateral femoral cutaneous nerve

- Obturator nerve
- Genitofemoral nerve
- Lumbar plexus nerves supply motor and sensory innervation to the anterior thigh, hip joint and sensory innervation to the medial leg.
- Posterior Lumbar Plexus (Psoas Compartment) Block: Useful for surgeries involving areas innervated by the femoral, lateral femoral cutaneous, and obturator nerves. Includes procedures on the hip, knee, and anterior thigh. Complete anesthesia of the knee can be achieved when combined with a proximal sciatic nerve block. Has one of the highest complication rates among peripheral nerve blocks. Complications: Retroperitoneal hematoma Intravascular local anesthetic injection with toxicity Intrathecal or epidural injections Renal capsular puncture
- Useful for surgeries involving areas innervated by the femoral, lateral femoral cutaneous, and obturator nerves.
- Includes procedures on the hip, knee, and anterior thigh.
- Complete anesthesia of the knee can be achieved when combined with a proximal sciatic nerve block.
- Has one of the highest complication rates among peripheral nerve blocks.
- Complications: Retroperitoneal hematoma Intravascular local anesthetic injection with toxicity Intrathecal or epidural injections Renal capsular puncture
- Retroperitoneal hematoma
- Intravascular local anesthetic injection with toxicity
- Intrathecal or epidural injections
- Renal capsular puncture
- Femoral nerve
- Lateral femoral cutaneous nerve
- Obturator nerve
- Genitofemoral nerve
- Useful for surgeries involving areas innervated by the femoral, lateral femoral cutaneous, and obturator nerves.
- Includes procedures on the hip, knee, and anterior thigh.
- Complete anesthesia of the knee can be achieved when combined with a proximal sciatic nerve block.
- Has one of the highest complication rates among peripheral nerve blocks.
- Complications: Retroperitoneal hematoma Intravascular local anesthetic injection with toxicity Intrathecal or epidural injections Renal capsular puncture
- Retroperitoneal hematoma

- Intravascular local anesthetic injection with toxicity
- Intrathecal or epidural injections
- Renal capsular puncture
- Retroperitoneal hematoma
- Intravascular local anesthetic injection with toxicity
- Intrathecal or epidural injections
- Renal capsular puncture



Sacral Plexus Block:

- Nerves of the Sacral Plexus:
- Supplies motor and sensory innervation to the posterior thigh, and most of the leg and foot. (Option B & C)
- Indications for Sciatic Nerve Block: Recommended for foot and leg surgeries Often combined with a saphenous nerve block to cover the medial aspect of the leg.
- Recommended for foot and leg surgeries
- Often combined with a saphenous nerve block to cover the medial aspect of the leg.
- Recommended for foot and leg surgeries
- Often combined with a saphenous nerve block to cover the medial aspect of the leg.

Ilioinguinal nerve block (Option D): Ilioinguinal nerve primarily provides sensory innervation to the proximal, medial thigh and is used for analgesia following inguinal hernia repair and for lower abdominal procedures, making it unsuitable for hip surgery, which requires more extensive anesthesia coverage.

Solution for Question 16:

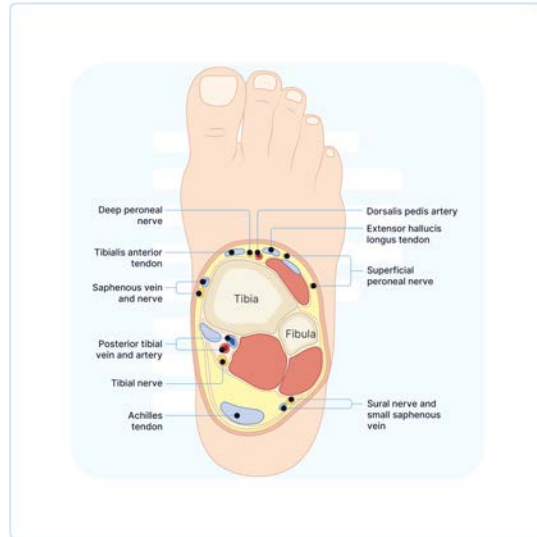
Correct Answer: D) Superficial peroneal nerve

Explanation: Superficial peroneal nerve innervates the dorsum of the foot, excluding the first interdigital cleft. This makes it ideal for covering the desired sensory area during the procedure.

Ankle Block:

- Relatively simple and offers adequate anesthesia for foot surgeries.
- Nerves Blocked: 4 of the 5 individual nerves blocked are terminal branches of the sciatic nerve. Tibial nerve Superficial peroneal nerve Deep peroneal nerve Sural nerve: Formed from both tibial and common peroneal contributions.
- 4 of the 5 individual nerves blocked are terminal branches of the sciatic nerve. Tibial nerve Superficial peroneal nerve Deep peroneal nerve Sural nerve: Formed from both tibial and common peroneal contributions.
- Tibial nerve
- Superficial peroneal nerve
- Deep peroneal nerve
- Sural nerve: Formed from both tibial and common peroneal contributions.
- 4 of the 5 individual nerves blocked are terminal branches of the sciatic nerve. Tibial nerve Superficial peroneal nerve Deep peroneal nerve Sural nerve: Formed from both tibial and common peroneal contributions.
- Tibial nerve
- Superficial peroneal nerve
- Deep peroneal nerve
- Sural nerve: Formed from both tibial and common peroneal contributions.
- Tibial nerve
- Superficial peroneal nerve
- Deep peroneal nerve
- Sural nerve: Formed from both tibial and common peroneal contributions.

Saphenous nerve: Major sensory branch of the femoral nerve.



- Procedure: A complete ankle block requires five injections to target all the nerves involved.

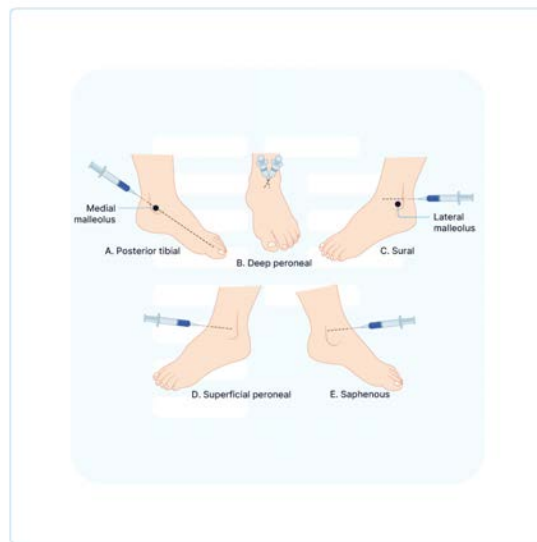
Nerve Blocked	Site of Block	Effect
Tibial nerve (Option B)	Behind the posterior tibial artery at the level of the medial malleolus.	Anesthetizes the heel, plantar portion of toes, and sole.
Sural nerve (Option A)	Lateral to the Achilles tendon and is directed towards the lateral malleolus.	Anesthetizes the lateral foot and lateral aspect of the proximal sole.
Deep peroneal nerve (Option C)	Groove between extensor hallucis longus and extensor digitorum longus tendon, lateral to dorsalis pedis artery.	Anesthetizes the skin between the first and second toes and the short toe extensors.
Superficial peroneal nerve (Option D)	Targeted by extending the injection laterally towards lateral malleolus from the deep peroneal block site.	Anesthetizes dorsum of the foot (excluding the first interdigital cleft).
Saphenous nerve	Redirecting the needle medially towards the medial malleolus from the same insertion point.	Anesthetizes the medial foot.

anesthesia produced by ankle block" data-author="NA" data-hash="693" data-license="NA" data-source="NA" data-tags="January2025" height="602" src="https://image.prepladder.com/notes/l1m3YnRpU71UUybEVcpc1736513246.png" width="570" />

- Limitations: An ankle block does not provide analgesia for below-the-knee tourniquet pain or allow for perineural catheter insertion.
- Disadvantages: Multiple injections may cause discomfort. Persistent paresthesias occur but are usually self-limited. Presence of edema or induration may complicate landmark

identification.

- Multiple injections may cause discomfort.
- Persistent paresthesias occur but are usually self-limited.
- Presence of edema or induration may complicate landmark identification.
- Multiple injections may cause discomfort.
- Persistent paresthesias occur but are usually self-limited.
- Presence of edema or induration may complicate landmark identification.



Oxygen Delivering Systems

1. A 60-year-old male patient undergoes general anaesthesia for abdominal surgery. After intubation, the anesthesiologist needs to confirm that the endotracheal tube (ETT) is correctly placed in the trachea. Which of the following is considered the most reliable method for confirming correct ETT placement?

- A. Auscultation of bilateral breath sounds
 - B. Visual condensation in the endotracheal tube
 - C. Waveform capnography
 - D. Symmetrical chest rise
-

2. A 70-year-old COPD patient with a history of COPD presents with type 2 respiratory failure and a SpO₂ of 86%. ABG results are normal. Which oxygen delivery device will be most appropriate for this patient?

- A. Nasal Prongs
 - B. Venturi Mask
 - C. Intubation
 - D. Non-Rebreathing Mask (NRBM)
-

3. A 4-year-old patient with respiratory distress is initiated on High-flow Nasal Cannula (HFNC). Which of the following statements is incorrect regarding HFNC?

- A. Provides warmed and humidified gas to the airway.
- B. It can generate low levels of positive airway pressure.
- C. Limits the maximum FiO₂ to 60%.
- D. Improved mortality rate in COVID patients

4. A 72-year-old female presents to the emergency department in severe respiratory distress, with an oxygen saturation of 78% on room air, tachypnea, and altered consciousness. She has a 3-day history of fever and cough, suggesting community-acquired pneumonia. What is the most appropriate intervention for ensuring adequate oxygenation?

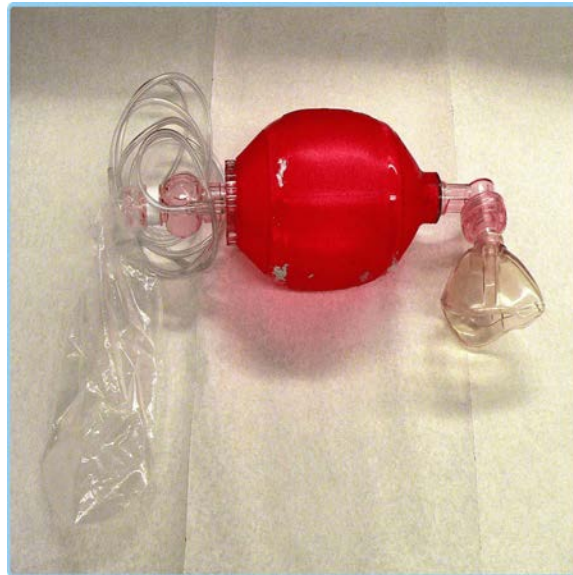
- A.
- B.
- C.
- D.

5. A 52-year-old male patient is brought to the emergency department after a motor vehicle accident. He is unconscious with an inadequate airway and requires immediate management. Which of the following is the most appropriate method for determining the correct size of the airway as shown in the image below?



- A. Angle of the mouth to the tragus
 - B. Angle of the mouth to the suprasternal notch
 - C. Tip of the nose to the suprasternal notch
 - D. Tip of the nose to the tragus
-

6. An intern in the ICU witnesses a patient suddenly become breathless and collapse, prompting a code blue. After resuscitation, the senior doctor asks the intern about the significance of the equipment shown in the image. Which of the following is NOT a significance of this equipment?



- A. Convenient for patient transport
 - B. Suitable for CPR
 - C. Can be used during ventilator failure.
 - D. Can deliver high-flow supplemental oxygen
-

7. A 70-year-old patient with a history of COPD and chronic smoking arrived at the emergency department in a semi-conscious state, showing an oxygen saturation of 78% on room air. Resuscitation efforts were initiated right away. What is the function of the component depicted in the image below?



- A. Deliver breaths to the patient
- B. Prevents over-inflation of lungs
- C. Provides additional oxygen supply
- D. Prevents rebreathing

8. A 30-year-old female scheduled for laparoscopic cholecystectomy under general anesthesia experiences a sudden increase in heart rate (130 bpm), rapidly rising end-tidal CO₂ levels, elevated body temperature, and severe muscle rigidity, especially in the extremities. Which of the following capnogram patterns is indicative of this condition?

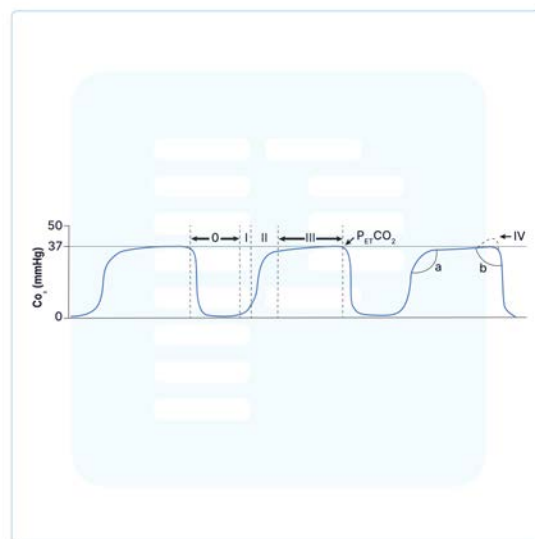
- A.
- B.
- C.

D.

9. A 65-year-old woman with a history of chronic obstructive pulmonary disease (COPD) presents to the emergency department in respiratory distress. The healthcare team decides to utilise capnography to monitor her end-tidal carbon dioxide (EtCO₂) levels as part of her assessment. Which of the following does EtCO₂ NOT assess?

- A. Ventilation-Perfusion (V/Q) matching
- B. Estimate adequacy of cardiac output
- C. BUN (blood urea nitrogen) determination
- D. Chest compression effectiveness during CPR

10. A 45-year-old patient is undergoing general anaesthesia for a laparoscopic cholecystectomy. The anesthesiologist is monitoring the patient's ventilation using capnography. The capnograph shows the following waveform. What does the plateau phases represent in the capnography waveform?



- A. Expiration of air from alveoli
 - B. Expiration of air from the trachea
 - C. Expiration of air from bronchi
 - D. Exhalation of air from dead space
-

11. Match the following Laryngeal mask airways (LMA) with their corresponding images. i. Classic LMA a. ii. Proseal LMA b. iii. Fast Track LMA c. iv. Supreme LMA d. v. I-gel LMA e.

i. Classic LMA	a.	
ii. Proseal LMA	b.	

iii. Fast Track LMA	c.	
iv. Supreme LMA	d.	
v. I-gel LMA	e.	

- A. i-e, ii-d, iii-b, iv-a, v-c
 B. i-c, ii-a, iii-d, iv-e, v-b
 C. i-d, ii-b, iii-a, iv-e, v-c
 D. i-b, ii-a, iii-e, iv-c, v-d

12. What is the primary advantage of using the endotracheal tube shown in the image below?



- A. Enhanced visibility during intubation
- B. Reduced risk of airway obstruction
- C. Improved patient comfort during long procedures
- D. Facilitated access to the surgical site in maxillofacial procedures

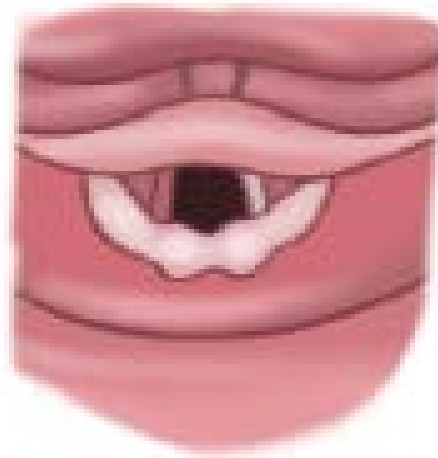
13. A 36-year-old with a suspected cervical spine injury is brought to the emergency department in respiratory distress, requiring urgent intubation. Which of the following endotracheal tubes is preferred for this patient?

- A. Armored endotracheal tube
- B. Double-lumen endotracheal tube
- C. Cuffed endotracheal tube
- D. Uncuffed endotracheal tube

14. A 4-year-old child is brought to the emergency department in Respiratory distress. What is the appropriate size of the endotracheal tube to intubate the child?

- A. 4 mm
 - B. 5 mm
 - C. 7 mm
 - D. 10 mm
-

15. A 55-year-old male is undergoing elective surgery under general anaesthesia. During laryngoscopy for intubation, the anaesthesiologist visualised the airway, as shown in the image below. According to the Cormack-Lehane classification, what grade is assigned to this view?



- A. Grade I
- B. Grade II
- C. Grade III

D. Grade IV

16. A 50-year-old male presents with gallstones and is scheduled for a cholecystectomy. During the preoperative assessment, the anaesthetist notes the patient's airway, as visualised in the image below. According to the Mallampati classification, which class does this fall into?



- A. Class I
 - B. Class II
 - C. Class III
 - D. Class IV
-

17. A 45-year-old obese female has been experiencing excessive vaginal bleeding for two months, and an ultrasound revealed uterine fibroids. She is scheduled for a hysterectomy. What is the preferred position for laryngoscopy in this case?

- A. Sniffing position

- B. Rose position
 - C. Lithotomy position
 - D. Ramped position
-

18. A 6-year-old child was brought in for an elective tonsillectomy in the operating theatre. During the procedure, the anaesthetist asked the junior resident about the appropriate laryngoscopy and its blades. Which type of laryngoscope is most preferred for this case?

- A. McCoy Laryngoscope
 - B. Miller Laryngoscope
 - C. Macintosh Laryngoscope
 - D. McGrath Laryngoscope
-

19. A 56-year-old male, weighing 62 kg and with a known case of COPD, was brought in with heart failure and in a semi-conscious state. Immediate resuscitation measures were initiated. What size of laryngeal mask would be preferred for this patient?

- A. 2.5
 - B. 3
 - C. 4
 - D. 6
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	3
Question 4	4
Question 5	4
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	1
Question 11	2
Question 12	4
Question 13	1
Question 14	2
Question 15	2
Question 16	3
Question 17	4
Question 18	2

Solution for Question 1:

Correct Answer: C) Waveform capnography

Explanation:

Waveform capnography is the most reliable method for confirming endotracheal tube placement. It provides continuous real-time monitoring of end-tidal CO₂, and the presence of a characteristic waveform over multiple breaths confirms tracheal intubation.

Signs of correct endotracheal tube placement

Detection of end-tidal CO₂ by waveform capnography (most reliable sign) Symmetrical chest rise (Option D)
Auscultation of bilateral breath sounds (Option A) Absence of breath sounds in the epigastrium Visible condensation in the endotracheal tube (Option B) Large exhales of tidal volume

endotracheal tube placement" data-author="NA" data-hash="481" data-license="NA" data-source="NA" data-tags="December2024" height="500" src="https://image.prepladder.com/notes/LnrLwO4vFvWGEz4EKVjt1734971884.png" width="500" />

Solution for Question 2:

Correct Answer: B) Venturi Mask

Explanation:

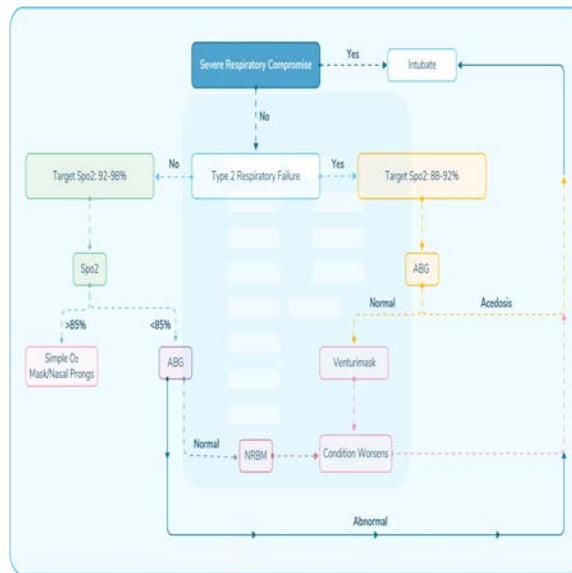
- The Venturi mask is the most appropriate oxygen delivery device for COPD patients with type 2 respiratory failure, as it is a fixed performance high flow device.
- It provides a precise concentration of oxygen (FiO₂), preventing over-oxygenation and minimising the risk of hypercapnia by allowing titration to maintain target SpO₂ levels (usually 88-92%).

Color coding of Venturi mask		
Color	FiO ₂	Require Flow rate (L/min)
Blue	24%	4
White	28%	4
Orange	31%	6
Yellow	35%	8
Red	40%	8
Green	60%	12



- If the ABG results were abnormal or if the patient's respiratory status was severely compromised, intubation (Option C) would be indicated.
- Nasal prongs (Option A) and Non-rebreathing masks (Option D) are low-flow devices and are not suitable for managing type 2 respiratory failure.

Criteria for selection of Oxygen Delivery System:



Solution for Question 3:

Correct Answer: C) Limits the maximum FiO2 to 60%

Explanation:

High-Flow Nasal Cannula (HFNC) can deliver FiO2 levels up to 100%, not limited to 60%.

High-Flow Nasal Cannula (HFNC) / High-Flow Humidified Nasal Cannula (HFHNC):

- It works by delivering warmed & humidified gas (Option A) , preventing nasal mucosa drying and overcoming the limitations of traditional cannulas.
- Provide flows up to 60 L/min , allowing for accurate delivery of high FiO2 levels
- Generates positive airway pressures of over 5 cm H2O, making it a potential substitute for low-level CPAP (Conitinous Positive Airway Pressure). (Option B).
- Can deliver 100% Fio2 (Option C)

Nasal Cannula" data-author="NA" data-hash="505" data-license="NA" data-source="NA" data-tags="December2024" height="500" src="https://image.prepladder.com/notes/5RQESHZOsHknLqoBkVoe1735012914.png" width="500" /> Nasal Cannula" data-author="NA" data-hash="507" data-license="NA" data-source="NA" data-tags="December2024" height="500" src="https://image.prepladder.com/notes/oBHsOeOpf1pex1UfYVPh1735013217.png"

width="500" />

- Advantages: Better tolerance compared to traditional face masks, allowing for greater comfort during oxygen therapy. Accurate delivery of high FiO₂ in various clinical situations. Improved mortality rates in patients experiencing acute hypoxemic respiratory failure due to Sepsis and SARS-CoV-2 infection. (Option D)
- Better tolerance compared to traditional face masks, allowing for greater comfort during oxygen therapy.
- Accurate delivery of high FiO₂ in various clinical situations.
- Improved mortality rates in patients experiencing acute hypoxemic respiratory failure due to Sepsis and SARS-CoV-2 infection. (Option D)
- Disadvantages: Higher cost compared to standard oxygen delivery devices. Limited availability outside of critical care settings
- Higher cost compared to standard oxygen delivery devices.
- Limited availability outside of critical care settings
- Better tolerance compared to traditional face masks, allowing for greater comfort during oxygen therapy.
- Accurate delivery of high FiO₂ in various clinical situations.
- Improved mortality rates in patients experiencing acute hypoxemic respiratory failure due to Sepsis and SARS-CoV-2 infection. (Option D)
- Higher cost compared to standard oxygen delivery devices.
- Limited availability outside of critical care settings

Solution for Question 4:

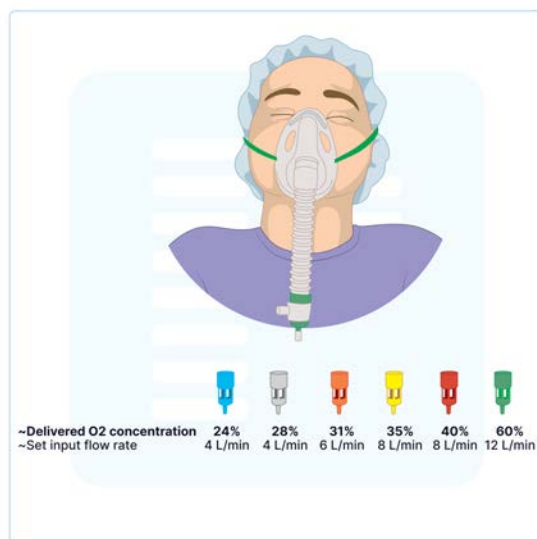
Explanation:

- The above image depicts a high-flow nasal cannula (HFNC) , which delivers inspired gas with a set concentration of oxygen (FiO₂) at flow rates of 40 to 60 L/min .

This high-flow device is effective for ensuring adequate oxygenation and is commonly used in conditions such as pneumonia, COPD, respiratory failure, post-operative care, and COVID-19.

- Venturi Mask is also a high-flow device with fixed performance that delivers a precise concentration of oxygen (FiO₂) by mixing oxygen with ambient air at specific flow rates, typically ranging from 2 to 15 L/min. It is often used in patients requiring controlled oxygen

therapy, such as those with COPD.



- In contrast, Variable performance devices, such as nasal prongs (Option A) , simple face masks (Option B) , and non-rebreather masks (NRBM) (Option C) , provide oxygen at a fixed flow rate, which constitutes only a portion of the total inspired gas. These devices are typically used in patients with minute ventilation defects to help maintain adequate oxygen saturation.

Type of mask	Nasal Prong	Face Mask	NRBM
Flow rate	1 to 6 L	6 to 10 L	10 to 15 L
FiO2	25 to 45%	35 to 60%	65 to 95 %

Learning Objective:

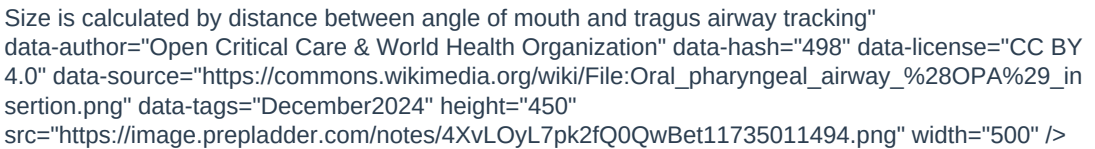
- High-Flow Devices: High-Flow Nasal Cannula (HFNC), Venturi mask
- Low-Flow Devices: Nasal Prongs, Simple Face Mask, Non-Rebreather Mask (NRBM)

Solution for Question 5:

Correct Answer: D) Tip of the nose to the tragus

Explanation:

The image depicts Nasopharyngeal airway , used to prevent back falling of the tongue, especially in unconscious patients.

Nasopharyngeal airway	Guedel airway (Oropharyngeal)
Placed through nose	Placed through mouth
Prevents back falling of tongue	Prevents back falling of tongue
Size is calculated by distance between tip of the nose and tragus	Size is calculated by distance between angle of mouth and tragus 
Contraindicated in skull fracture and bleeding disorders	Contraindicated in absence of gag reflex

Solution for Question 6:

Correct Answer: D) Can deliver high-flow supplemental oxygen

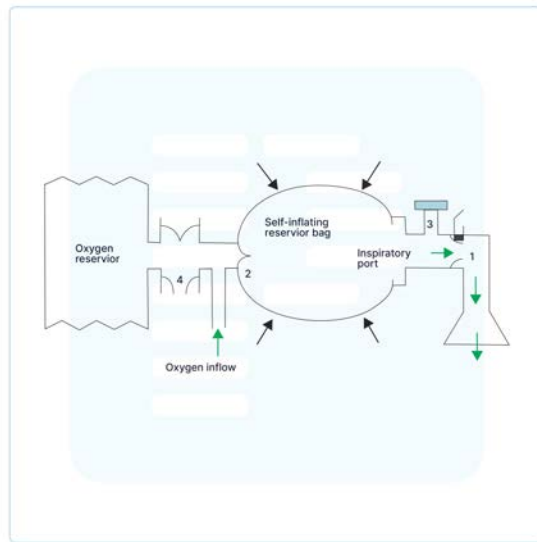
Explanation:

The image depicts an Ambu bag or Bag mask ventilation , which is used during CPR, patient transport, and also as an emergency backup during ventilatory failure.

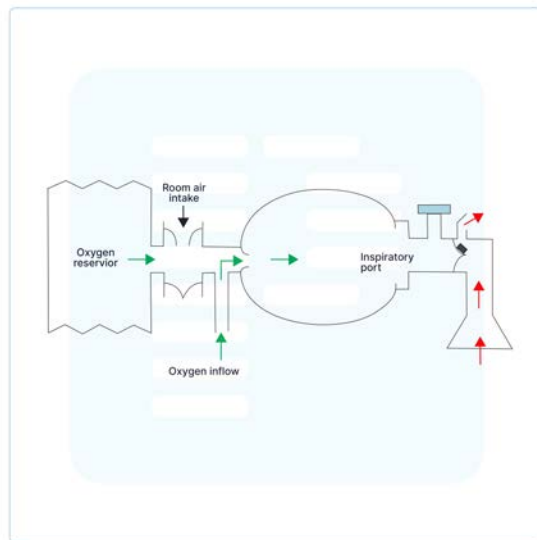
It is designed to provide assisted ventilation, not to deliver high-flow oxygen directly like a non-rebreather mask or high-flow nasal cannula.

Mechanism of Bag mask ventilation:

- During inspiration: A T-shaped non-rebreathing valve sits between the bag and the patient, directing gas flow. It opens during inspiration to allow gas from the bag to the patient while blocking the expiratory port.



- During exhalation, the inspiratory port is blocked, and the expiratory port opens to vent alveolar gas. Various valve types are available, including spring-disc and fishmouth designs.
- An inlet valve allows the bag to refill with reservoir gas or room air.
- A pop-off valve may be included to limit peak inspiratory pressure (PIP), which can become excessively high with these devices.



Advantages and disadvantages of Bag mask ventilation:

Advantages	Disadvantages
------------	---------------

Compressible reservoir automatically expands upon release. Ubiquitous for patient transport (Option A) . Suitable for CPR (cardiopulmonary resuscitation) (Option B) Can be utilized as emergency back-up (anesthesia machine, ventilator or oxygen supply failure). (Option C)

Require high fresh gas flows to achieve a high Fio₂
Exhaled moisture can cause valve sticking
Exhaled gas into the atmosphere can lead to local contamination (contaminated expiratory gases)
Can cause barotrauma or gastric insufflation (high inspiratory pressures)
Increase the work of breathing during spontaneous ventilation (generate resistance).

Solution for Question 7:

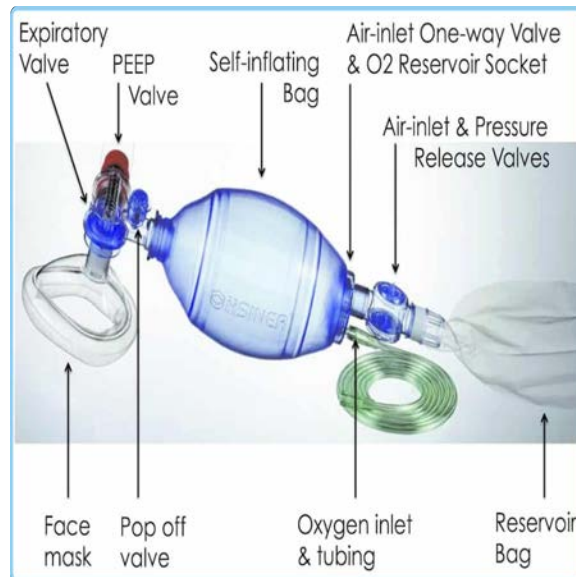
Correct Answer: D) Prevents rebreathing

Explanation:

The image illustrates the unidirectional valve/ fishmouth valve/ Ruben's valve found in an Ambu bag , which prevents rebreathing.



Ambu Bag (or) Bag mask ventilation:

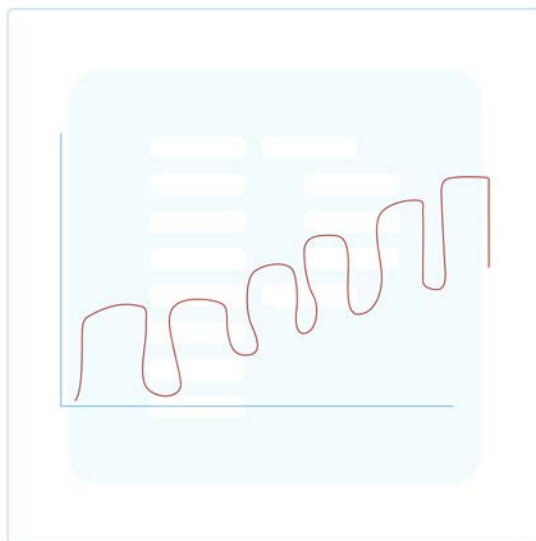


Parts	Function
Reservoir bag (self-inflating)	Provides additional oxygen supply (Option C ruled out)
Mask	Creates a seal over the patient's mouth and nose to ensure effective delivery of air from the bag
Oxygen port	Allows for the connection of supplemental oxygen
One way valve (Fish mouth or unidirectional or Ruben's valve)	Prevents rebreathing (prevents exhaled air from re-entering the bag)
Pressure valve	Prevents over-inflation of the lungs (Option B ruled out)
Ventilation bag	Deliver breaths to the patient, providing ventilation (Option A ruled out)
Inlet valve	Allows ambient air to enter the ventilation bag
Outlet valve	Vents oxygen if fresh gas flow is excessive

Solution for Question 8:

Explanation:

The scenario describes a case of Malignant hyperthermia (MH), which is characterized by a step-ladder pattern (increased metabolic rate causes a marked rise in ETco₂) on the capnogram.



• Few specific patterns of capnogram

Capnogram pattern	Description	Seen in
fin pattern" data-author="NA" data-hash="485" data-license="NA" data-source="NA" data-tags="December2024" height="200" src="https://image.prepladder.com/notes/b18QkghradMde31RGDud1735008974.png" width="300" />	Shark fin pattern: Due to increased upslope of phase III (Option A)	Bronchospasm COPD
soda lime" data-author="NA" data-hash="489" data-license="NA" data-source="NA" data-tags="December2024" height="200" src="https://image.prepladder.com/notes/1wHi1X8VO0QzwgaWtskl1735009388.png" width="300" />	Above the baseline (high base value)	Exhausted soda lime
	Abrupt decrease in EtCO ₂ values.	Venous air embolism
	Reduced height of EtCO ₂ values	Hyperventilation

	Curare note h: clefts during Phase III. (Option C)	Resumption of spontaneous breathing. (Next step: Muscle relaxation)
	Flat capnograph: ETCO ₂ range: zero (Option D)	Esophageal intubation Cardiac arrest Tube displacement Disconnect circuit

Solution for Question 9:

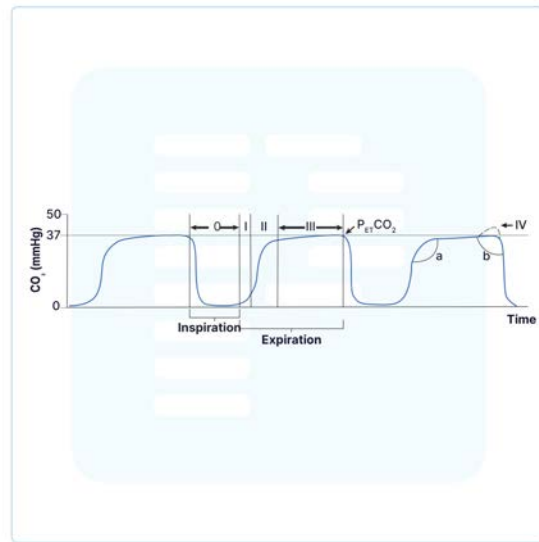
Correct Answer: C) BUN (blood urea nitrogen) determination

Explanation:

BUN (blood urea nitrogen) is determined through a blood sample, while EtCO₂ evaluates ventilation-perfusion (V/Q) matching, estimates cardiac output adequacy, and assesses the effectiveness of chest compressions during cardiopulmonary resuscitation (CPR).

Capnography

- The time capnogram is a display of exhaled CO₂ that illustrates both the inspiratory and expiratory phases.



- The capnograph segments are divided into the following: α : Angle between phase II and III
- β : Angle between phase III and 0
- α : Angle between phase II and III
- β : Angle between phase III and 0
- α : Angle between phase II and III
- β : Angle between phase III and 0

Phases	Description
Phase I	Exhalation of dead space gas
Phase II	Expiratory upstroke , where transitional gas between the airways and alveoli is sampled, leading to an increase in CO ₂ levels..
Phase III	Alveolar Plateau , characterized by a stable pCO ₂ level that reflects the gas exchange occurring in the alveoli.
Phase 0	Inspiratory downstroke , where there is a sharp decrease in pCO ₂ as fresh, inspired gas washes out the remaining CO ₂ at the sampling site.

Clinical significance of capnography:

- Confirmation of ET placement (Option D ruled out)
- CPR Assessment (adequacy of chest compressions)
- Ventilation-Perfusion (V/Q) matching (Option A ruled out)
- Estimate cardiac output adequacy (Option B ruled out)
- Early detection of respiratory depression before O₂ desaturation

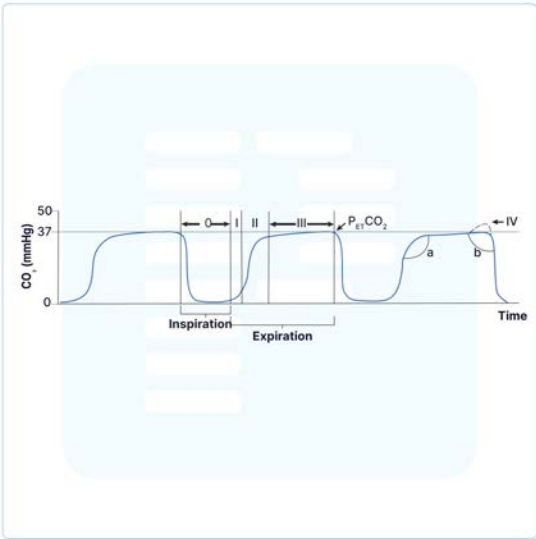
- The ratio of physiologic dead space (VD) to tidal volume (VT) by using the Bohr equation:
 physiologic dead space (VD) to tidal volume (VT)" data-author="NA" data-hash="484" data-license="NA" data-source="NA" data-tags="December2024" height="370" src="https://image.prepladder.com/notes/NcydcPIS0j45dWyXr3xV1734972344.png" width="370" />

Solution for Question 10:

Correct Answer: A) Exhalation of air from alveoli

- Plateau phase or Alveolar plateau is phase III of the capnography waveform, which represents the exhalation of air from alveoli rich in CO₂
- The simplest and most widely used form of display for exhaled CO₂ is time capnography, which shows both inspiratory and expiratory phases.

Plateau phase or Alveolar plateau is phase III of the capnography waveform, which represents the exhalation of air from alveoli rich in CO₂



Phase	Description	Corresponds to
Phase 0	Baseline	Inspiration and end of expiration.

Phase I	Initial part of exhalation	Exhalation of dead space gas.
Phase II	Transitional phase	Exhalation of transitional gas between airways and alveoli.
Phase III	Plateau phase	Exhalation of alveolar CO ₂ .
Alpha Angle (α)	Transition from Phase II to Phase III	Change from mixed gas to alveolar CO ₂ .
Beta Angle (β)	Transition from Phase III to Phase 0	End of exhalation and beginning of inspiration.

Exhalation of air from trachea (Option B) , Exhalation of air from bronchi (Option C) and Exhalation of air from dead space (Option D) occur during Phase I of capnography.

Solution for Question 11:

Correct Answer: B) i-c, ii-a, iii-d, iv-e, v-b

Explanation:

Laryngeal mask airways (LMAs) are classified as:

First Generation	Second Generation
Classic LMA Flexometallic LMA unique	Proseal LMA Supreme LMA I-gel LMA

Types of LMA's:

Type of LMA	Description
Classic LMA	Silicone mask (oval) Inflatable (sits in the hypopharynx) Forms a seal around the periglottic tissues. Has a risk of aspiration.

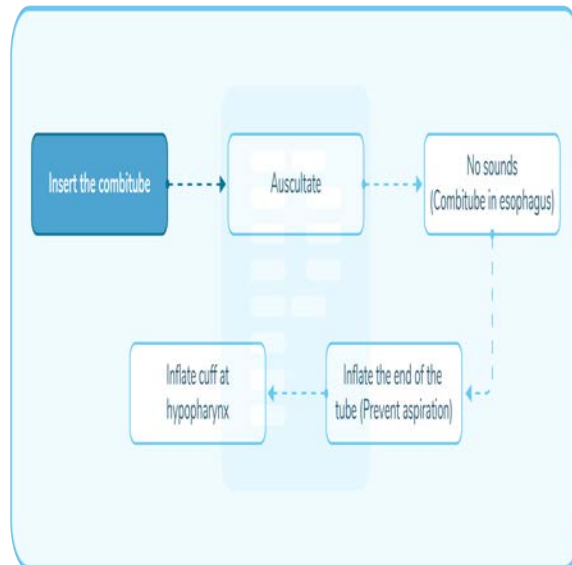
Proseal LMA LMA " data-author="Ignis" data-hash="451" data-license="CC BY SA 3.0" data-source="https://commons.wikimedia.org/wiki/File:ProSeal_Laryngeal_Mask_Airway_inflated_001.jpg" data-tags="December2024" height="205" src="https://image.prepladder.com/notes/f8XWvICxq2pBuMNGwwU61734962233.png" width="276" />	Components: - Posterior Cuff : Enhances the seal around the larynx for adequate positive pressure ventilation (PPV). - Gastric Drainage Tube : Allows placement of an orogastric tube to isolate the respiratory and gastrointestinal tracts. - Bite Block : Protects the airway device from being bitten, maintaining airway patency.
Fast Track or Intubating LMA	Facilitate endotracheal intubation through the LMA device. We can pass the endotracheal tube through this to secure difficult airway

Supreme LMA	Material: PVC (polyvinyl chloride) It fits easily and offers superior sealing (compared to other LMAs) It prevents aspiration (as it contains a gastric drainage tube) Integrated bite block (Protects the airway device from being bitten, ensuring airway patency)
I-gel LMA LMA " data-author="CYP3A42D6" data-hash="694" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Masques_laryng%C3%A9s-_LMA_Unique_et_I-gel.jpg" data-tags="January2025" height="500" src="https://image.prepladder.com/notes/bOWz2Nt1C6N6pJee20bd1736513718.png" width="200" />	Material: Silicon (Helps to grip the pharynx) Balloon and cuff are absent (gel occluder) Temperature regulated LMA Prevent aspiration (gastric port) and Bite block.
LMA unique LMA (1)" data-author="CYP3A42D6" data-hash="695" data-license="CC BY SA 4.0" data-source="https://commons.wikimedia.org/wiki/File:Masques_laryng%C3%A9s-_LMA_Unique_et_I-gel.jpg" data-tags="January2025" height="434" src="https://image.prepladder.com/notes/tyj1KUIh1X4YhPrGBrL01736513825.png" width="166" />	Material: PVC (polyvinyl chloride)

EsophagealTracheal Combitube:

- Consists of two fused tubes, each with a connector on its proximal end. The longer tube has an occluded distal tip (forces the gas to exit through side perforations). The shorter tube has an open tip and no side perforations. Used in emergency procedures.
- The longer tube has an occluded distal tip (forces the gas to exit through side perforations).
- The shorter tube has an open tip and no side perforations.

- Used in emergency procedures.
- The longer tube has an occluded distal tip (forces the gas to exit through side perforations).
- The shorter tube has an open tip and no side perforations.
- Used in emergency procedures.



Solution for Question 12:

Correct Answer: D) Facilitated access to the surgical site in maxillofacial procedures

Explanation:

- The given image depicts Ring-Adair-Elwyn ET tubes, nasotracheal tubes designed to position the tube and breathing circuit out of the surgical field, allowing enhanced surgical access and reduced interference with instruments.
- Uses: Oral and Maxillofacial Surgery : Minimizes surgical field interference. Pediatric Patients : Secures airway while facilitating positioning. Congenital or acquired facial Deformities : Reduces manipulation during intubation. Head or Neck Surgery : Ensures clear surgical access. Restricted Mouth Opening : Effective airway option with limited access.
- Oral and Maxillofacial Surgery : Minimizes surgical field interference.
- Pediatric Patients : Secures airway while facilitating positioning.
- Congenital or acquired facial Deformities : Reduces manipulation during intubation.
- Head or Neck Surgery : Ensures clear surgical access.
- Restricted Mouth Opening : Effective airway option with limited access.
- Oral and Maxillofacial Surgery : Minimizes surgical field interference.
- Pediatric Patients : Secures airway while facilitating positioning.
- Congenital or acquired facial Deformities : Reduces manipulation during intubation.
- Head or Neck Surgery : Ensures clear surgical access.
- Restricted Mouth Opening : Effective airway option with limited access.
- Types:
-

North Ring-Adair-Elwyn	South Ring-Adair-Elwyn
Used in maxillofacial or dental surgery and as an alternative to tracheostomy for weaning from respiratory support .	Used in Head and neck surgery, ophthalmic surgery.

Solution for Question 13:

Correct Answer: A) Armored endotracheal tube

Explanation:

- In adult patients with suspected cervical spine injuries, armoured endotracheal tubes are preferred due to their spiral metal reinforcement, which resists kinking and provides greater flexibility.
- This design is particularly beneficial in patients requiring neck stabilisation, as it helps secure the airway while minimising the risk of further injury to the cervical spine.
- Hence are preferred for surgeries requiring a prone position, such as head, neck, and spine procedures.

endotracheal tube" data-author="Edward A. Luce Sonia M. Alvarez" data-hash="474" data-license="CC BY 4.0" data-source="https://www.researchgate.net/figure/armored-endotracheal-tube-Coviden-Inc-dublin-Ireland_fig1_327520558" data-tags="December2024" height="337" src="https://image.prepladder.com/notes/OwZjYRqUsgDpJFO7sKTR1734971003.png" width="500" />

Double-lumen endotracheal tubes (Option B) are specialised tubes that allow single-lung ventilation. They are commonly used in thoracic surgeries like pneumonectomy, but they are not preferred in trauma due to their complexity and placement challenges.

endotracheal tubes" data-author="bigomar2" data-hash="475" data-license="CC BY SA 3.0" data-source="https://en.wikipedia.org/wiki/Double-lumen_endobronchial_tube" data-tags="December2024" height="376" src="https://image.prepladder.com/notes/MPa2UhpjG1sp0DCE74B1734971023.png" width="500" />

Cuffed endotracheal tubes (Option C) are the most commonly used ET tubes but may not offer the flexibility needed for cervical injury management.

The cuff creates a seal in the trachea to allow higher ventilation pressures and prevent aspiration.

endotracheal tubes" data-author="Meimei Tao, Xinxia Wang, Qian Sun, Hui Li, Hang Zou, Guangfa Zhu" data-hash="476" data-license="CC BY 4.0" data-source="https://onlinelibrary.wiley.com/doi/10.1111/crj.13539" data-tags="December2024" height="500" src="https://image.prepladder.com/notes/upG0W5gB7VQAa8Qyl70r1734971045.png" width="500" />

Uncuffed endotracheal tubes (Option D) are often used in pediatric patients because of concerns of necrosis caused by cuff pressure at the level of the cricoid cartilage.



Solution for Question 14:

Correct Answer: B) 5 mm

Explanation:

For pediatric patients with age >1 year, the formula used to calculate the appropriate size of an uncuffed endotracheal tube (ETT) is:

$$\text{ETT size (mm)} = \text{Age (years)} / 4 + 4$$

For a 4-year-old child:

$$\text{ETT size} = 4(\text{years}) / 4 + 4 = 5 \text{ mm}$$

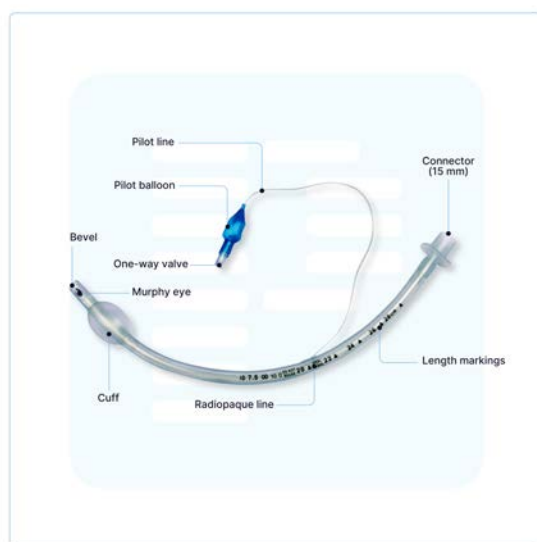
- The ETT size is described in terms of internal diameter:

Age	Internal Diameter	length
Term-Neonate	3 mm (uncuffed)	9 cm
1 Year Old	4 mm (uncuffed)	10 cm
Child (>1 Year)	Age/4 + 4 mm (uncuffed)	Age/2 + 12 cm

Adult	> 7 mm	18-22 cm
-------	--------	----------

- The cuff is positioned 3-4 cm above the carina and 2-2.5 cm below the glottis.

Murphy's eye is a side hole at the tip, which allows continued ventilation if the end abuts the tracheal wall.



Solution for Question 15:

Correct Answer: B) Grade II

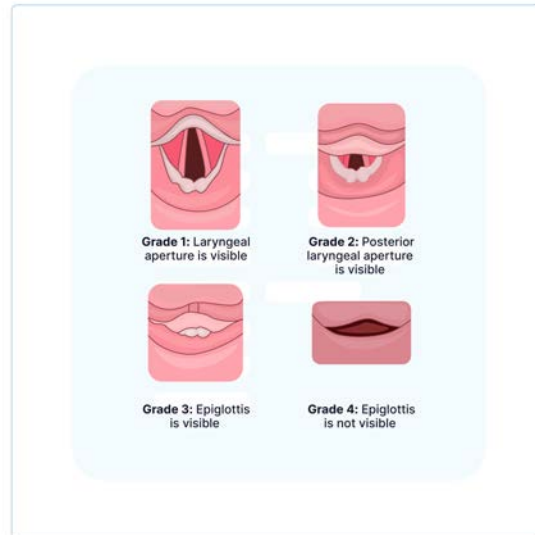
Explanation: Grade II is visualization of only the posterior portion of the laryngeal aperture

Cormack-Lehane grading system

- It is used to grade the laryngoscopic view based on the number of glottic openings being visualised.
- Higher the grade more challenging is the intubation.

Grade	Description
Grade I	The entire laryngeal aperture is visible
Grade II	Only the posterior portion of the laryngeal aperture is visible

Grade III	Only the epiglottis is visible
Grade IV	Neither the epiglottis nor the larynx is visible



Solution for Question 16:

Correct Answer: C) Class III

Explanation:

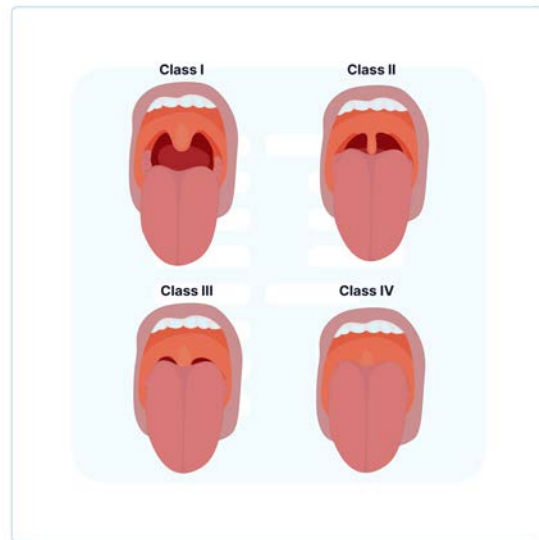
In Mallampati Class III , only the base of the uvula and the soft palate are visible.

Mallampati classification: It evaluates the size of the tongue in relation to the oral cavity. The greater the obstruction of the view of the pharyngeal structures by the tongue, the more challenging intubation may be.

Class	description
Class I	The P illars, u vula, s oft and h ard palate are visible. (mnemonic PUSH)
Class II	The soft palate, faucial pillars (upper part) and portion of uvula is visible.
Class III	The soft palate and base of uvula is visible.

Class IV

Only the hard palate is visible.

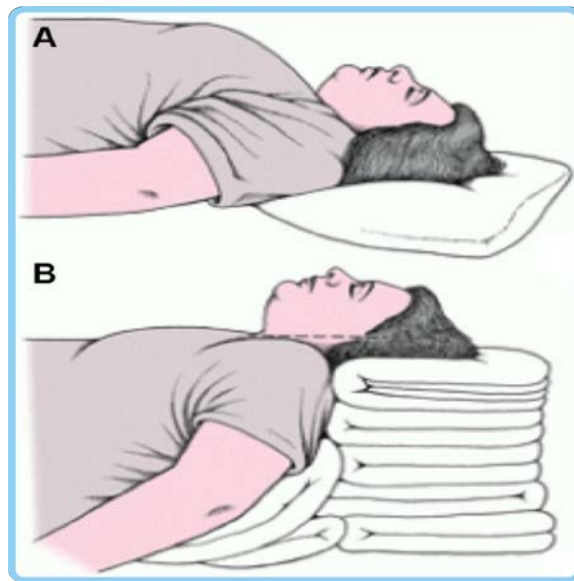


Solution for Question 17:

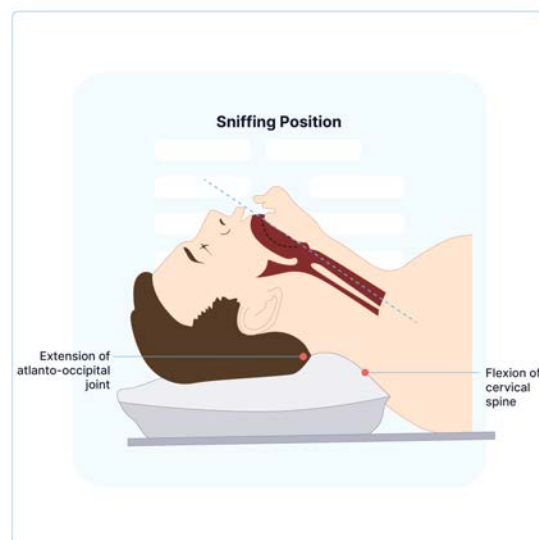
Correct Answer: D) Ramped position

Explanation:

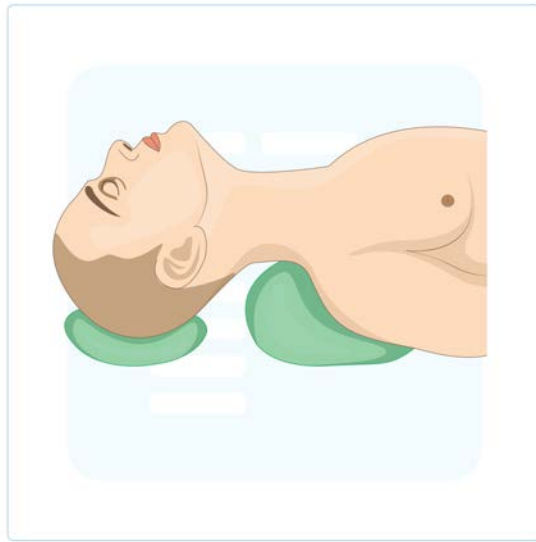
- The preferred position for intubation in obese patients is referred to as the Ramped Position, in which the patient's head, neck, and shoulders are elevated with pillows or padding, aligning the external auditory meatus with the sternal notch.
- This alignment improves the visualisation of the glottis during intubation, optimising airway management and reducing the risk of difficult intubation.



Sniffing position (Option A) is used for intubation in non-obese patients, but in obese patients, it may not provide sufficient alignment for a clear view of the vocal cords. The patient is positioned supine with the head elevated on a firm cushion (patients with shorter necks may require less elevation) while flexing the neck and extending the atlanto-occipital joint.



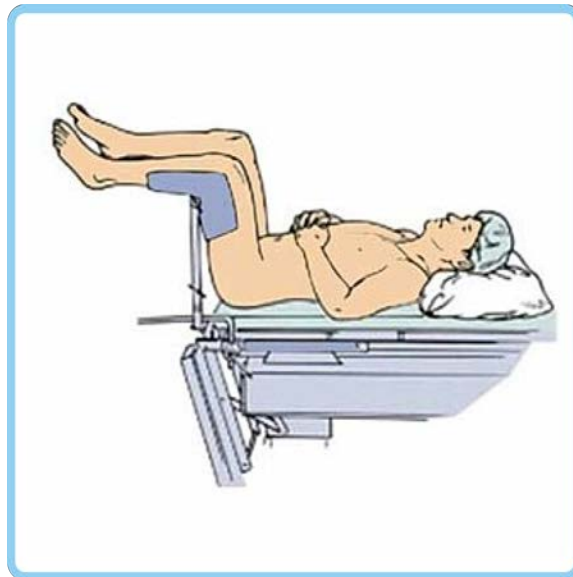
Rose position (Option B) : is used during tonsillectomy. The patient lies supine with the head extended by placing a pillow under the shoulders. A rubber ring is placed under the head to stabilise it.



Lithotomy position (Option C) is preferred during labour, as well as for examinations or surgeries involving the pelvis, female genitalia, rectum, prostate, and other urethral procedures.

The patient lies in a supine position with the buttocks at the lower break of the table.

- The legs are flexed at the hips to 90° and abducted to 30° , with the knees bent at 70° to 90° and the lower legs supported by padded shells.



Solution for Question 18:

Correct Answer: B) Miller Laryngoscope

Explanation:

The Miller laryngoscope is the preferred choice for pediatric patients, especially for children under 8 years of age. It has a straight blade, which is advantageous in lifting the epiglottis directly, providing a better view of the vocal cords in younger patients.

Types of Laryngoscopy

- Direct Laryngoscopy: Direct visualisation of the glottis with the assistance of a laryngoscope Following laryngoscopes are most commonly used:
- Direct visualisation of the glottis with the assistance of a laryngoscope
- Following laryngoscopes are most commonly used:
- Direct visualisation of the glottis with the assistance of a laryngoscope
- Following laryngoscopes are most commonly used:

	Macintosh Laryngoscope (Option C)	Miller Laryngoscope (Option B)	McCoy Laryngoscope (Option A)
Blade	Curved shape	Straight shape	Curved with levering tip
Use	In adults	In children (large epiglottis)	It can be used in both adults and children.

- Indirect Laryngoscopy: Indirect visualisation of the glottis with the help of optical aids, such as fiberoptic bundles, video cameras, mirrors, prisms, or lenses like: Flexible Intubation Scopes
- Flexible Intubation Scopes
- Flexible Intubation Scopes



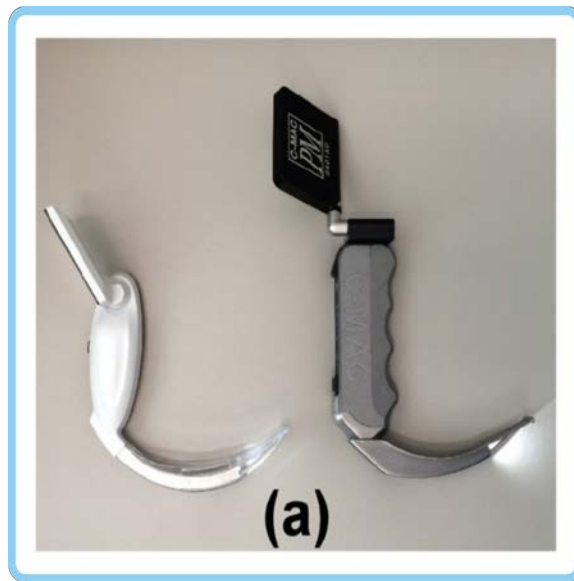
- Rigid Indirect Laryngoscopes
- Lighted Optical Stylets



- Video Laryngoscopes (C-MAC laryngoscope, McGrath laryngoscope, GlideScope)
- Airtraq
- Ambu ascope



McGrath laryngoscope (Option D) : It is a video laryngoscope used in cases of difficult intubation.



Solution for Question 19:

Correct Answer: C) 4

Explanation:

For an adult weighing 62 kg, a size 4 Laryngeal Mask Airway (LMA) is recommended.

• Total 8 sizes of LMA are available as:

Size	Weight of the patient
1	<5 kg
1.5	5-10 kg
2	10-20 kg
2.5 (Option A)	20-30 kg
3 (Option B)	30-50 kg
4 (Option C)	50-70 kg
5	70-90 kg

6 (Option D)	>100 kg
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- Most commonly used sizes in infants: 1 and 1.5
 - Most commonly used sizes for adults: 4 and 5.
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